

(12) **United States Patent**
Radtke

(10) **Patent No.:** **US 12,414,494 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **WHEEL FOR CLOSING SYSTEM**

(56) **References Cited**

(71) Applicant: **Precision Planting LLC**, Tremont, IL (US)

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(72) Inventor: **Ian Radtke**, Washington, IL (US)

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(73) Assignee: **Precision Planting LLC**, Tremont, IL (US)

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 288 days.

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(21) Appl. No.: **17/594,400**

(22) PCT Filed: **Apr. 16, 2020**

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(86) PCT No.: **PCT/IB2020/053599**

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§ 371 (c)(1),

(2) Date: **Oct. 14, 2021**

(Continued)

(87) PCT Pub. No.: **WO2020/225623**

PCT Pub. Date: **Nov. 12, 2020**

Primary Examiner — Jamie L McGowan

(65) **Prior Publication Data**

US 2024/0251698 A1 Aug. 1, 2024

Related U.S. Application Data

(60) Provisional application No. 62/985,094, filed on Mar. 4, 2020, provisional application No. 62/967,272, filed (Continued)

(51) **Int. Cl.**
A01C 5/06 (2006.01)

(52) **U.S. Cl.**
CPC **A01C 5/068** (2013.01)

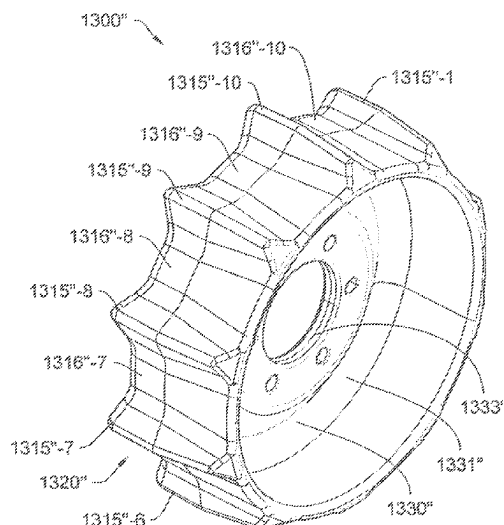
(58) **Field of Classification Search**
CPC A01C 5/068; A01C 5/064; B60B 3/02; B60B 25/004

See application file for complete search history.

(57) **ABSTRACT**

A wheel adapted for use with a trench closing assembly. The wheel has a circumferential tread coaxial with a central axis of the wheel, the circumferential tread extending axially an axial width. A continuous wall member joined with the circumferential tread extends radially inward to a central opening and axially outward from an inner edge of the circumferential tread by an axial distance, the axial distance not extending beyond a vertical plane that is perpendicular to the central axis at a midpoint of the axial width, whereby the circumferential tread and the wall member together defining a shape in the form of an uppercase Greek letter sigma.

20 Claims, 100 Drawing Sheets



Related U.S. Application Data

on Jan. 29, 2020, provisional application No. 62/866,700, filed on Jun. 26, 2019, provisional application No. 62/843,037, filed on May 3, 2019.

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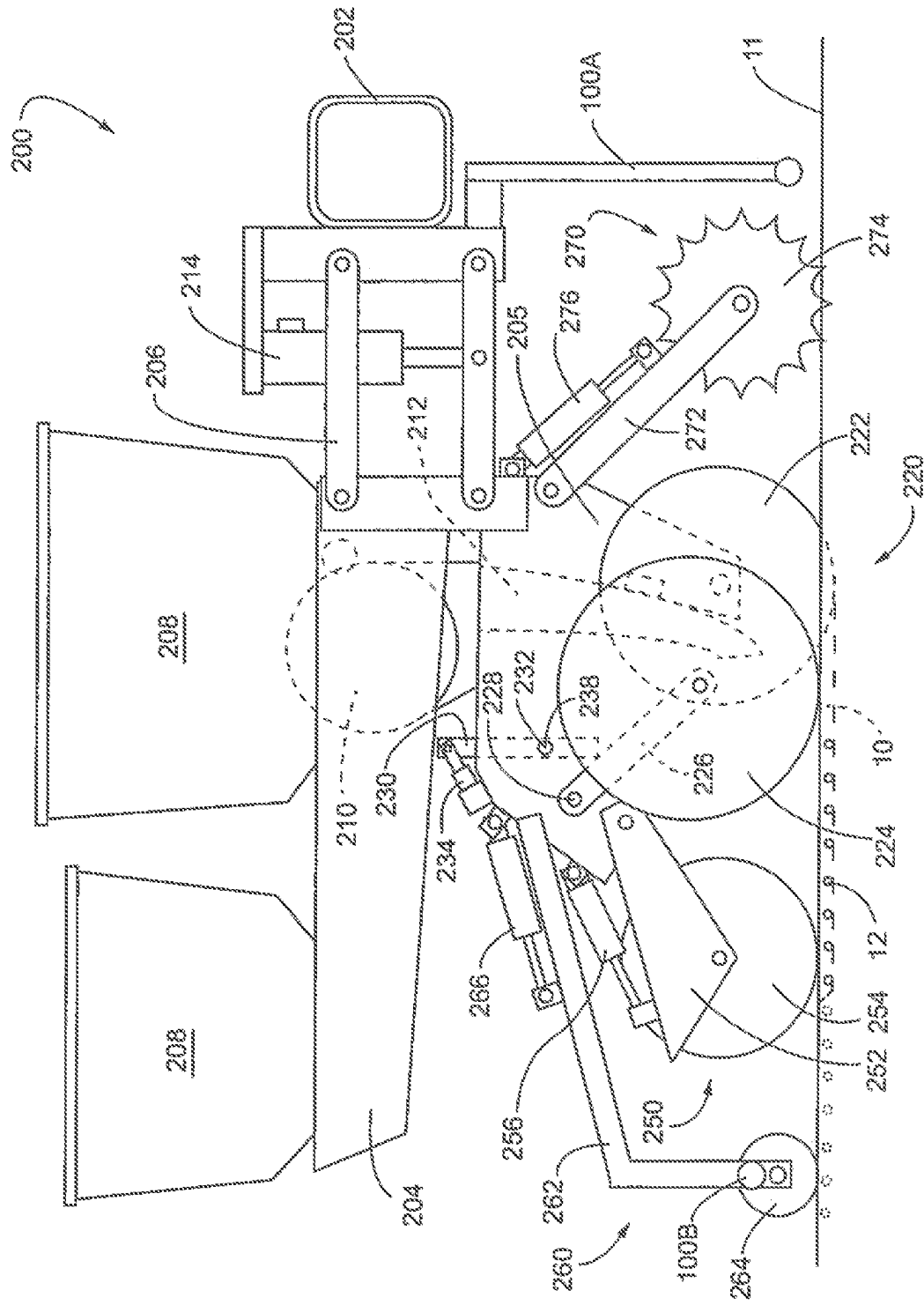


FIG. 1

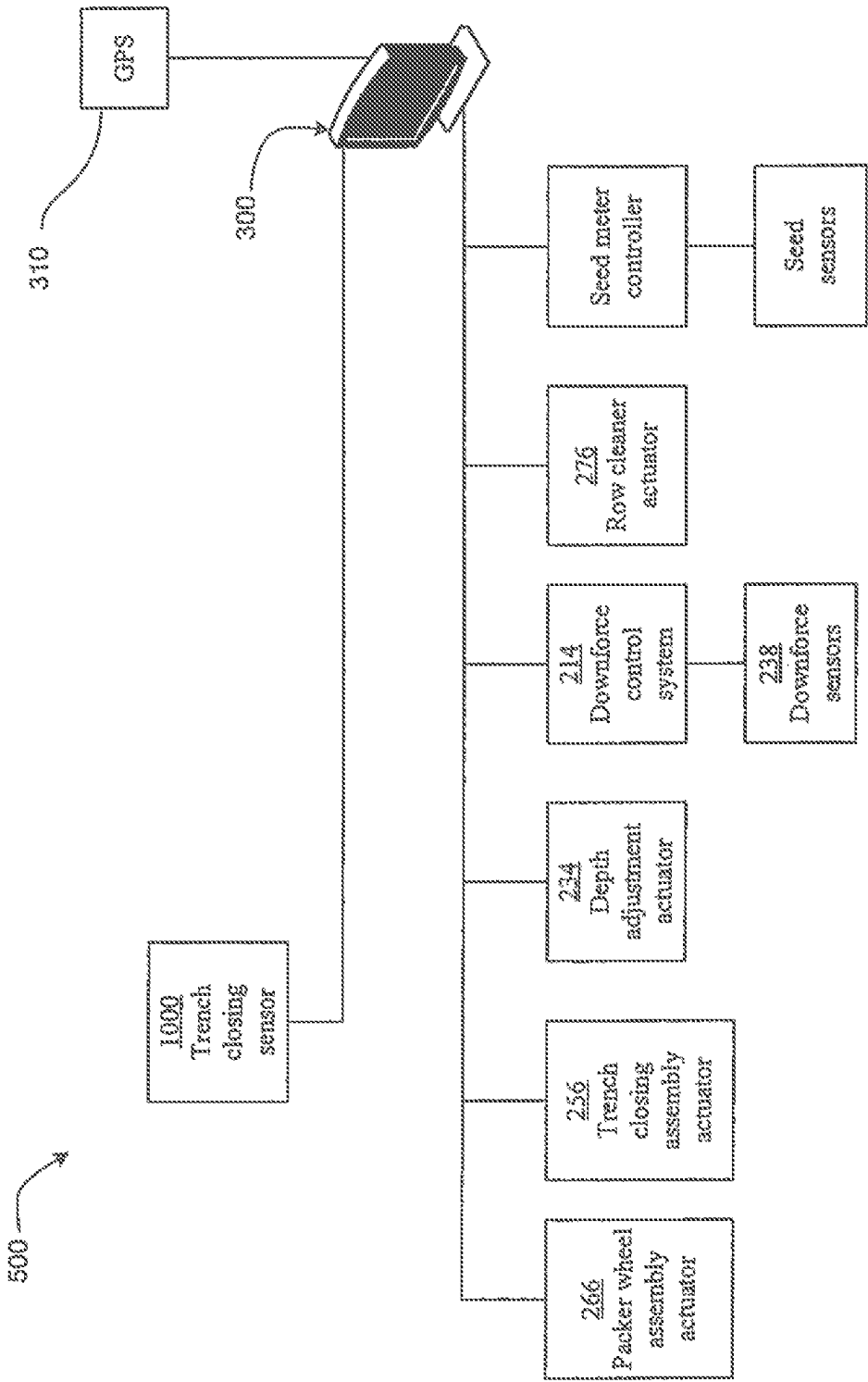
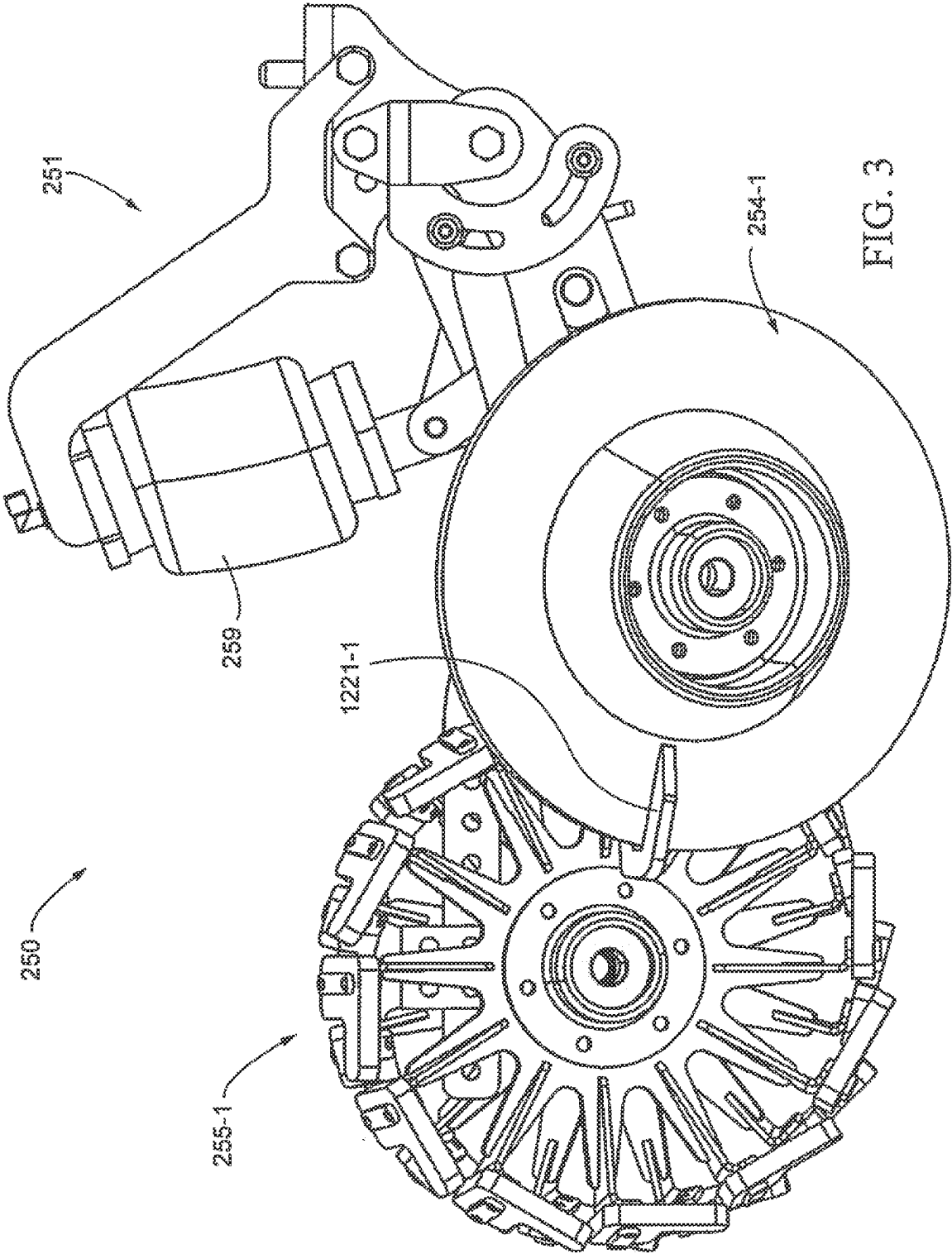


FIG. 2



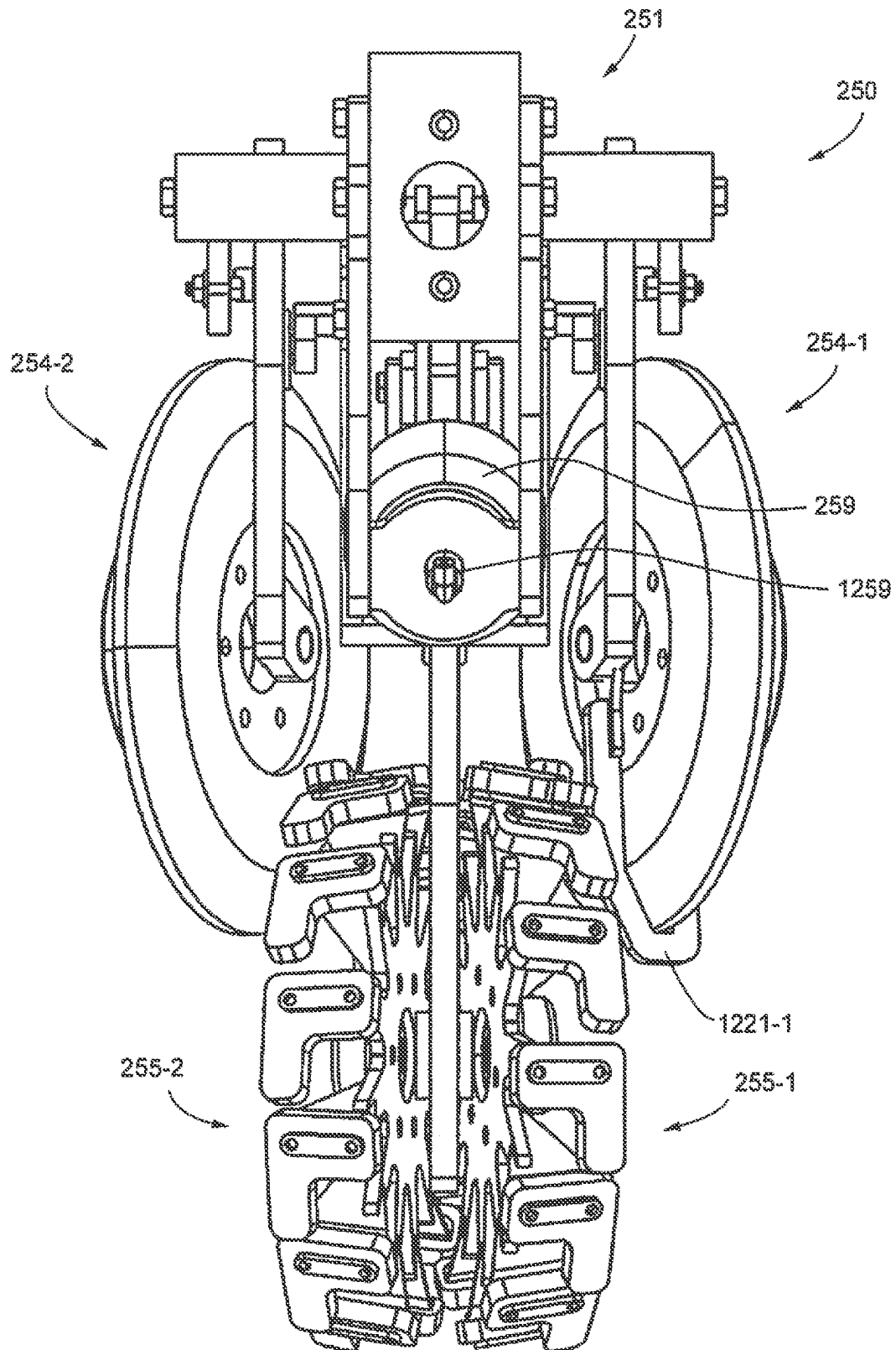


FIG. 4

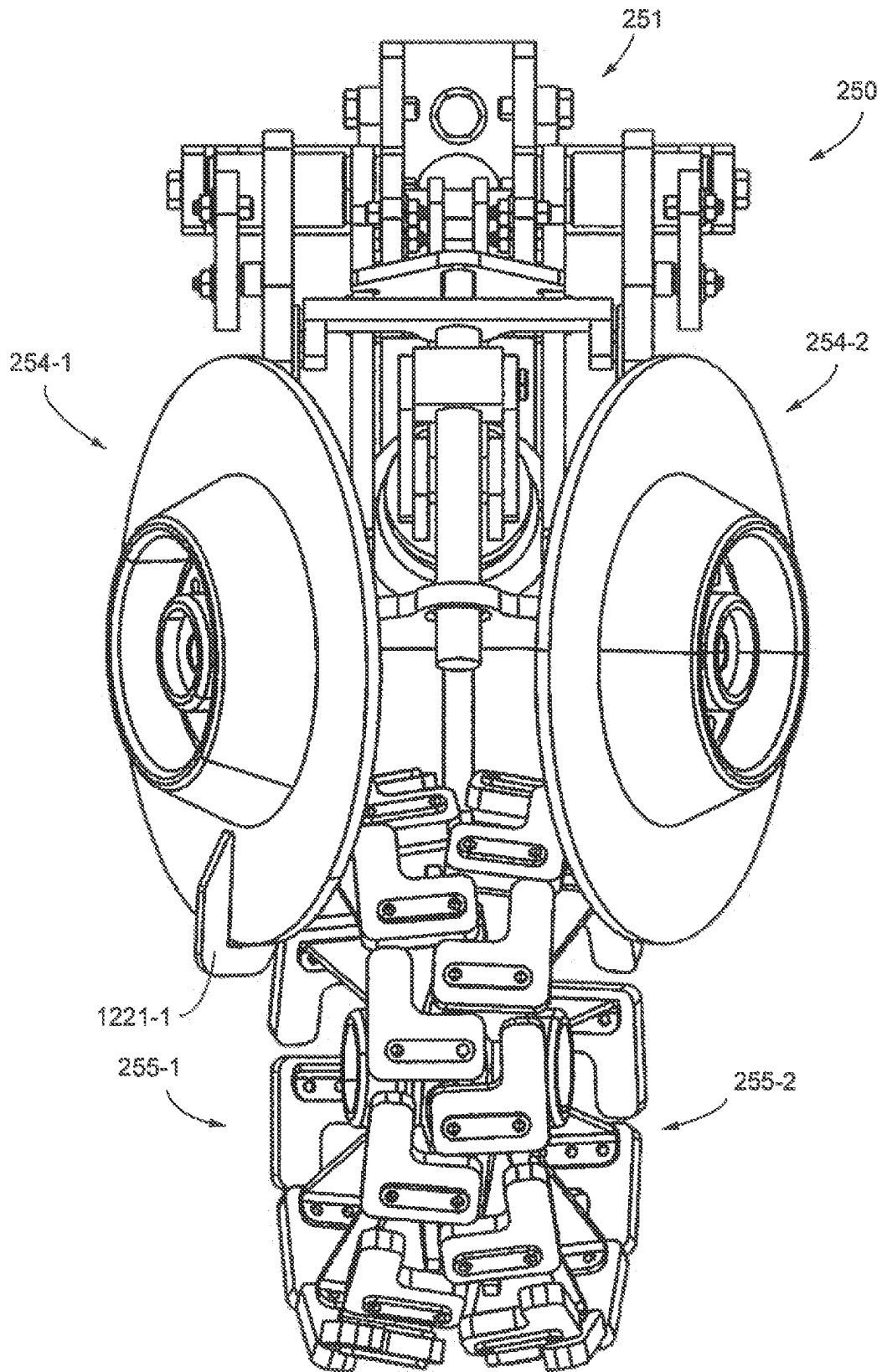


FIG. 5

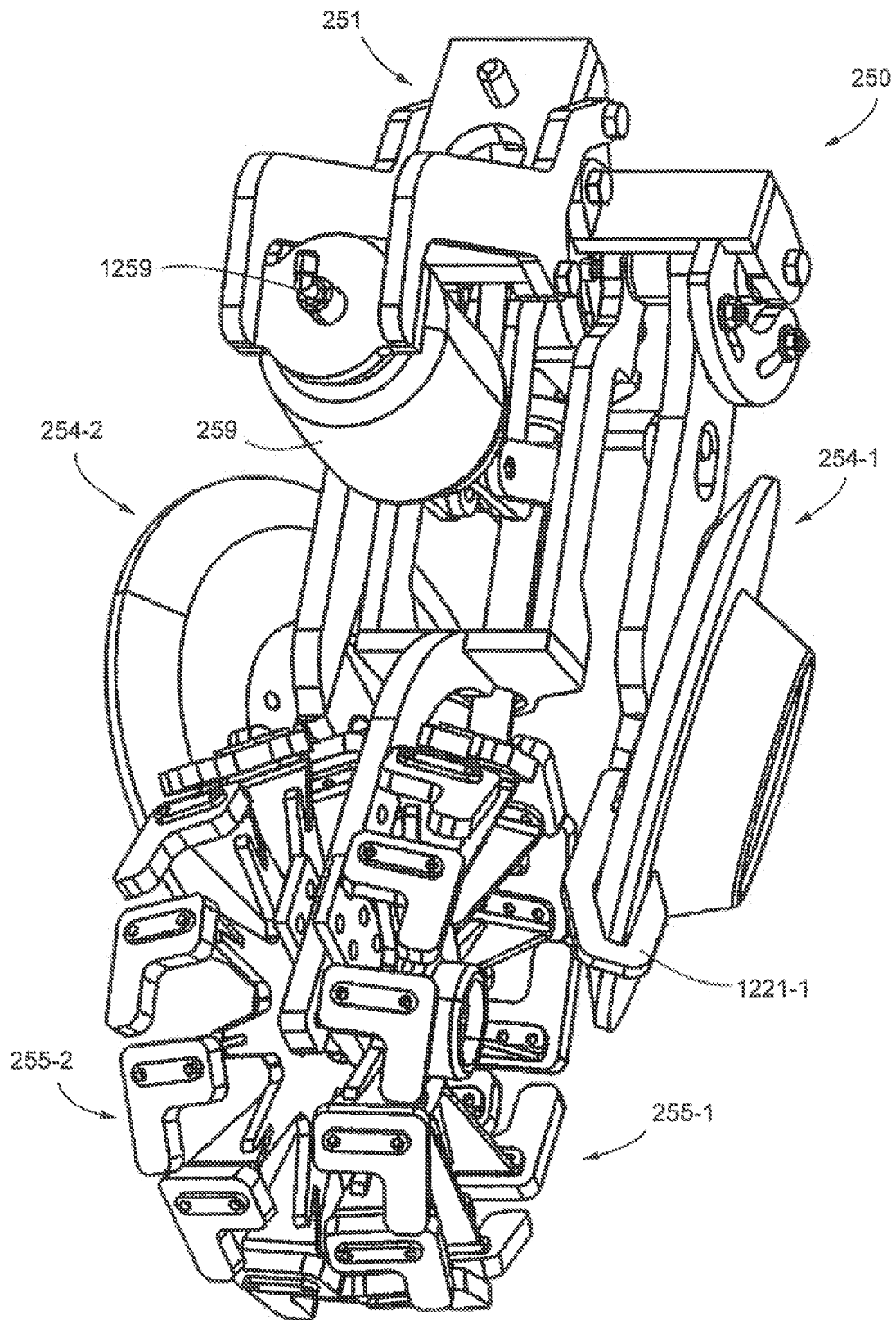


FIG. 6

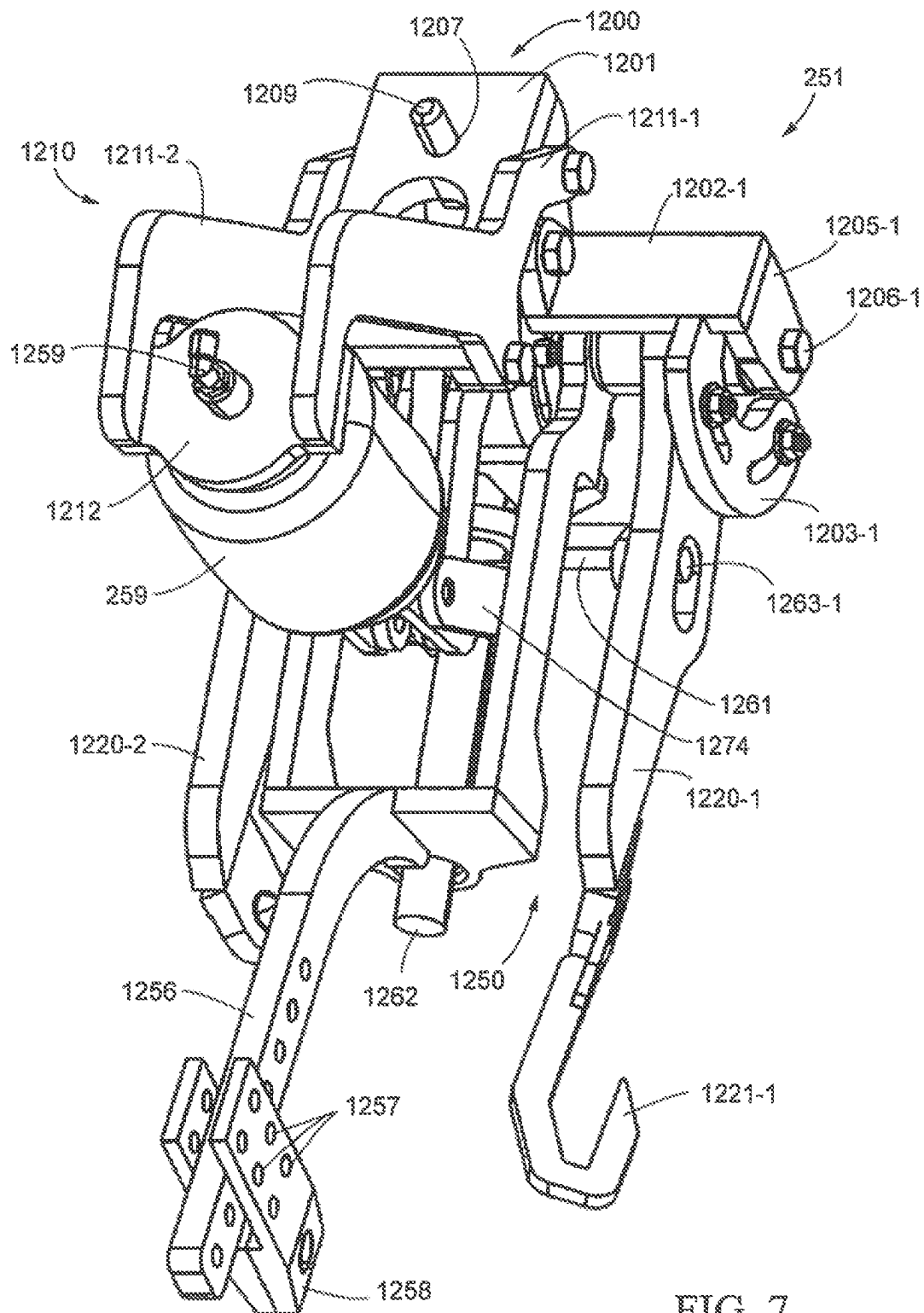


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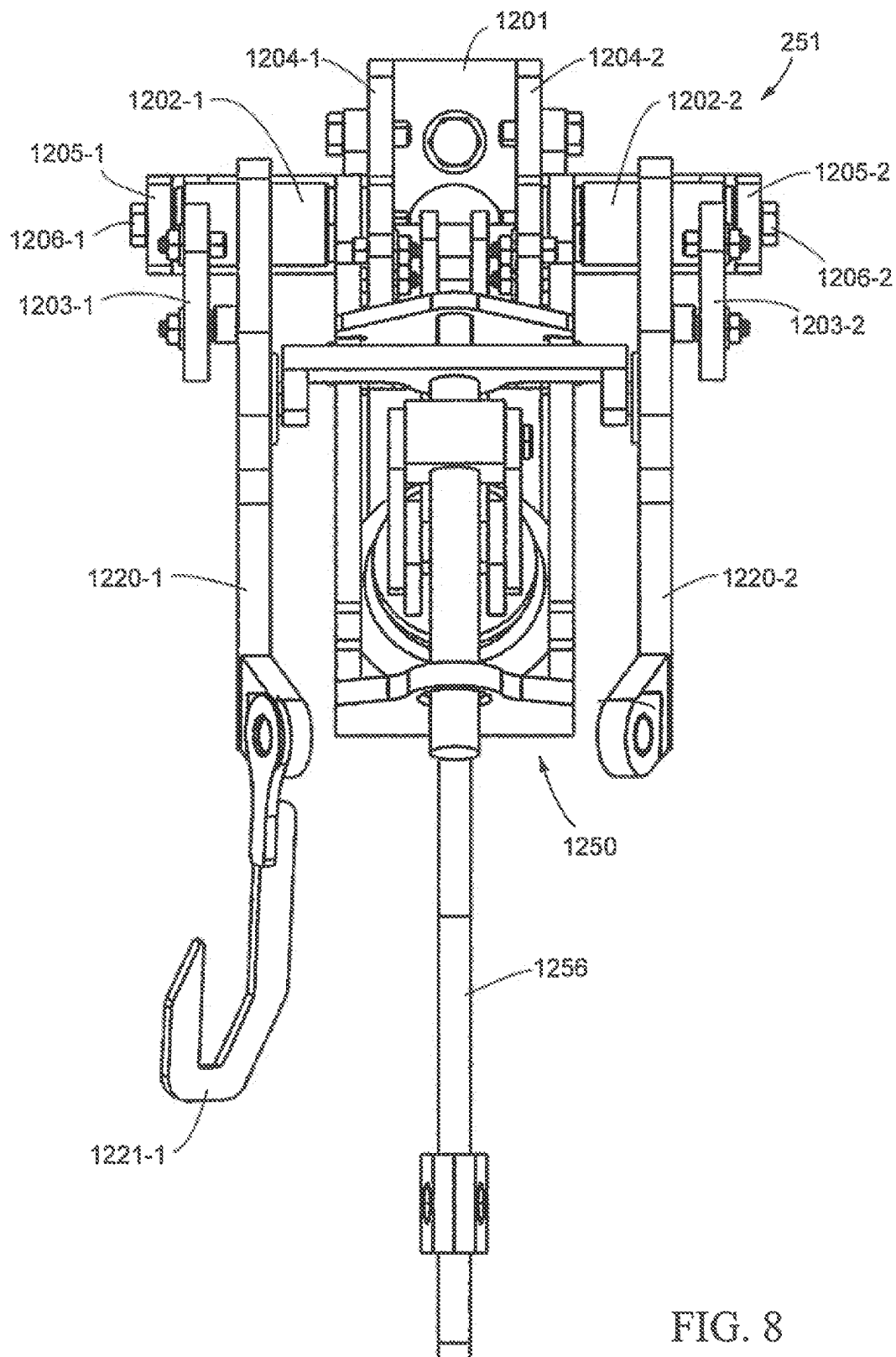


FIG. 8

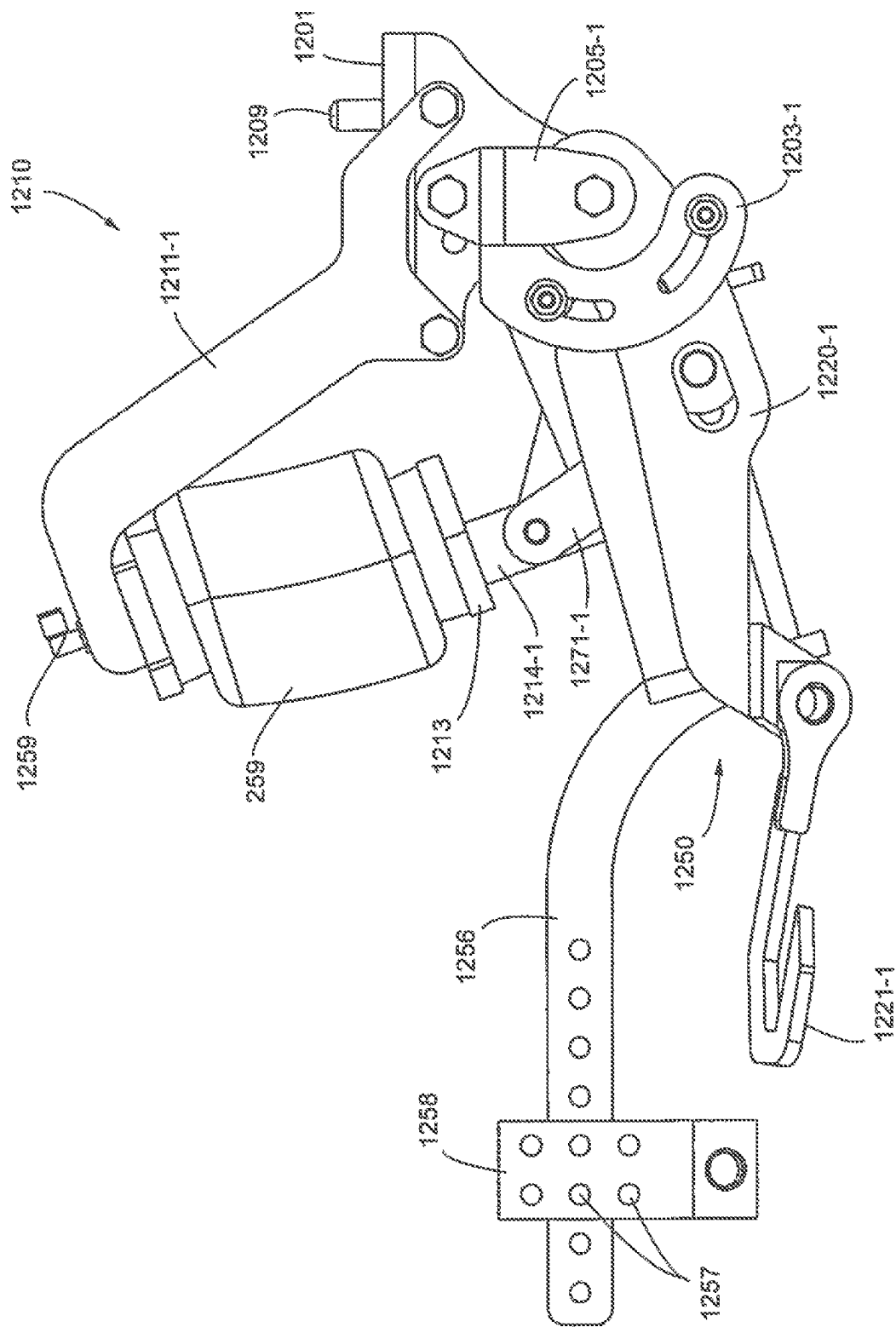


FIG. 9

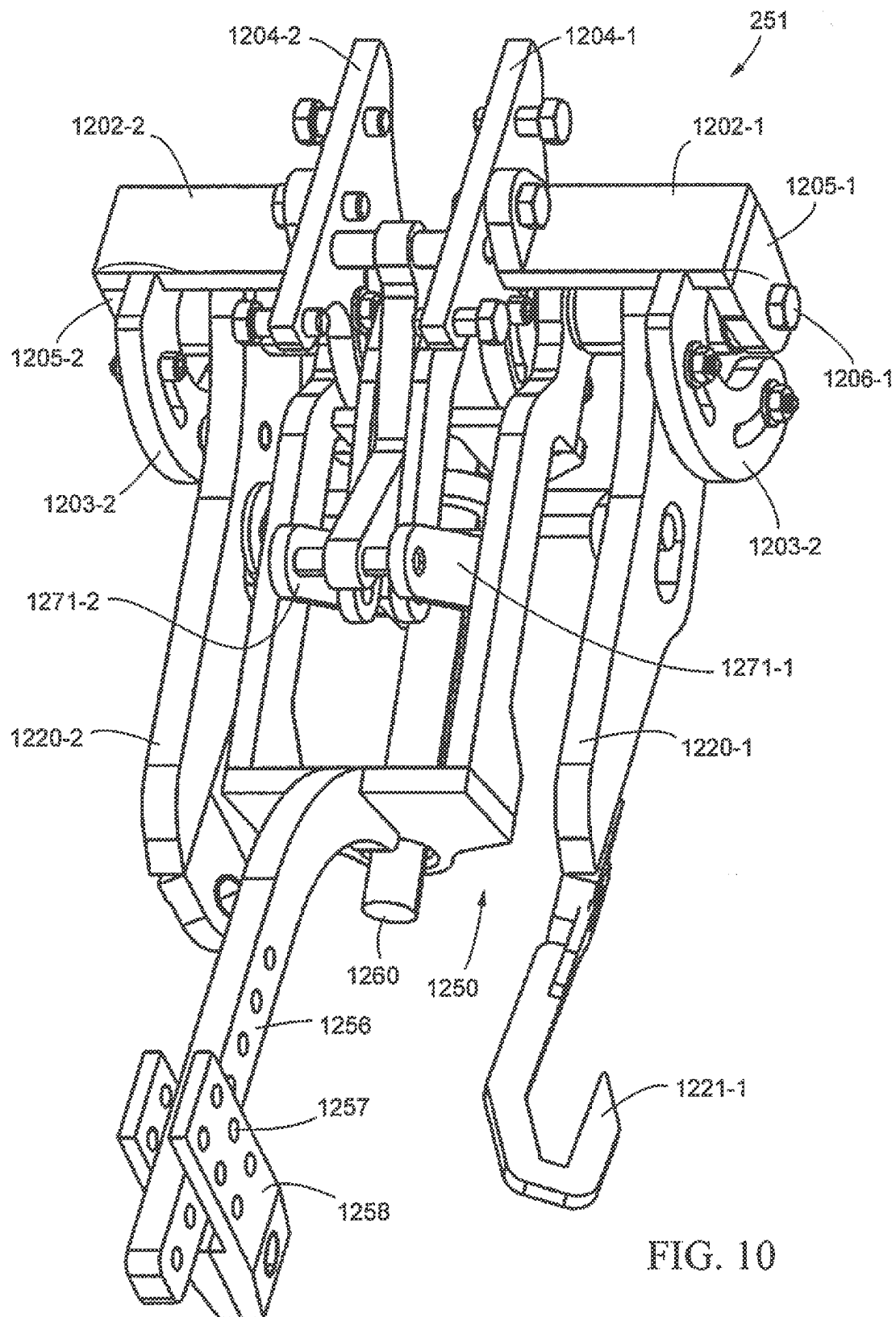


FIG. 10

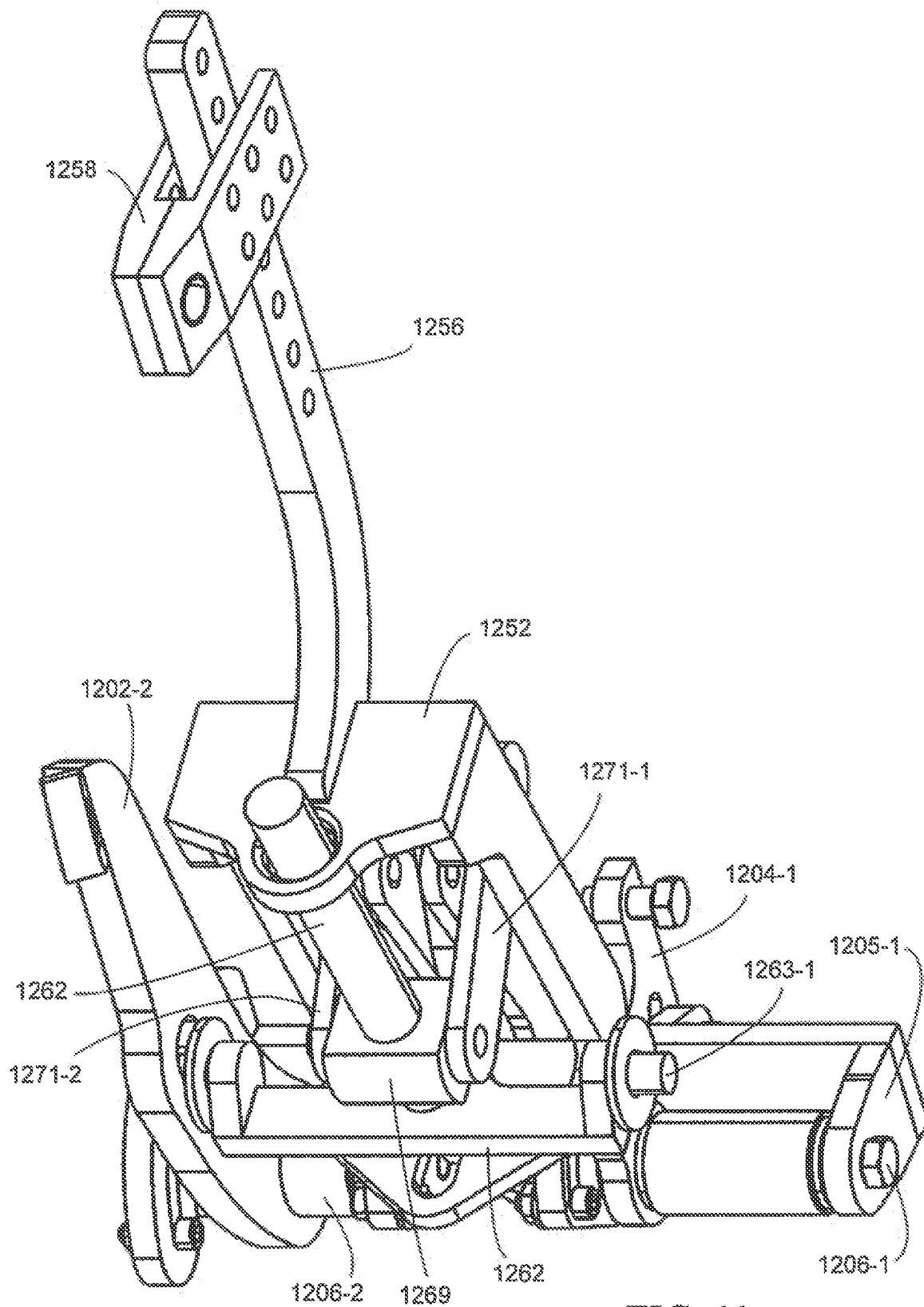
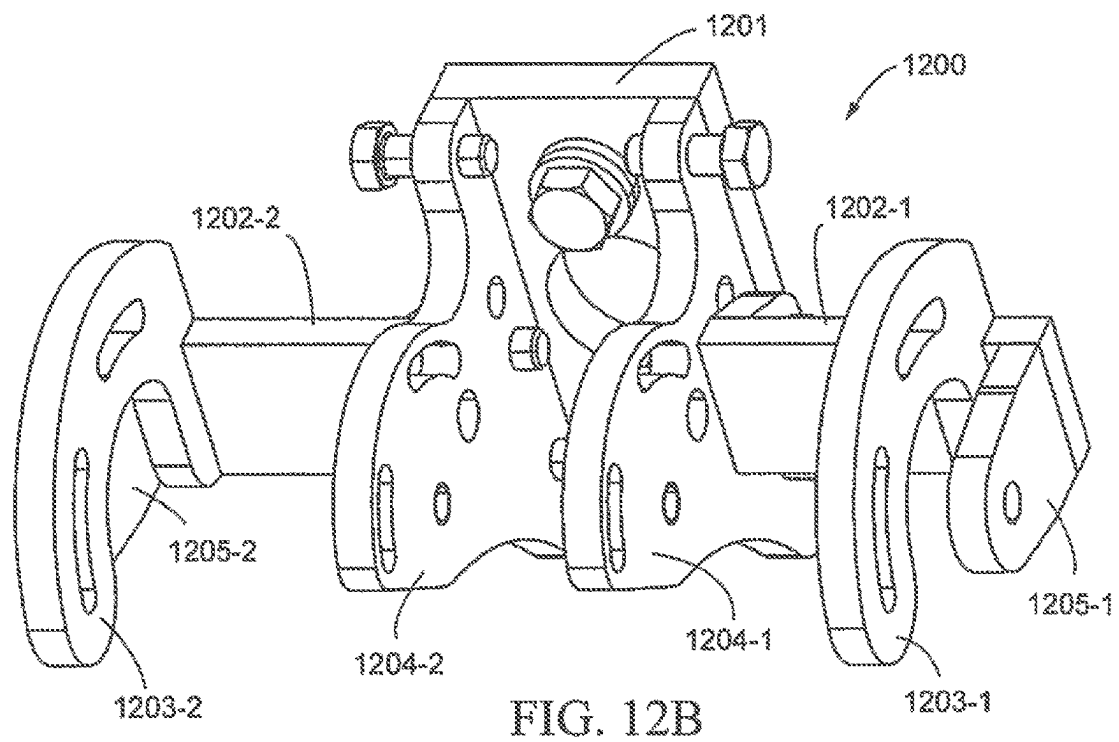
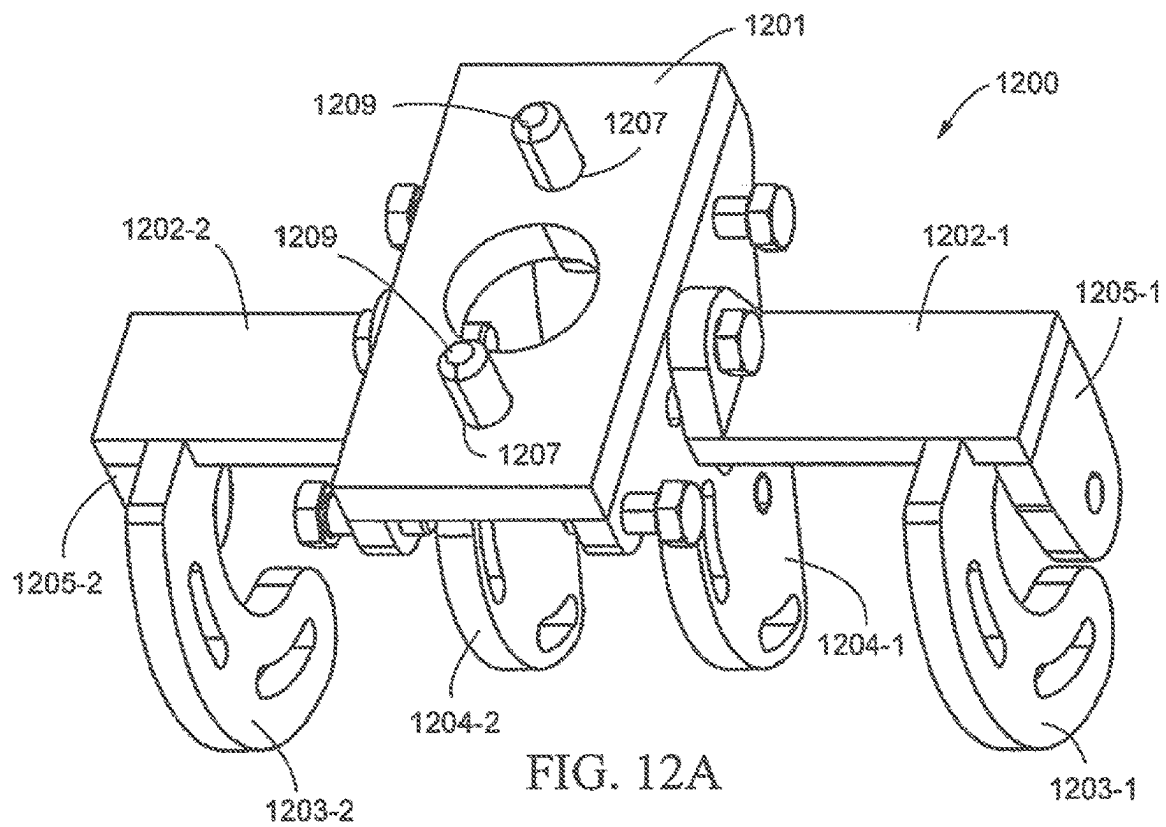


FIG. 11



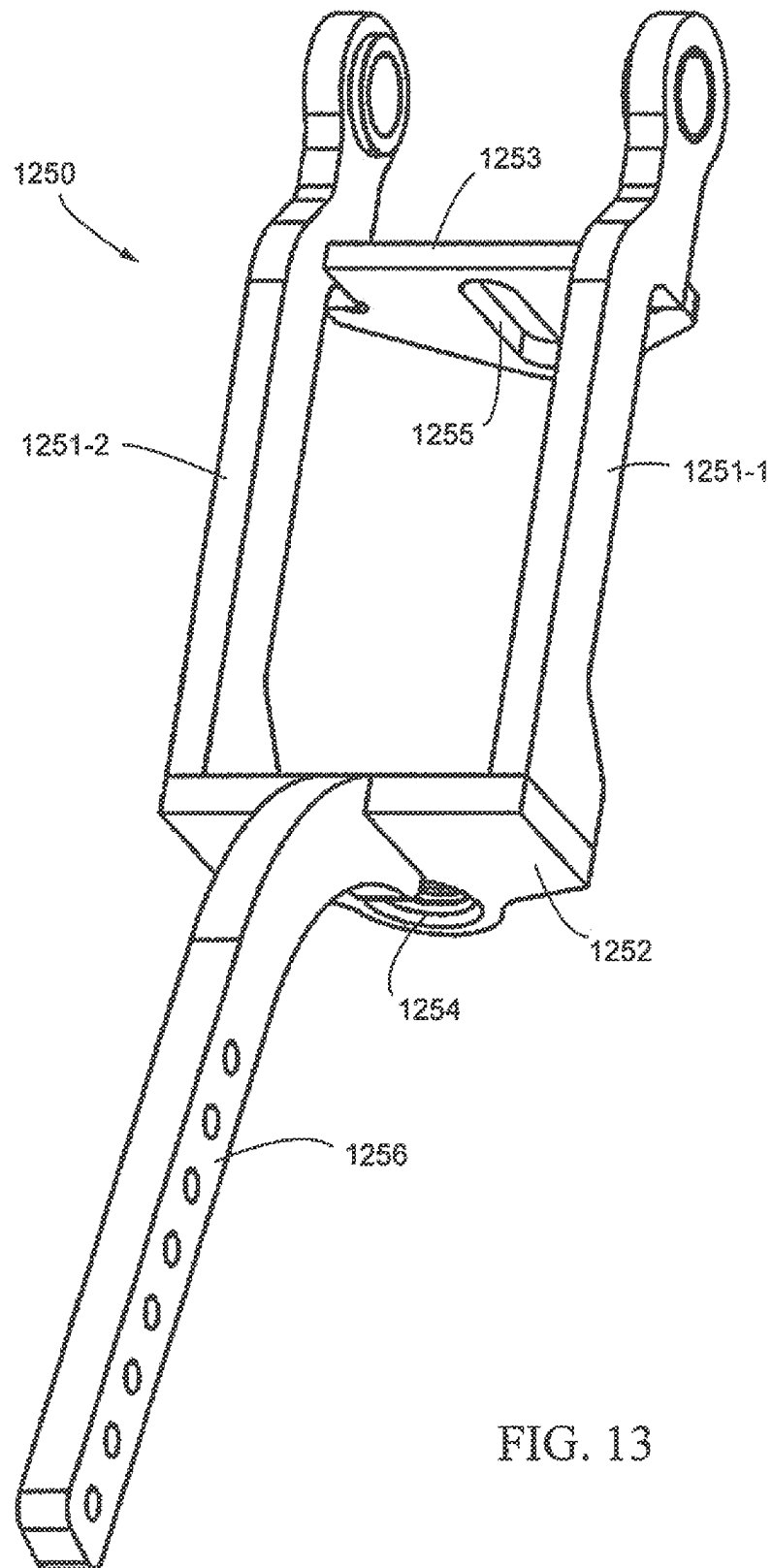


FIG. 13

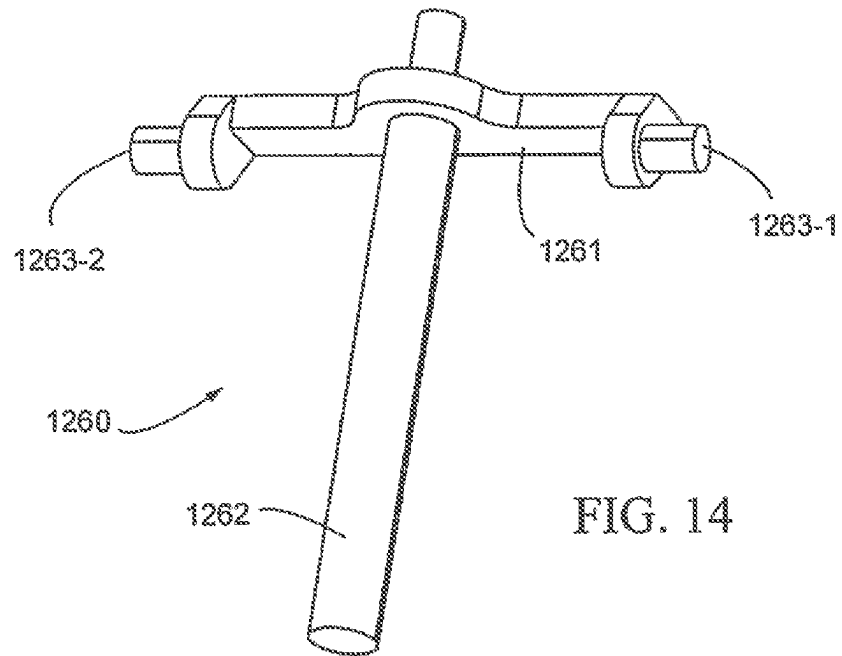


FIG. 14

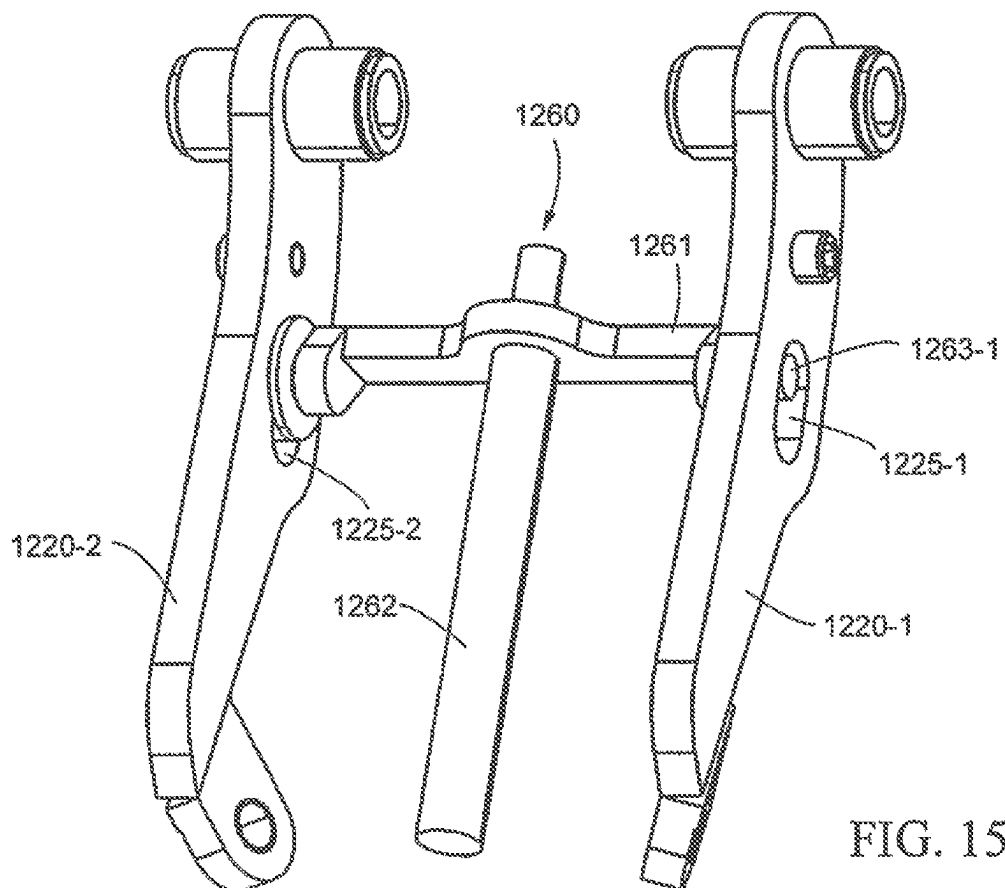


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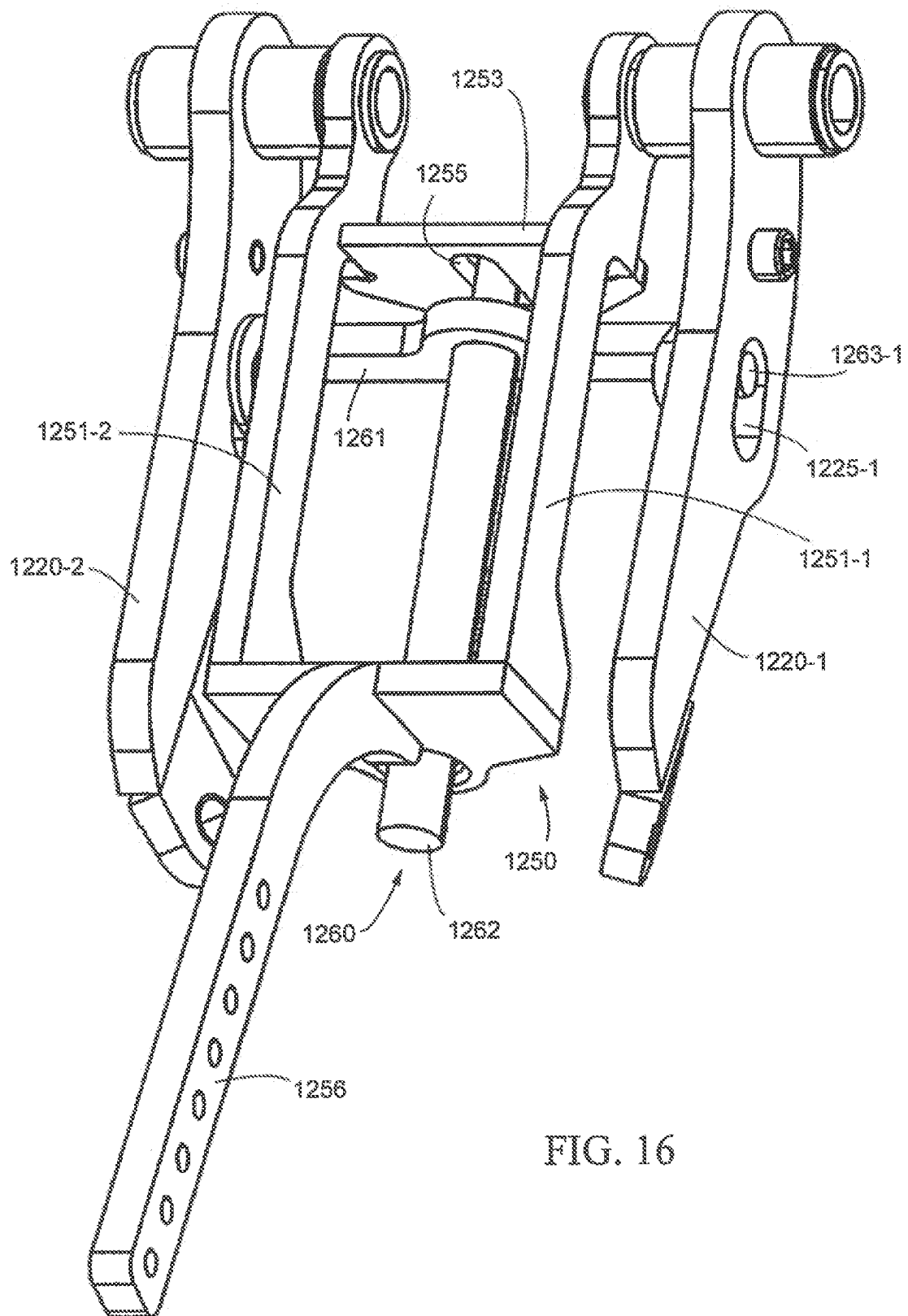


FIG. 16

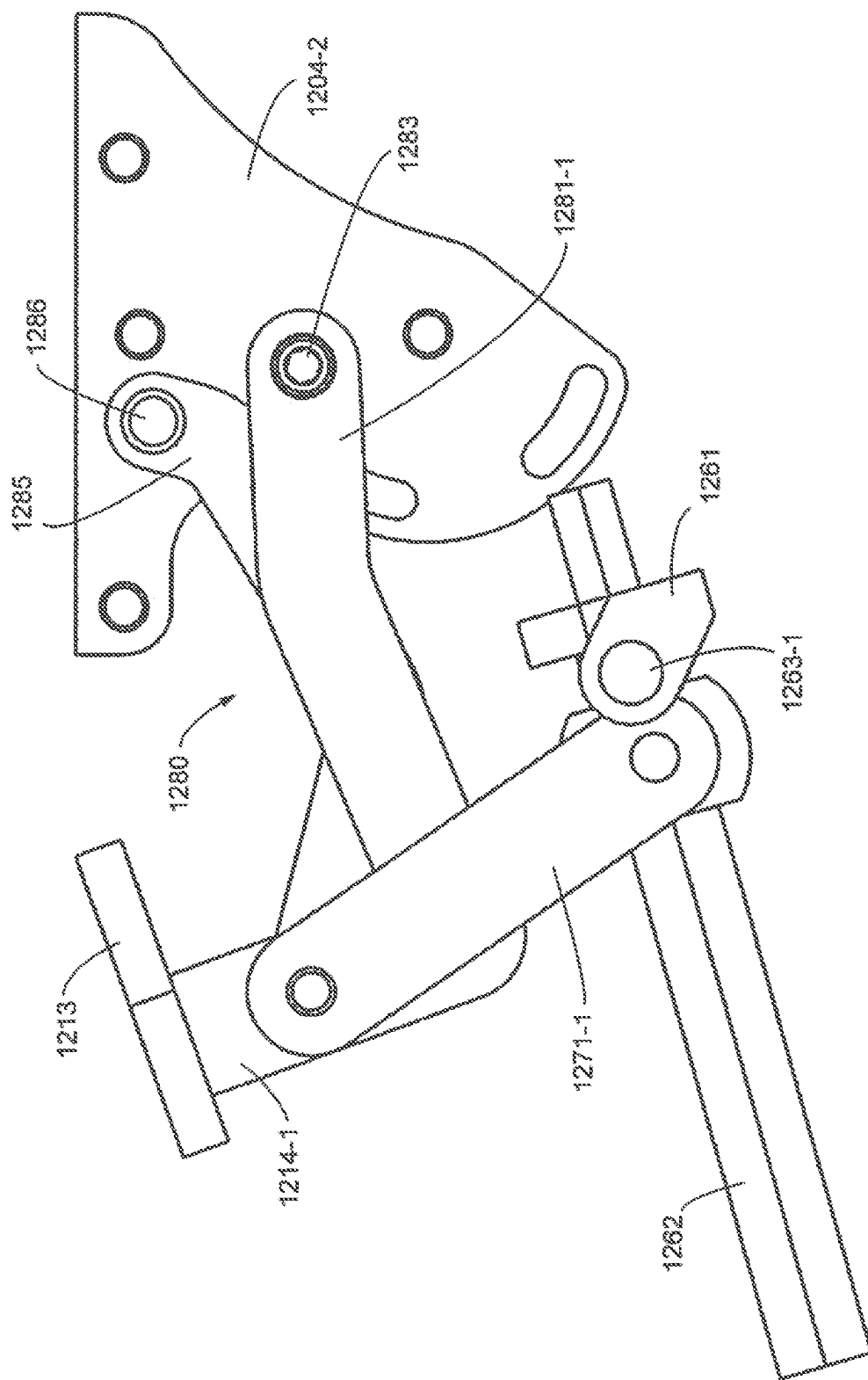


FIG. 17

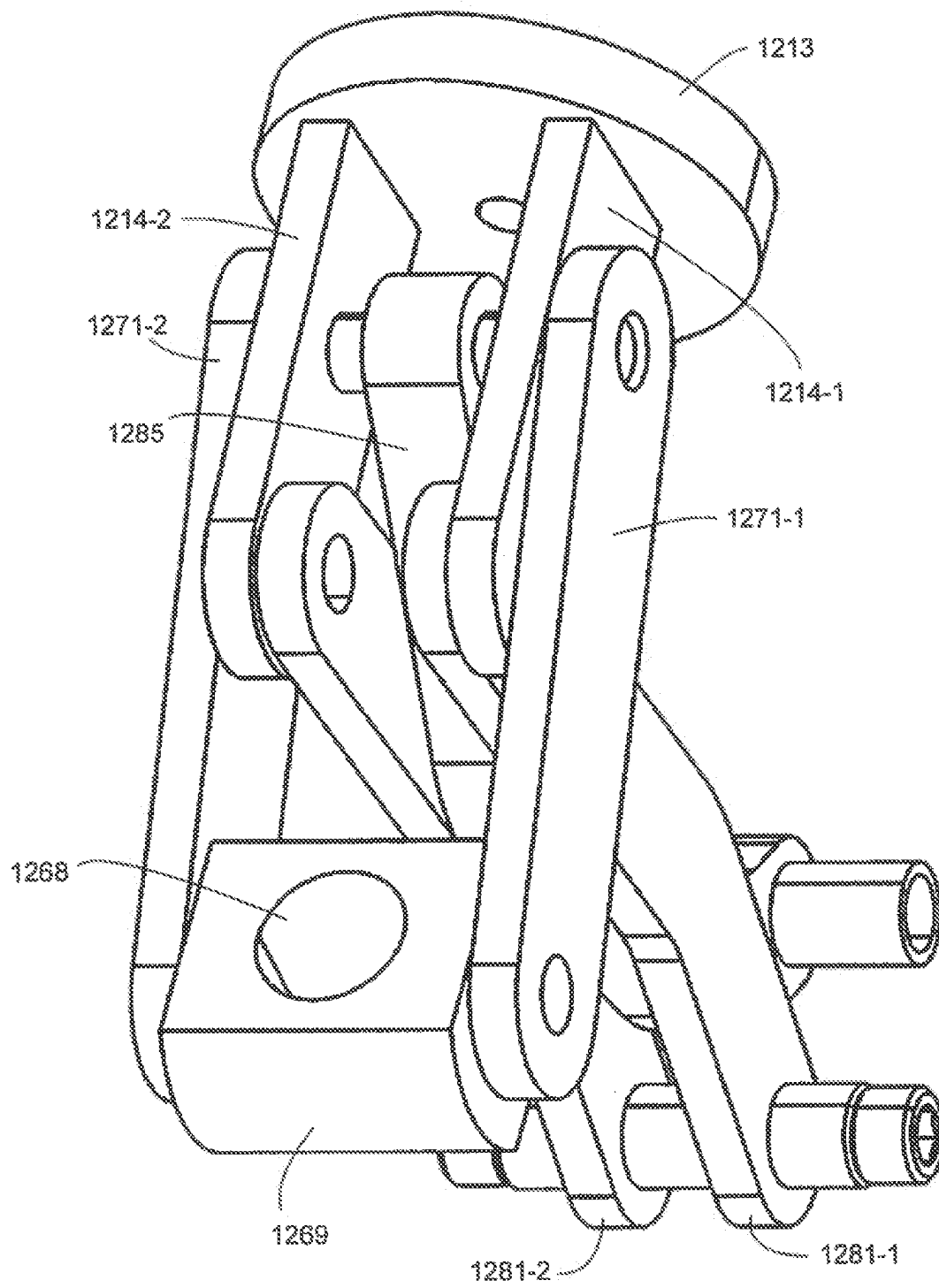


FIG. 18

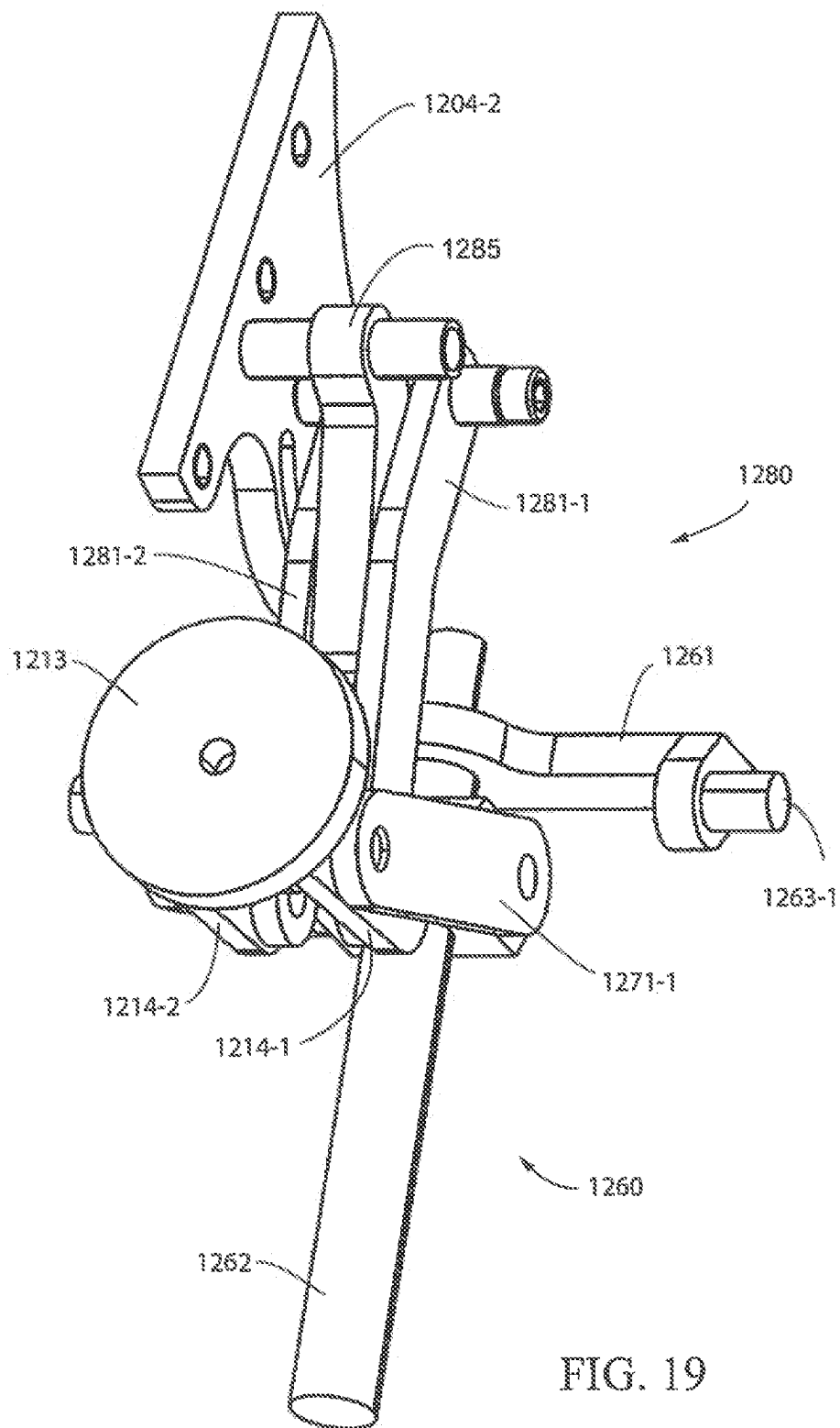


FIG. 19

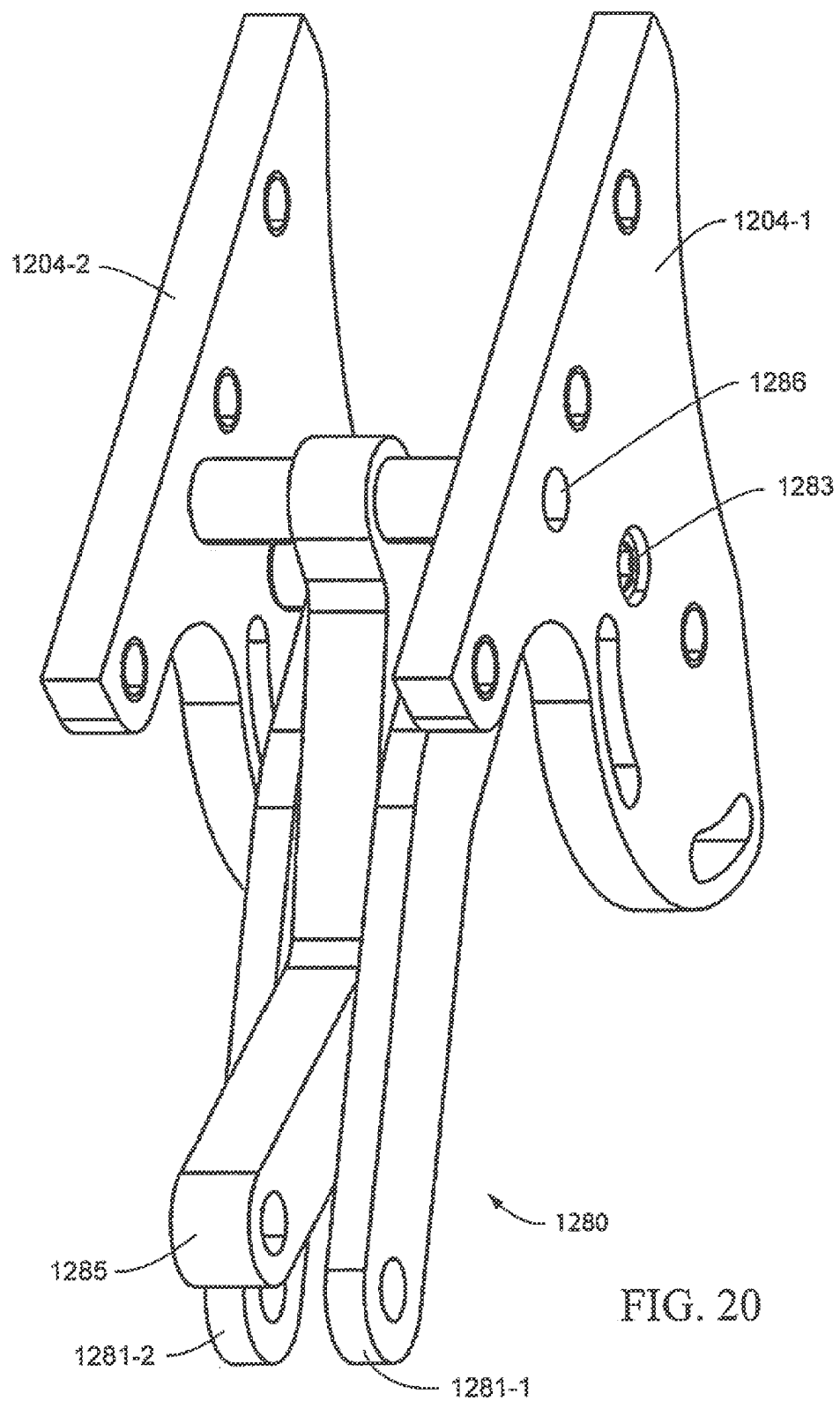


FIG. 20

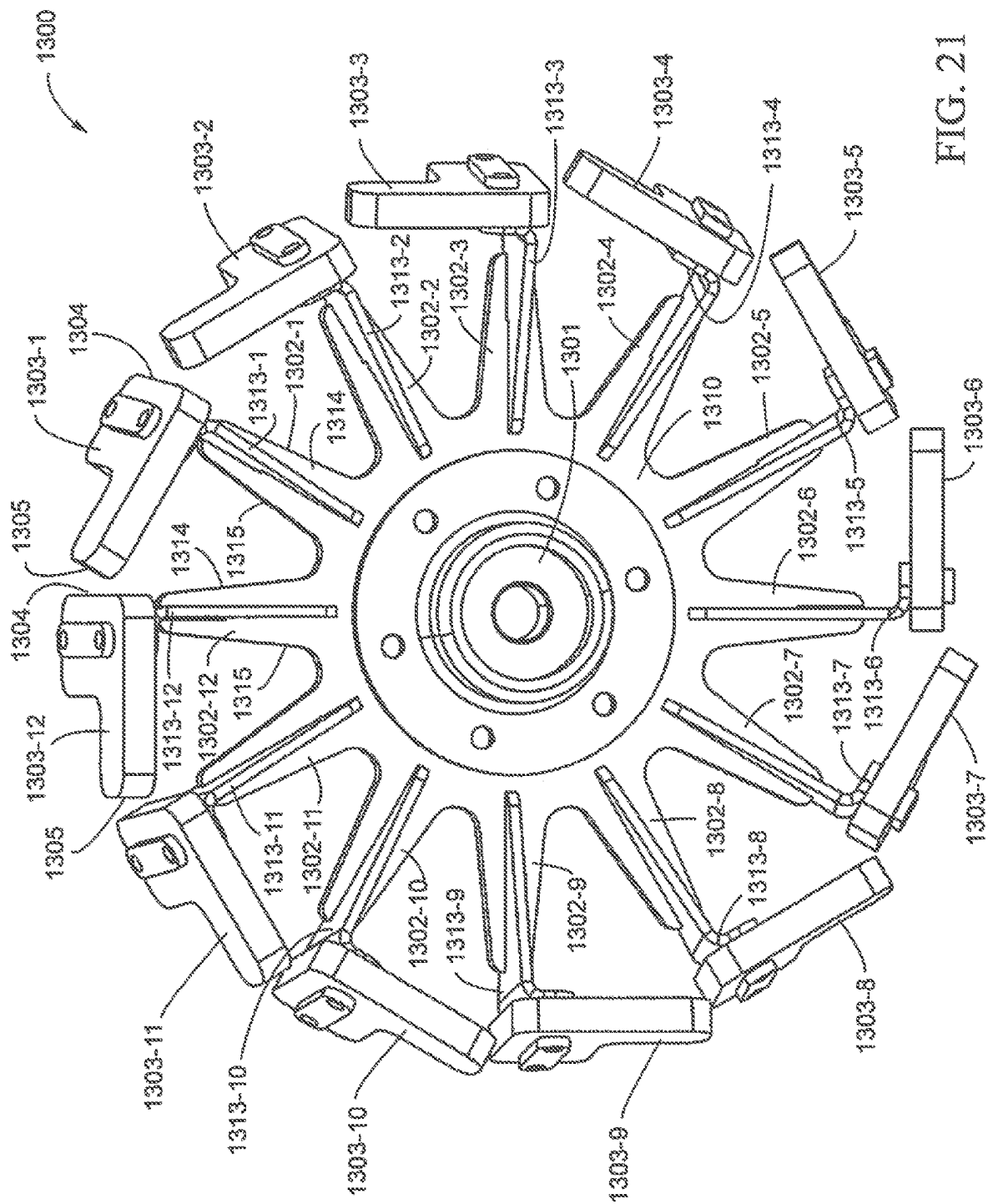


FIG. 21

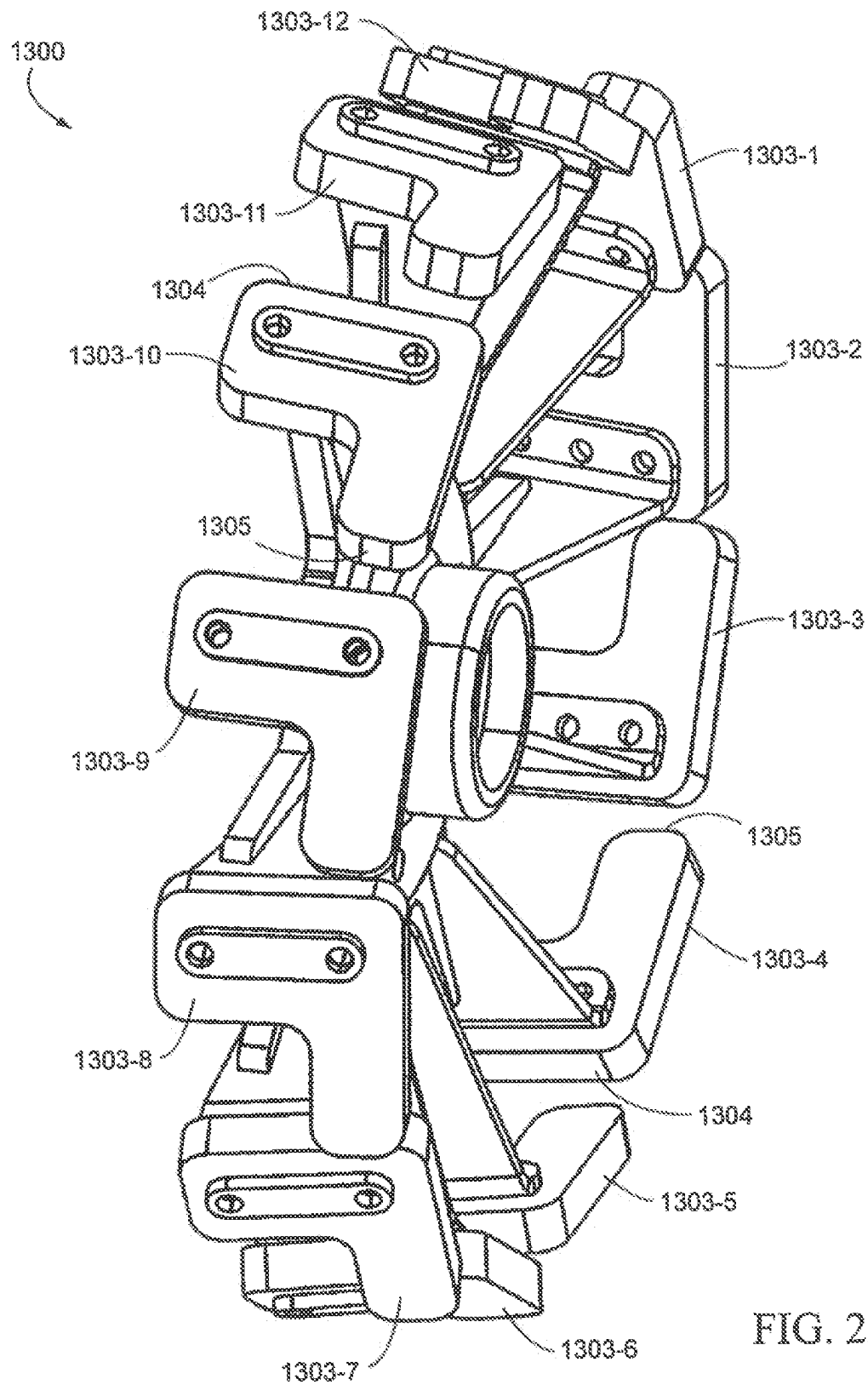
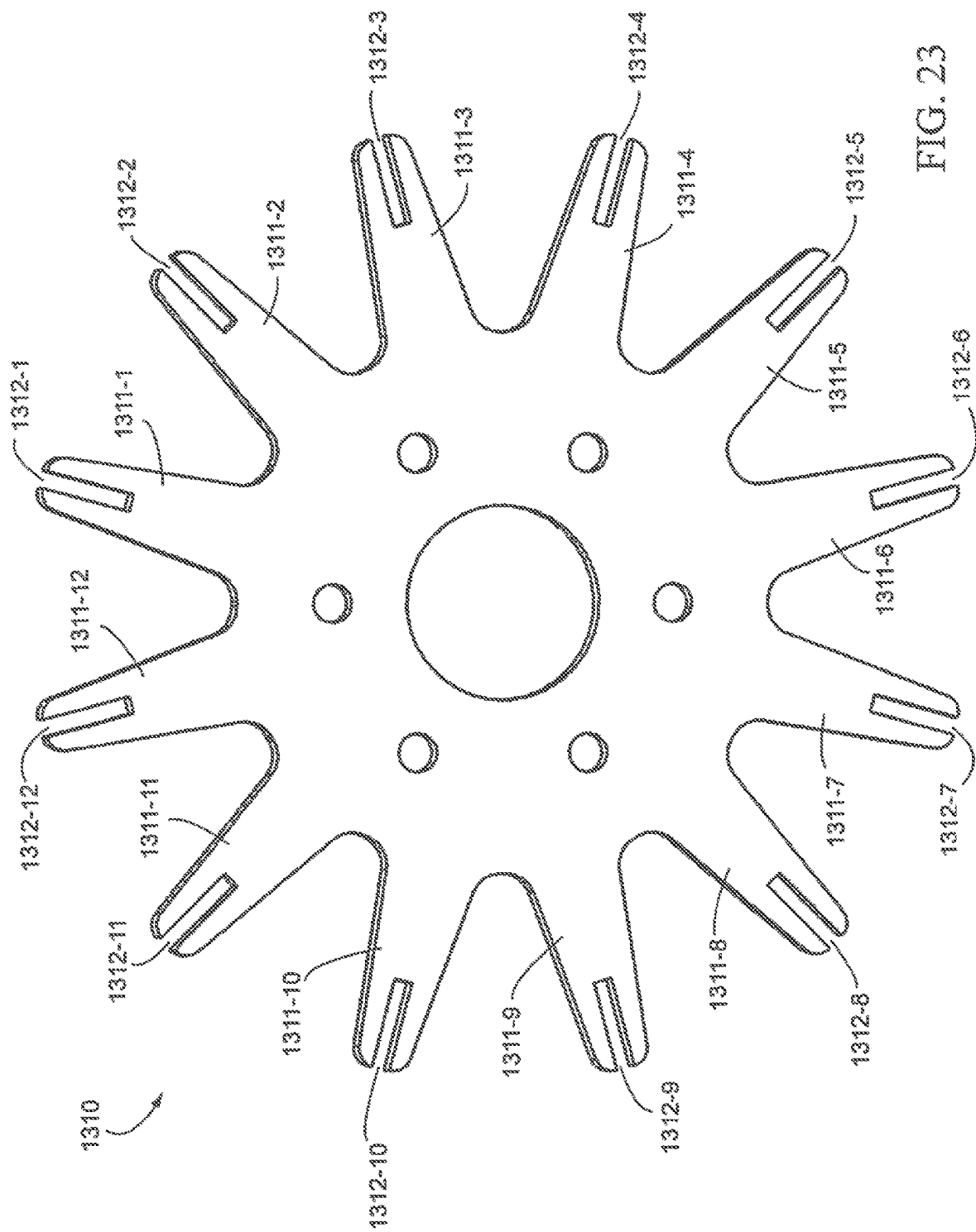


FIG. 22



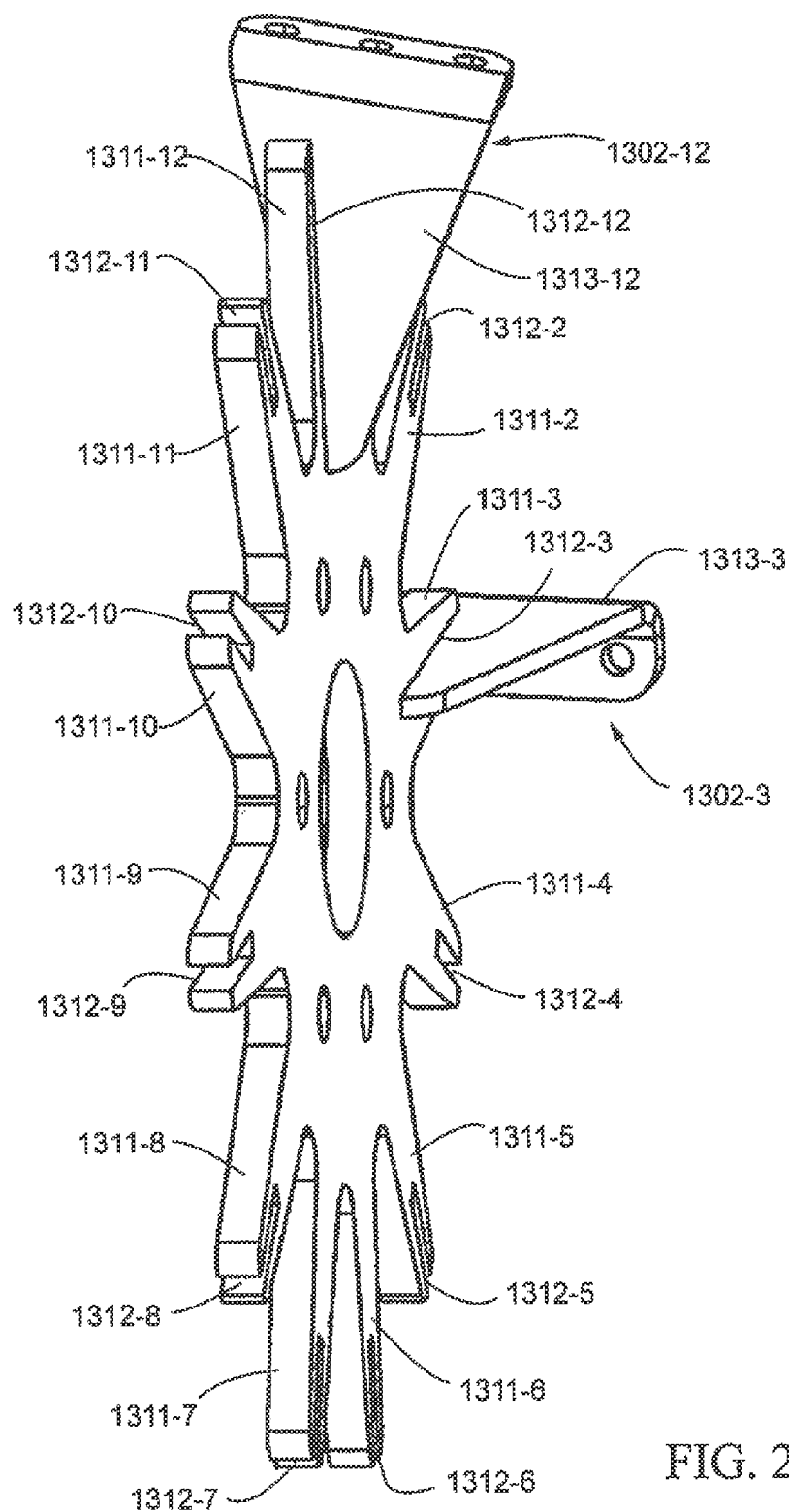


FIG. 24

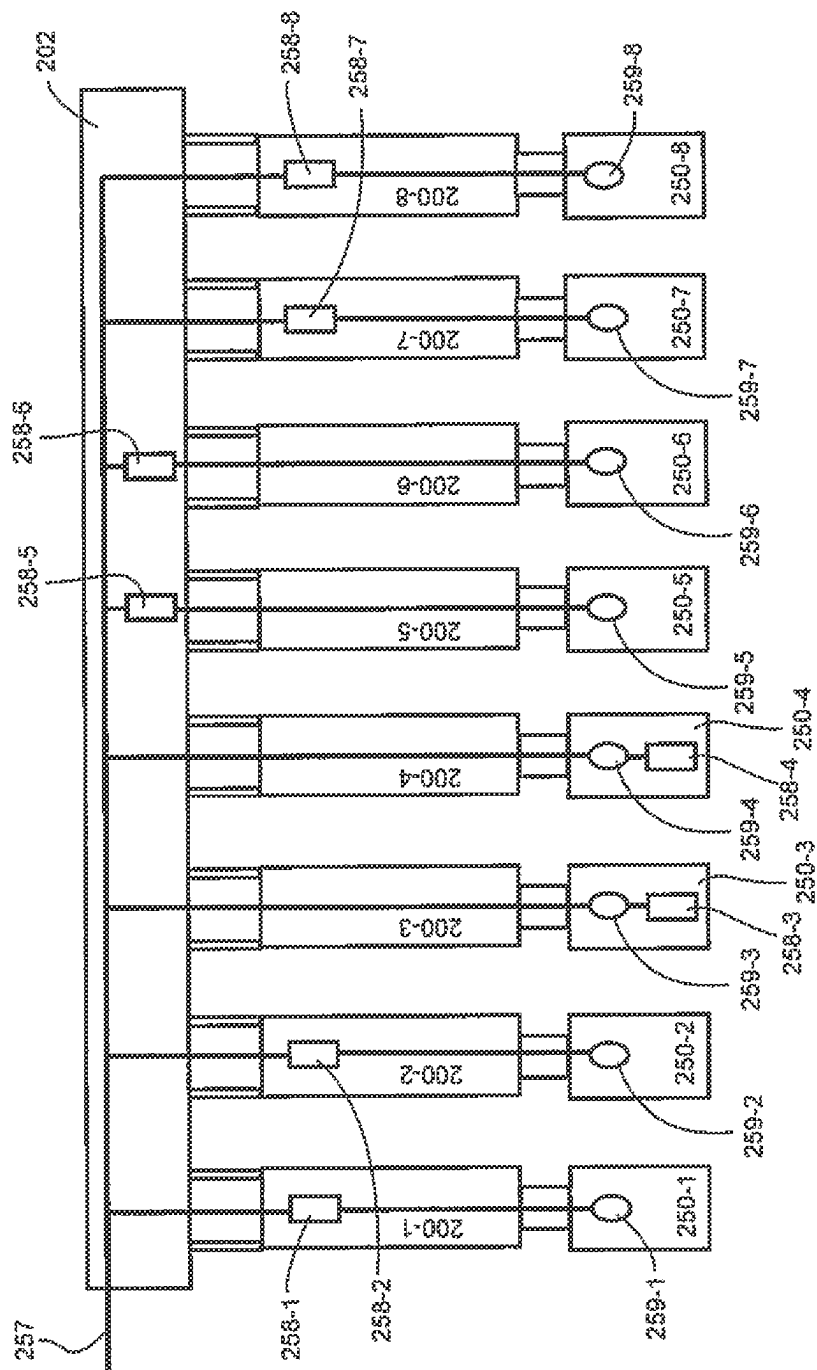


FIG. 25

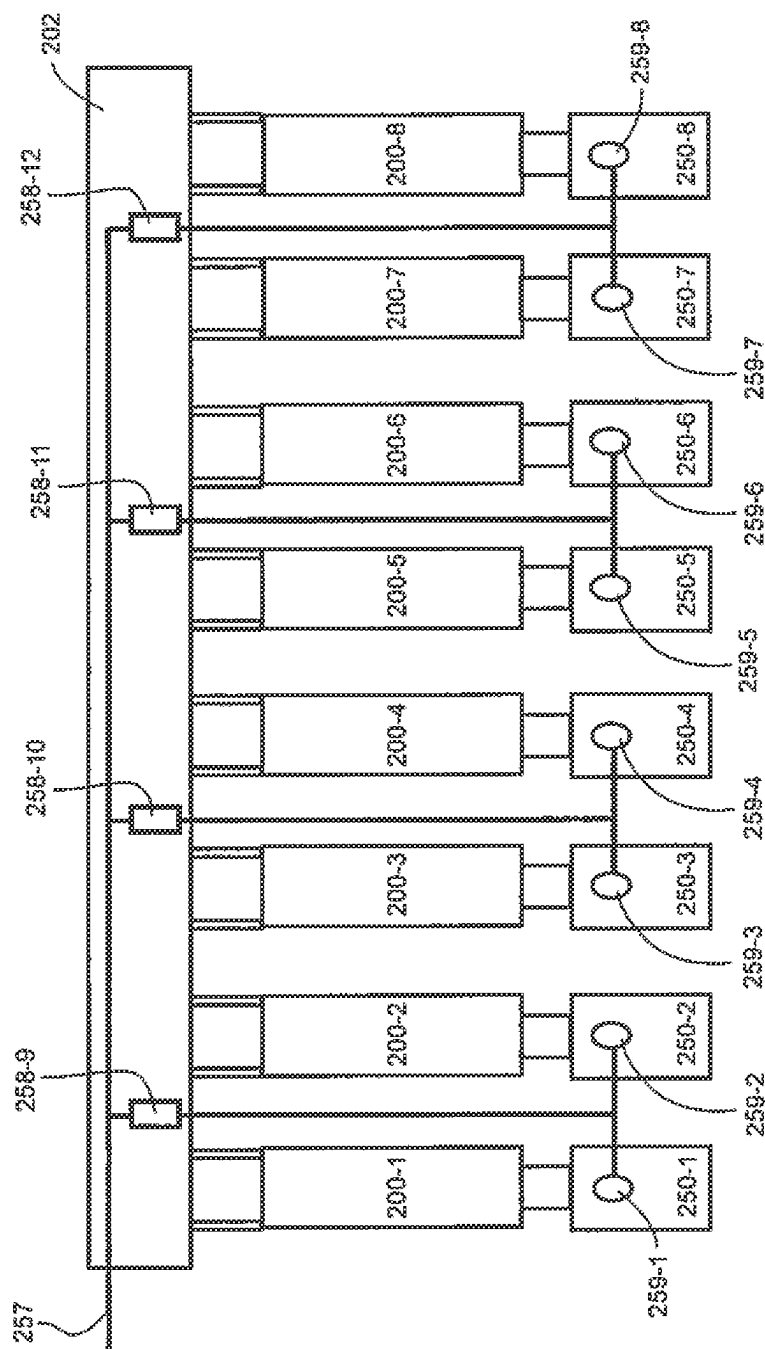


FIG. 26

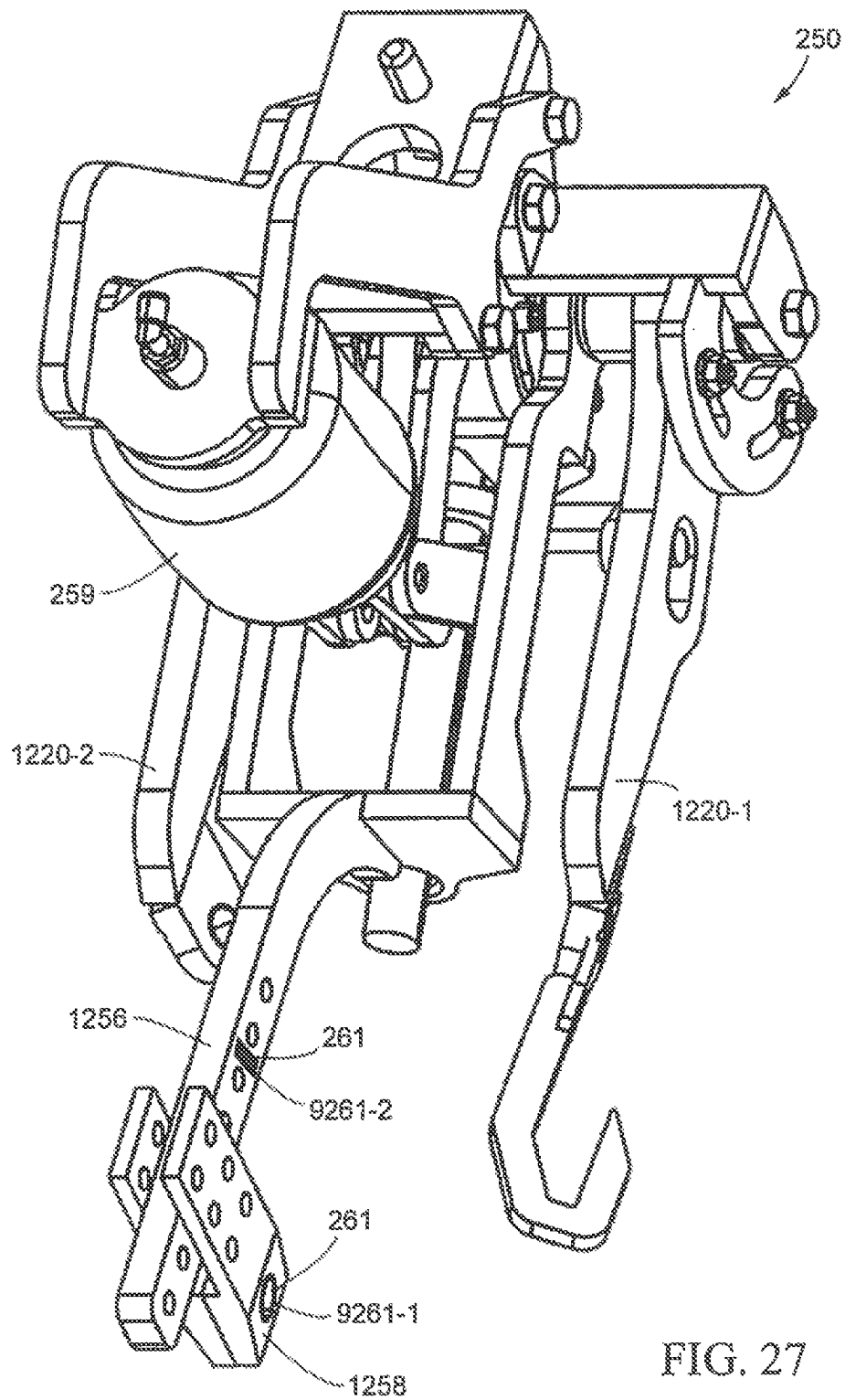


FIG. 27

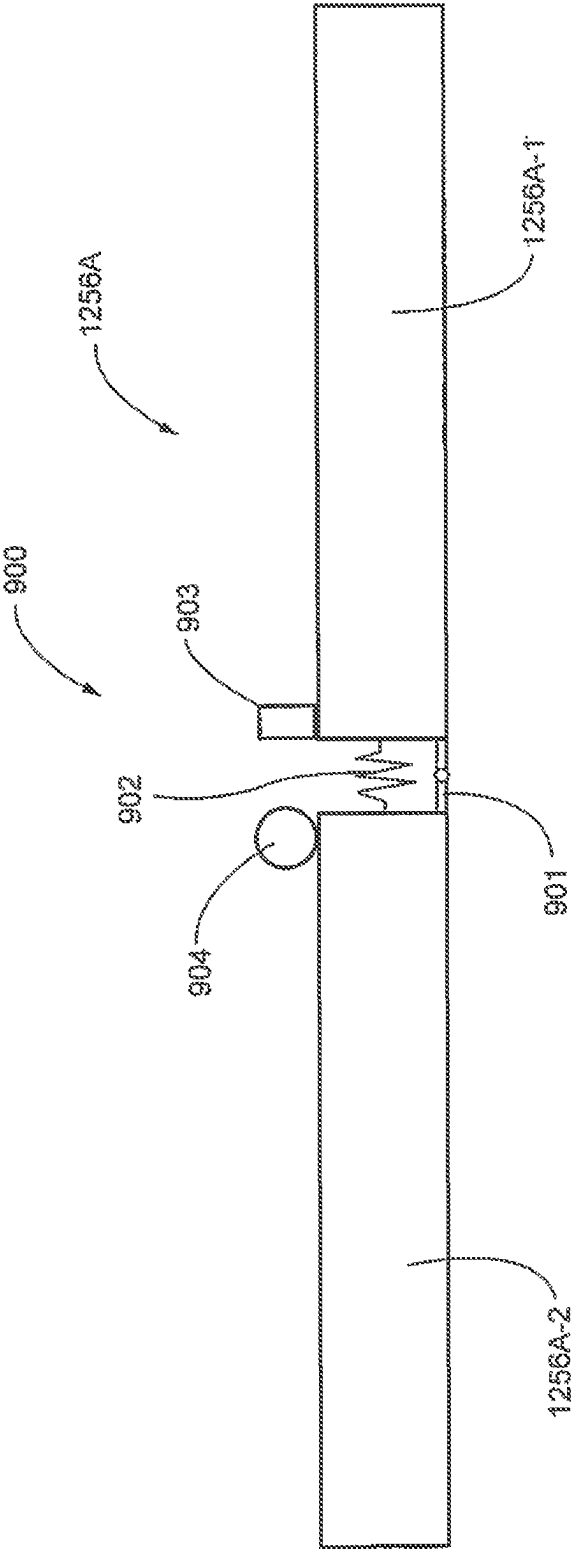


FIG. 28

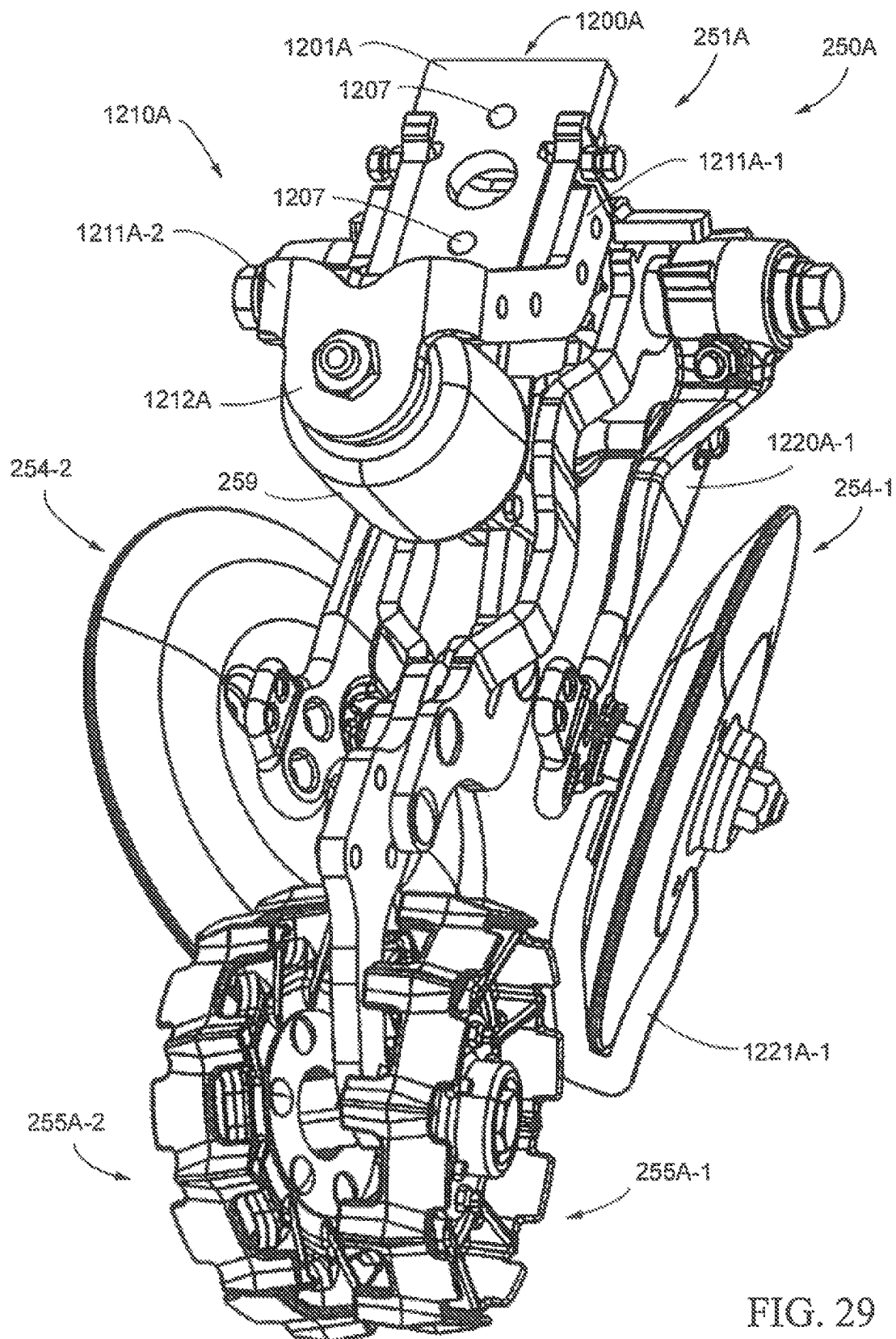


FIG. 29

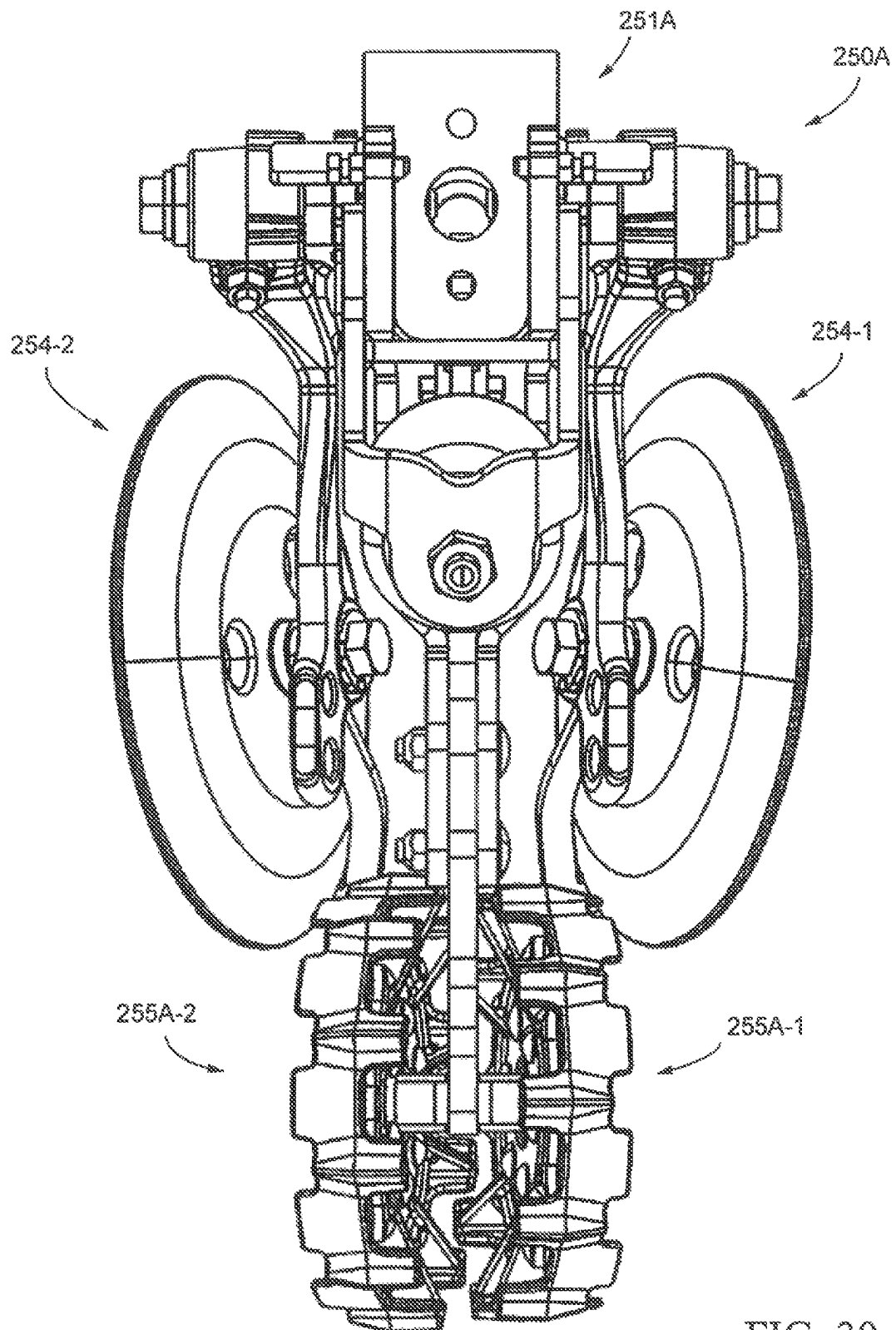


FIG. 30

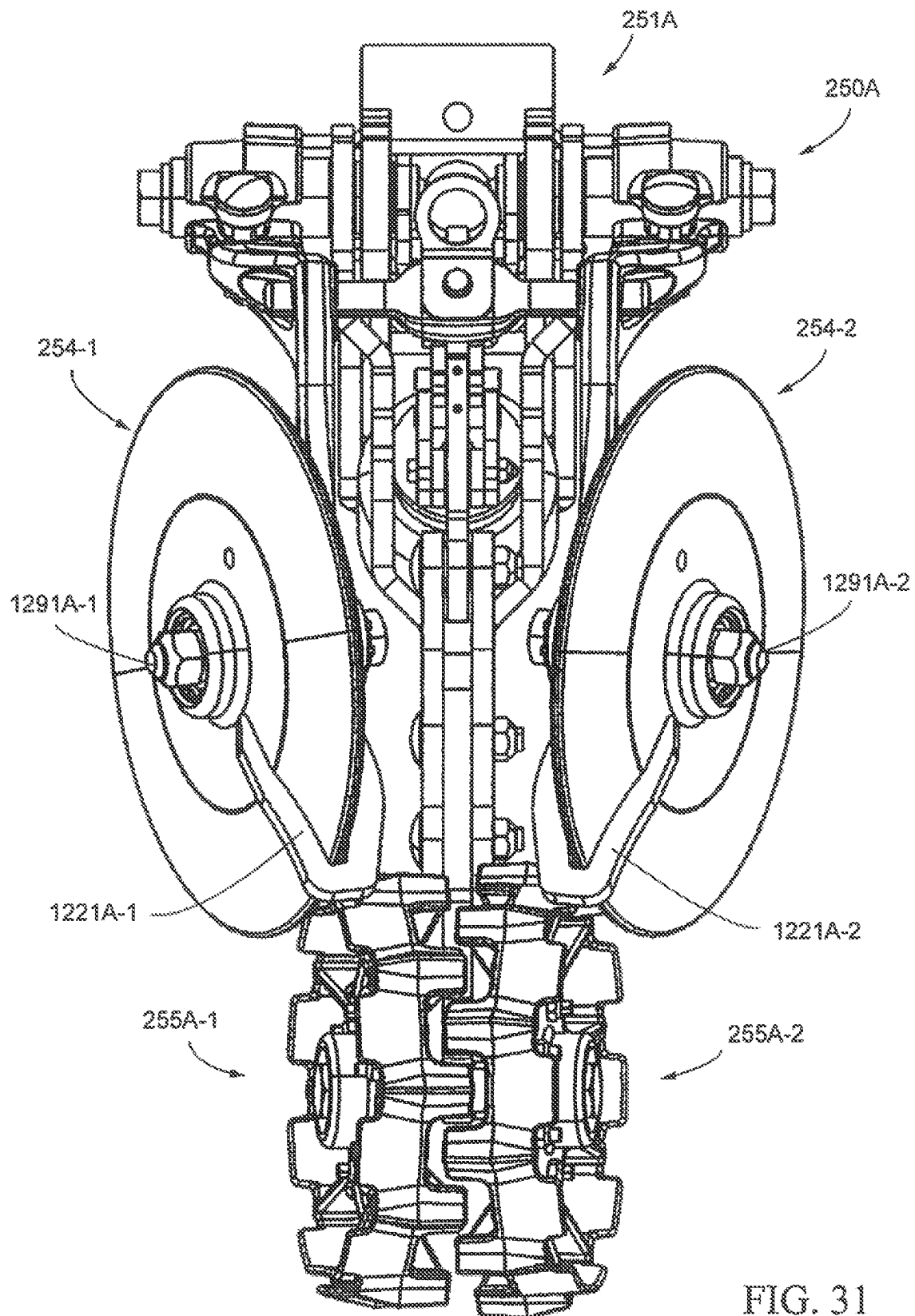


FIG. 31

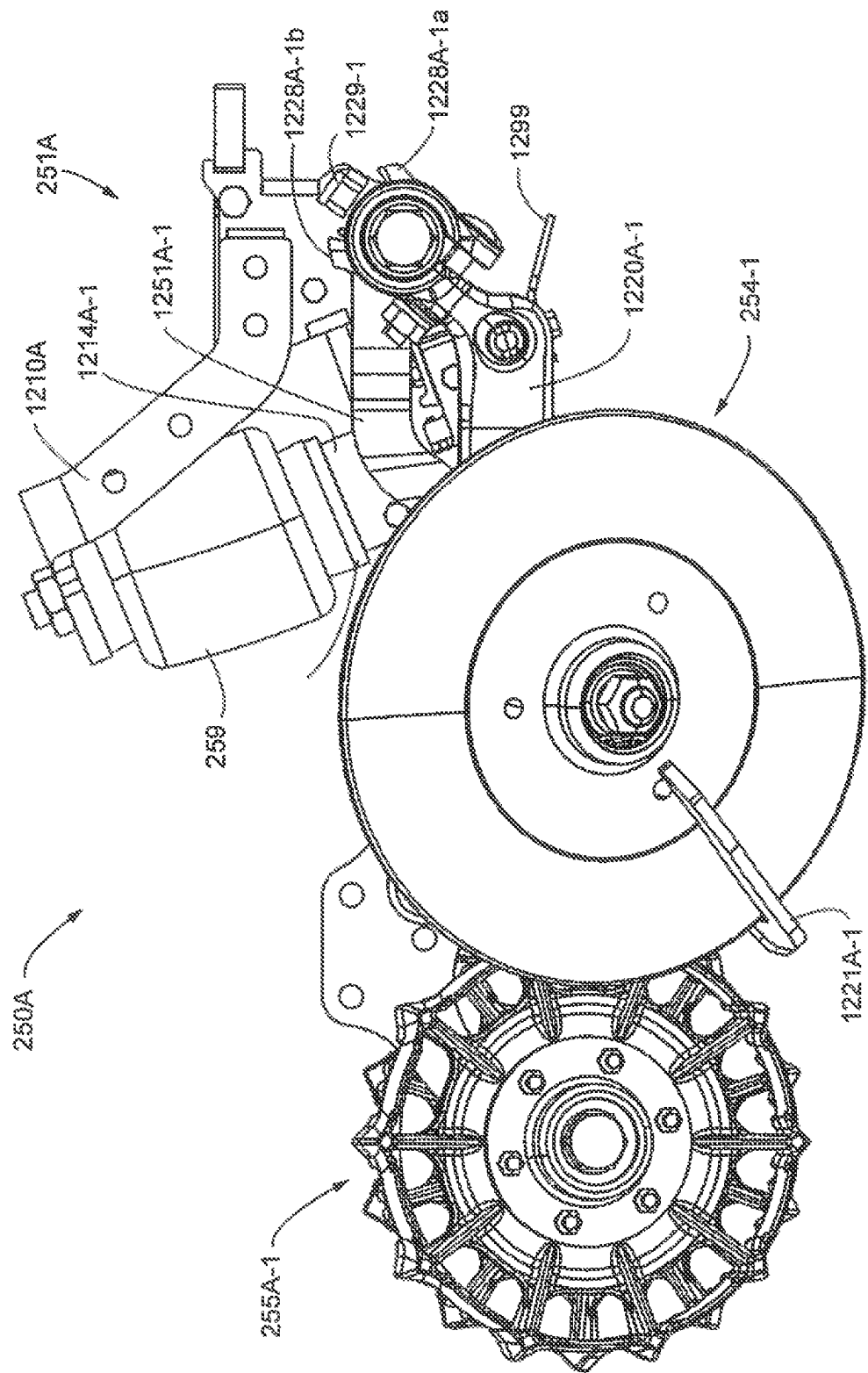


FIG. 32

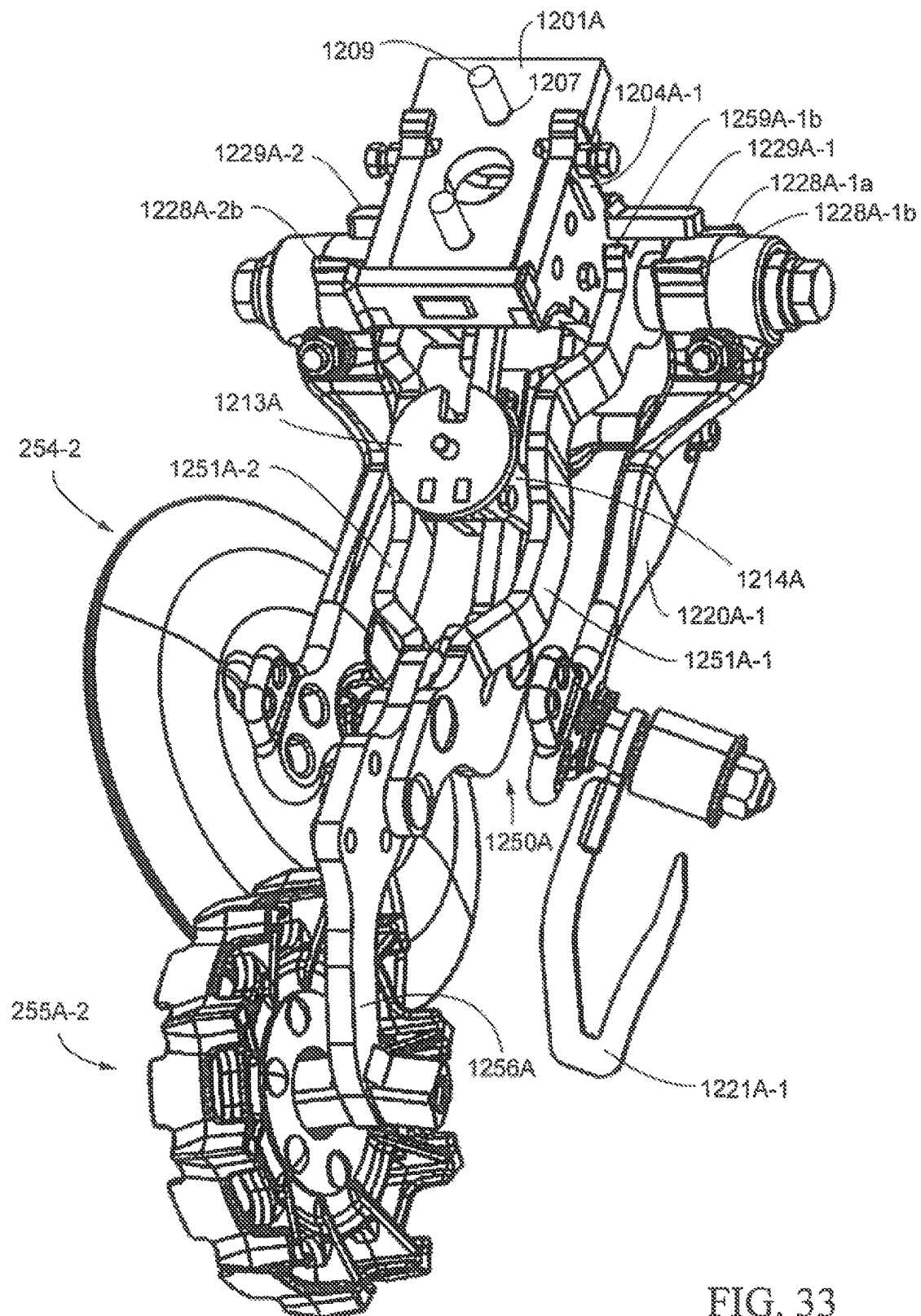


FIG. 33

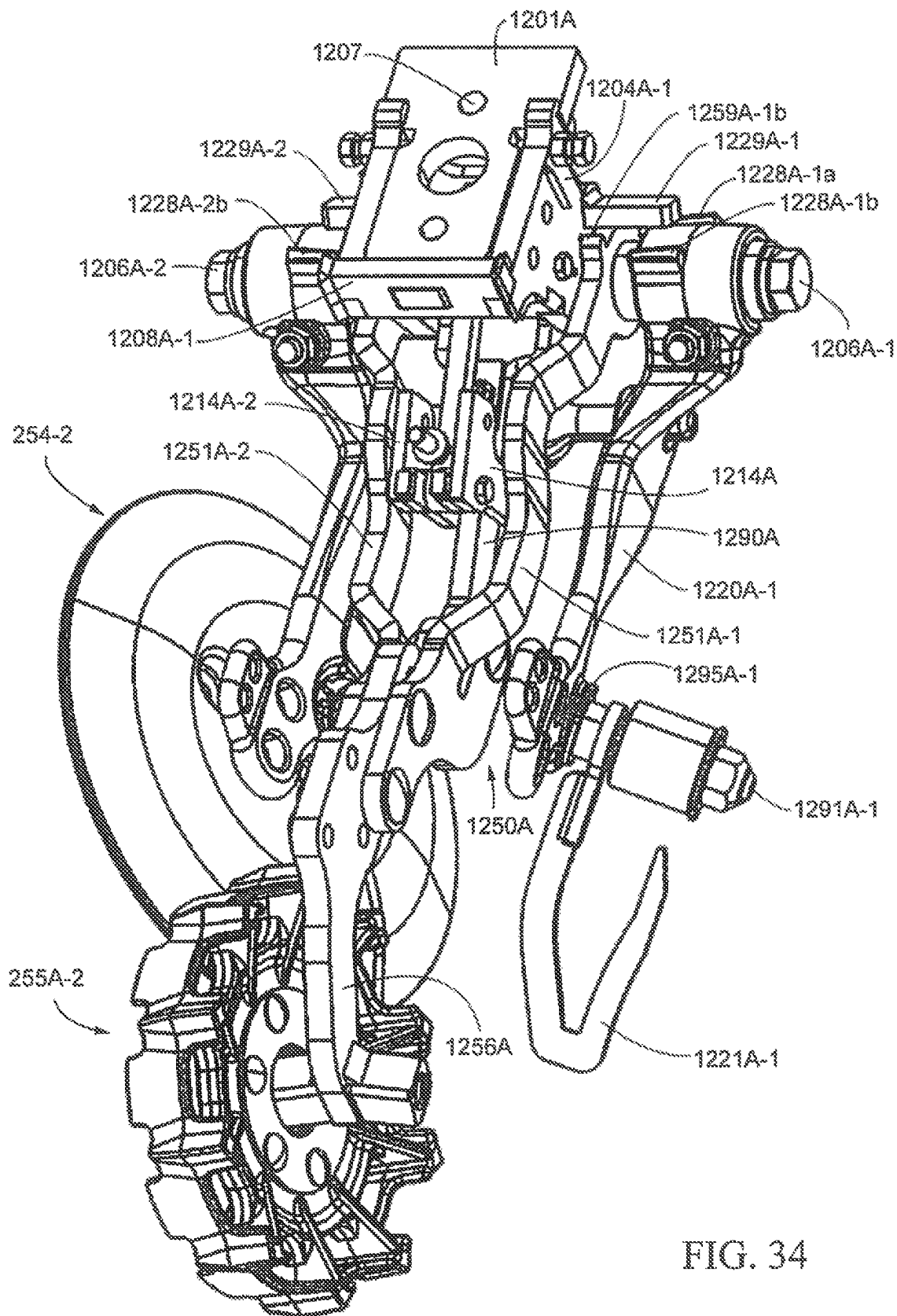
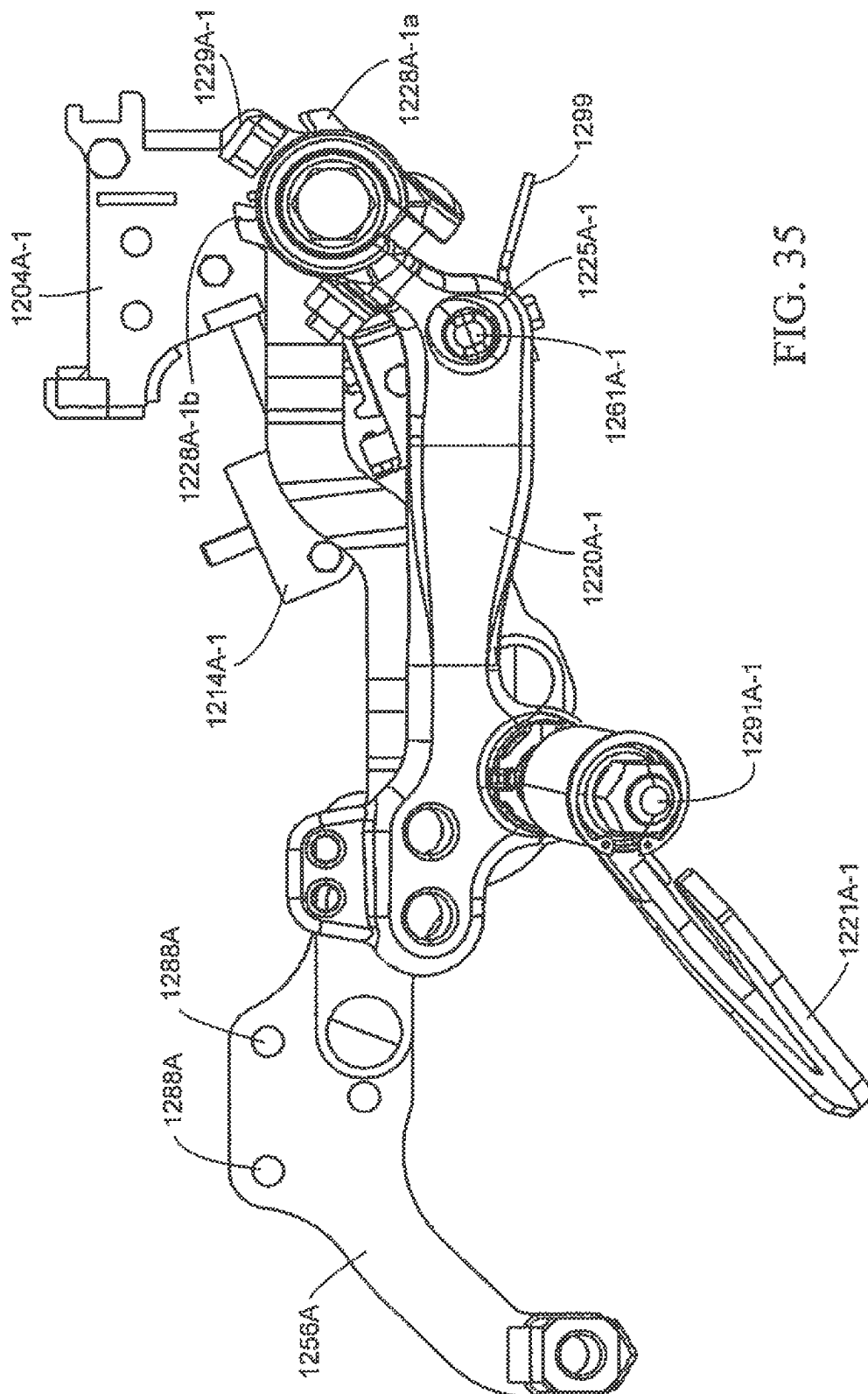


FIG. 34



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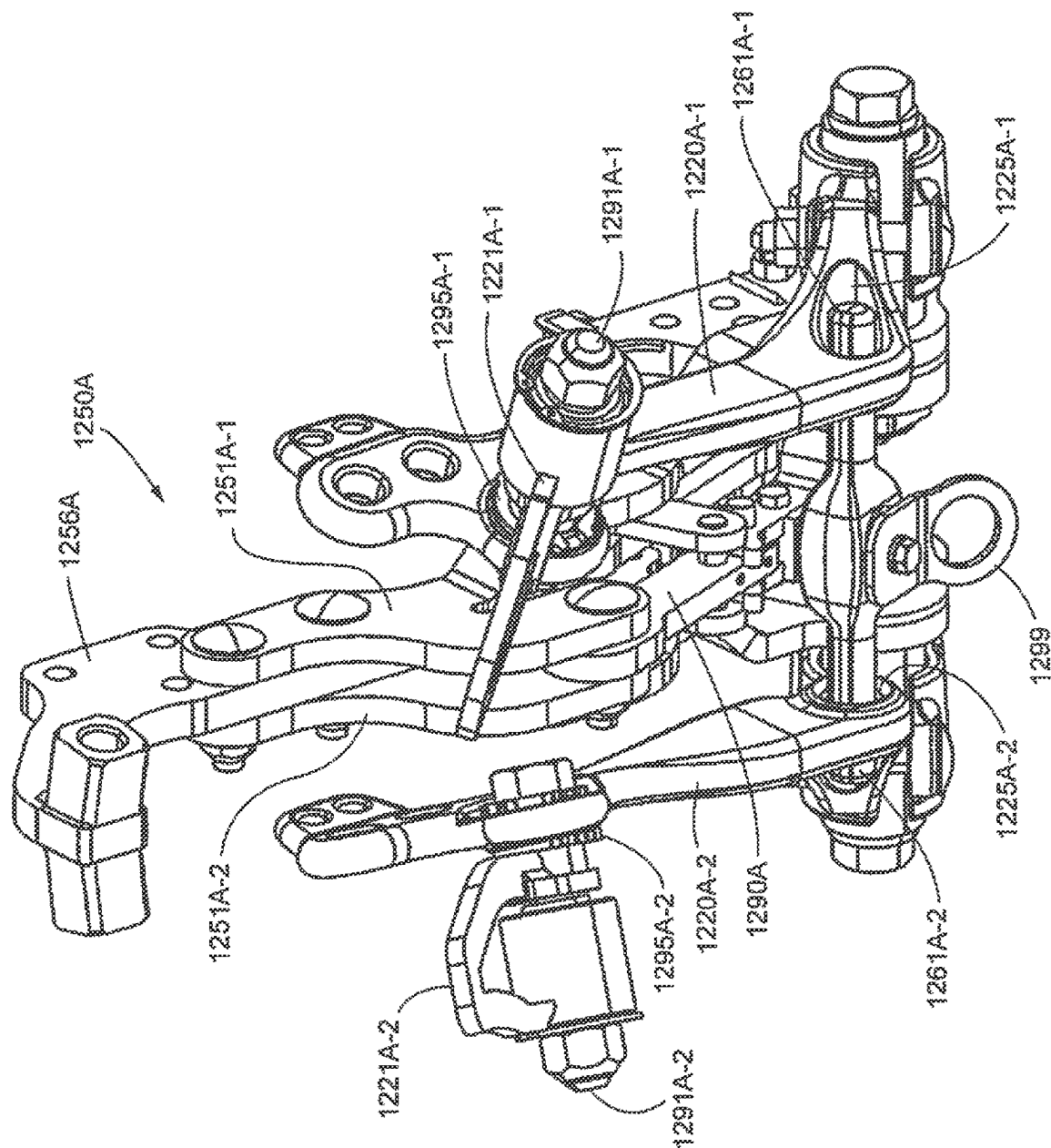


FIG. 36

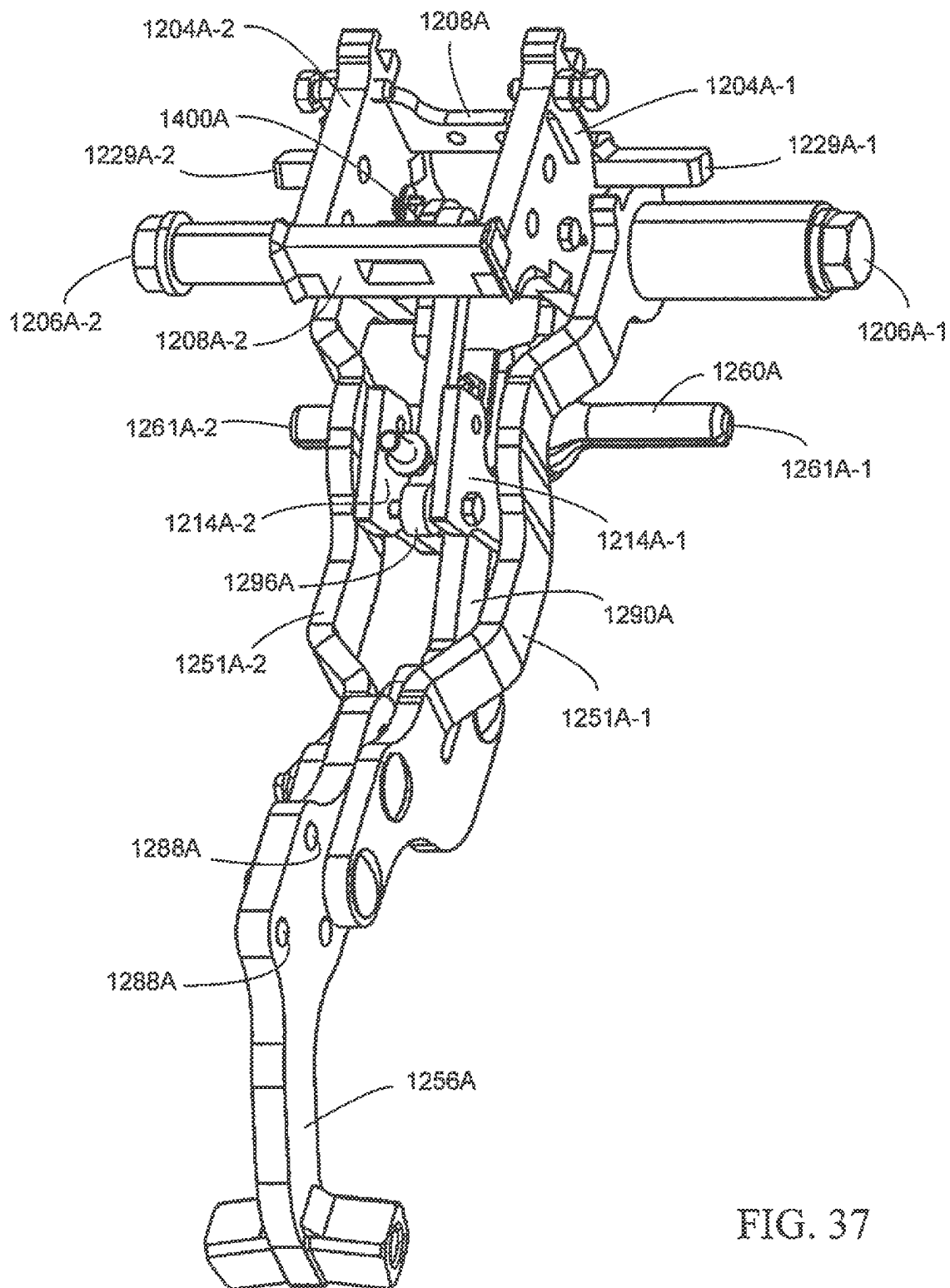


FIG. 37

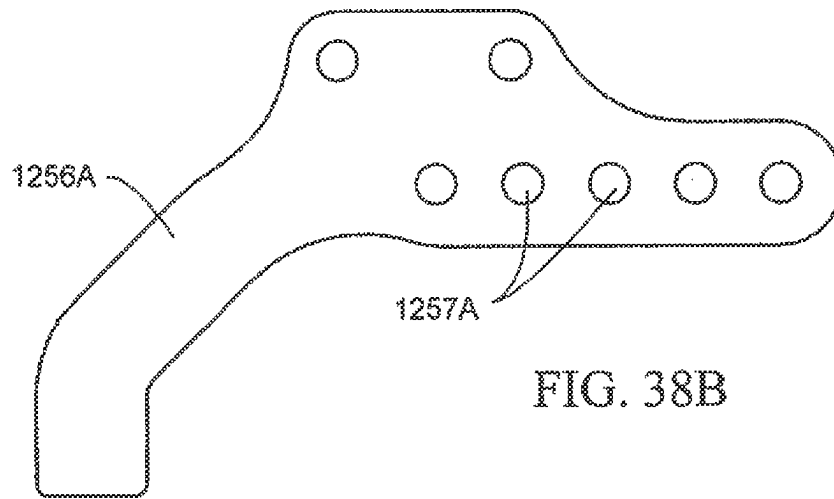


FIG. 38B

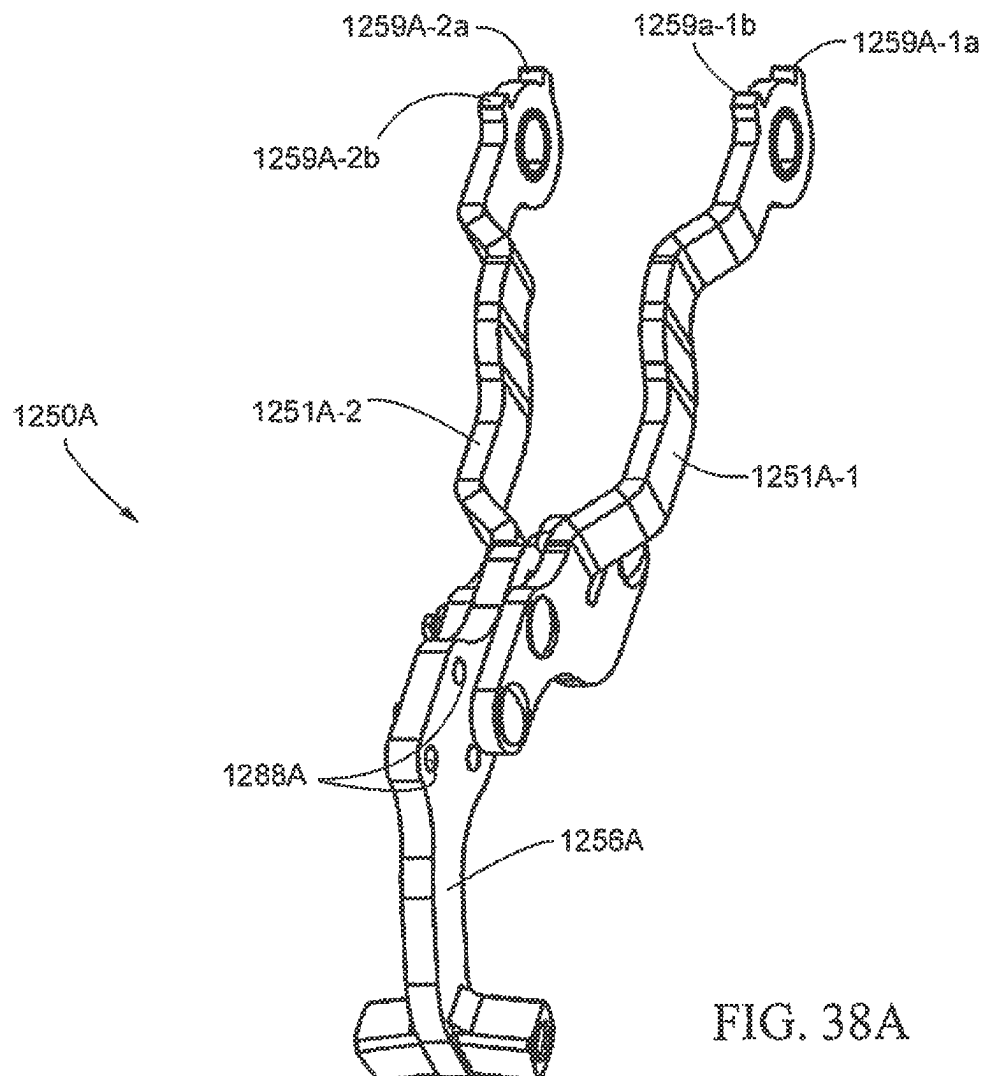


FIG. 38A

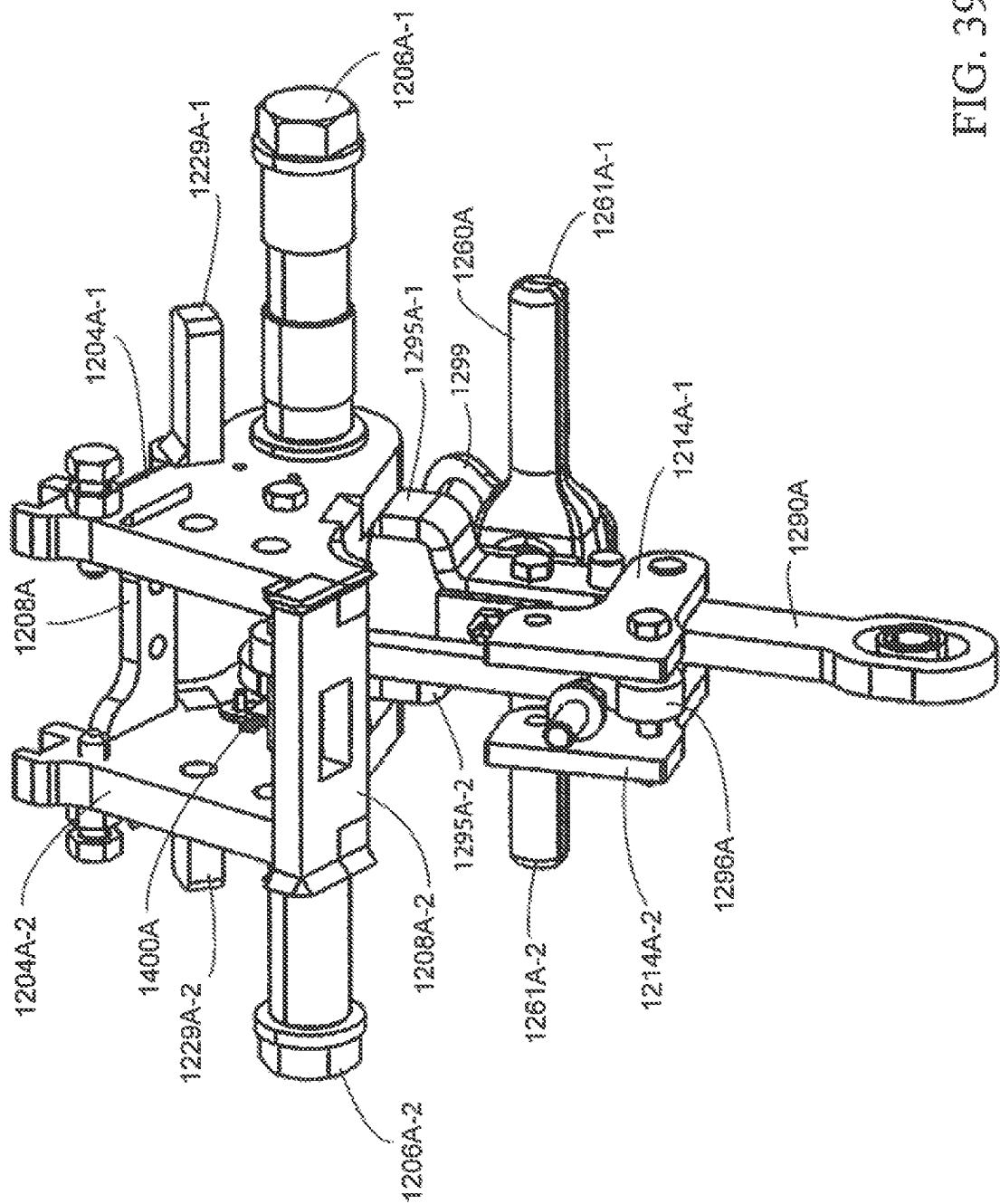


FIG. 39

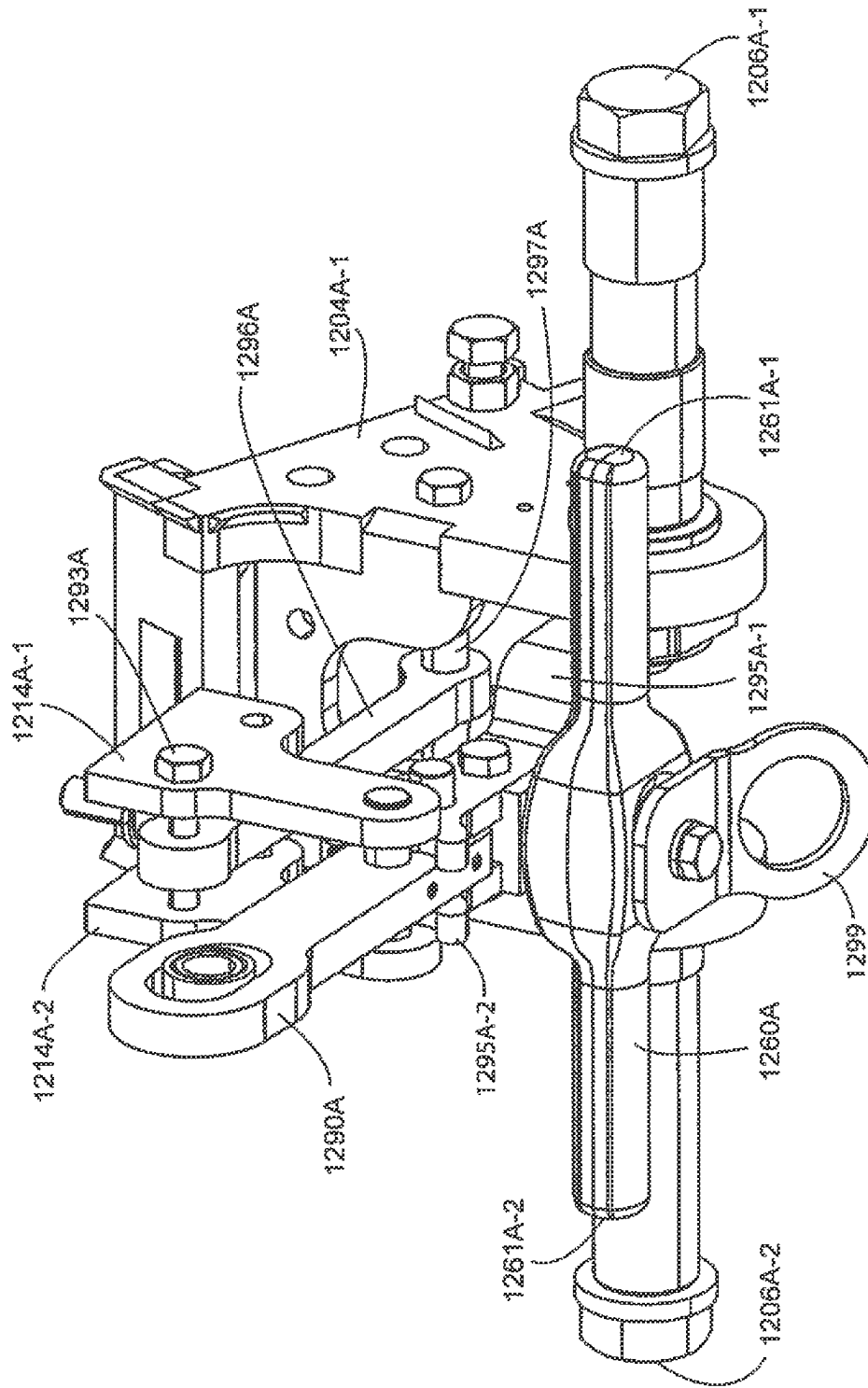
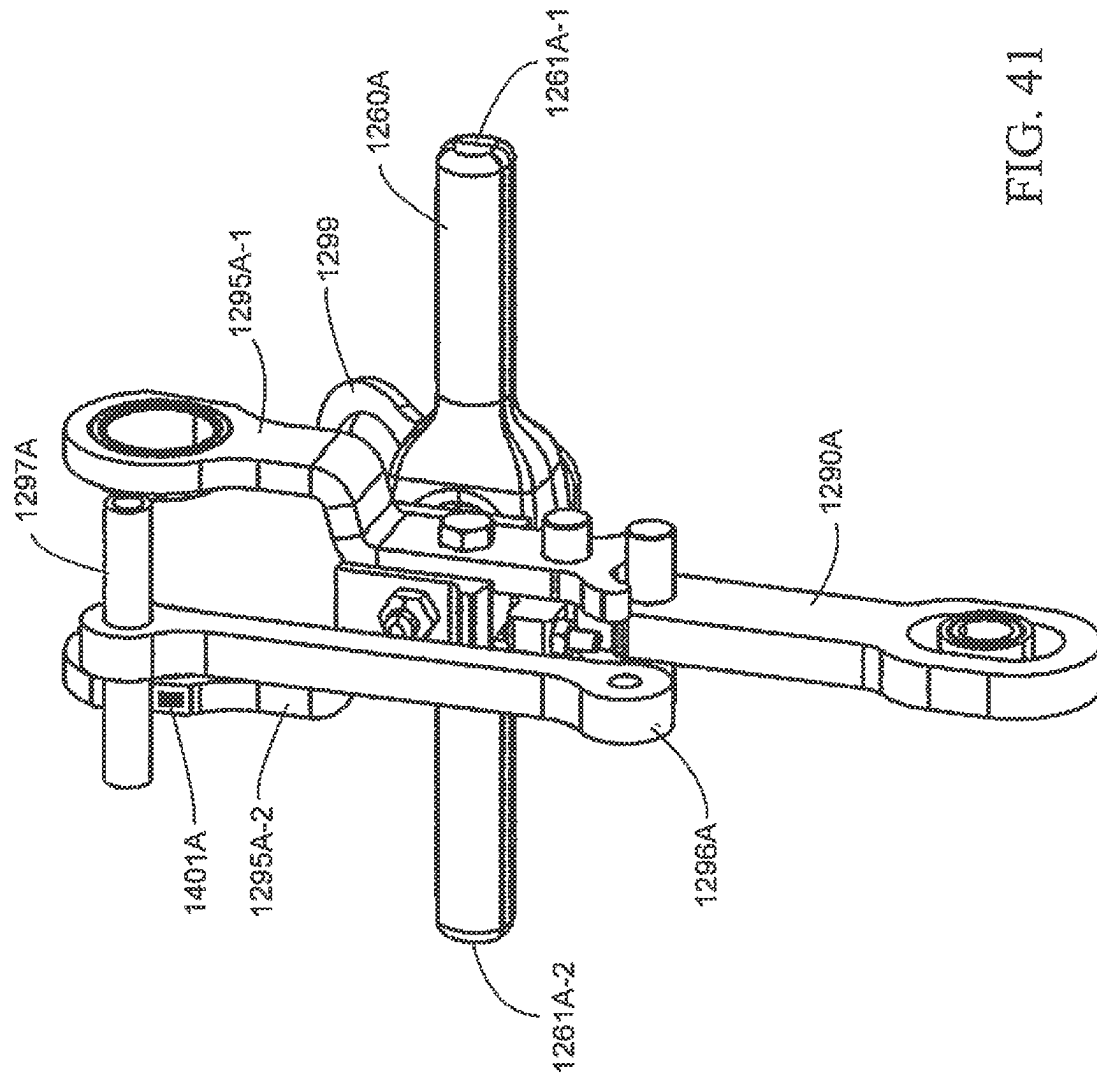


FIG. 40



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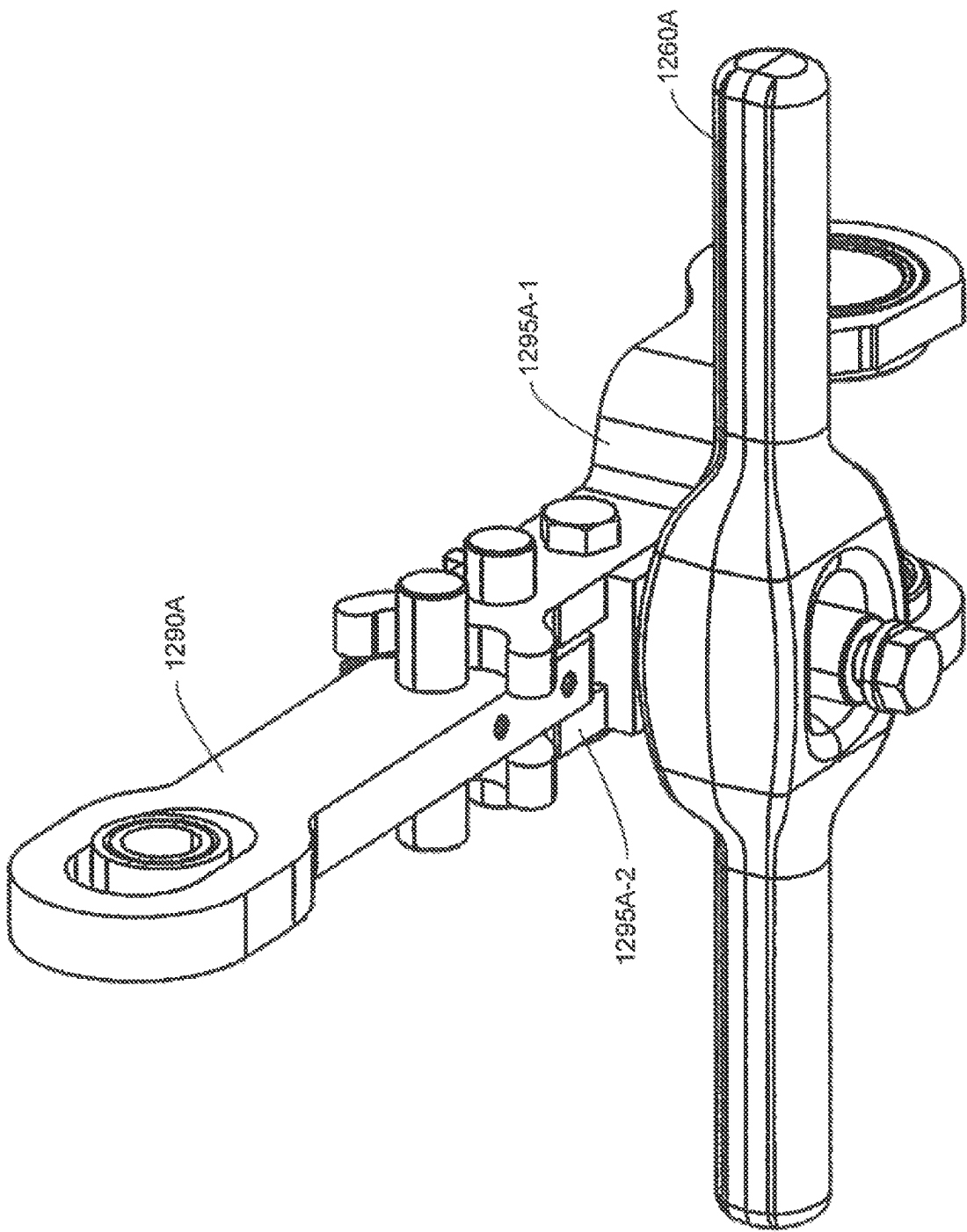


FIG. 42

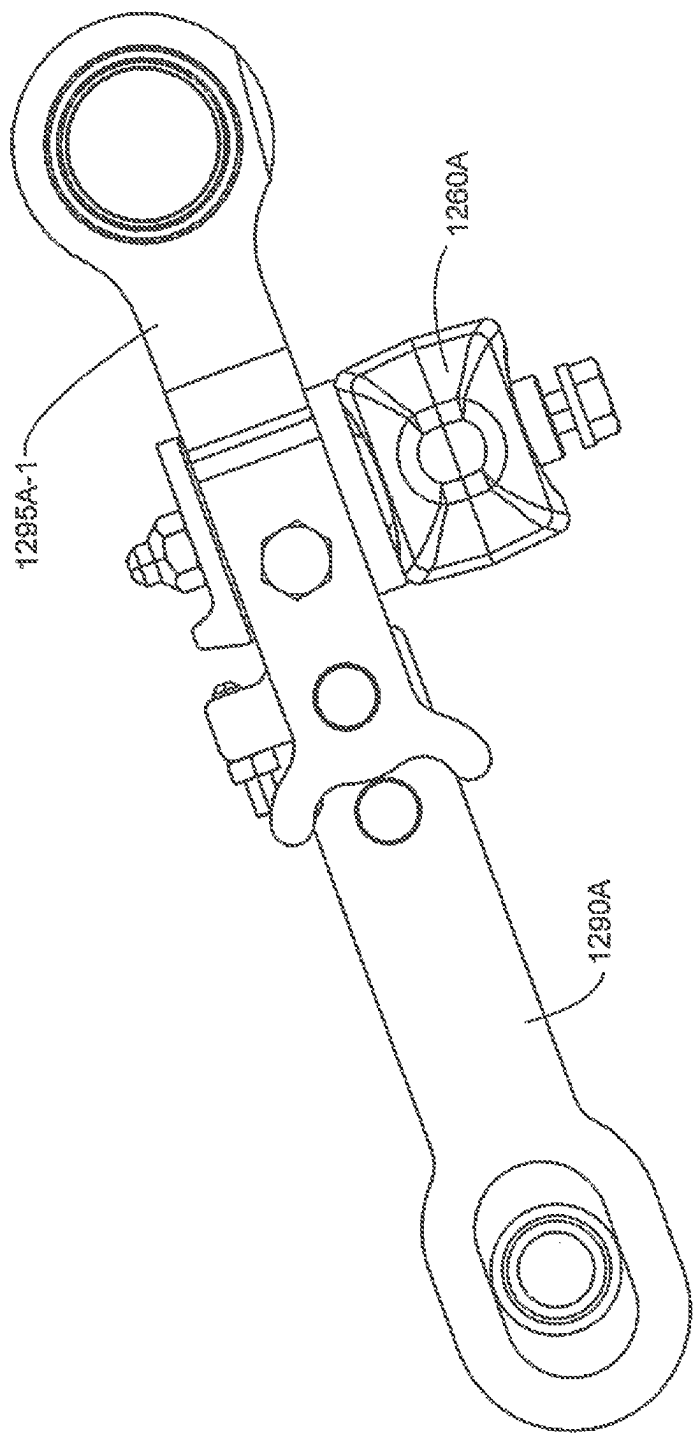


FIG. 43

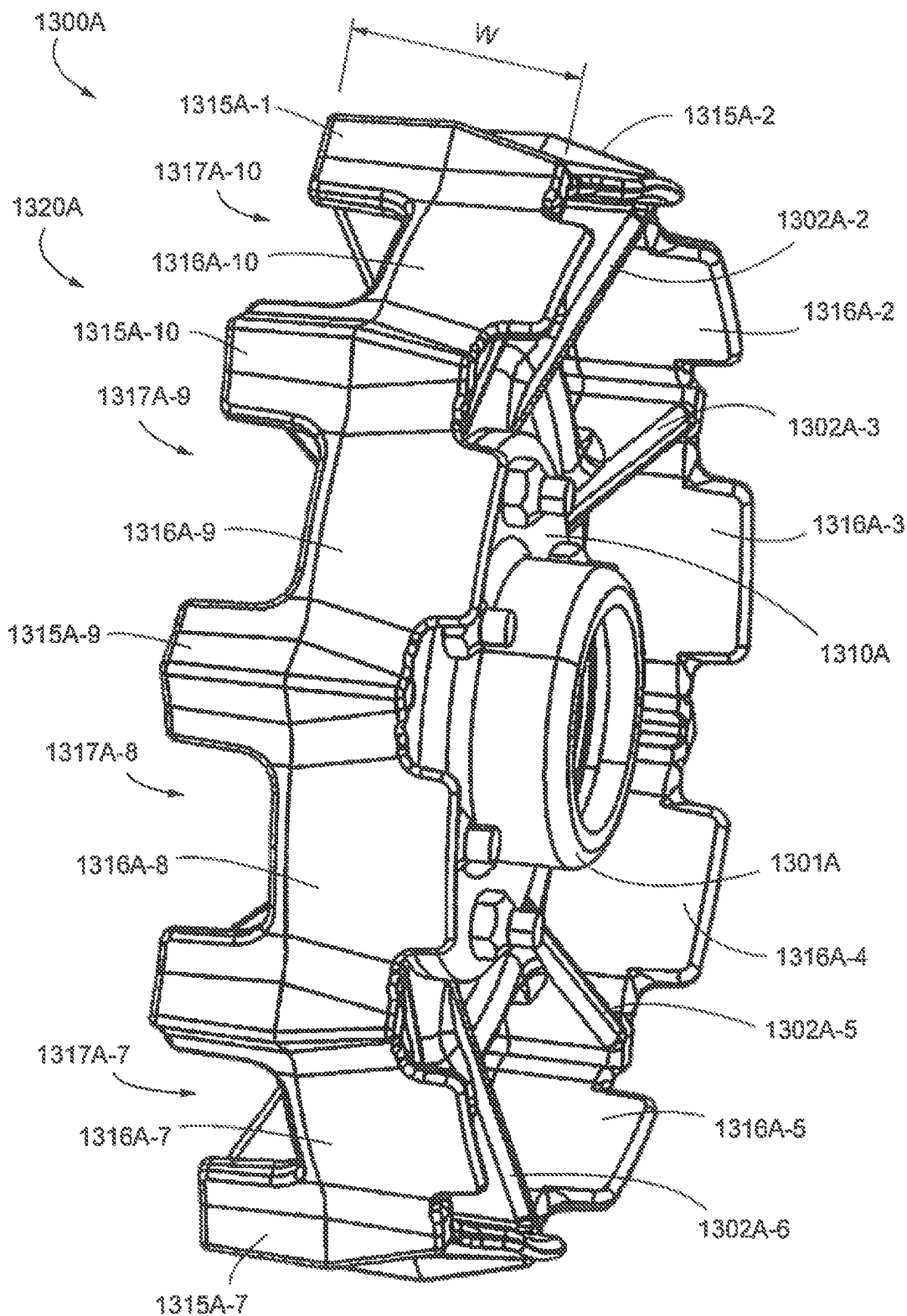


FIG. 44

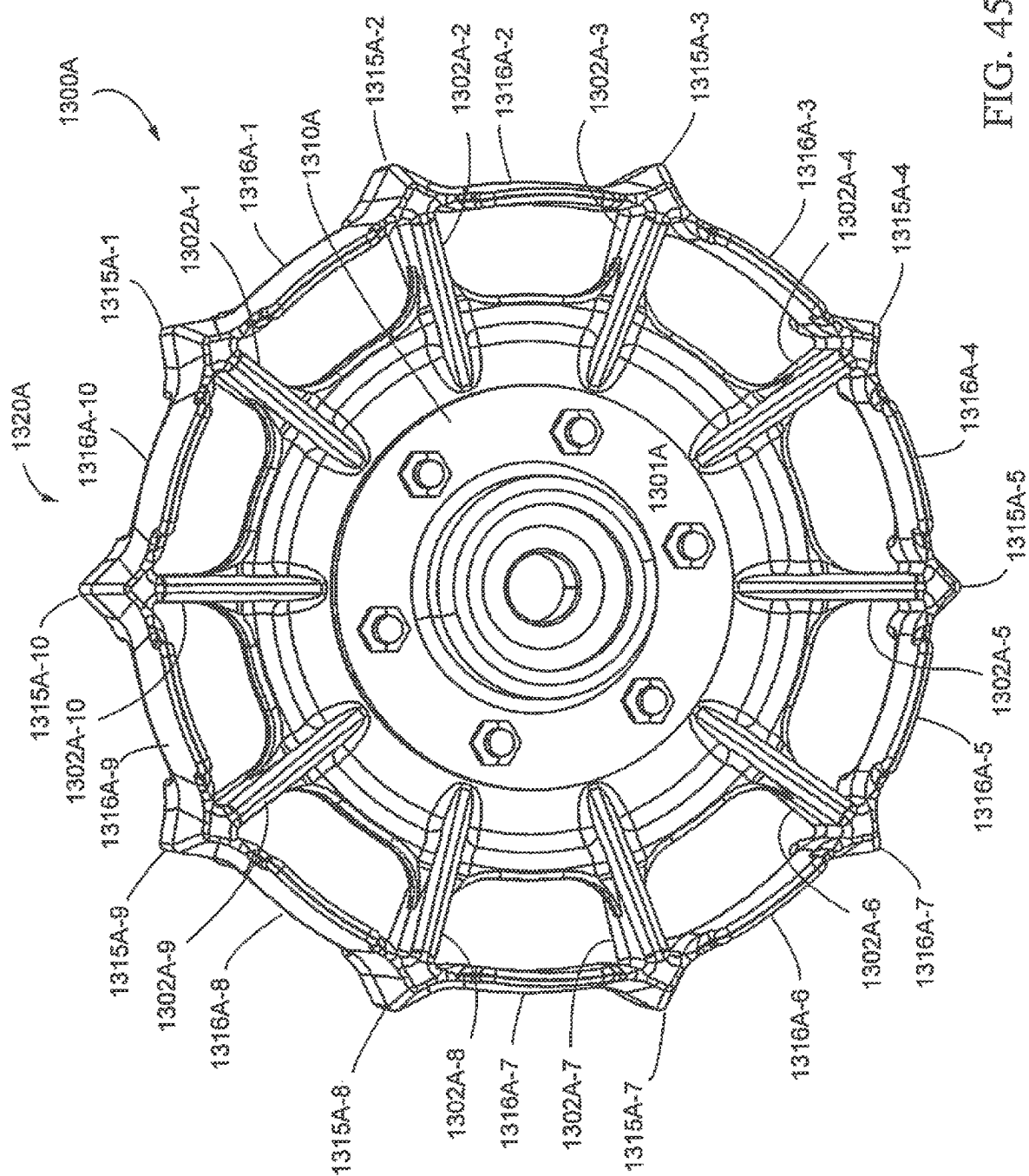
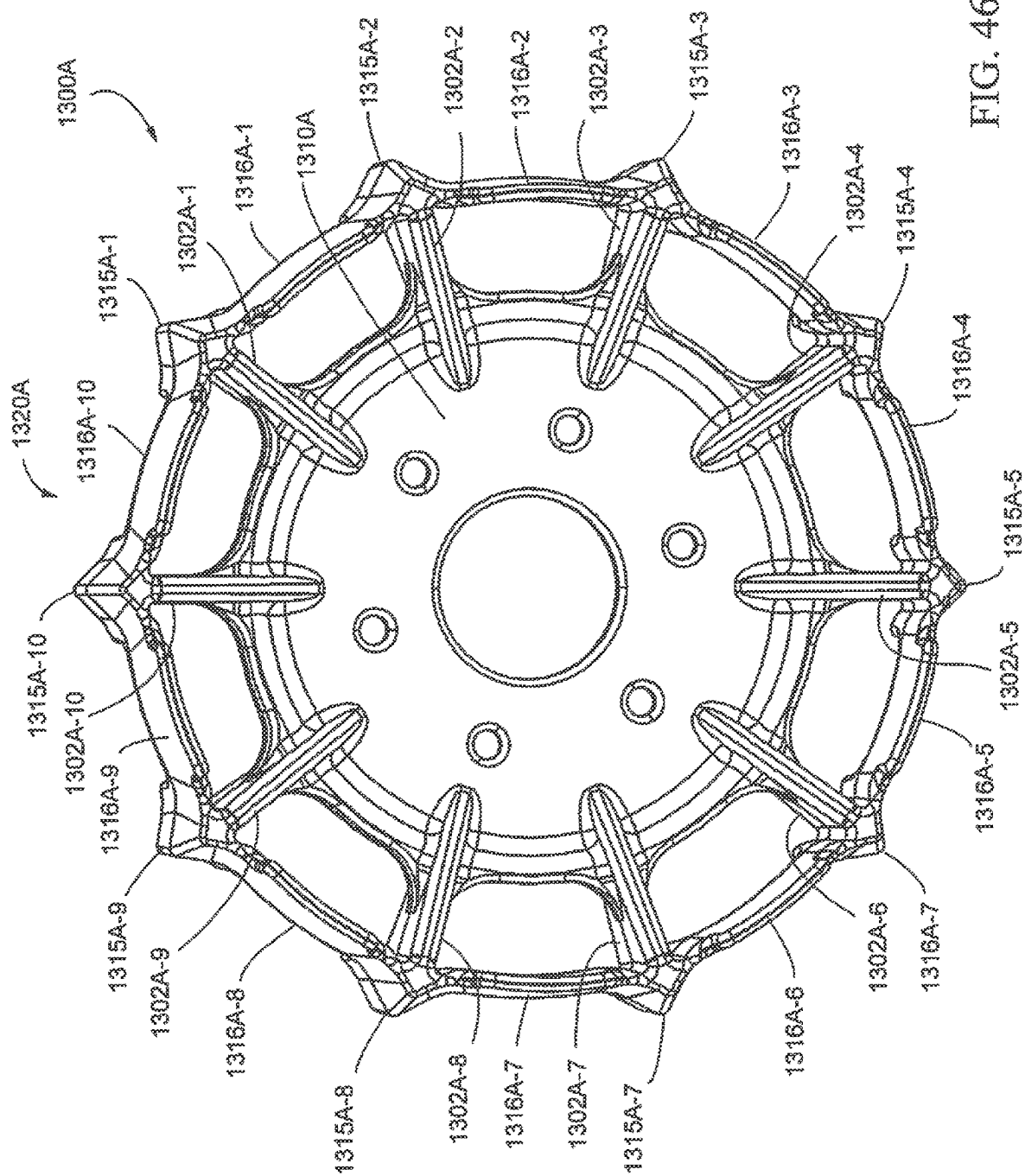
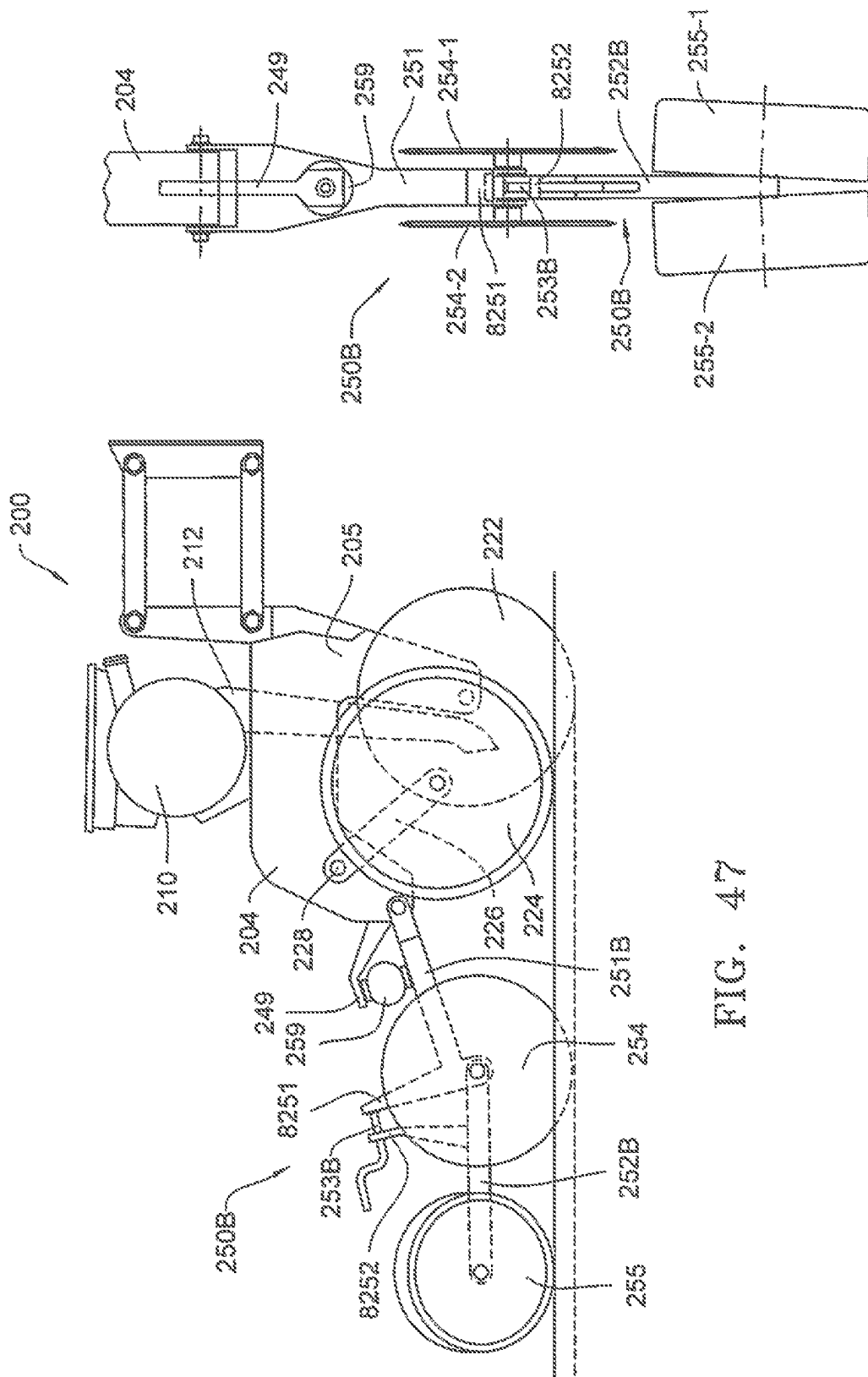


FIG. 45





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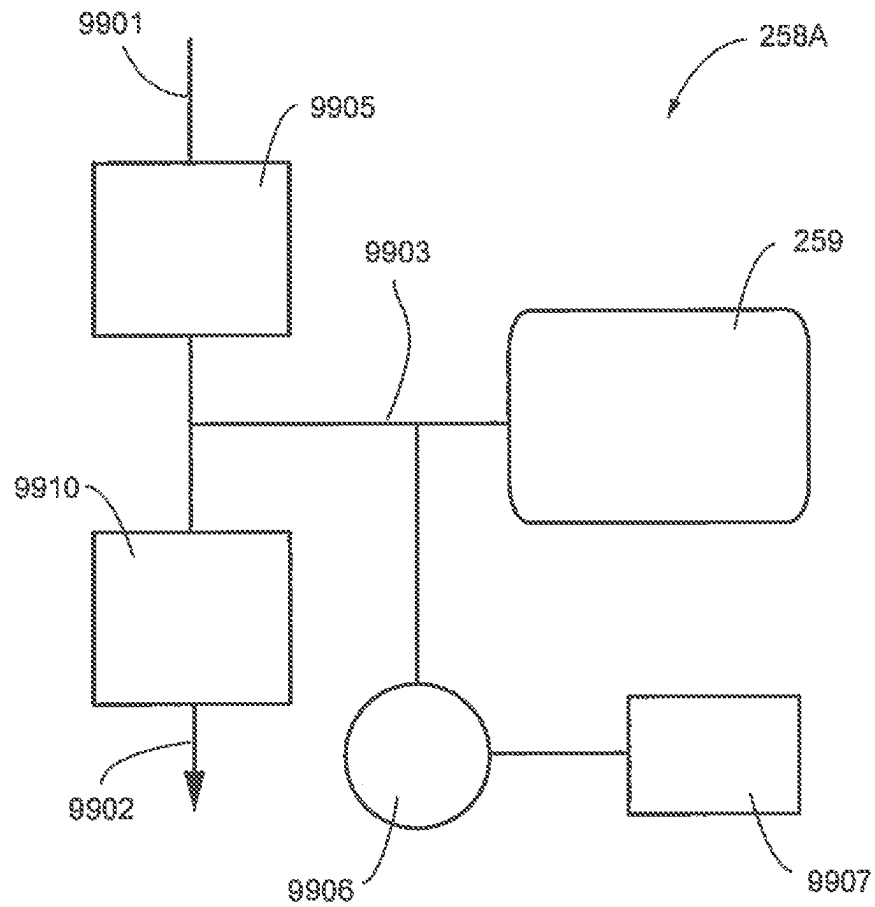


FIG. 49

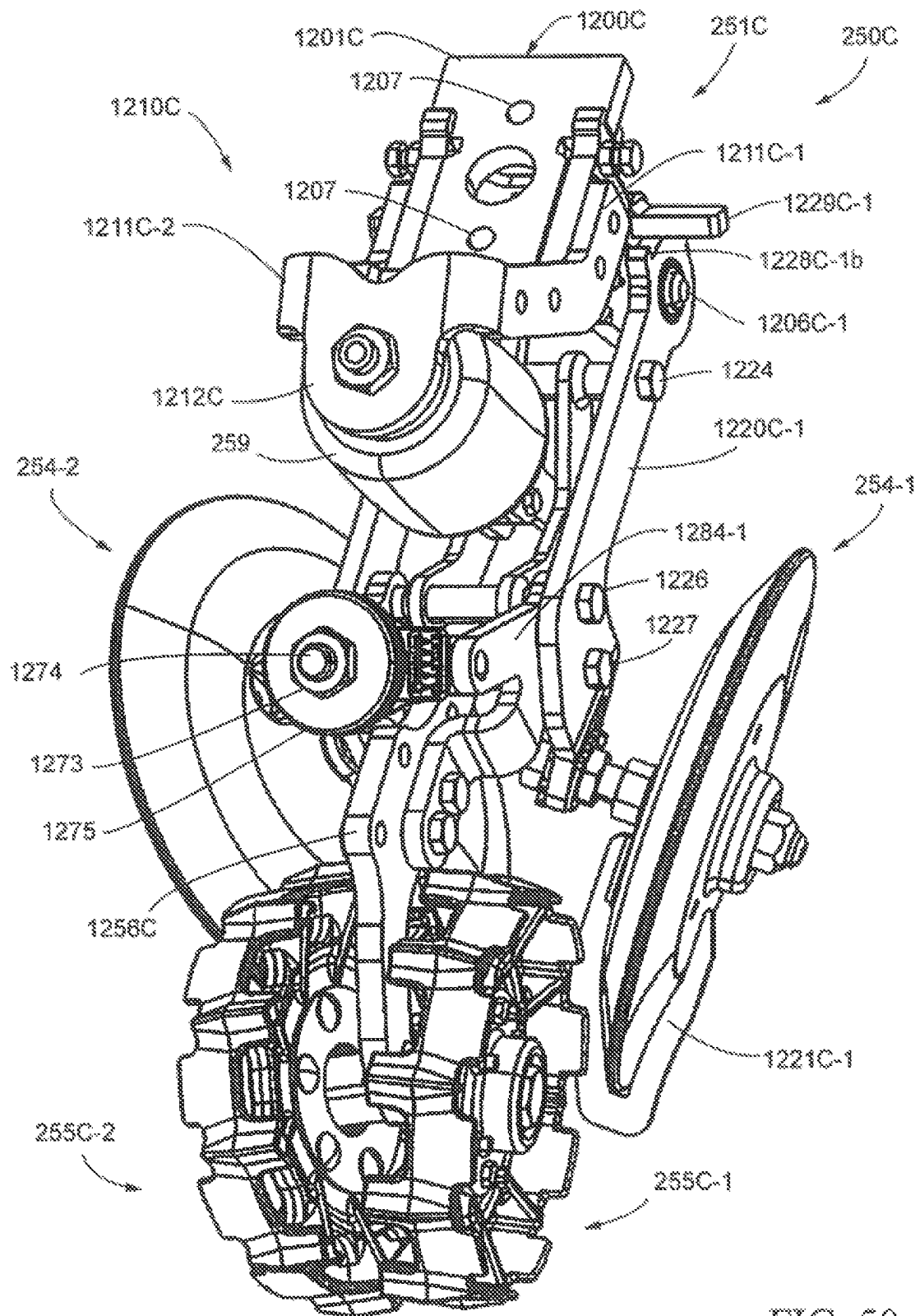


FIG. 50

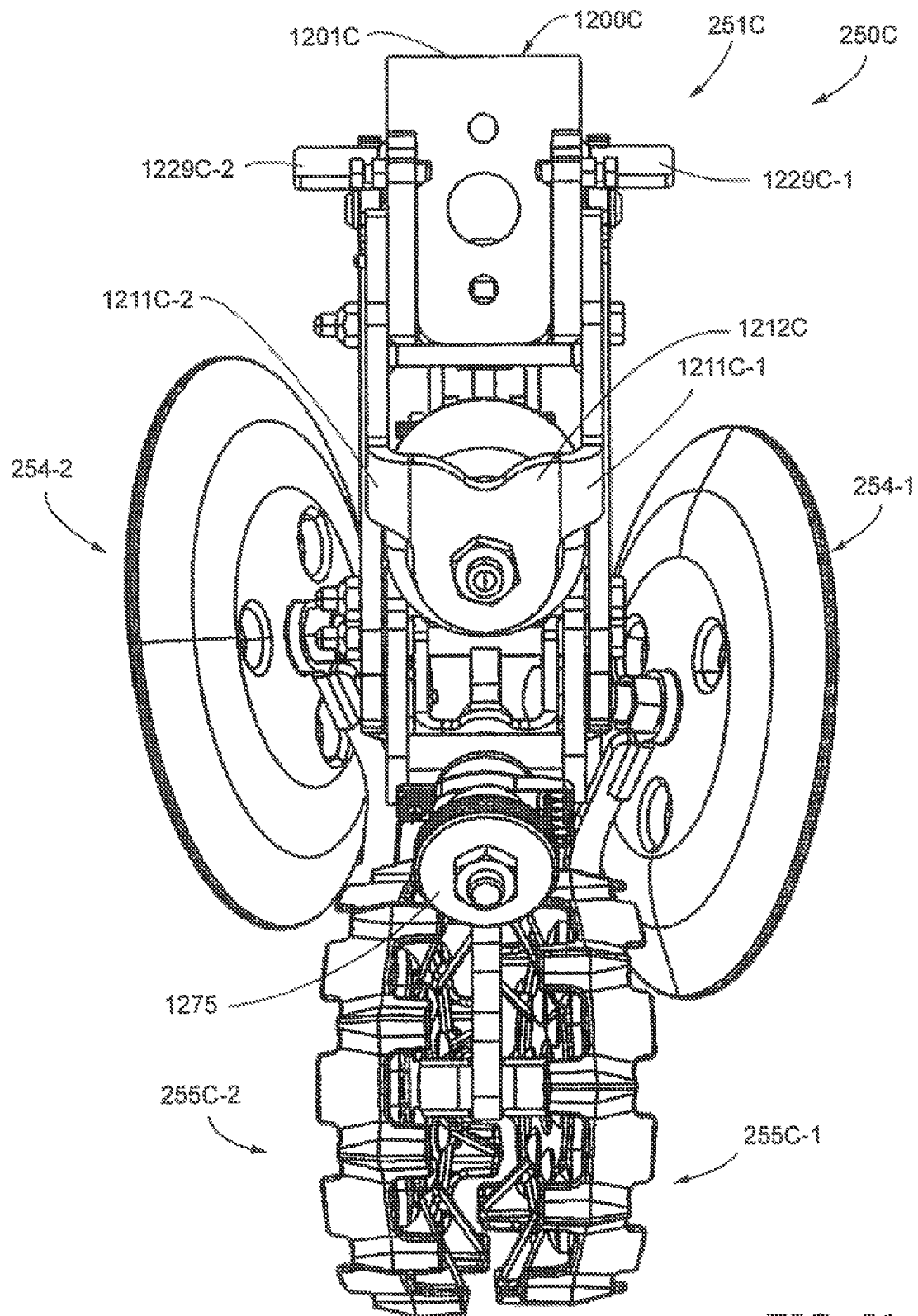
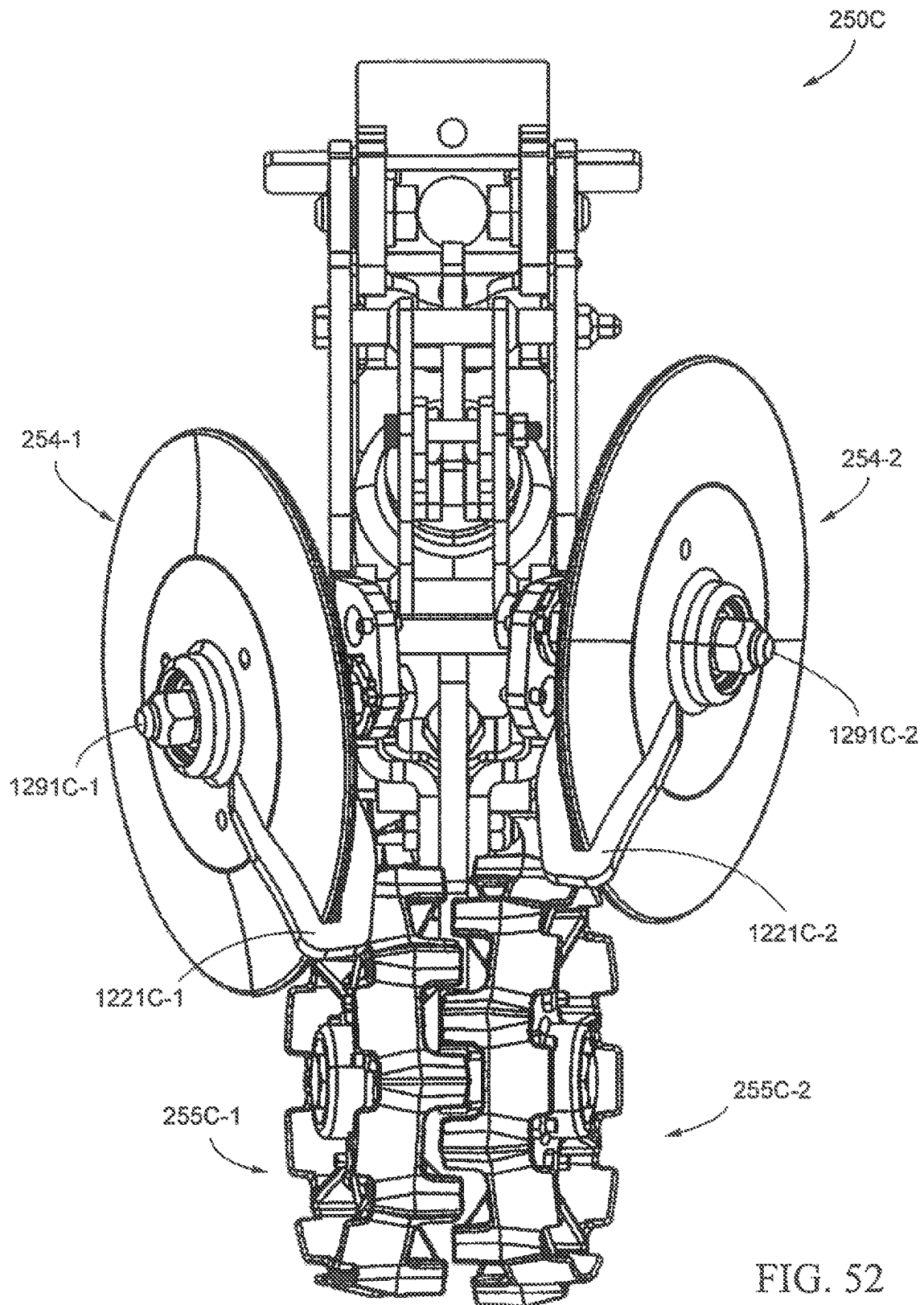


FIG. 51



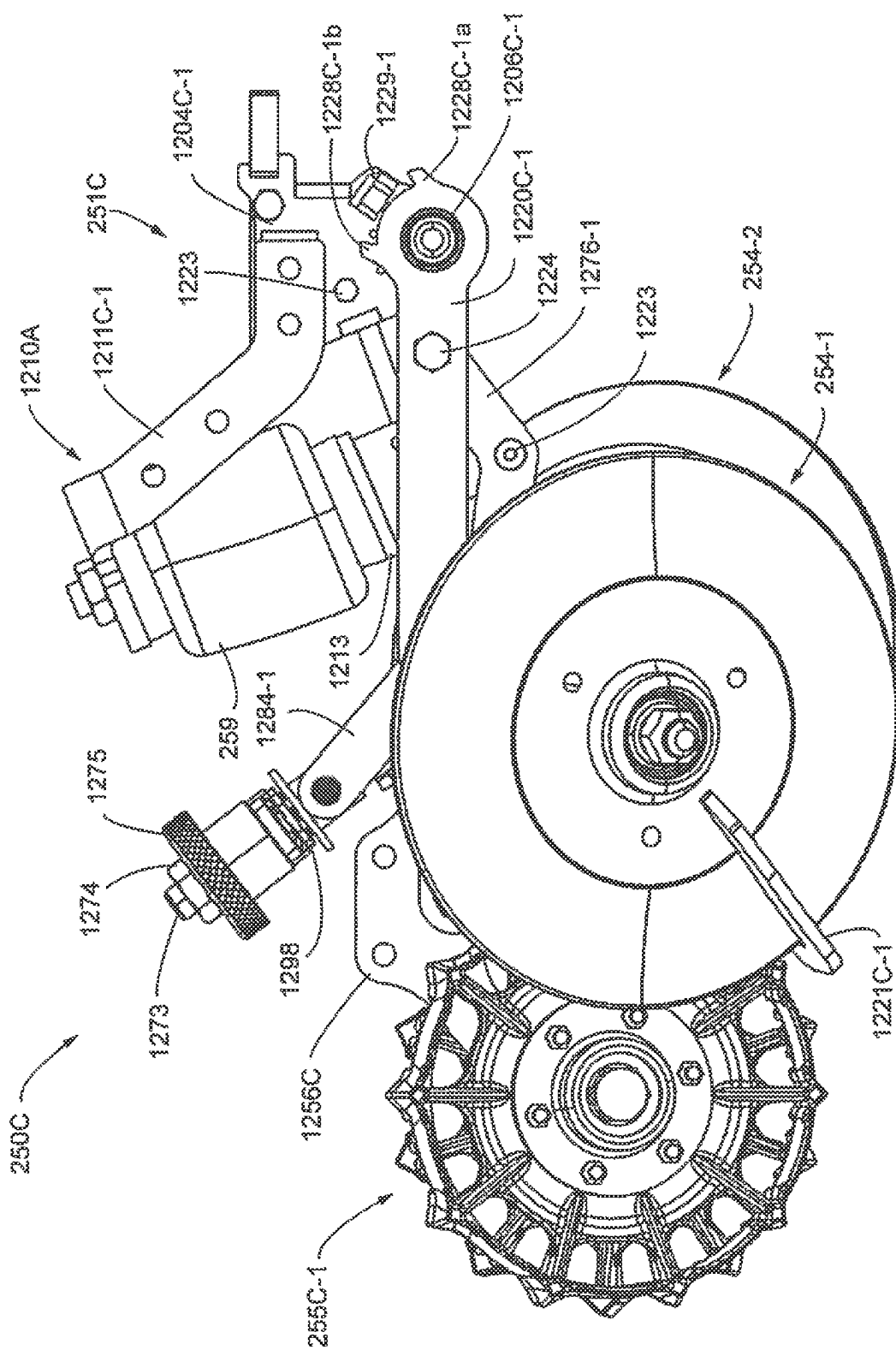


FIG. 53

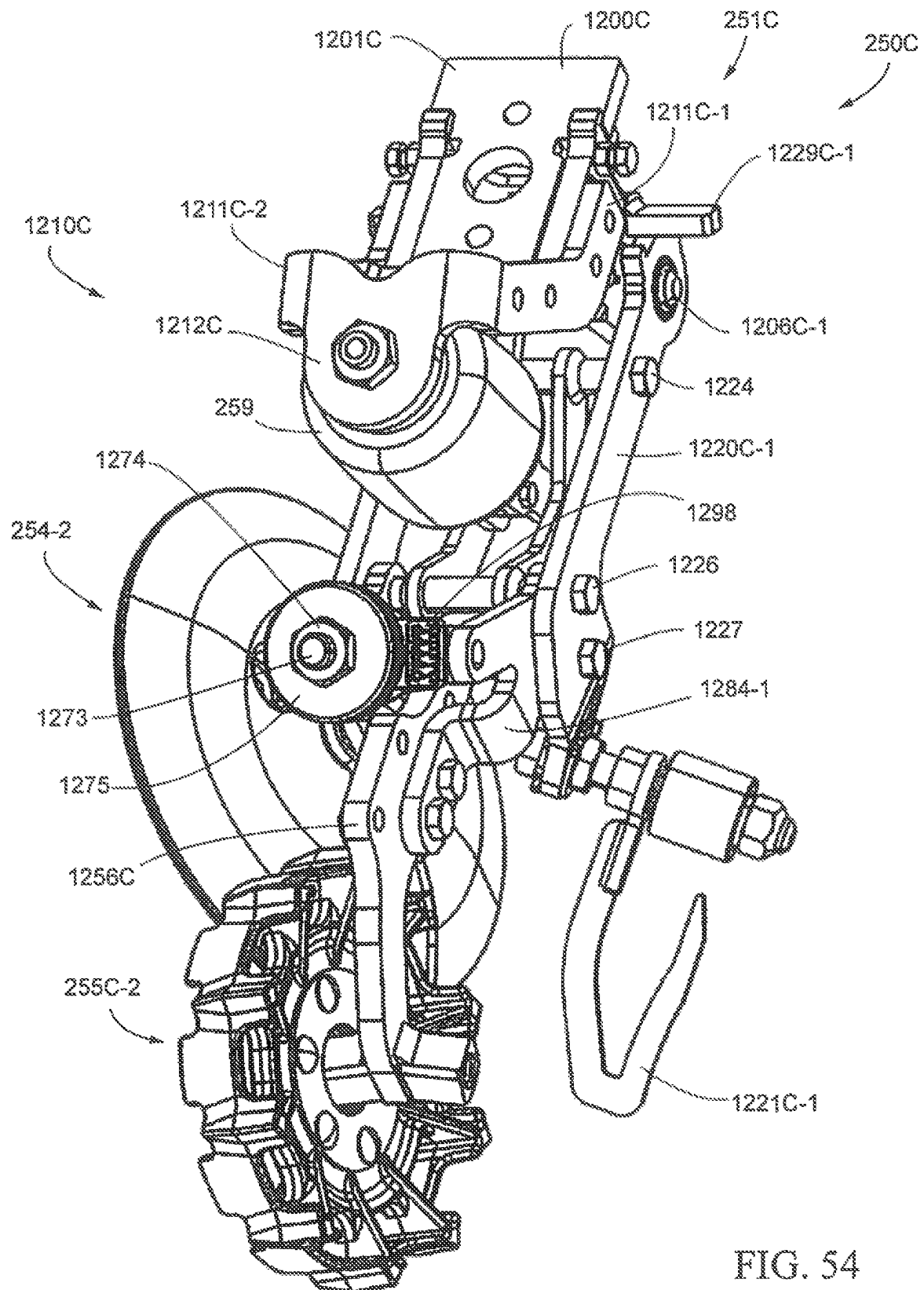


FIG. 54

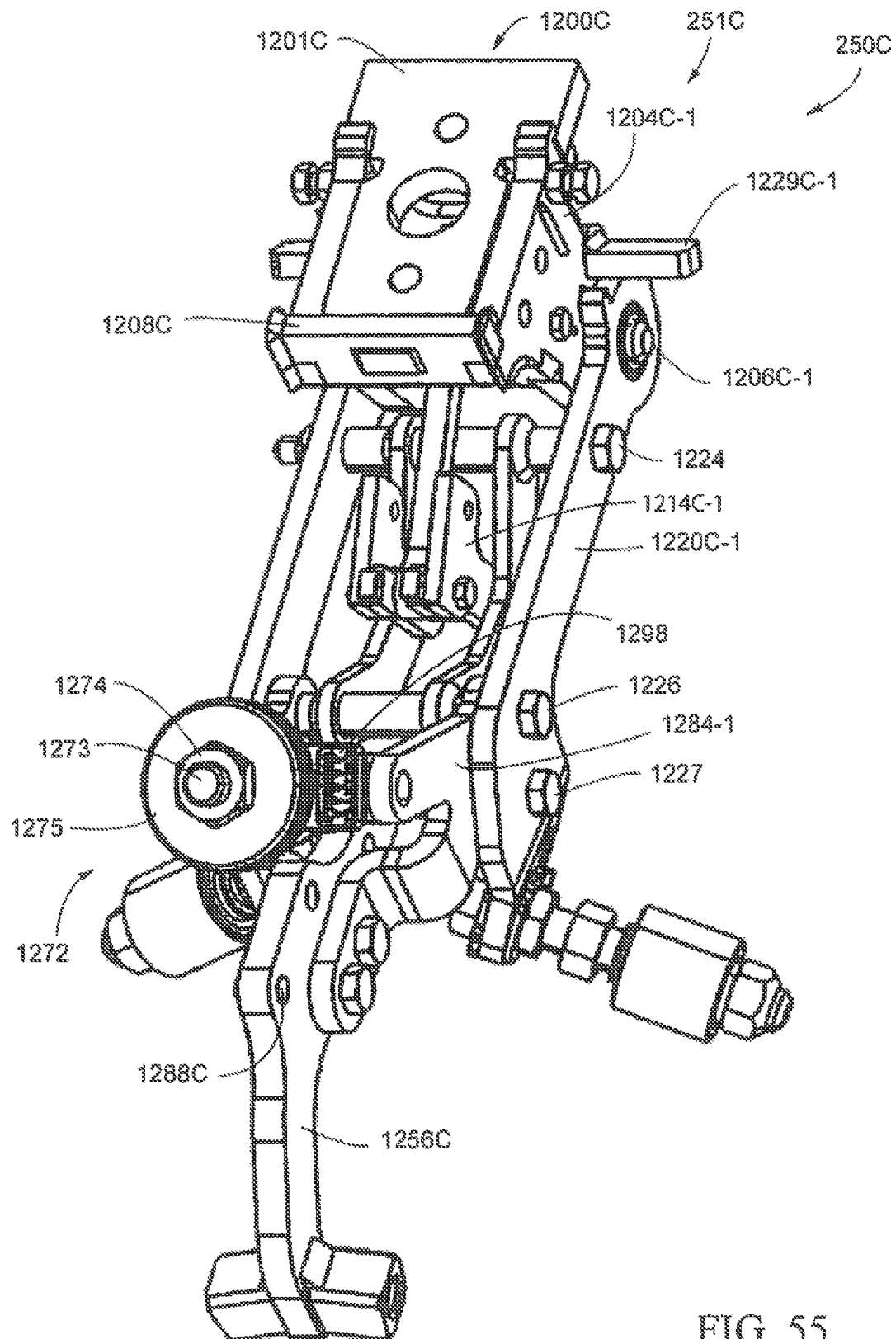


FIG. 55

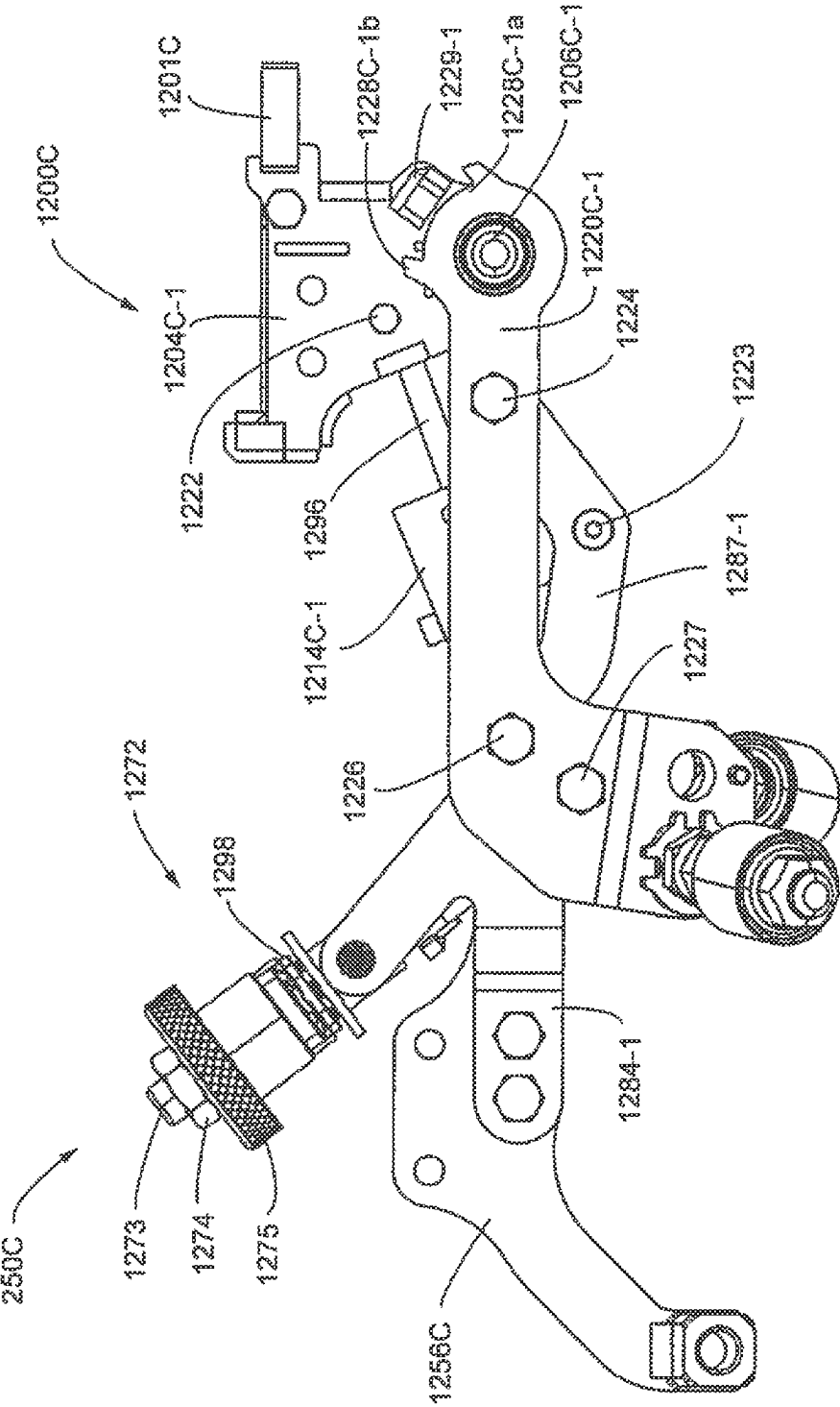


FIG. 56

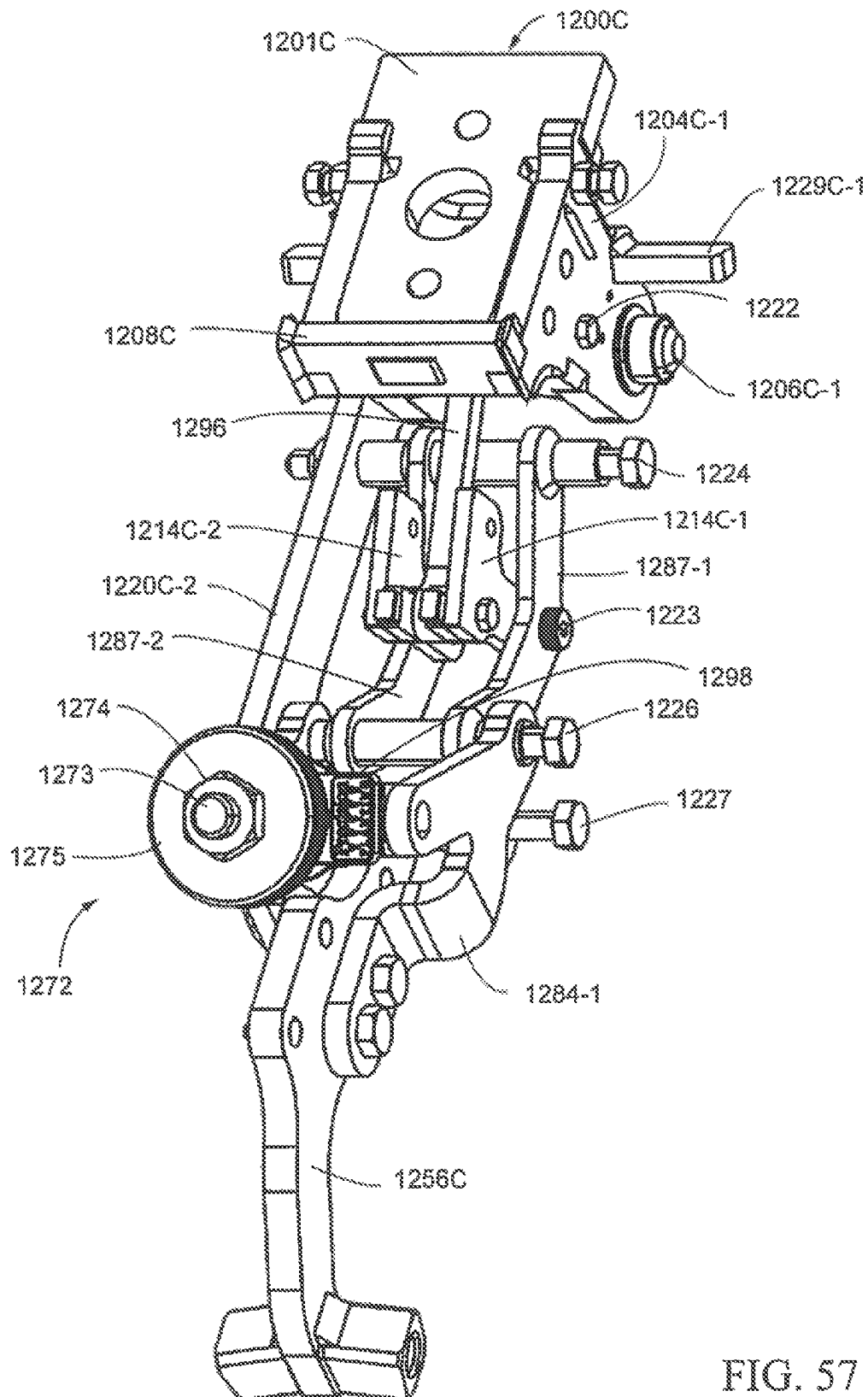


FIG. 57

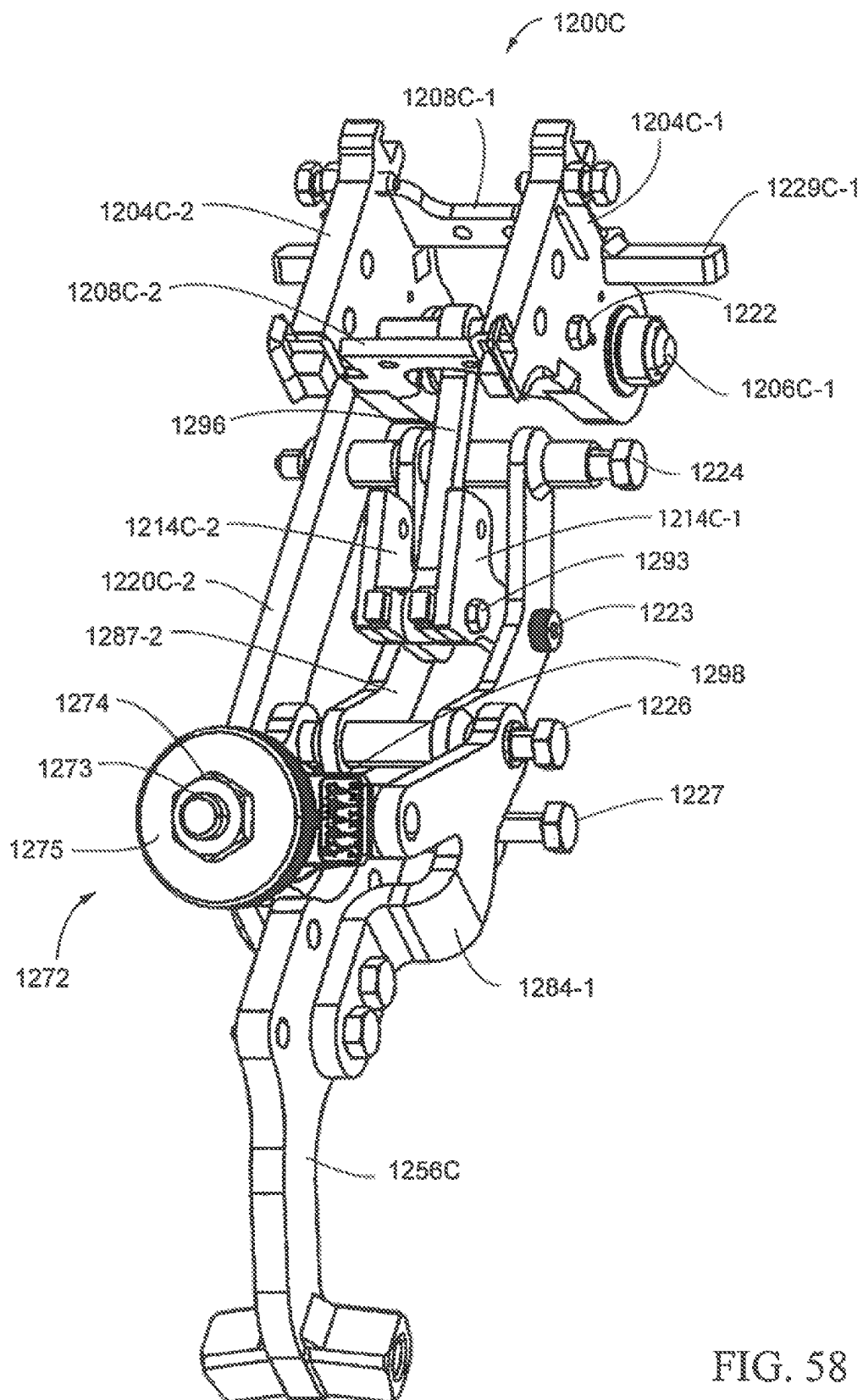


FIG. 58

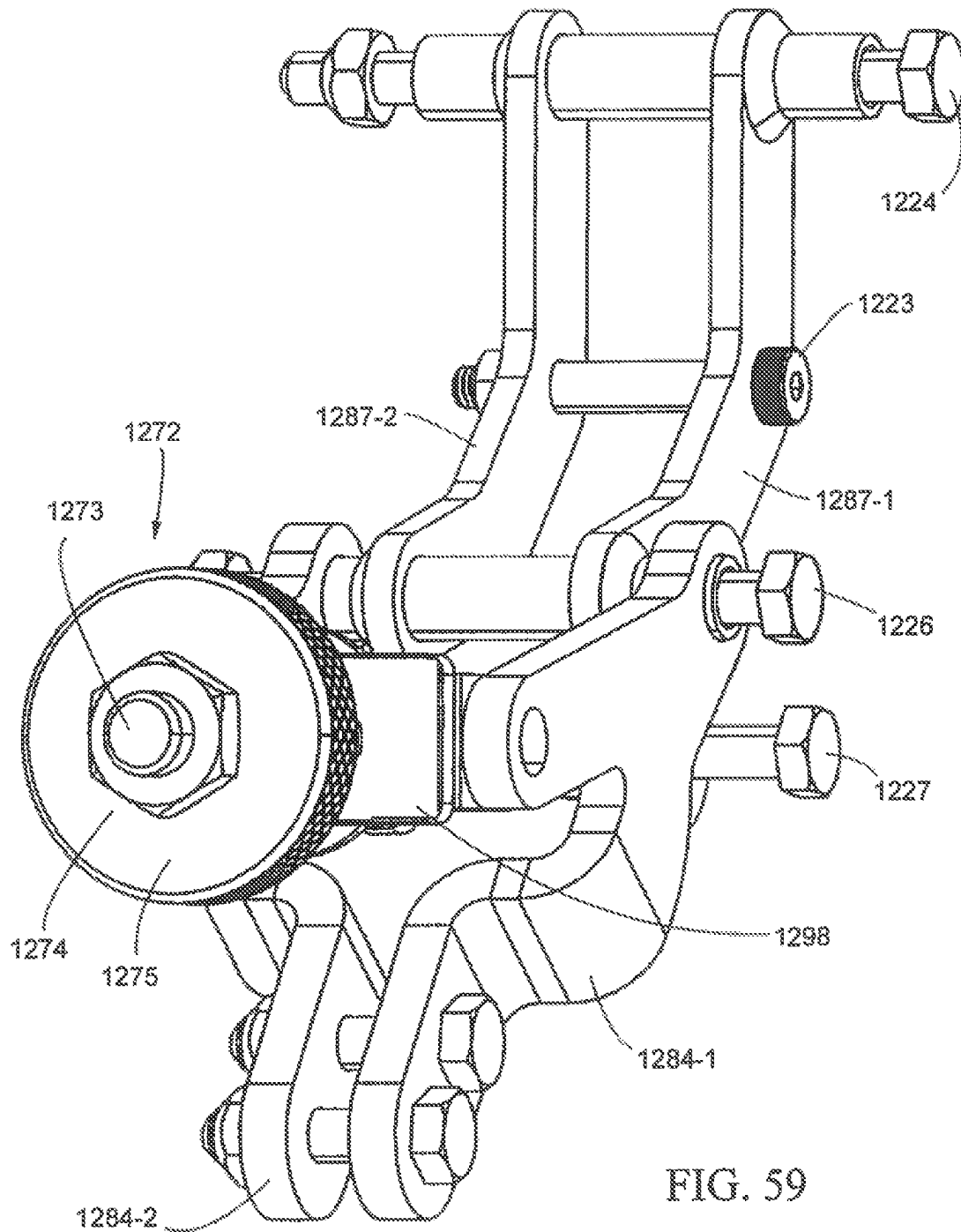


FIG. 59

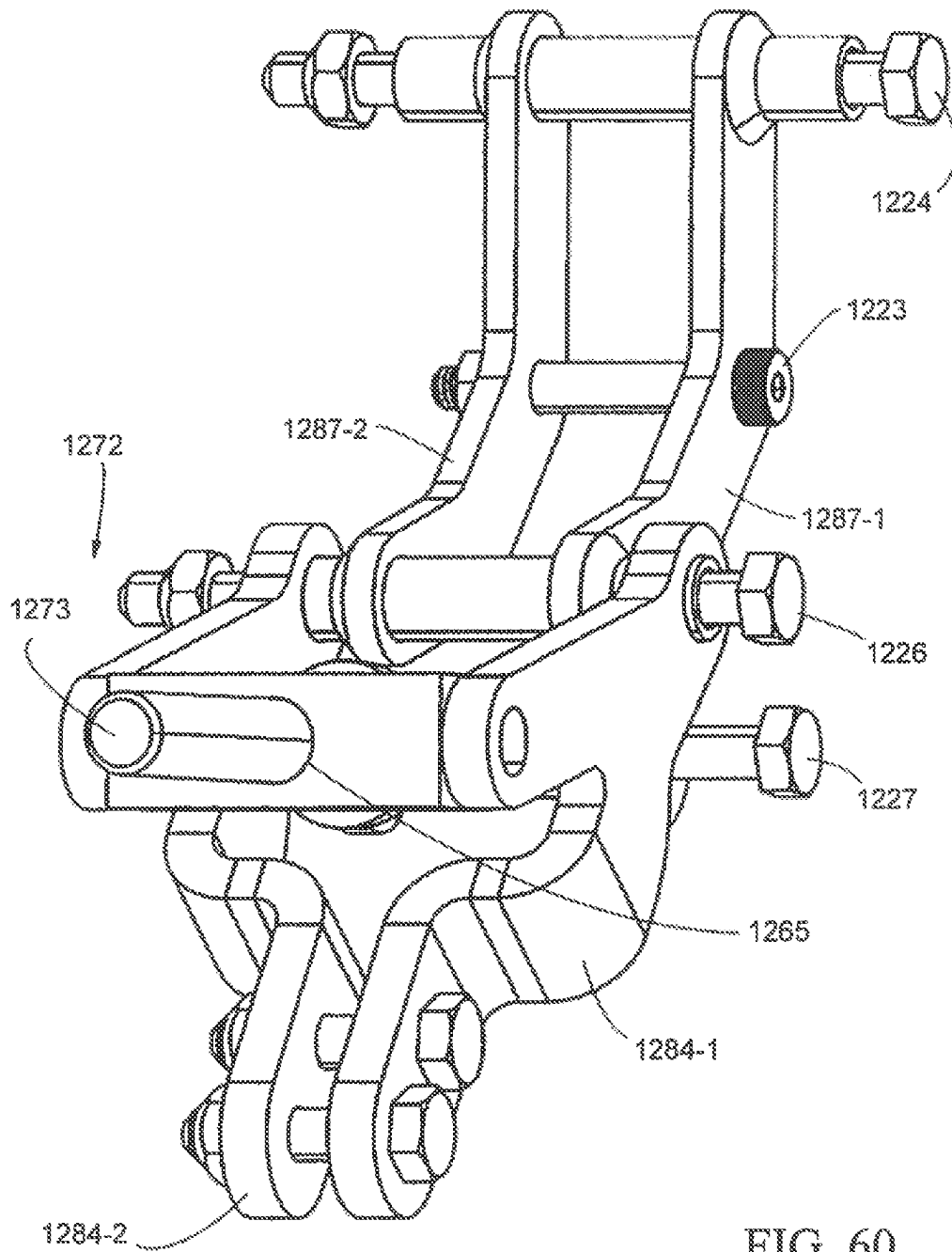


FIG. 60

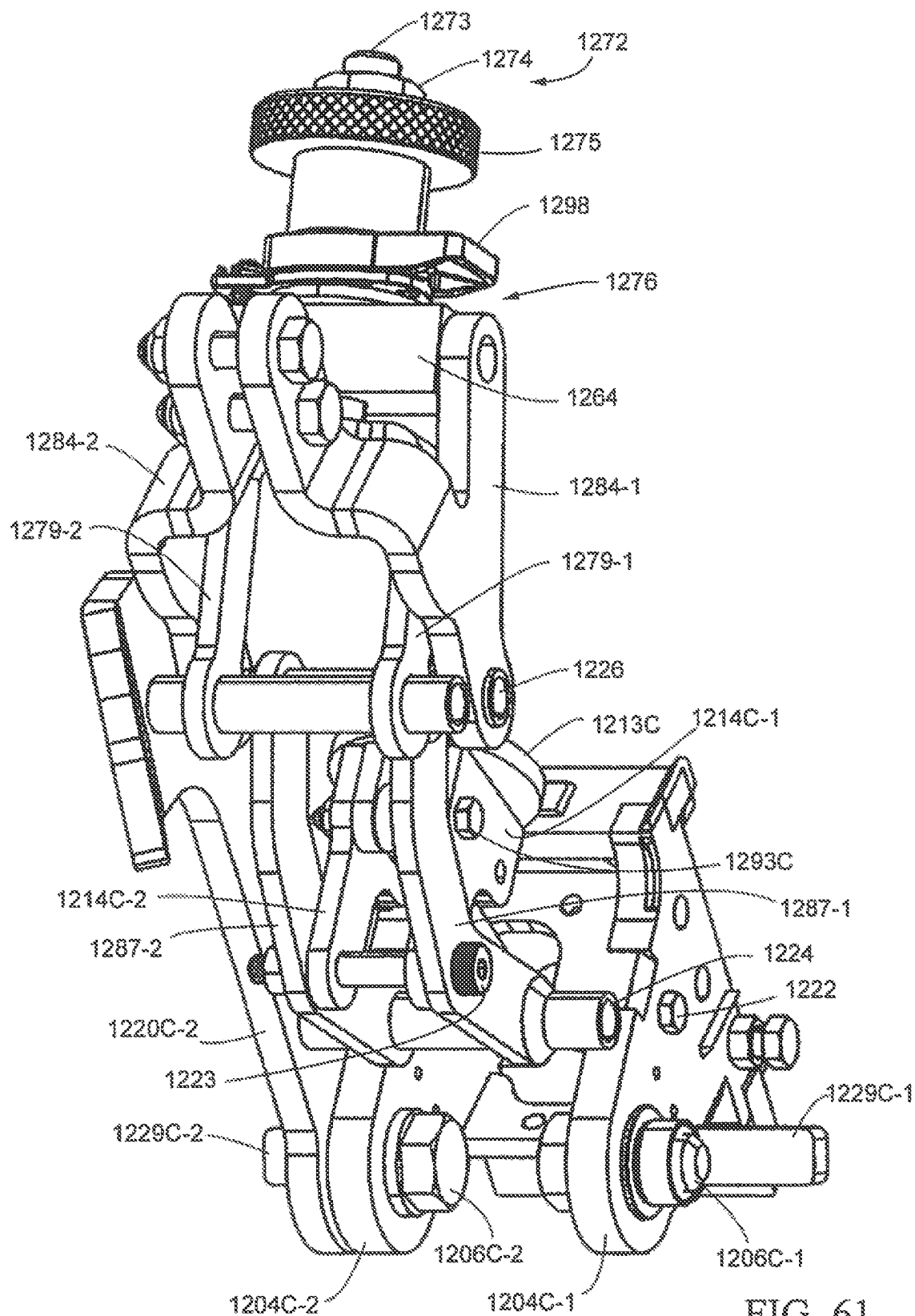


FIG. 61

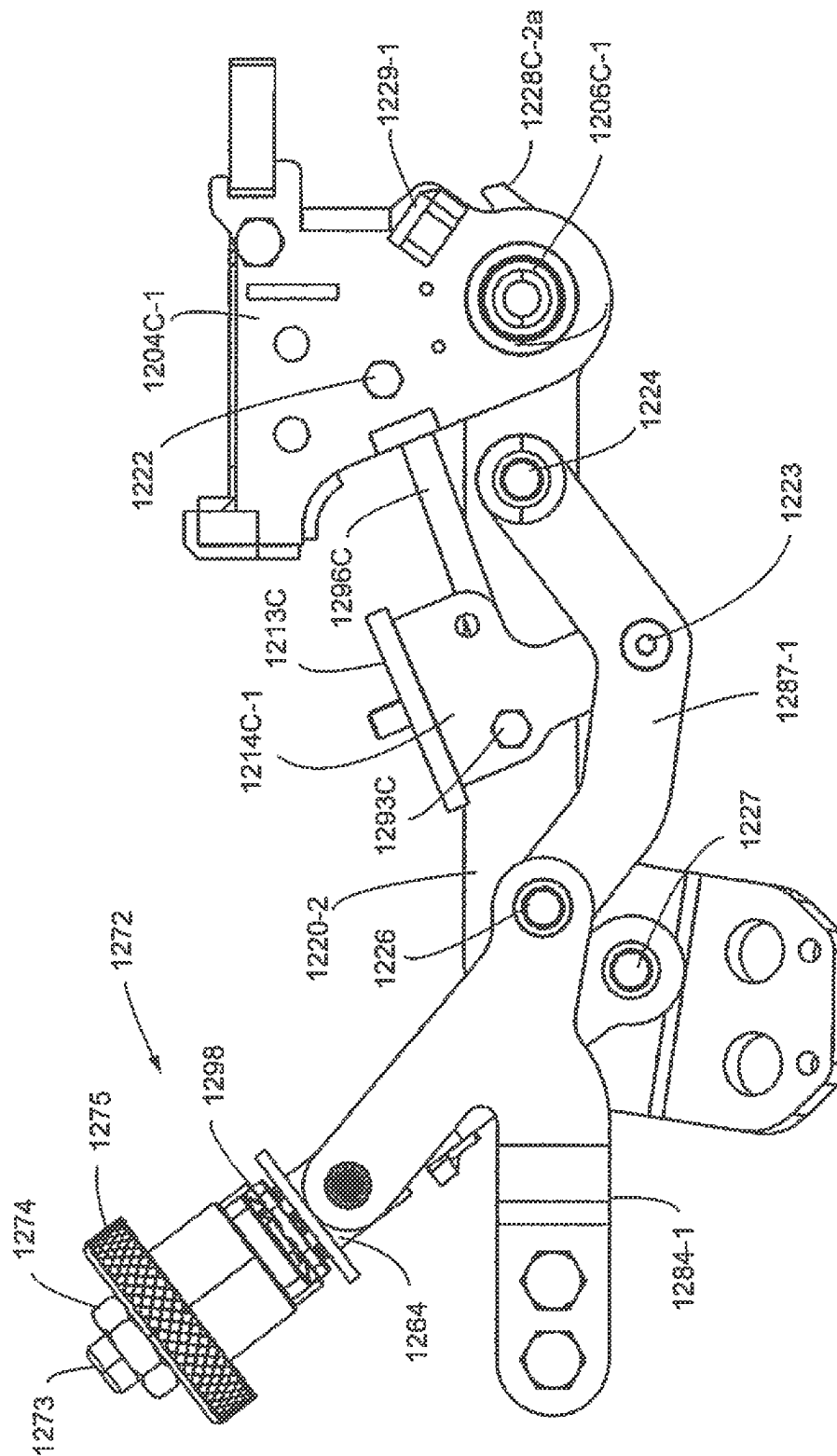


Fig. 2



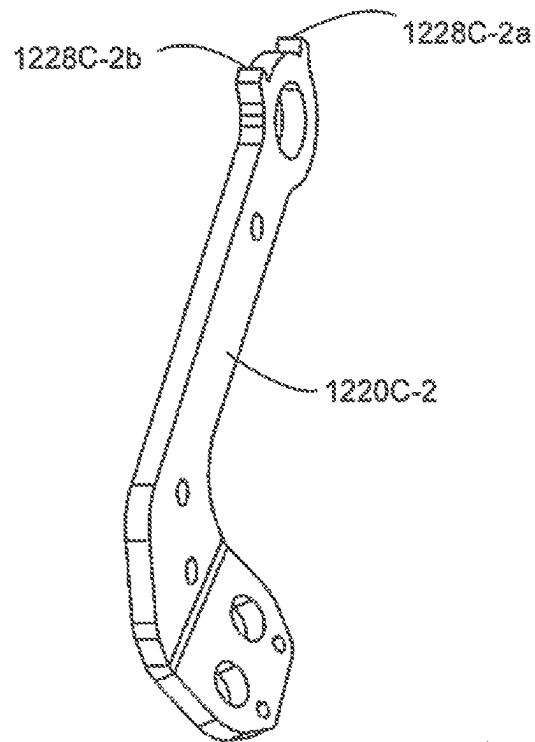


FIG. 64

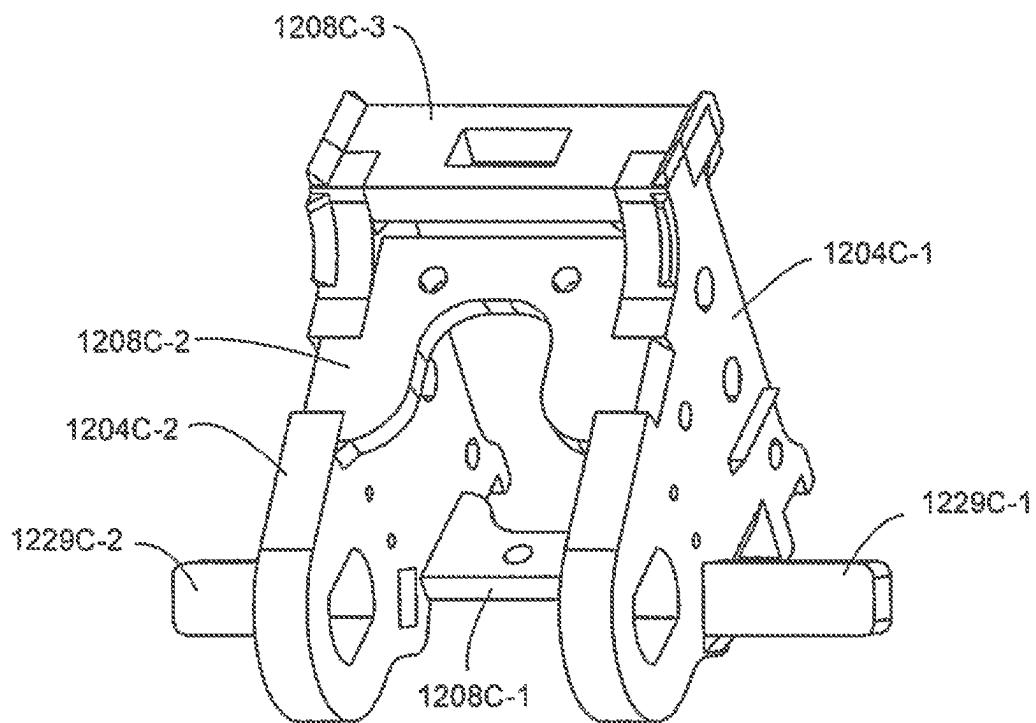


FIG. 65

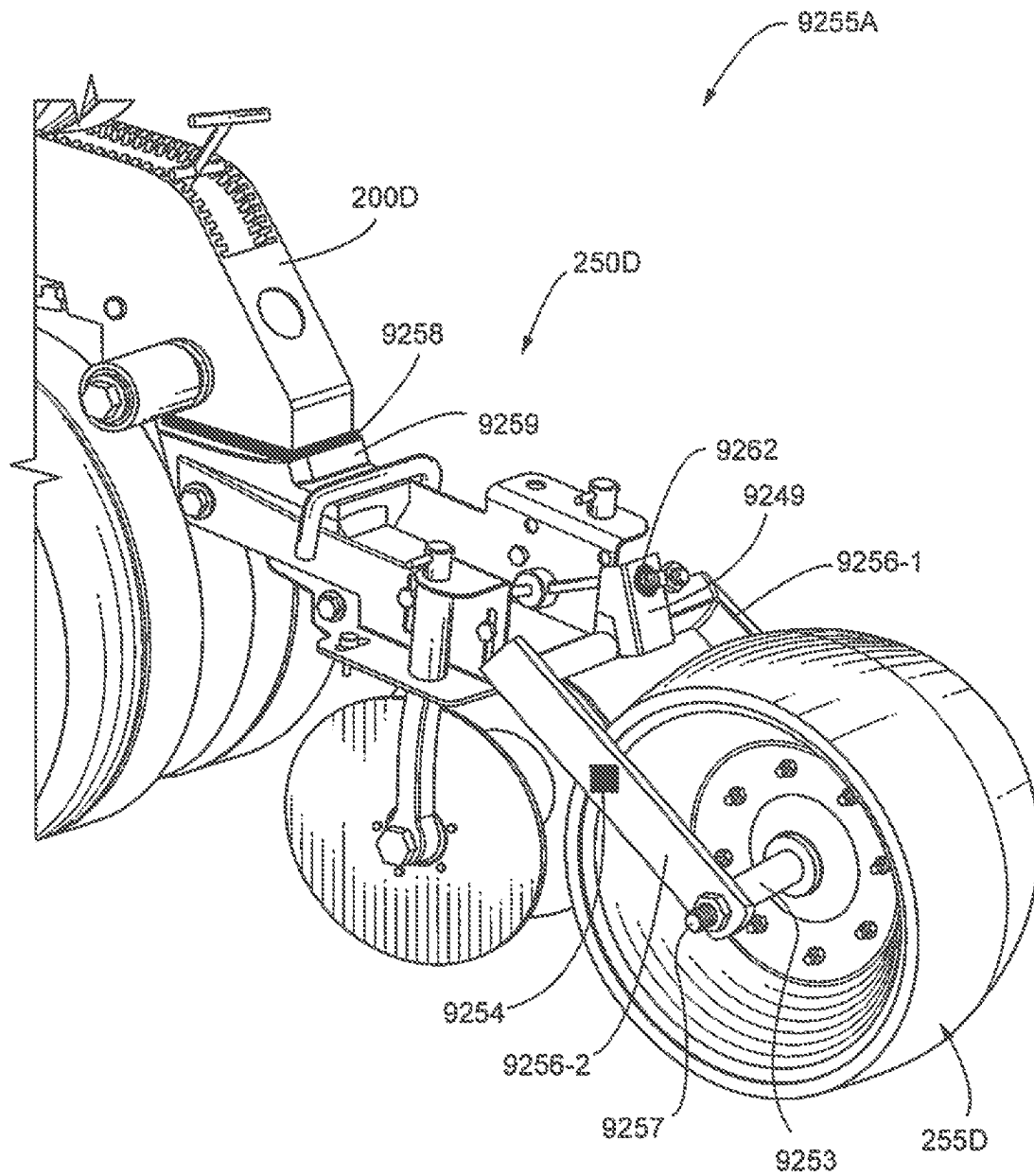


FIG. 66

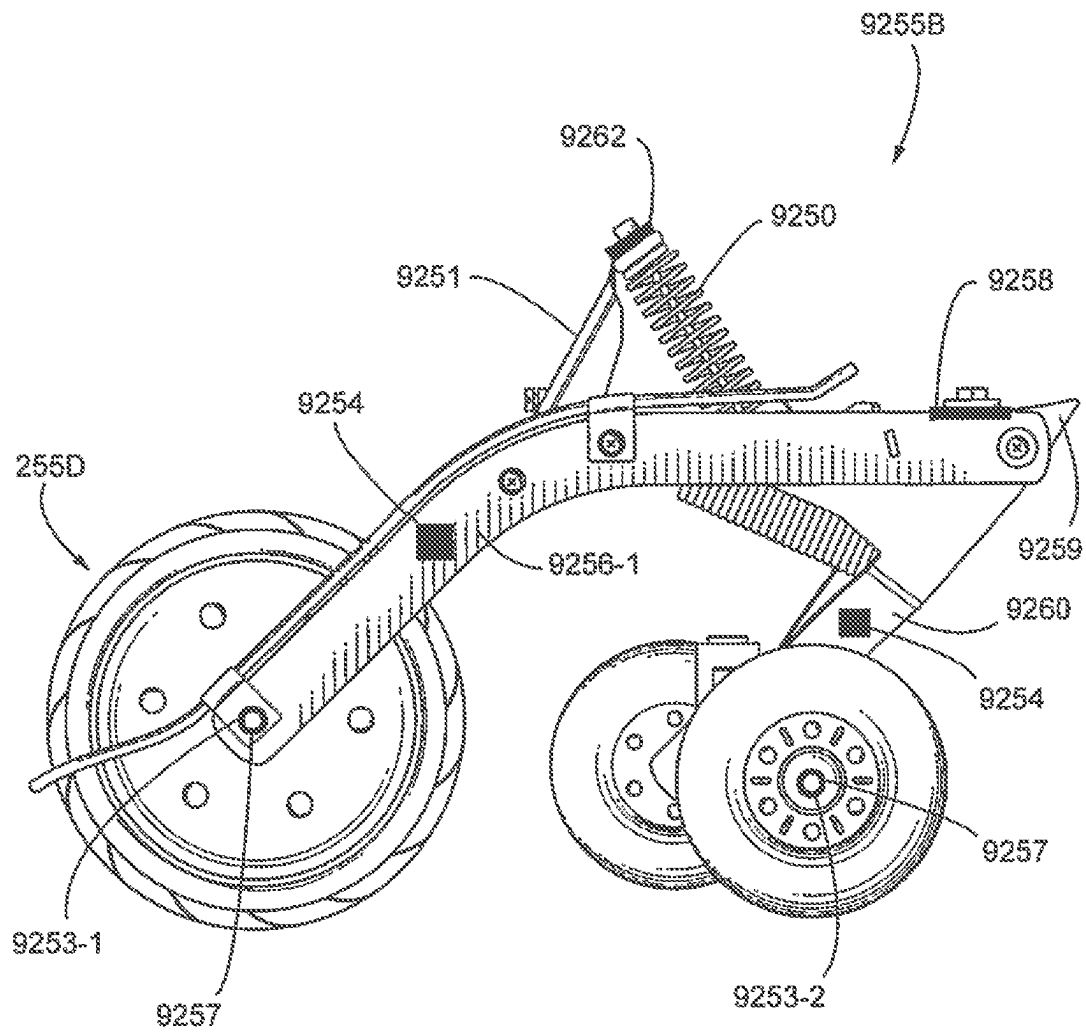


FIG. 67

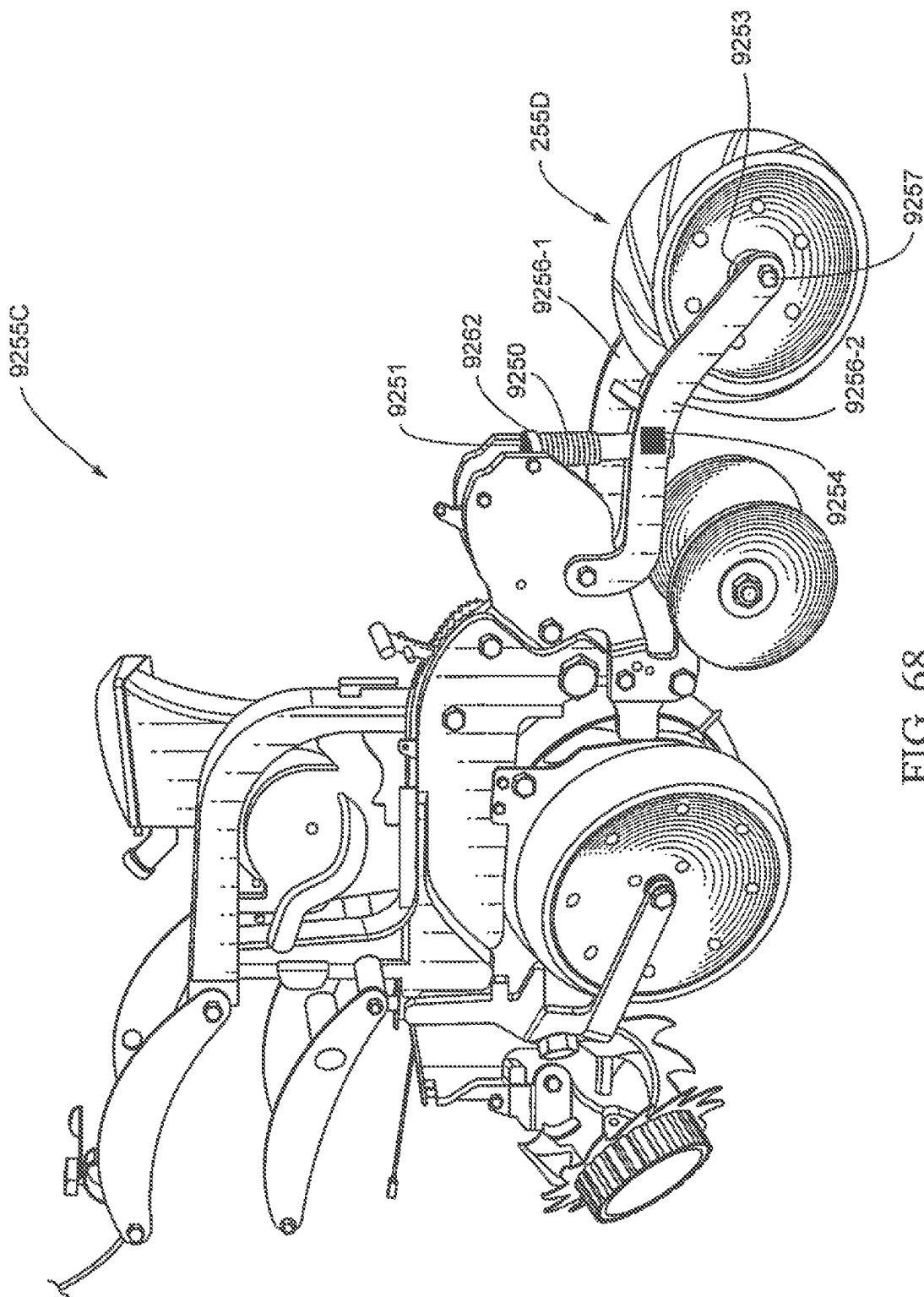


FIG. 68

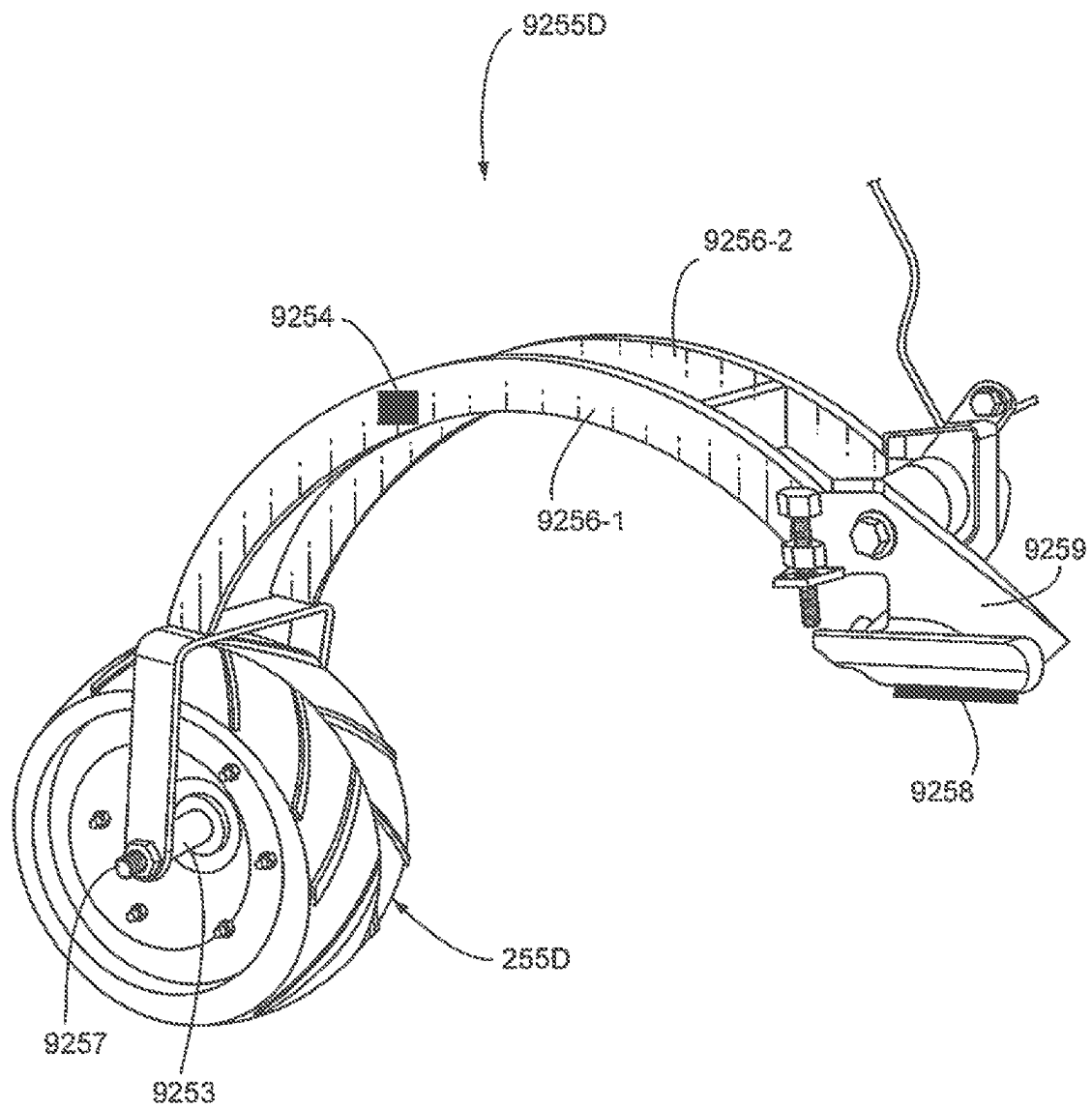


FIG. 69

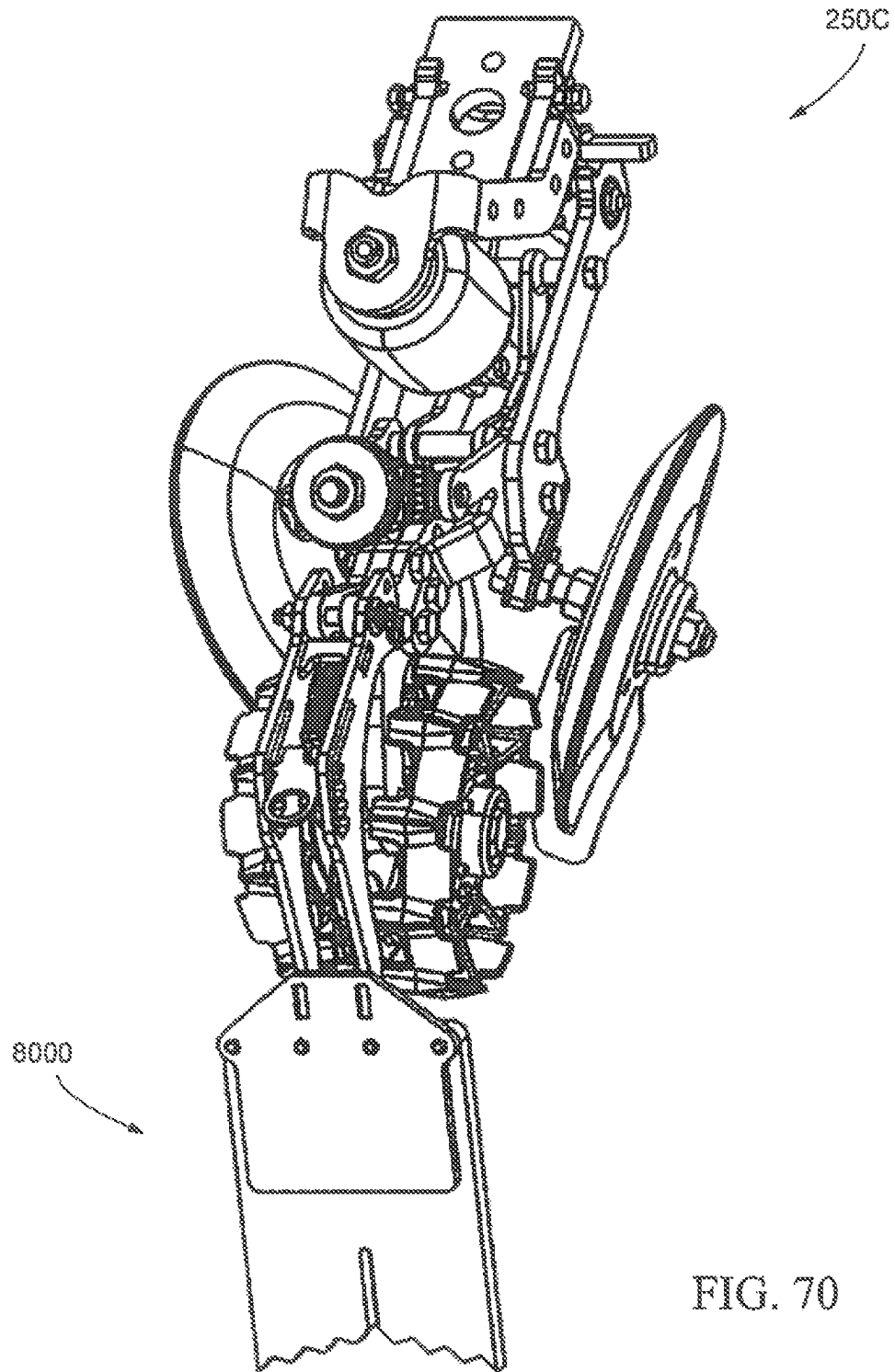


FIG. 70

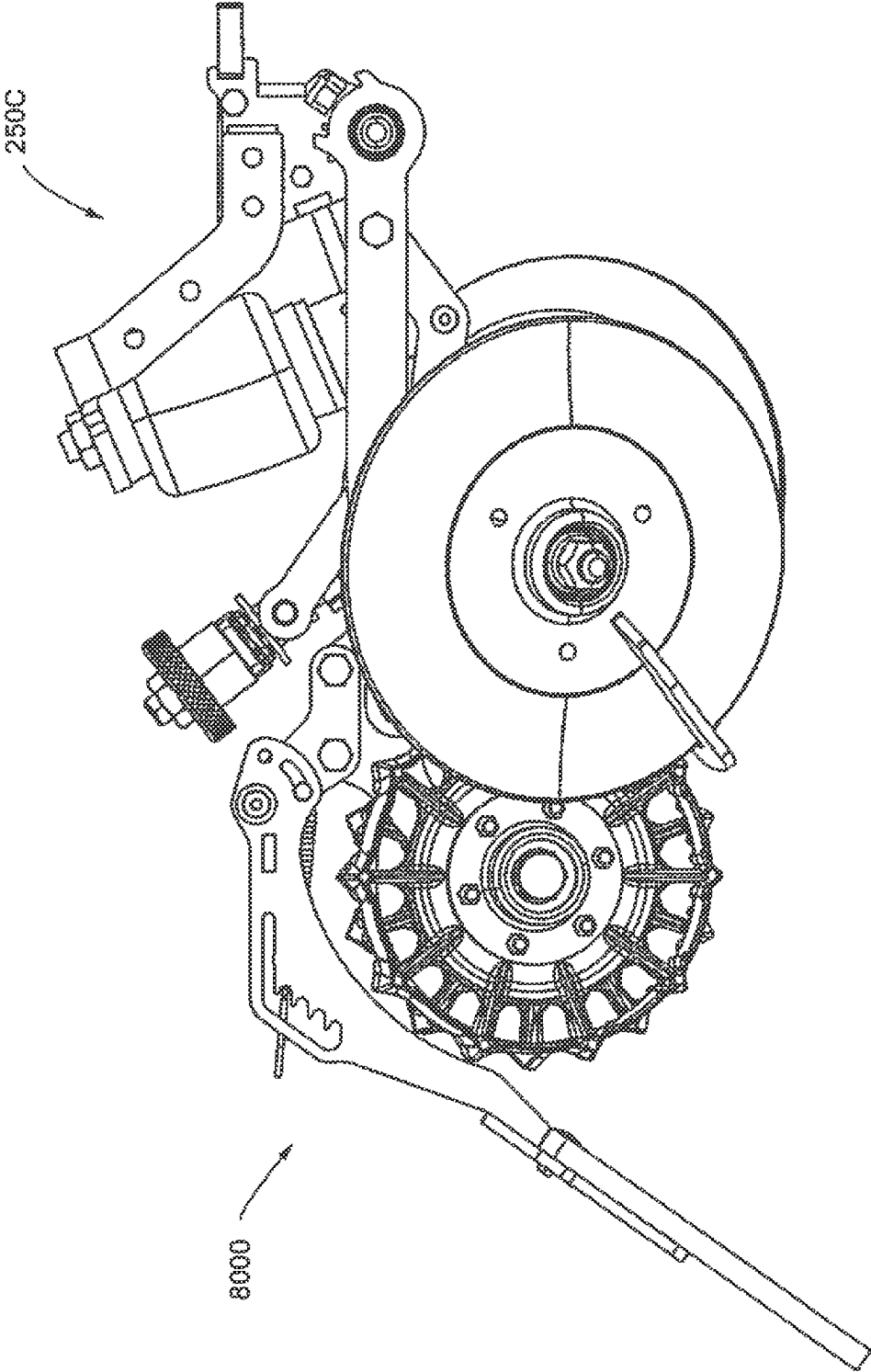
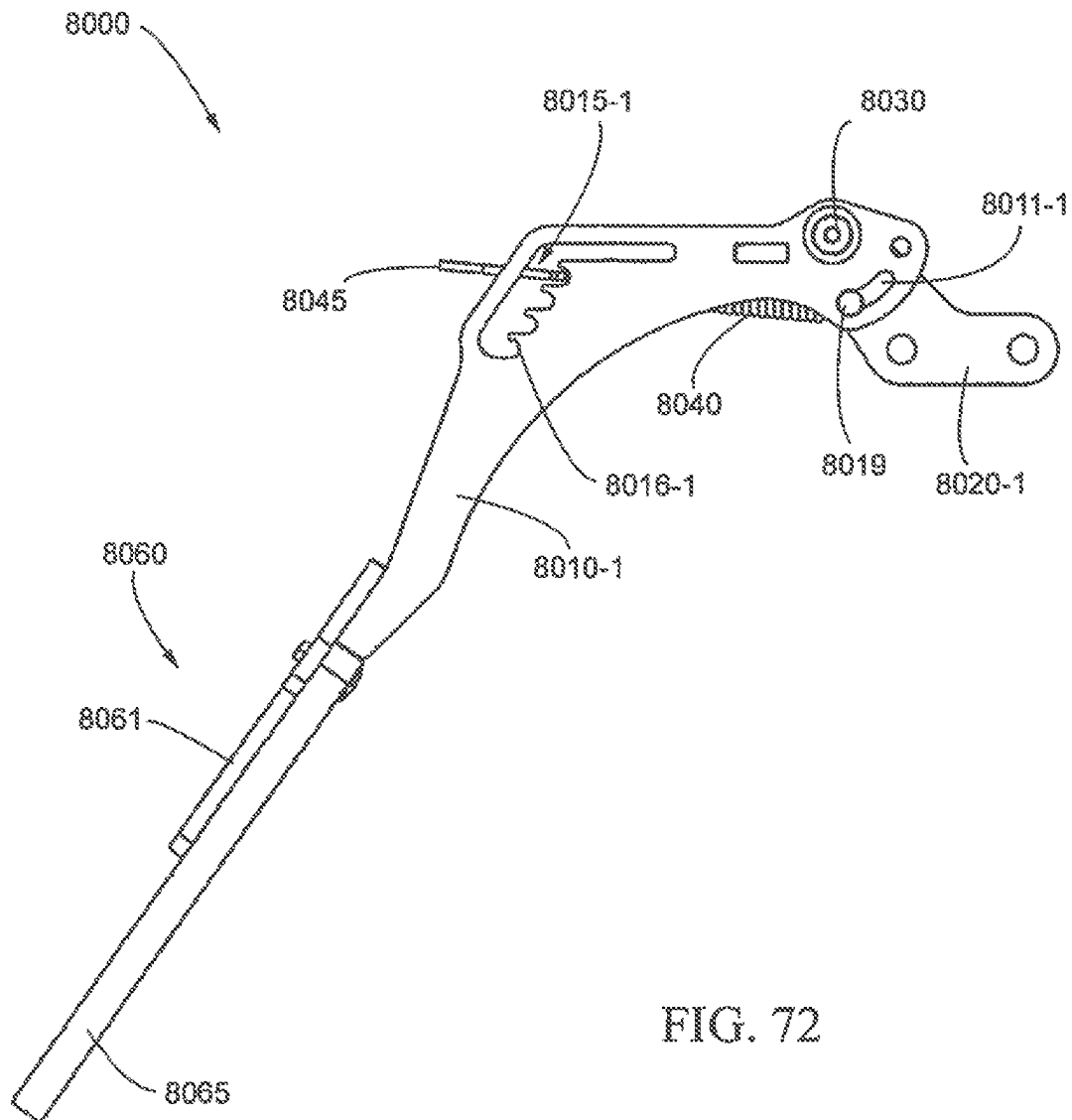
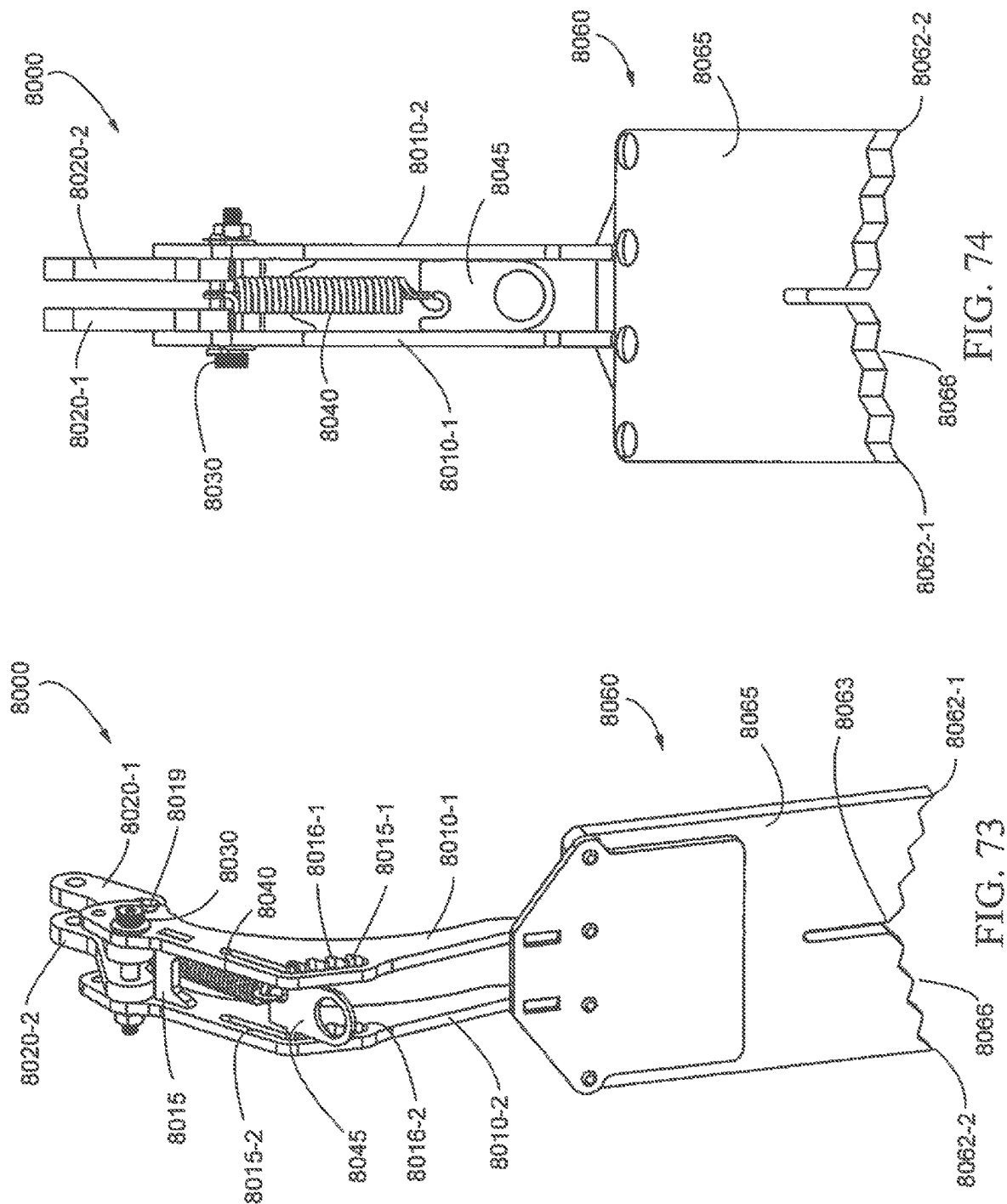


FIG. 71





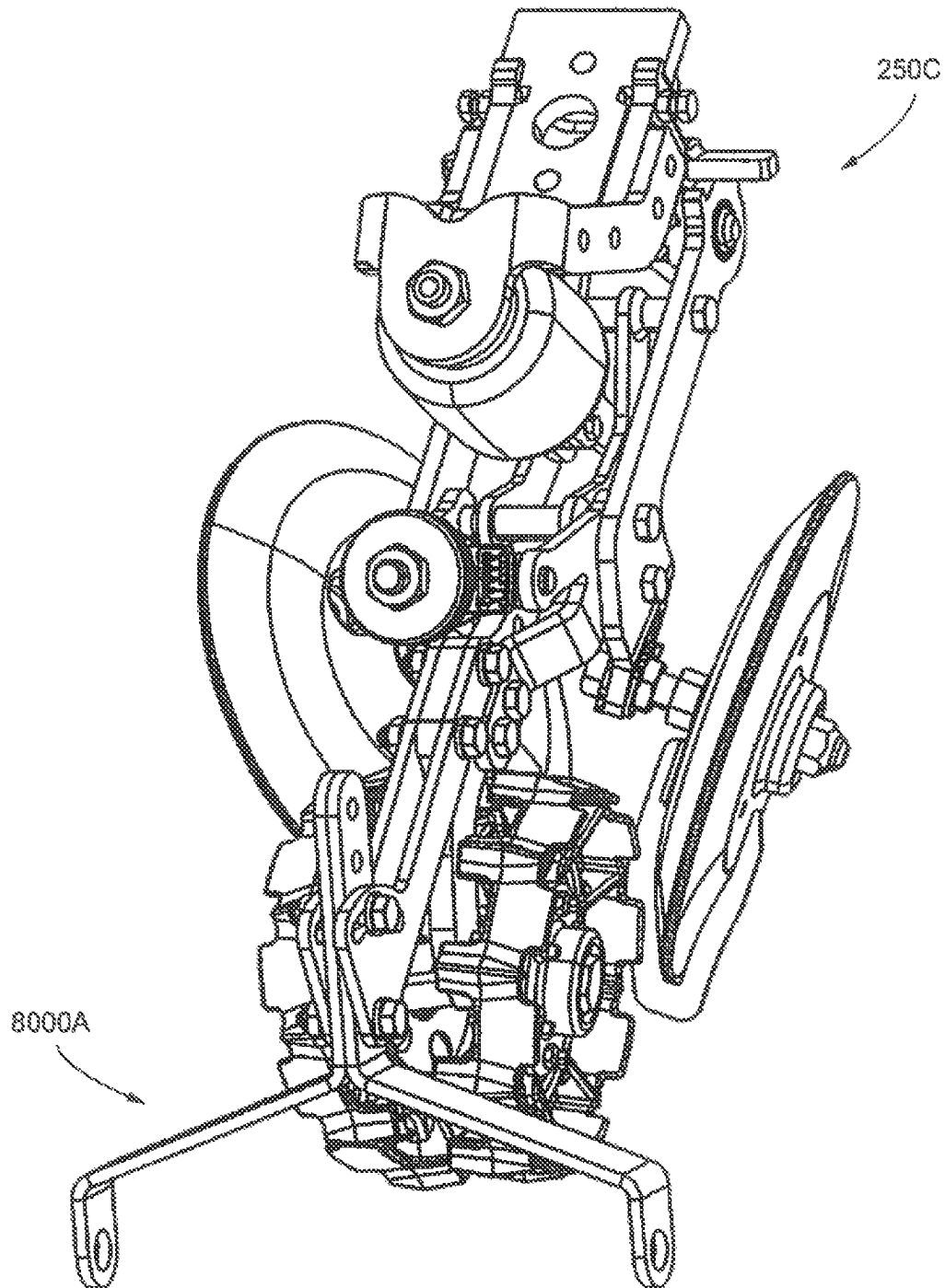


FIG. 75

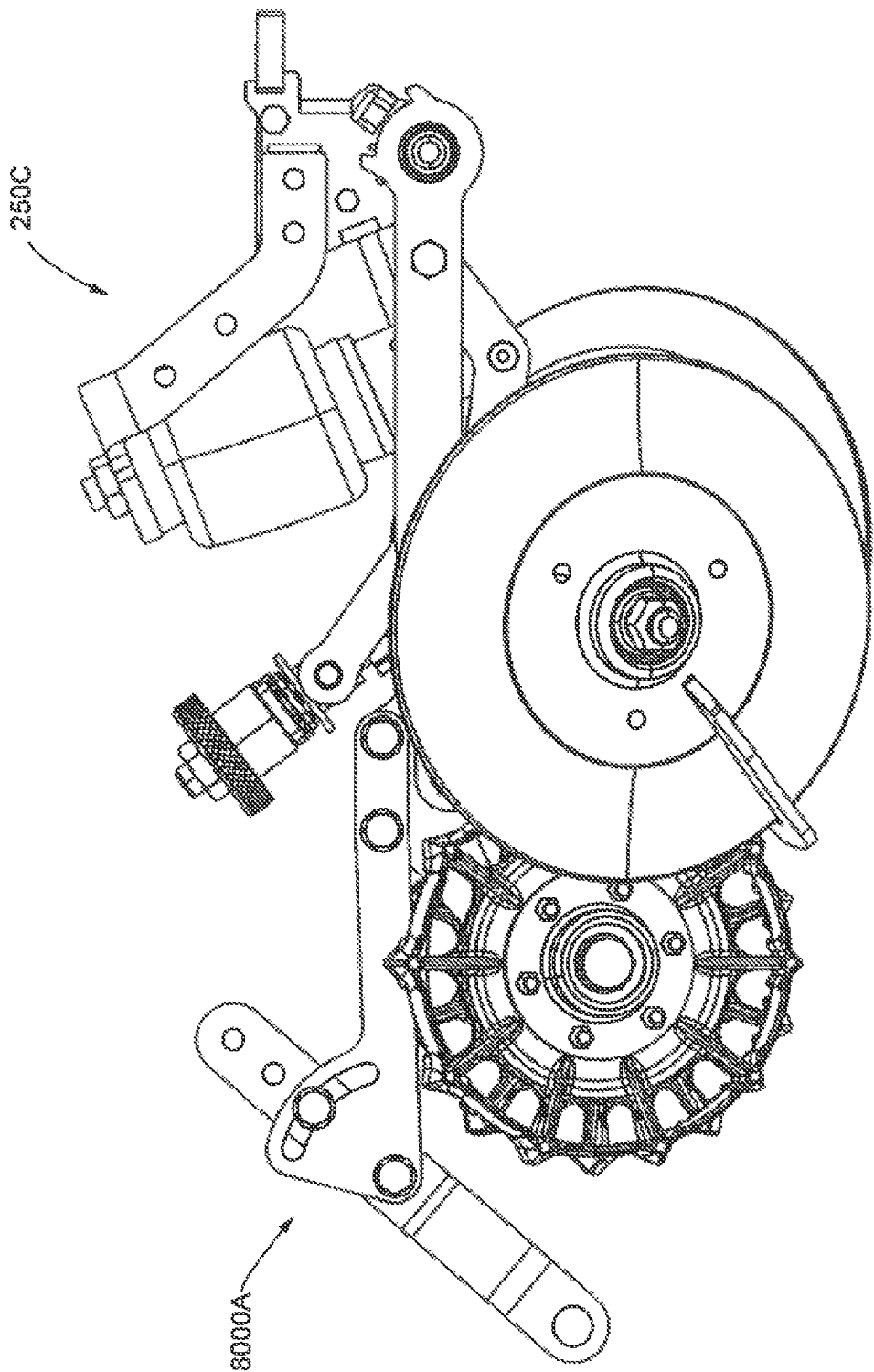


FIG. 76

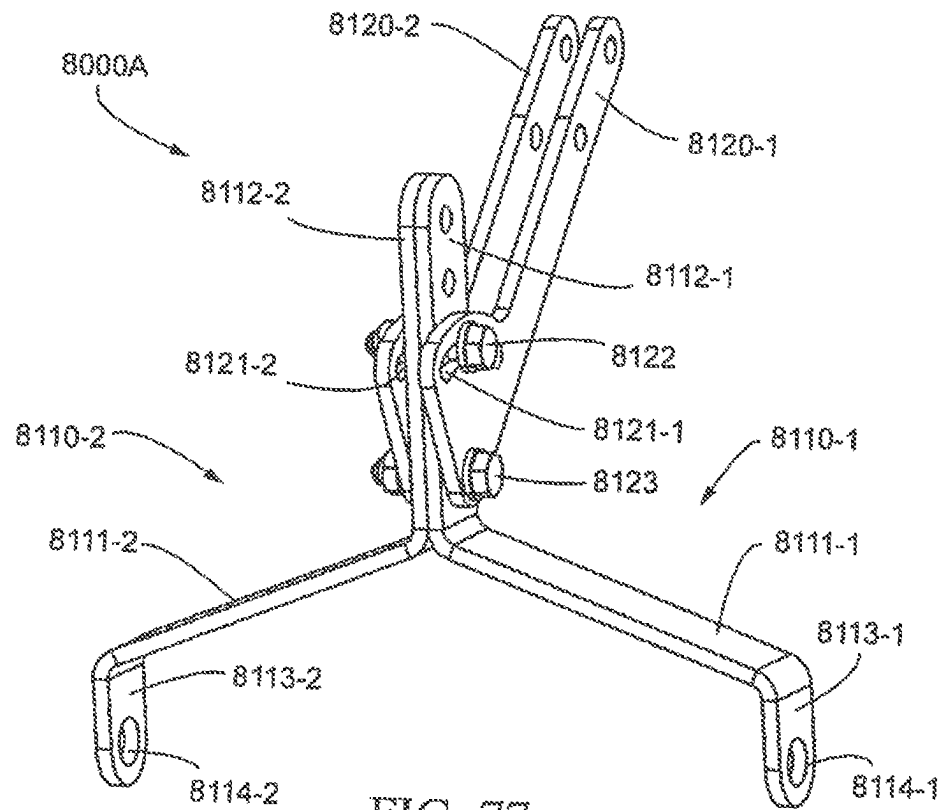


FIG. 77

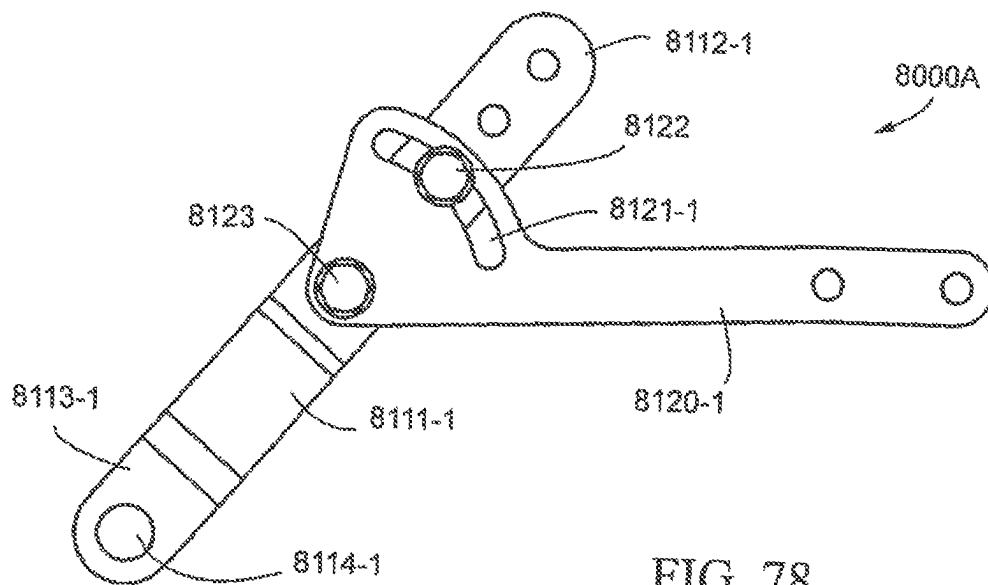


FIG. 78

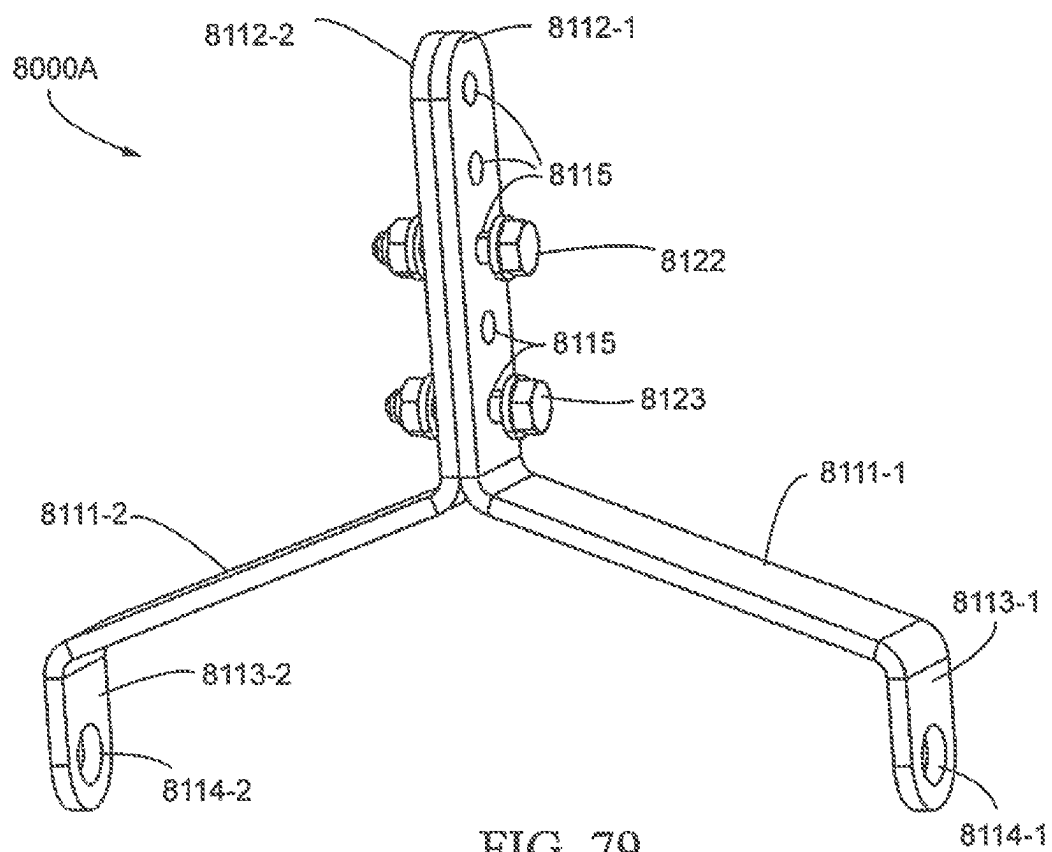


FIG. 79

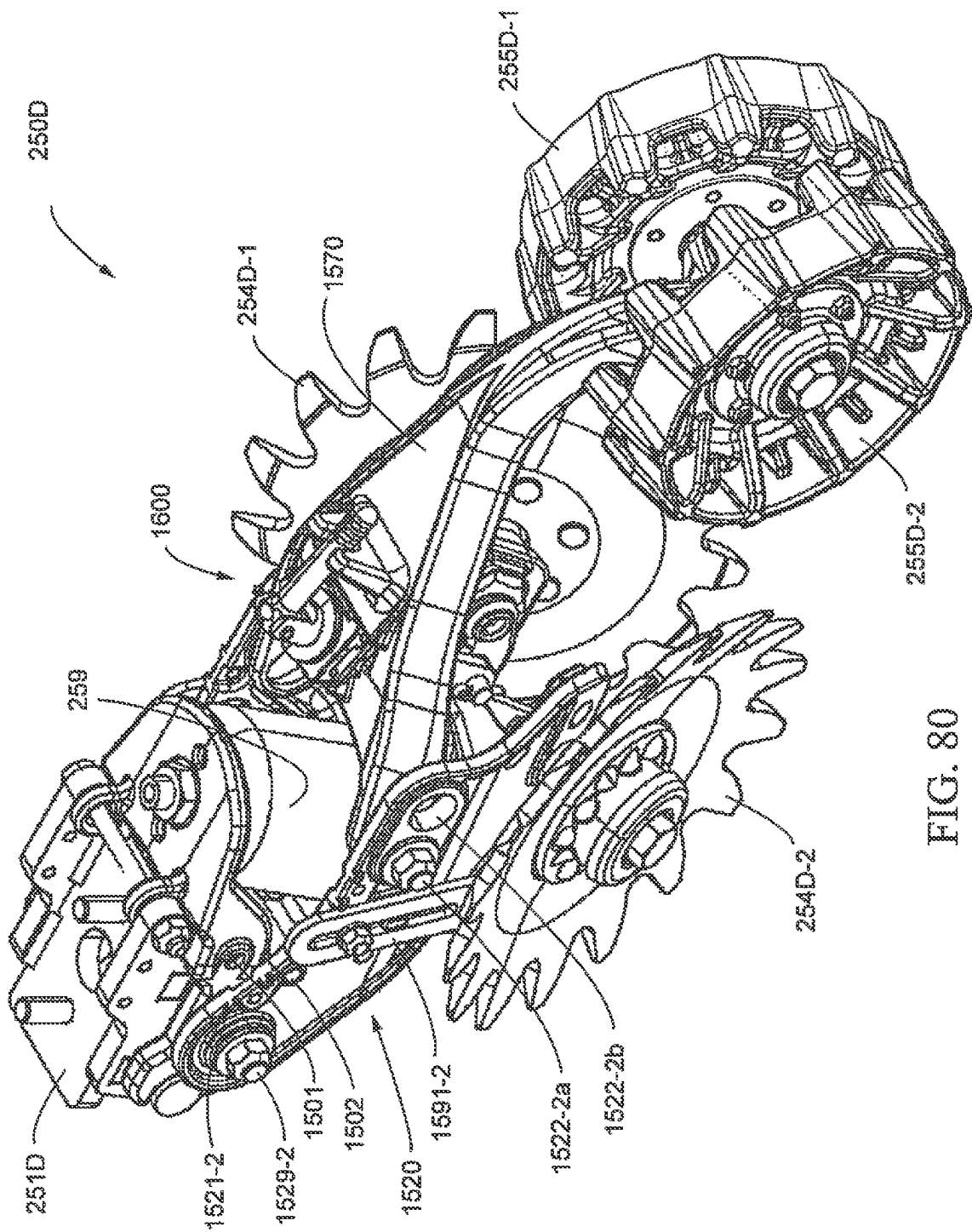


FIG. 80

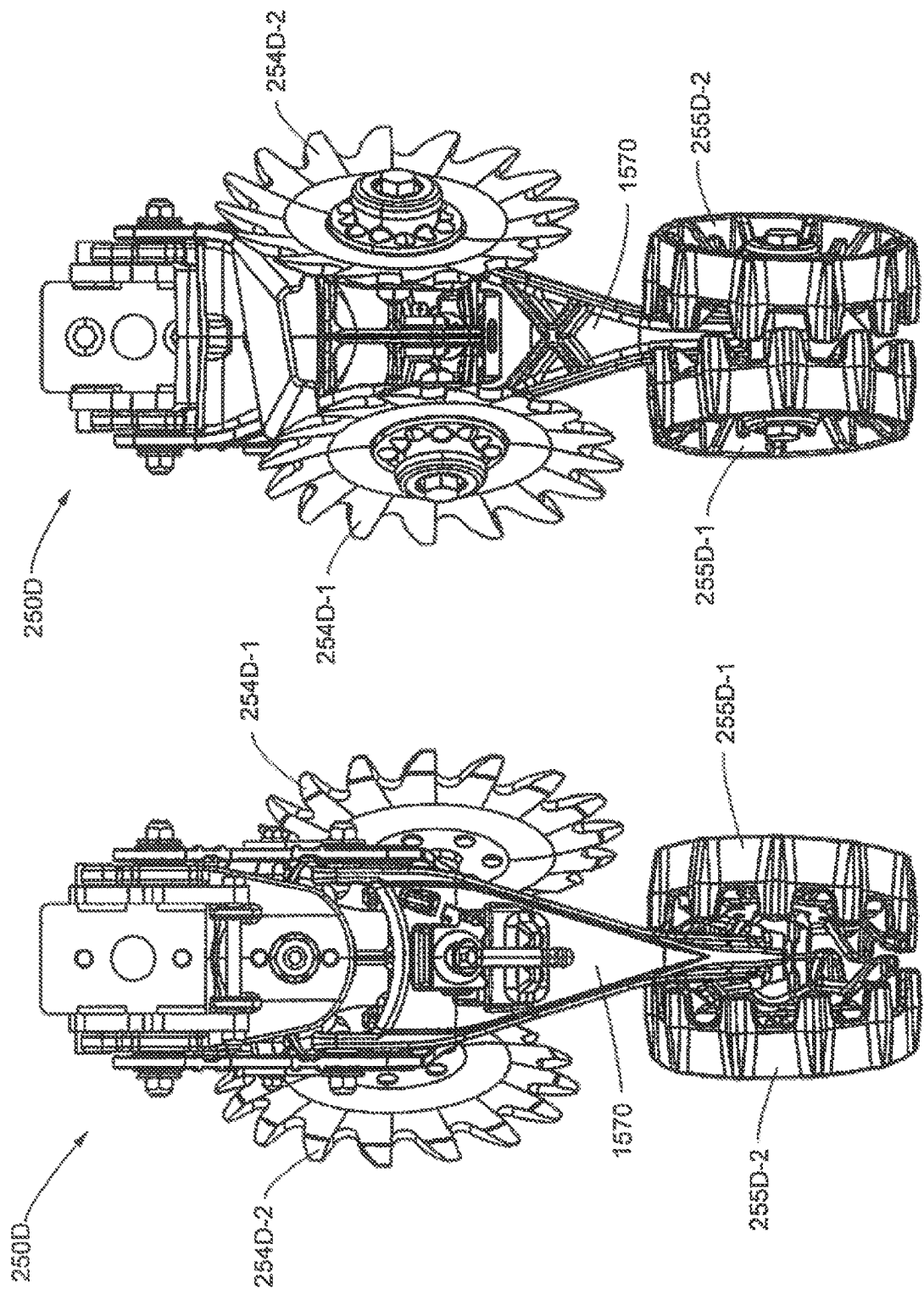
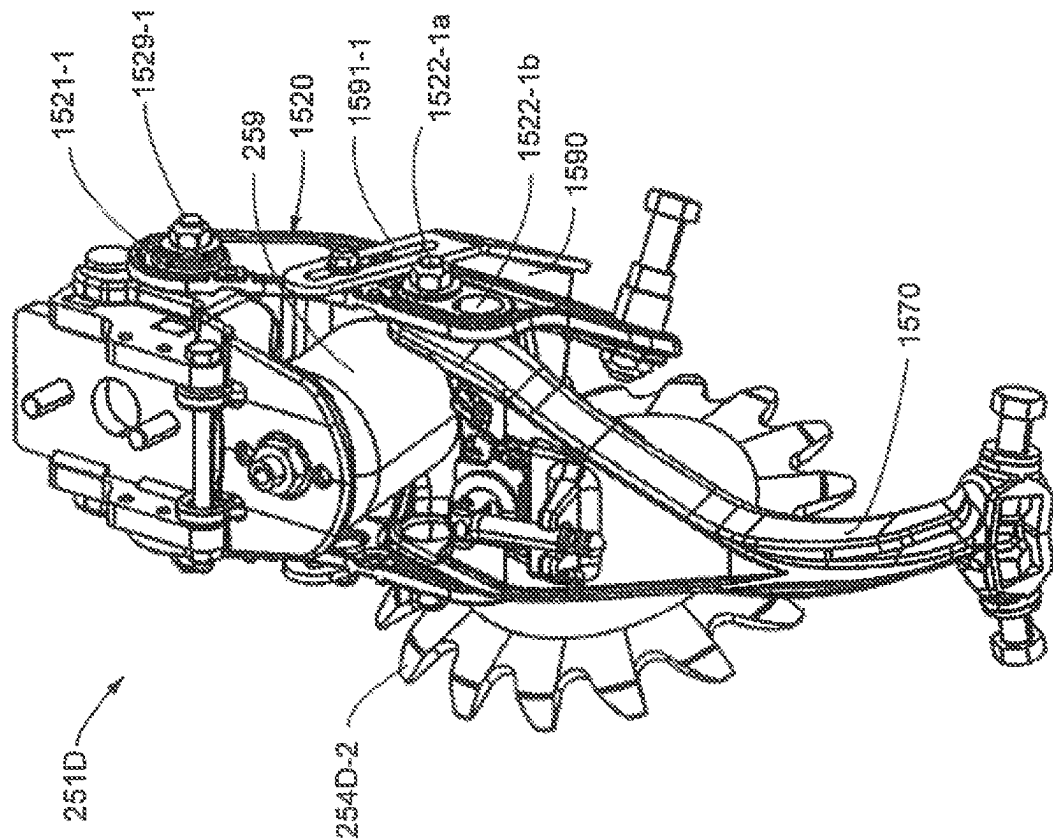
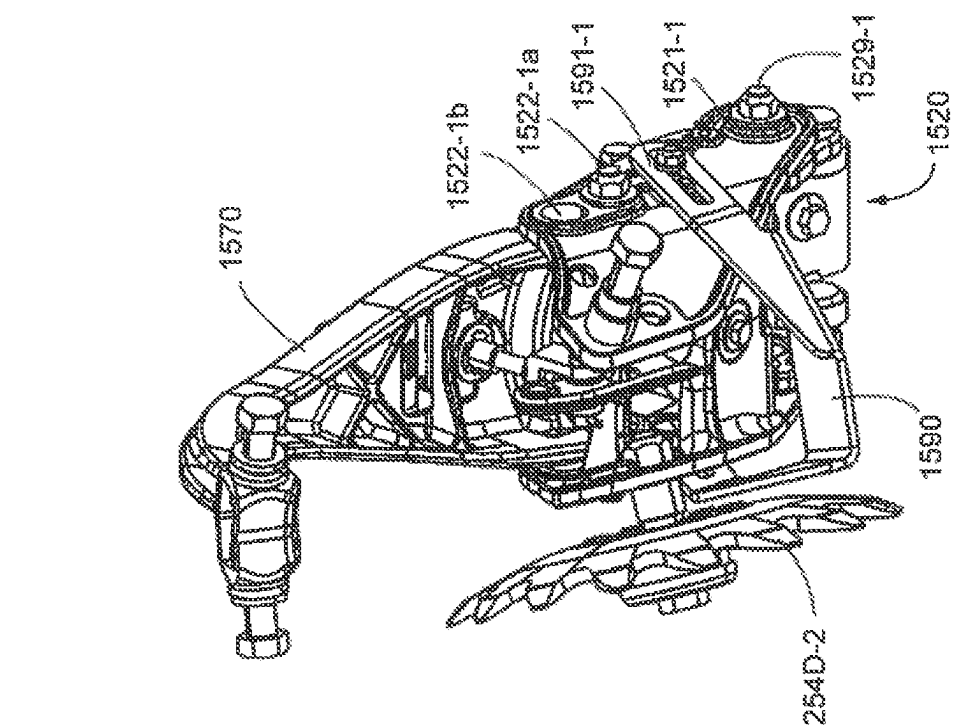


FIG. 82

FIG. 81



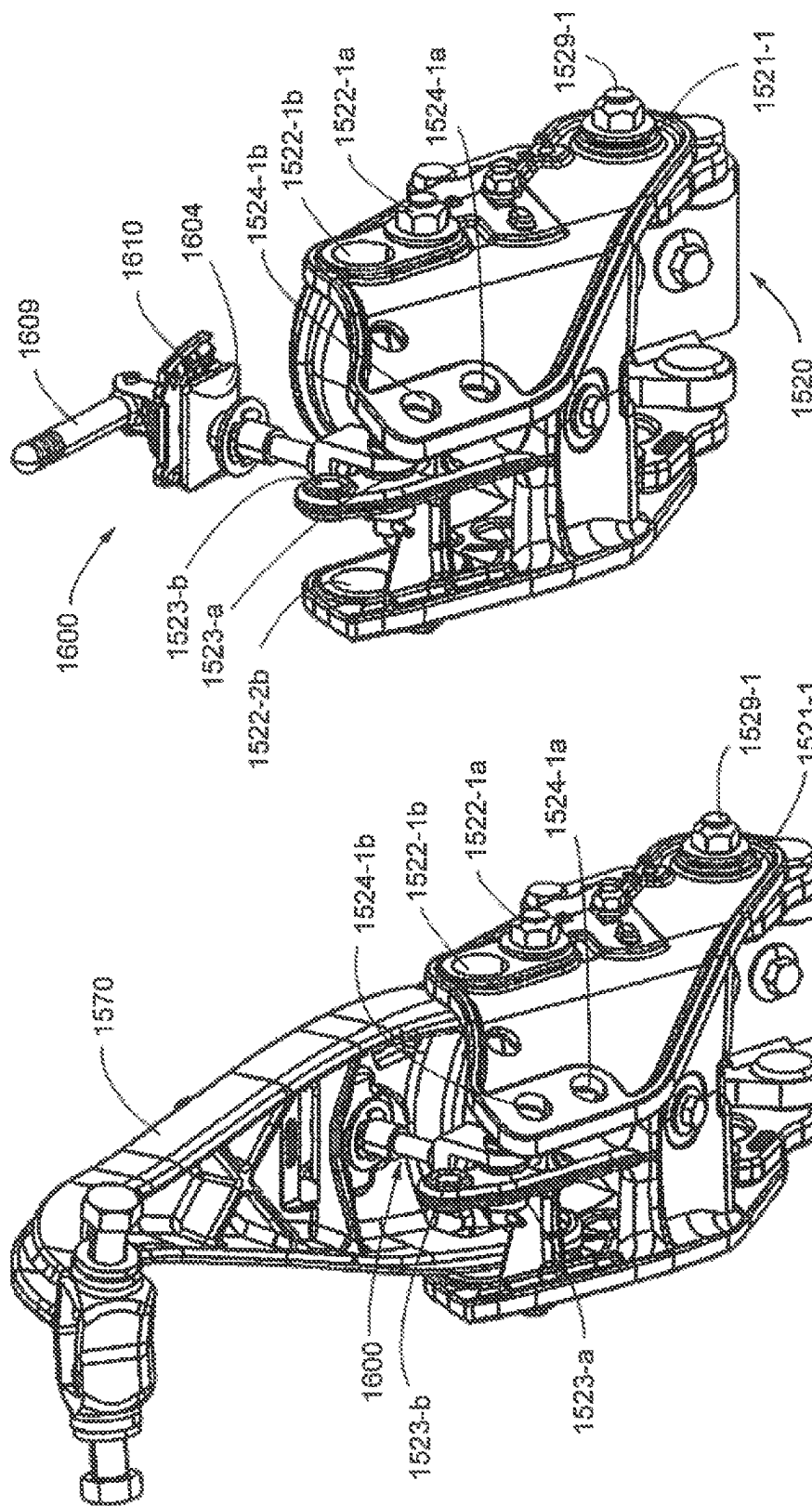


FIG. 86

FIG. 85

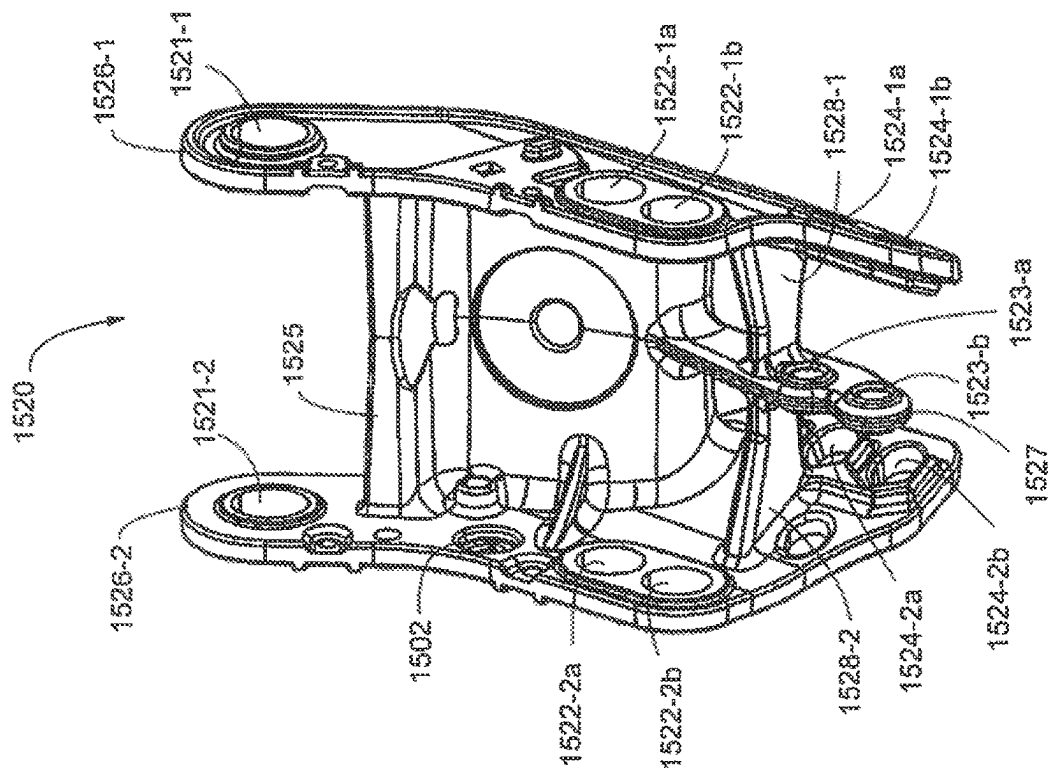


FIG. 88

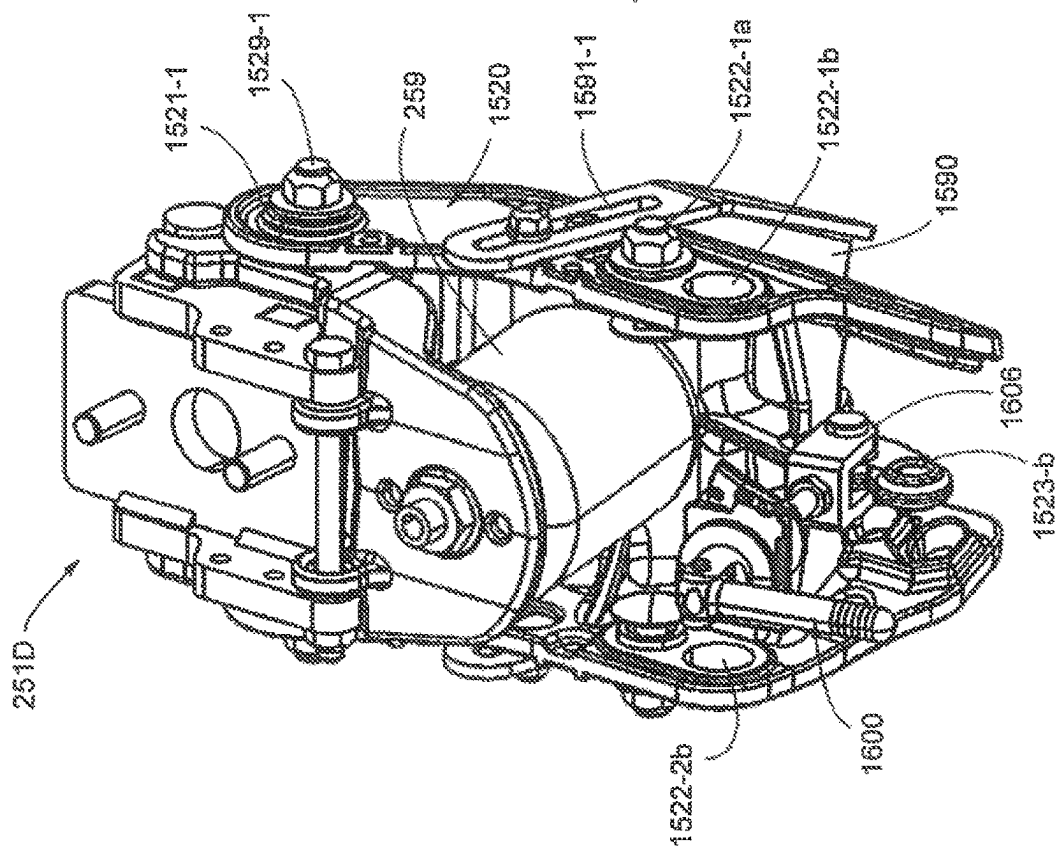


FIG. 87

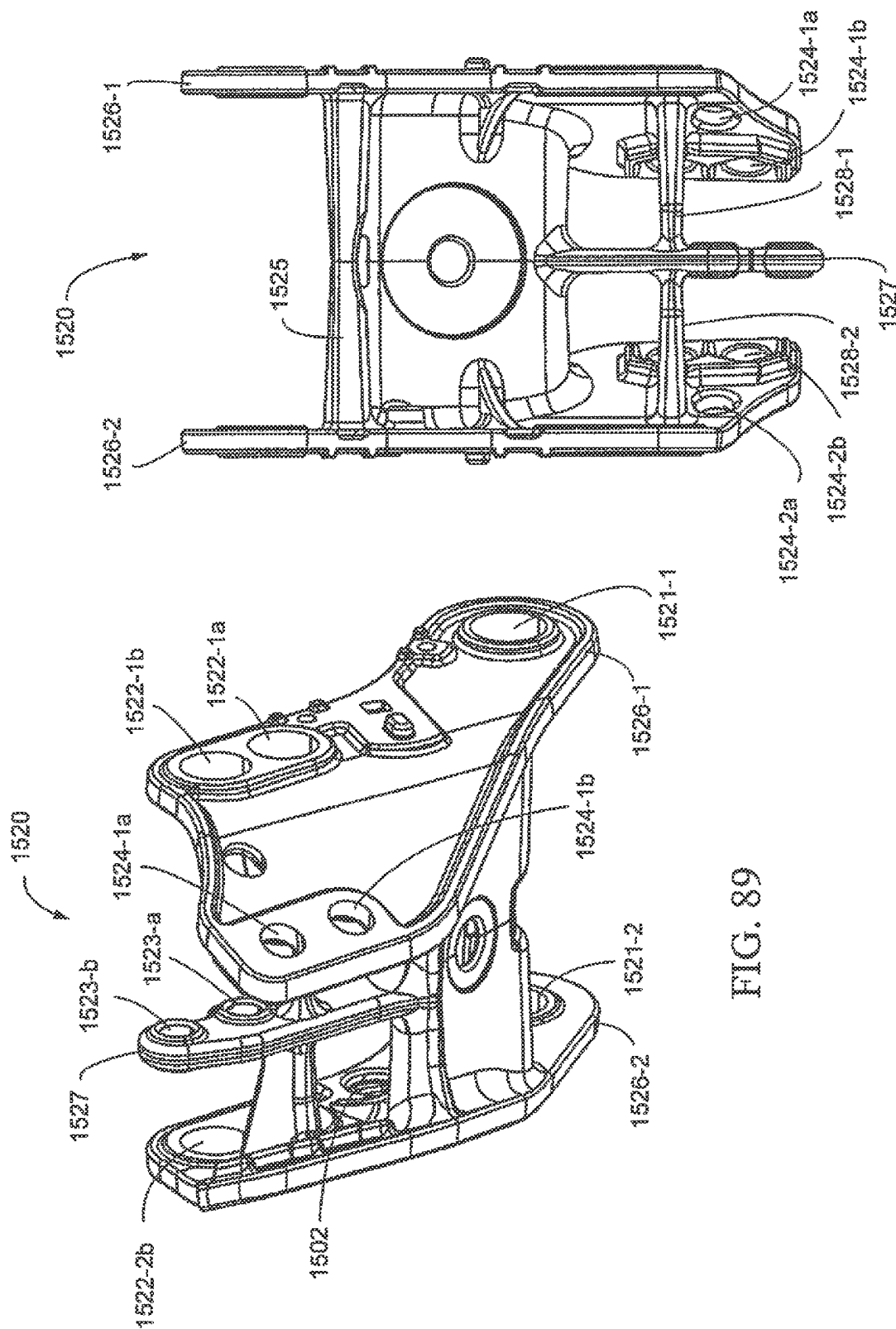


FIG. 89

FIG. 90

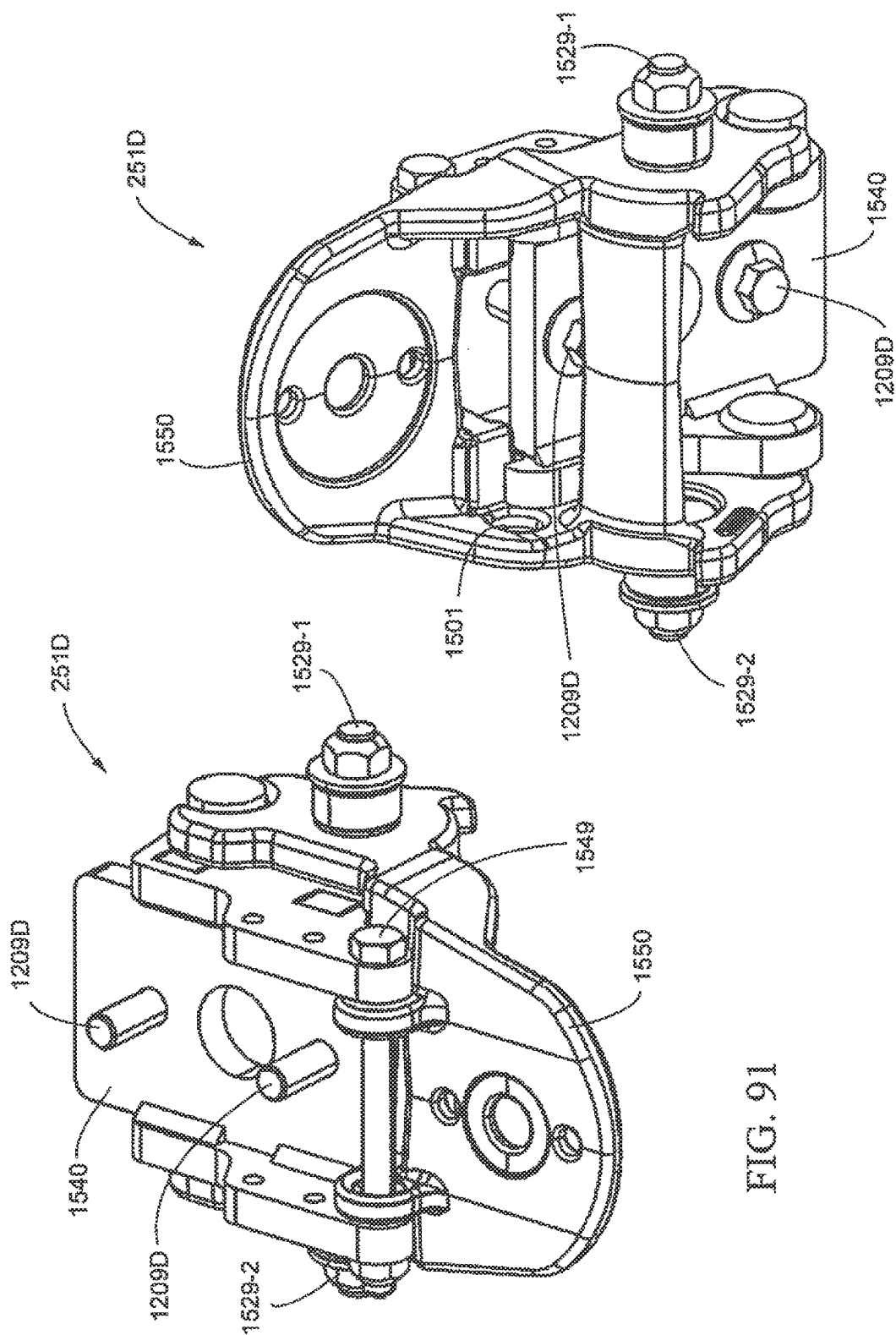


FIG. 92

FIG. 91

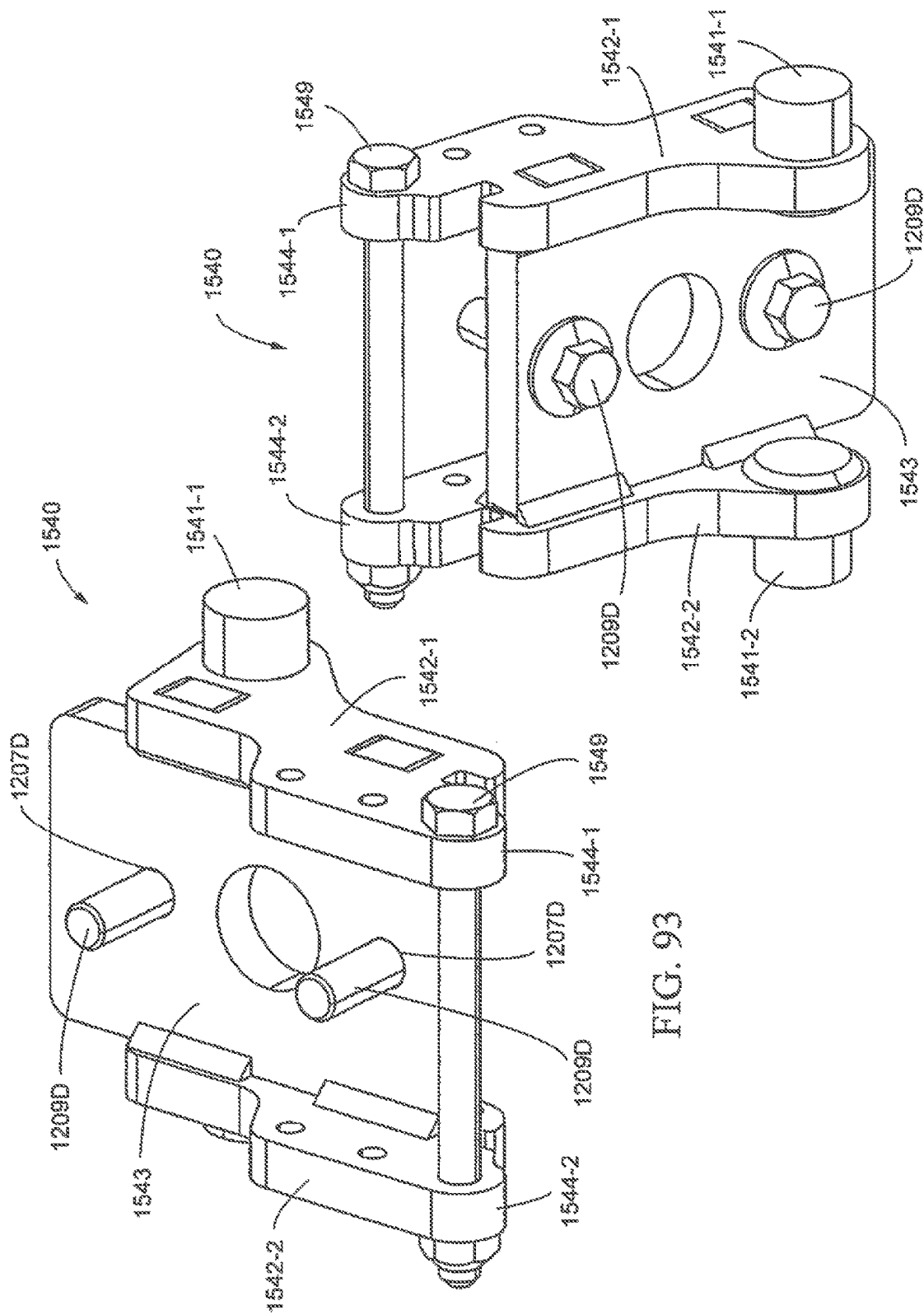


FIG. 94

FIG. 93

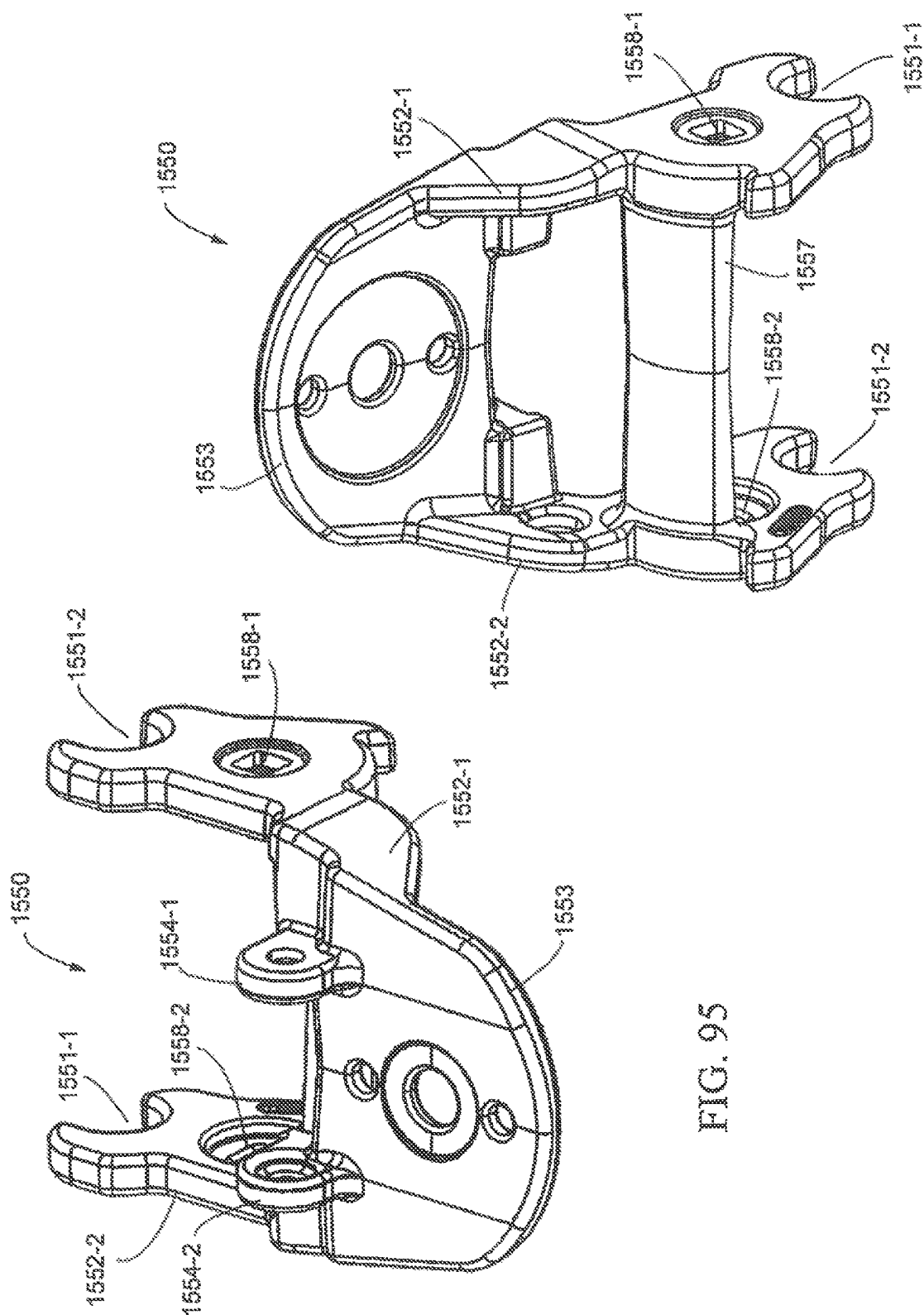
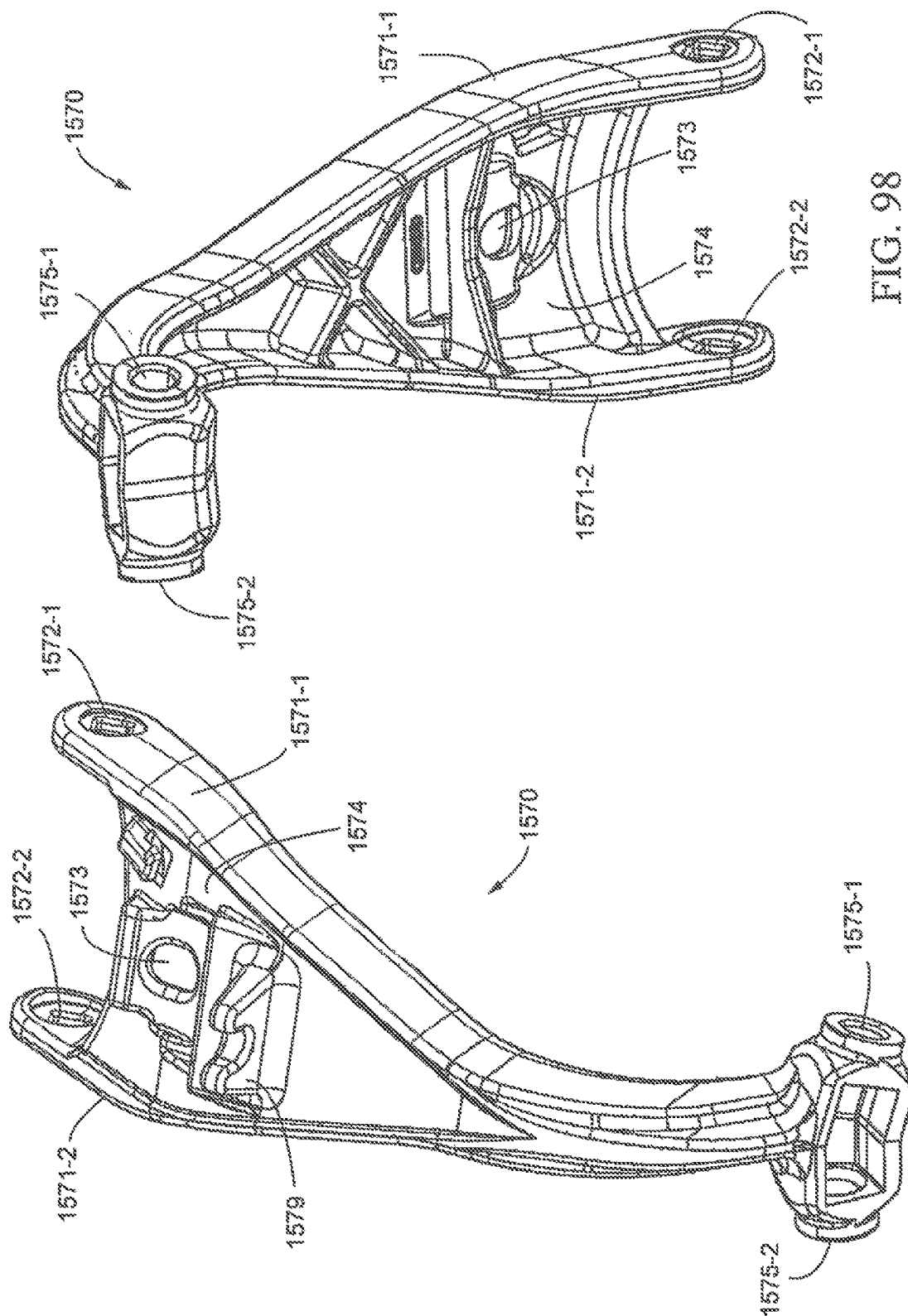
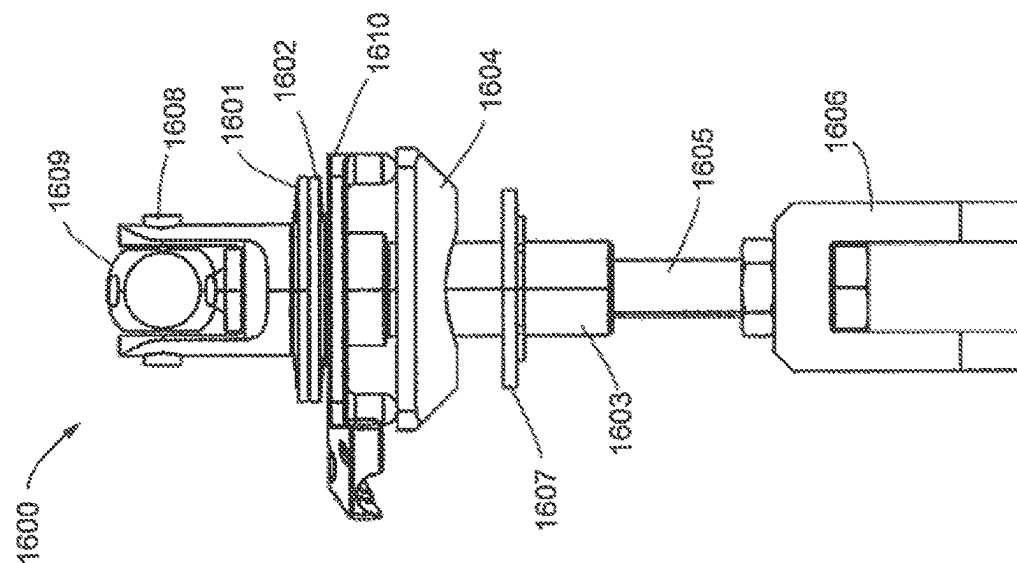
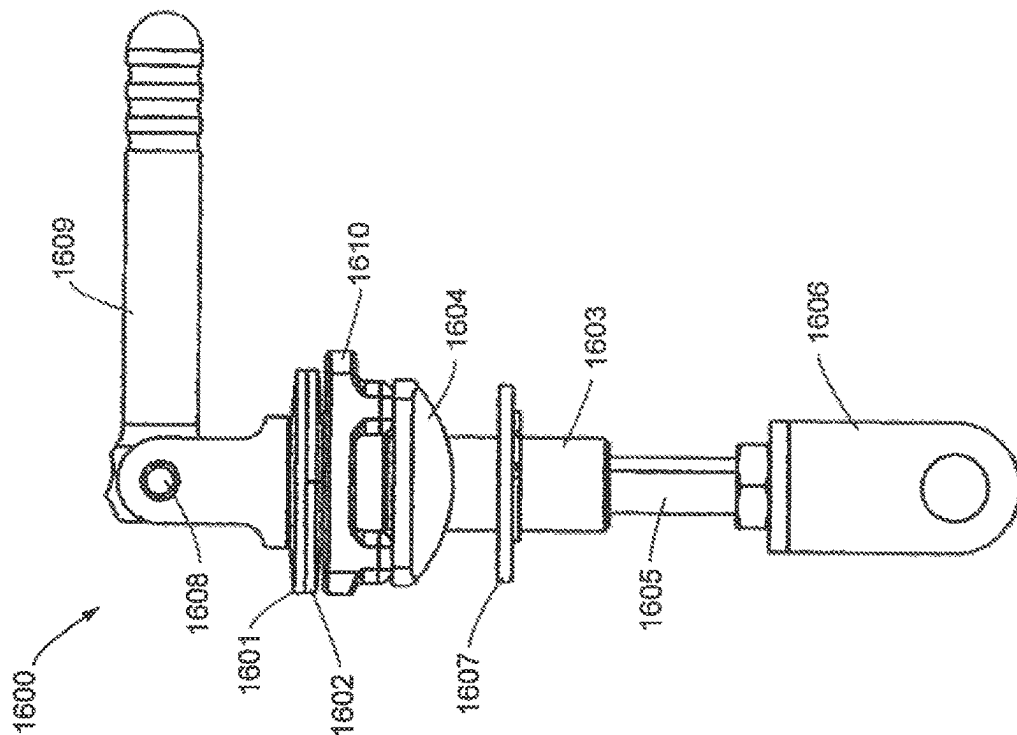


FIG. 95

FIG. 96





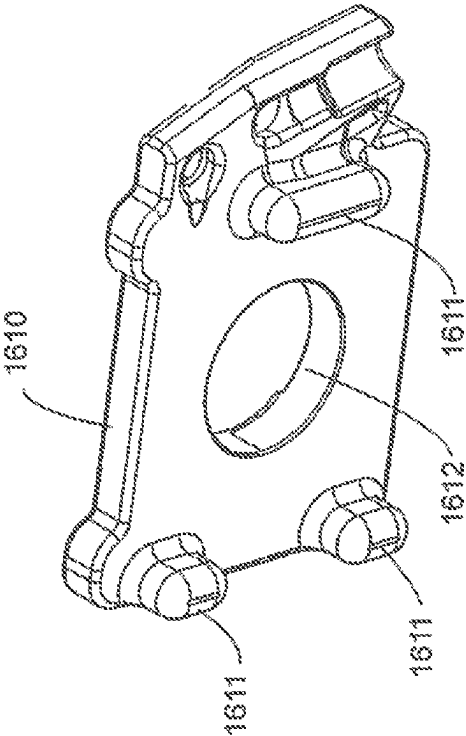


FIG. 103

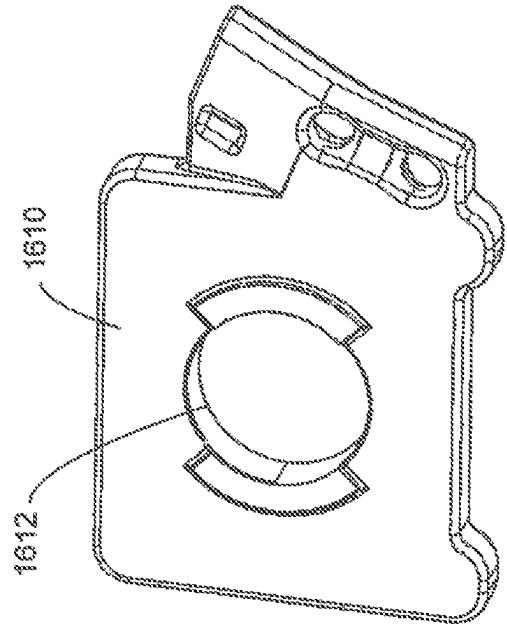


FIG. 102

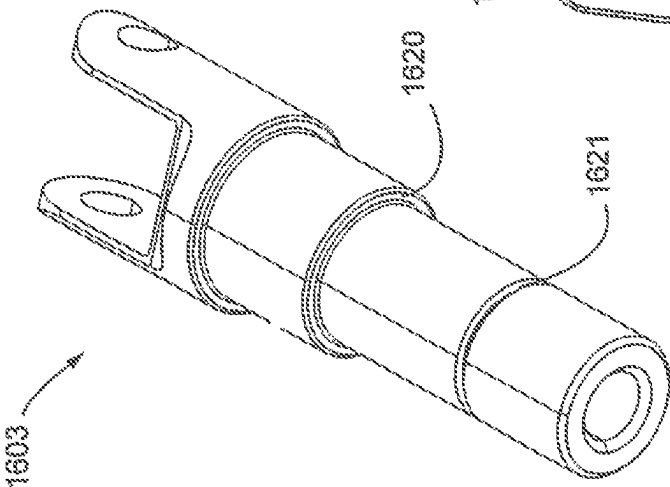


FIG. 101

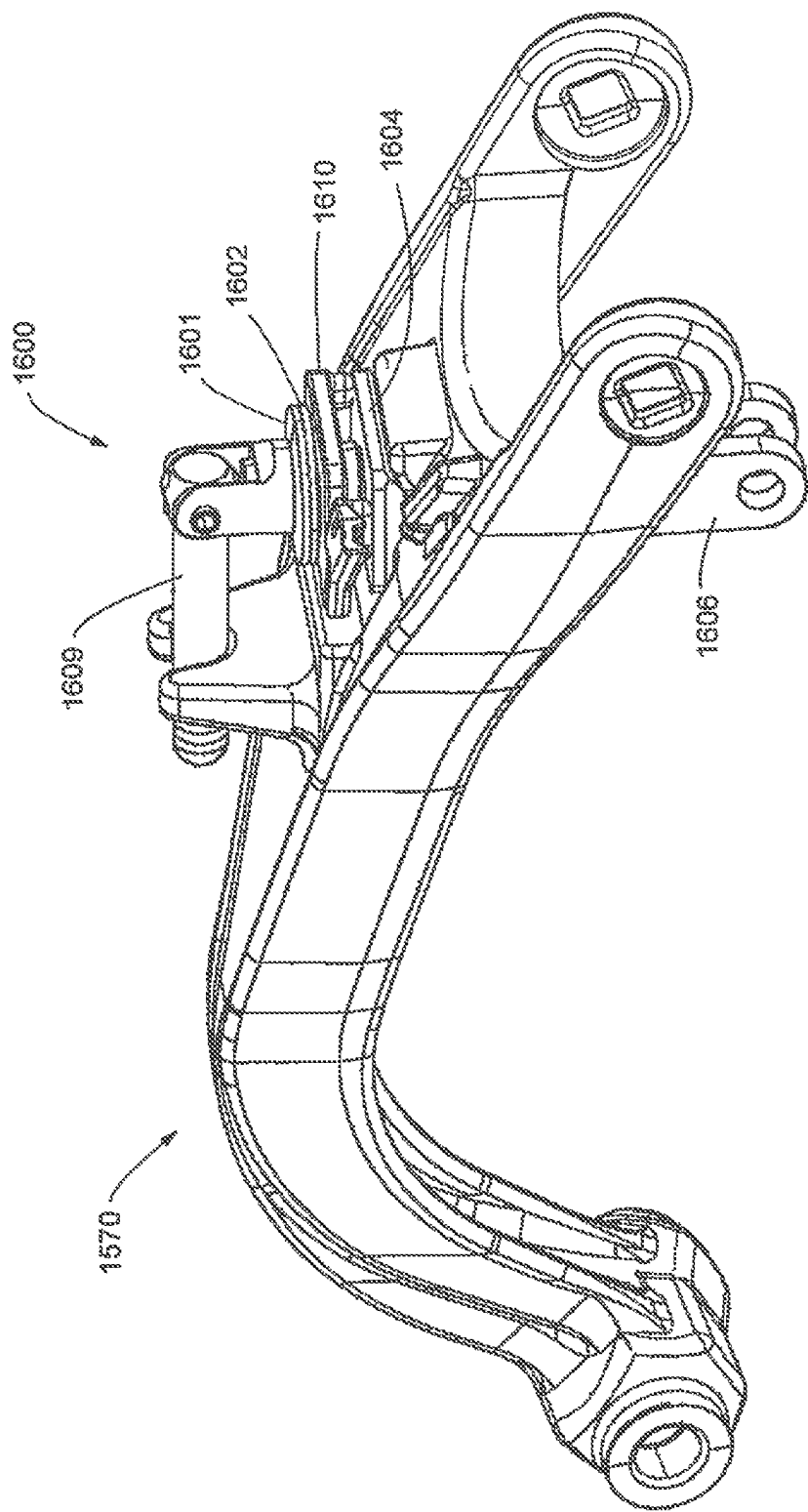


FIG. 104

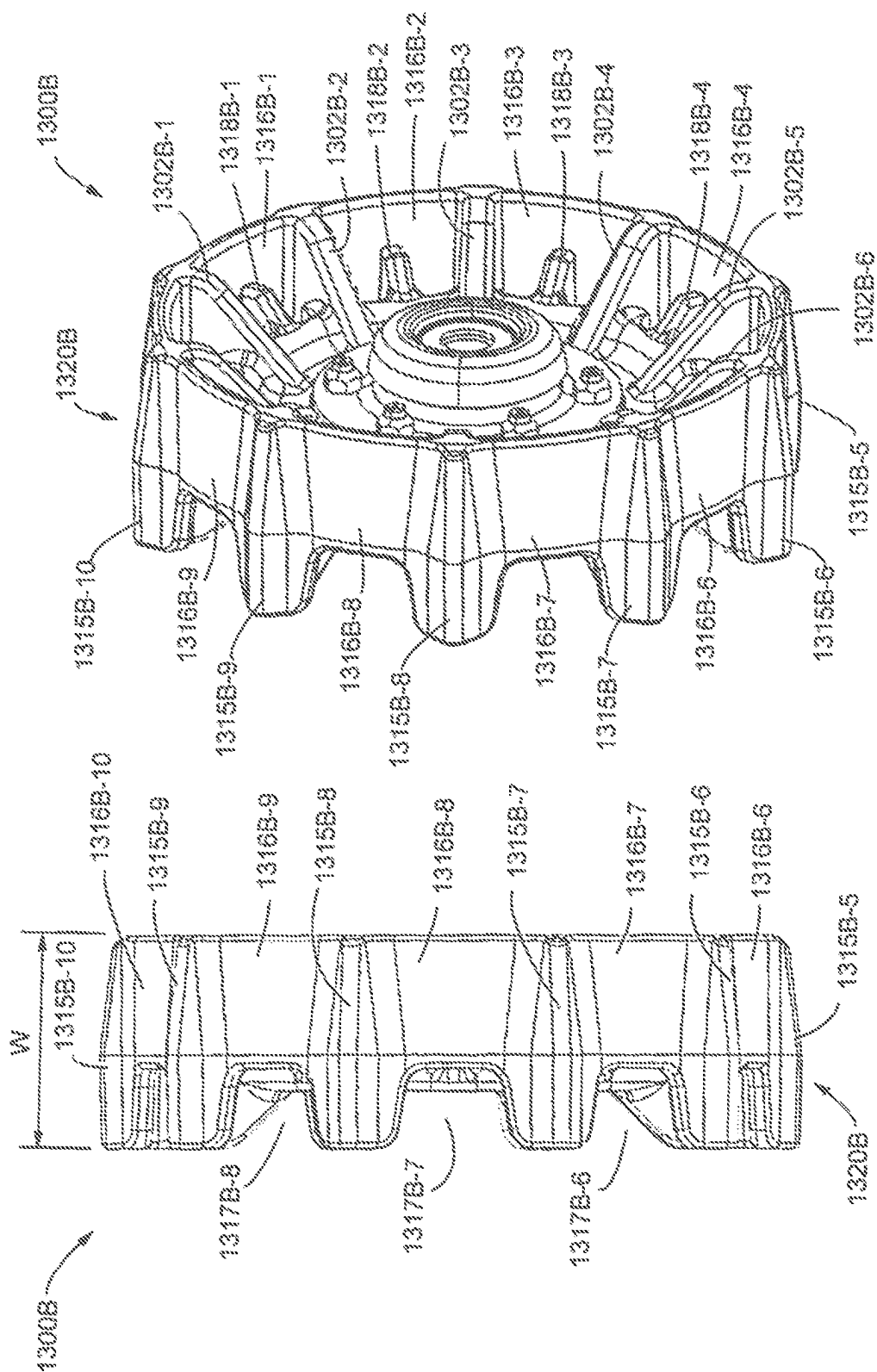
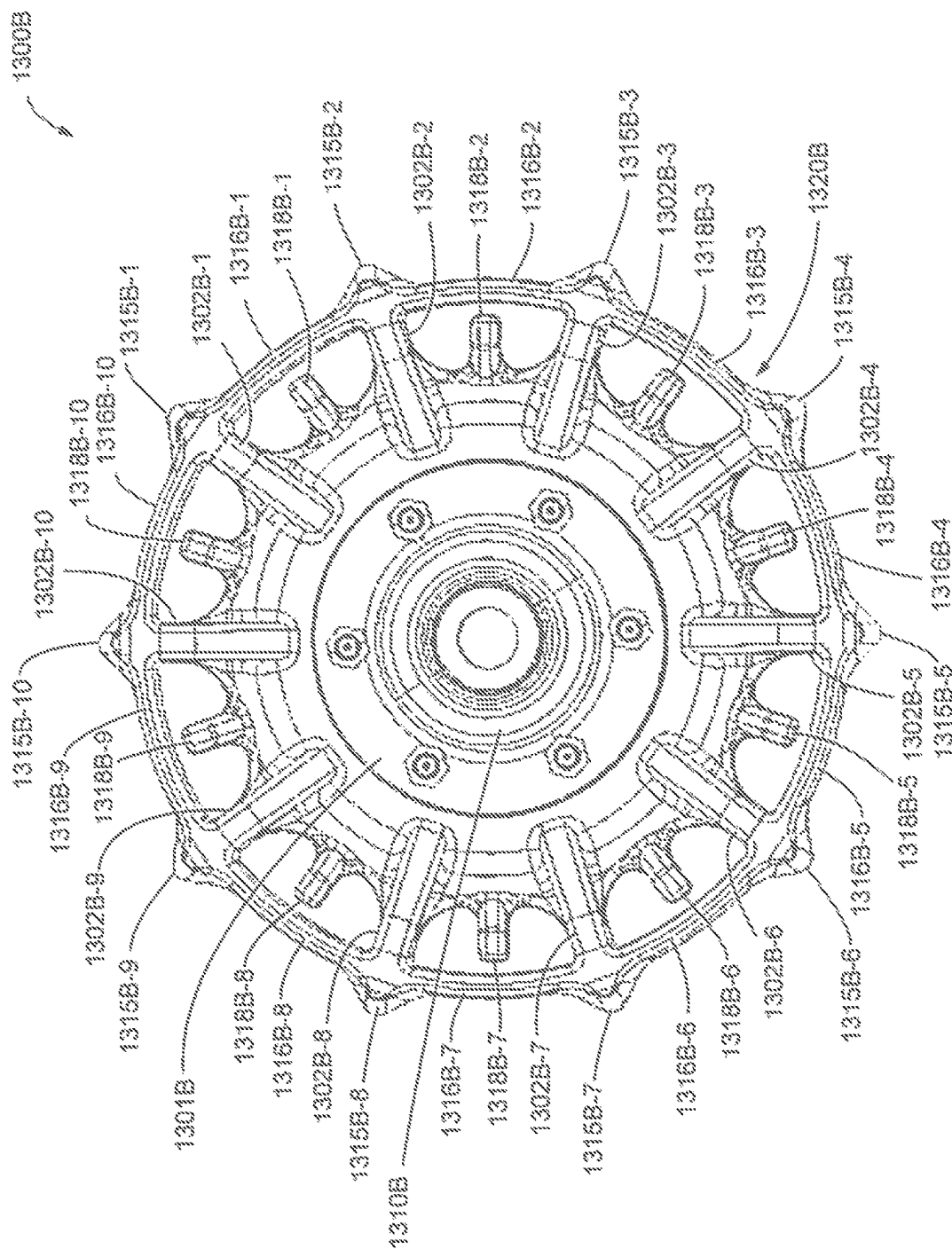


FIG. 105

FIG. 106



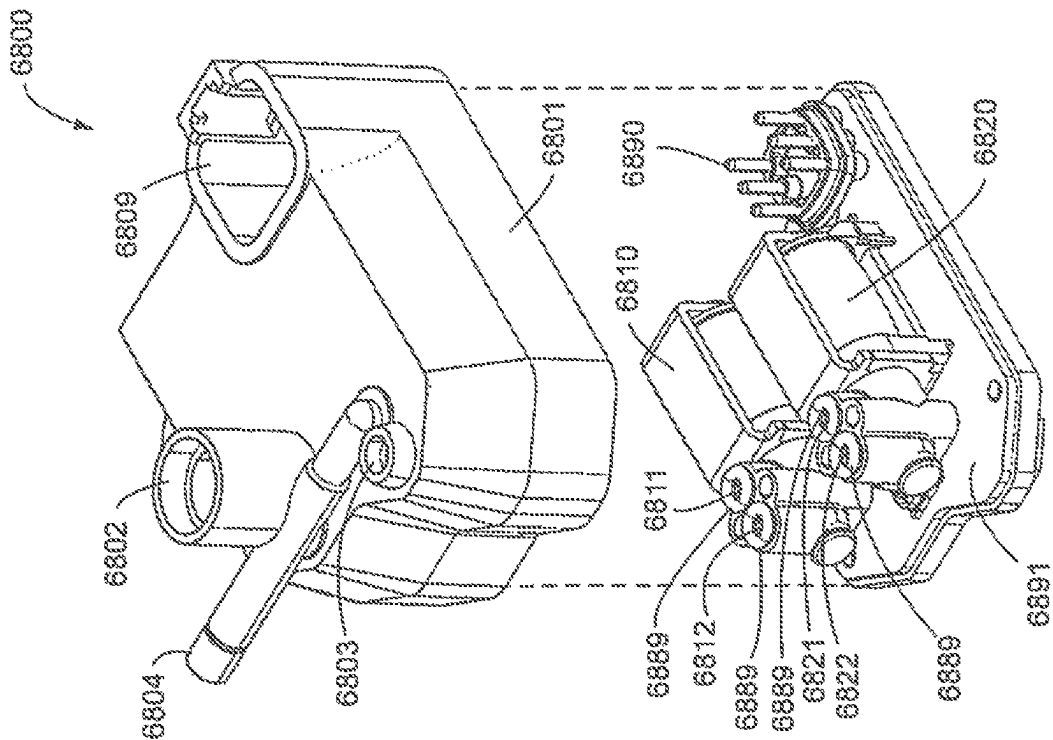


FIG. 108

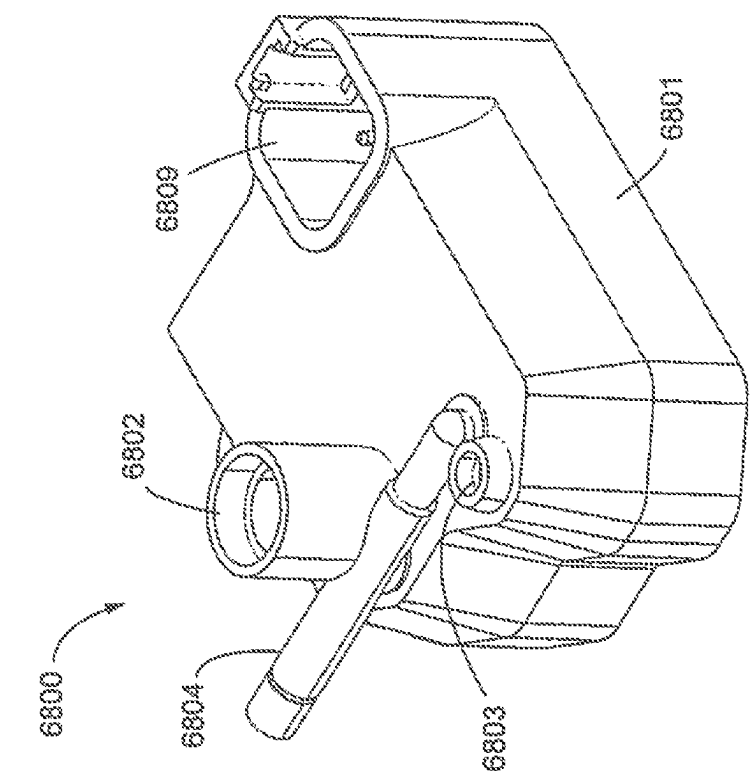


FIG. 109

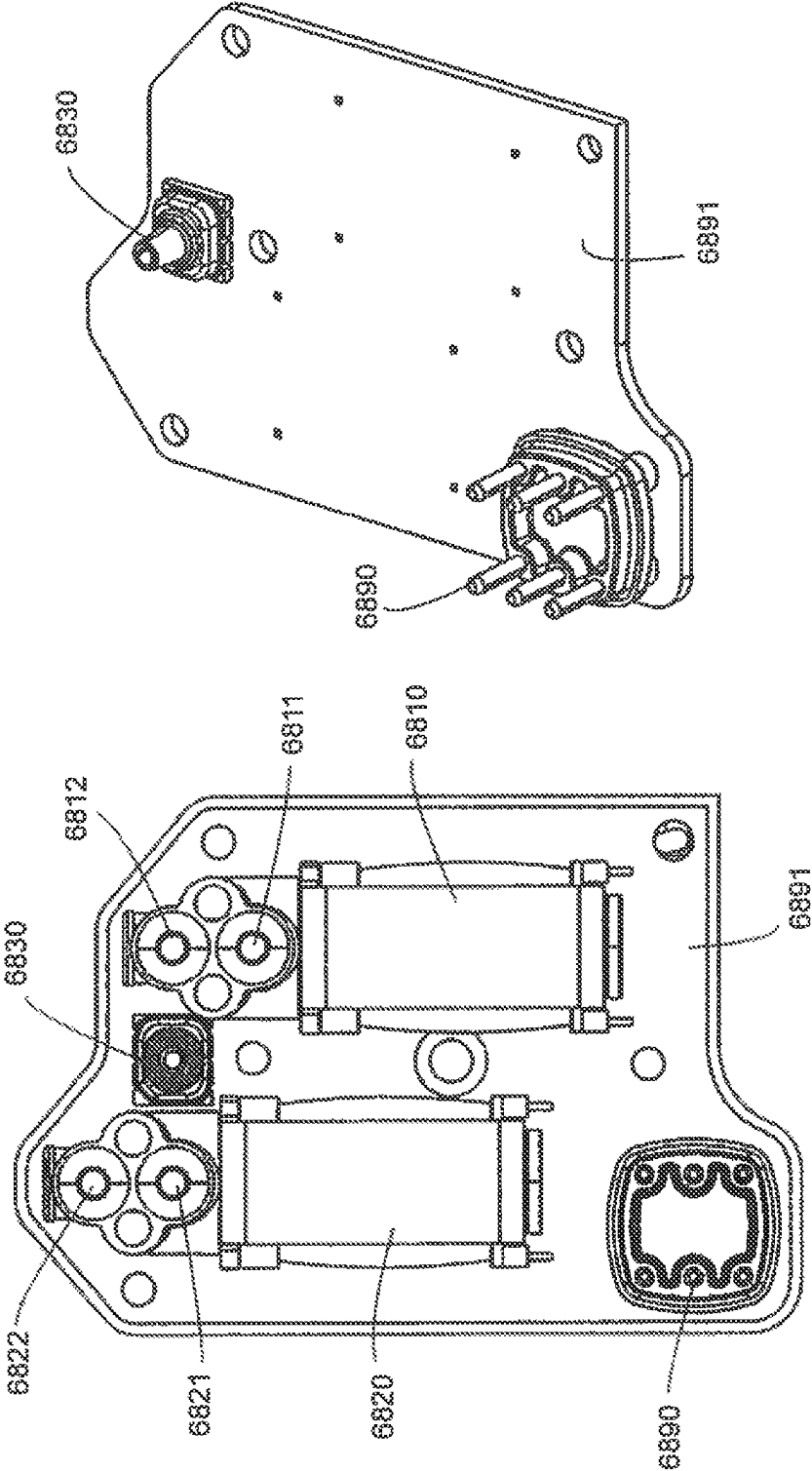


FIG. 111

FIG. 110

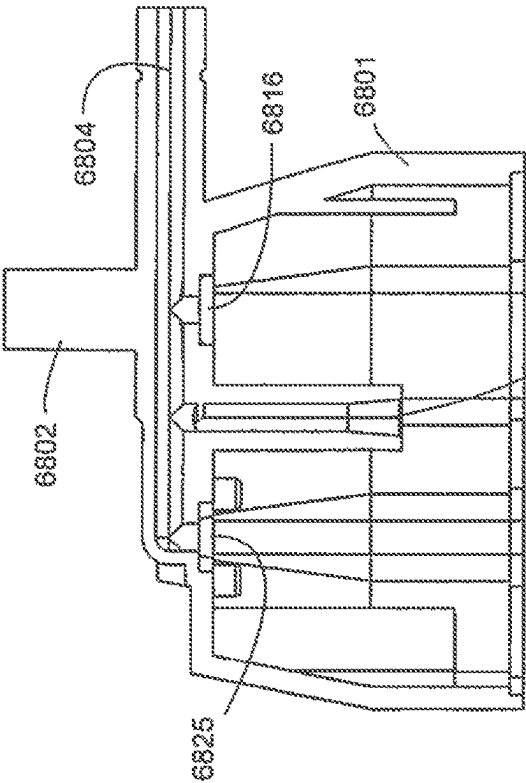


FIG. 113

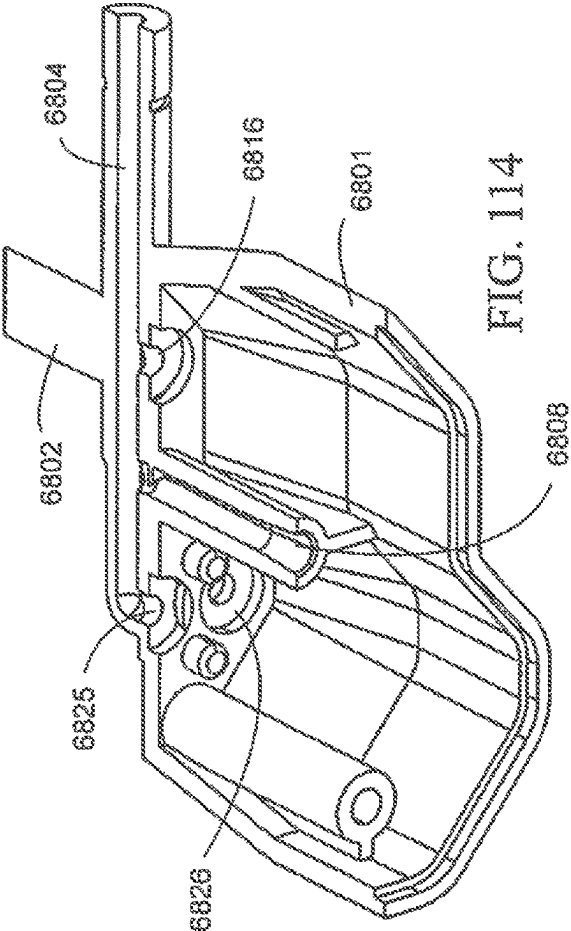


FIG. 114

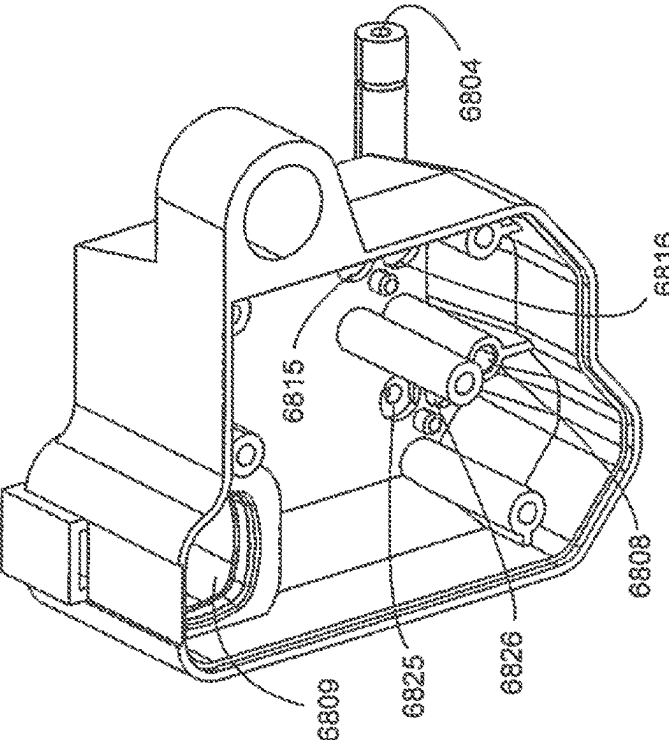


FIG. 112

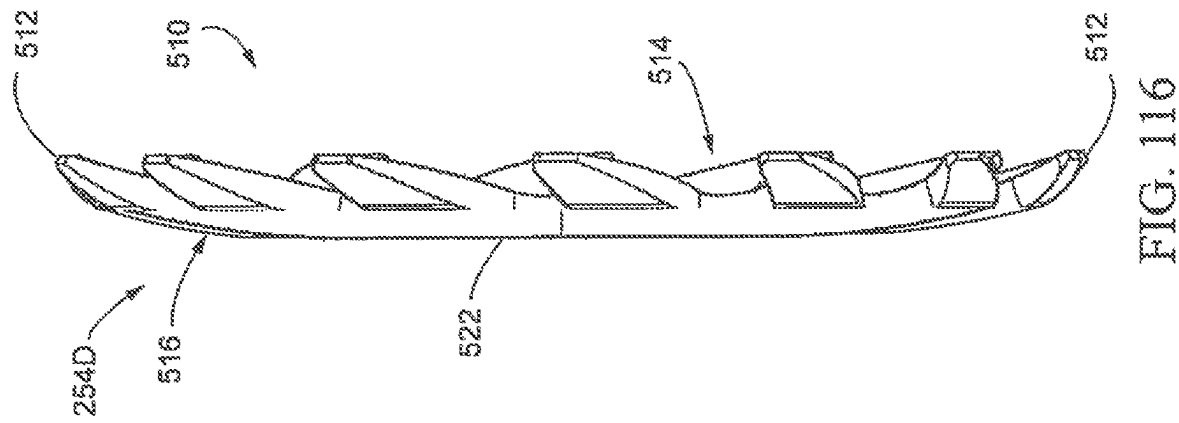


FIG. 115

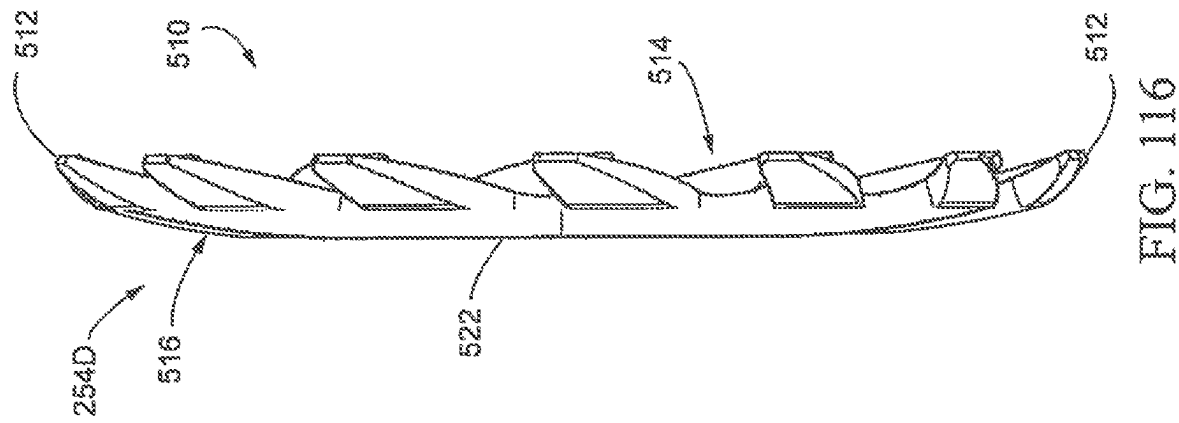


FIG. 116

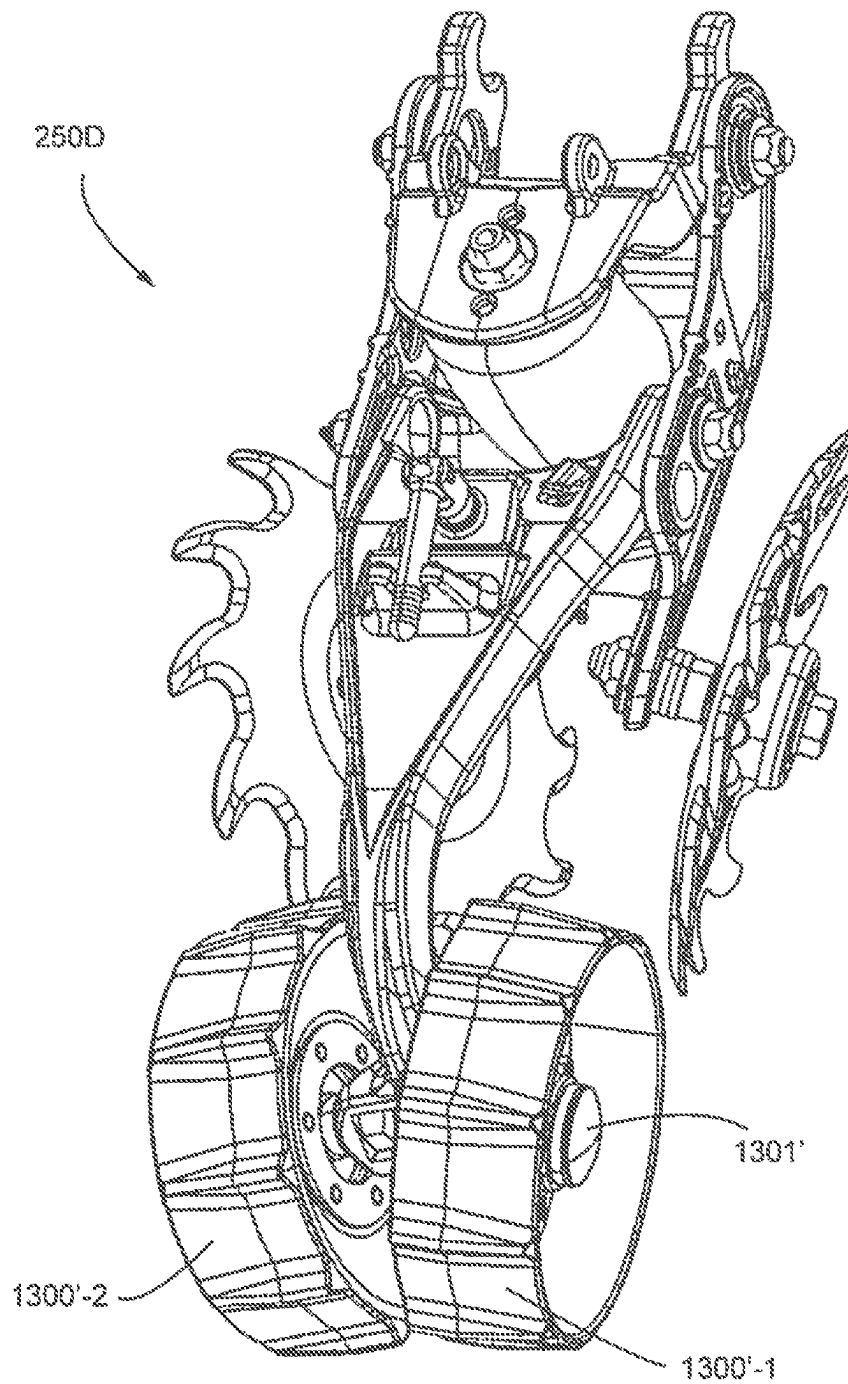


FIG. 117

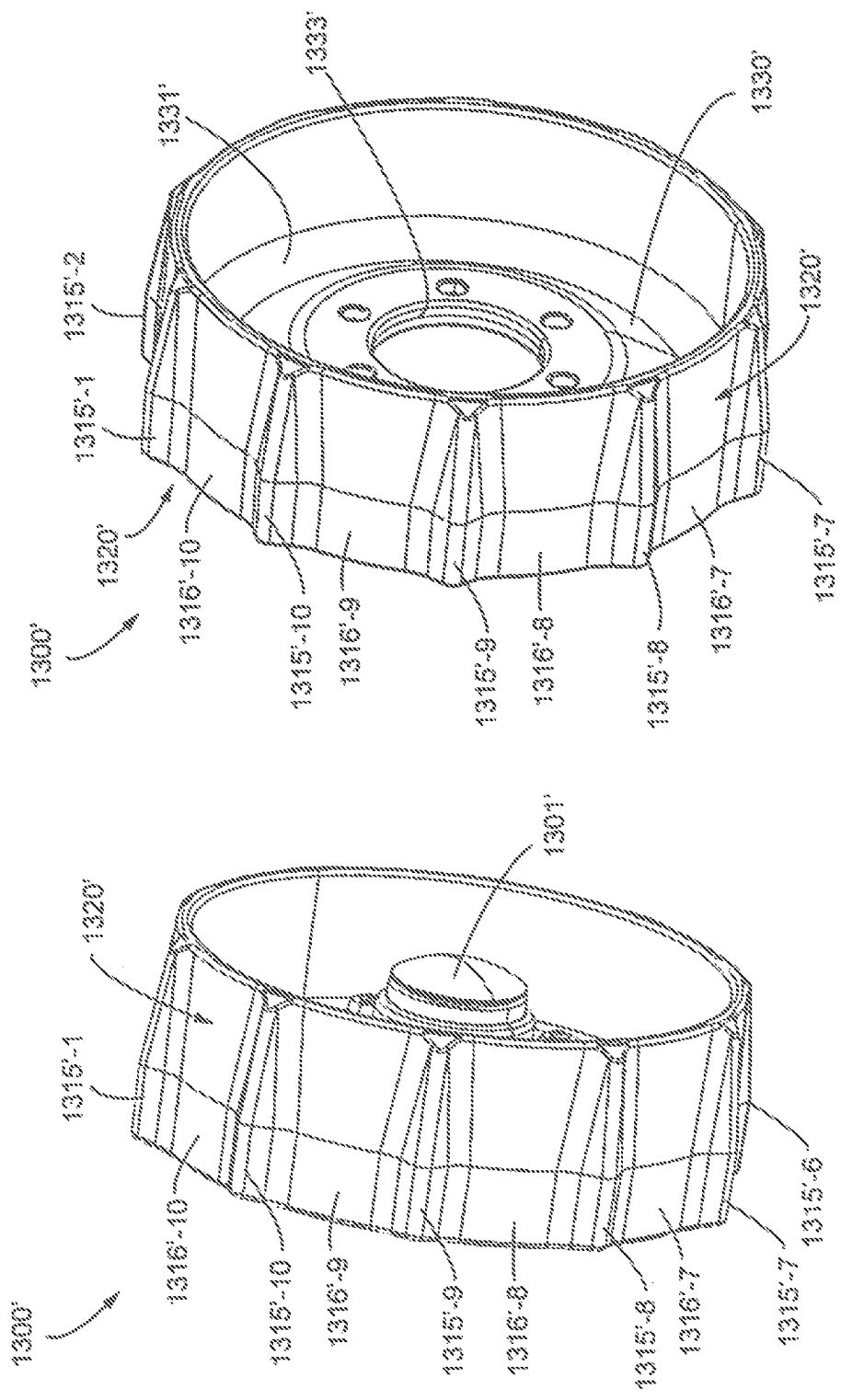


FIG. 121

FIG. 118

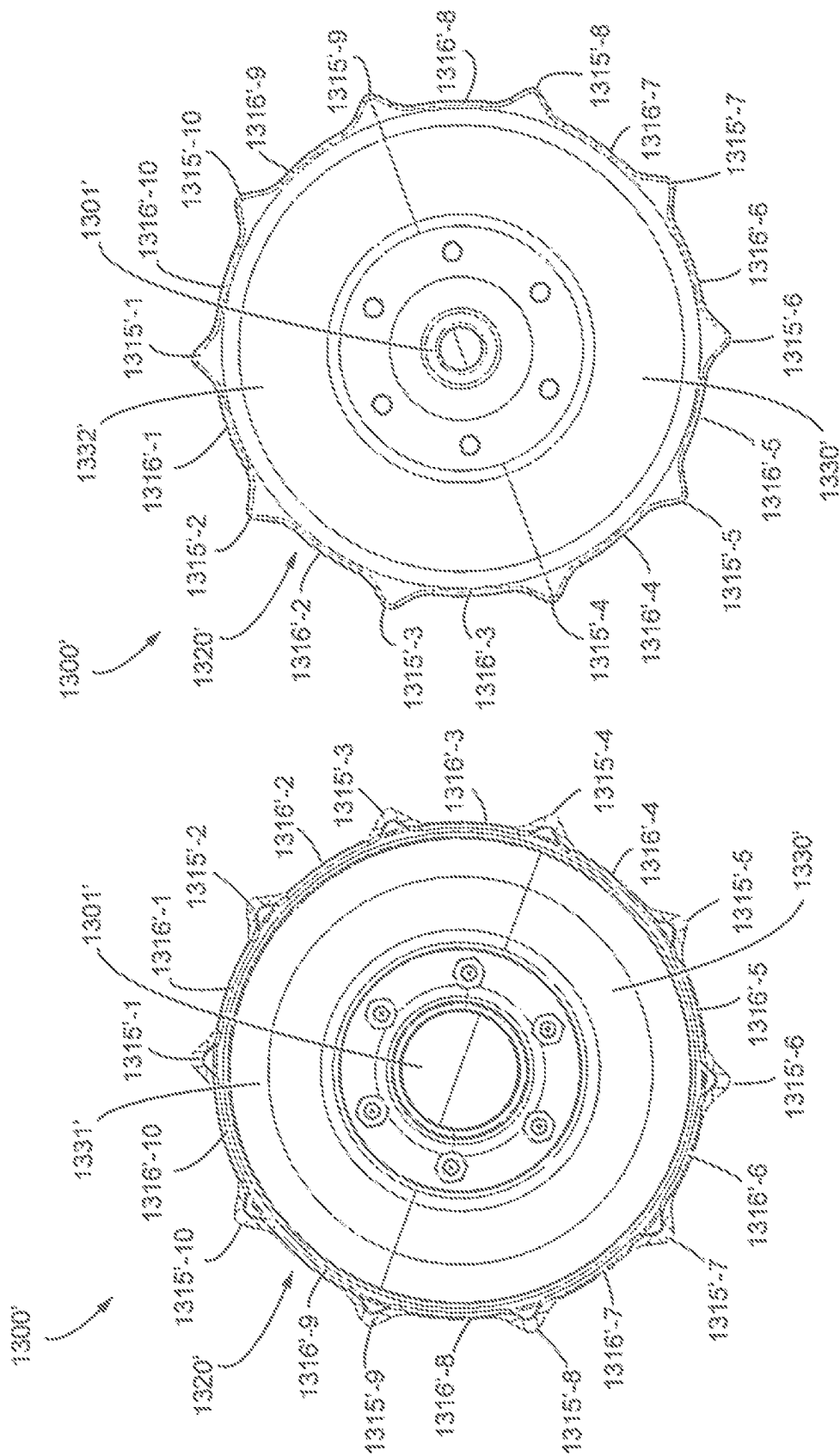


FIG. 119

FIG. 120

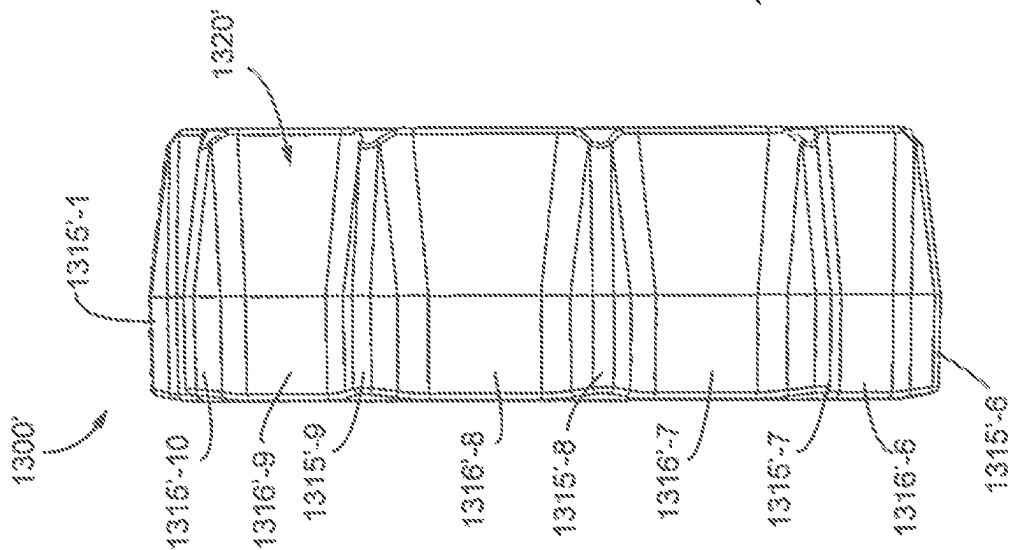


FIG. 122

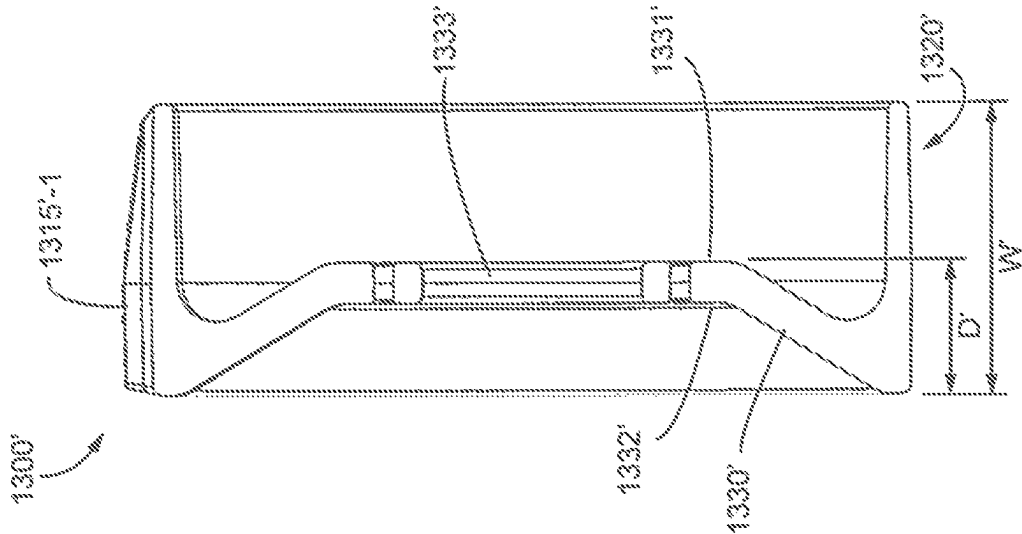


FIG. 123

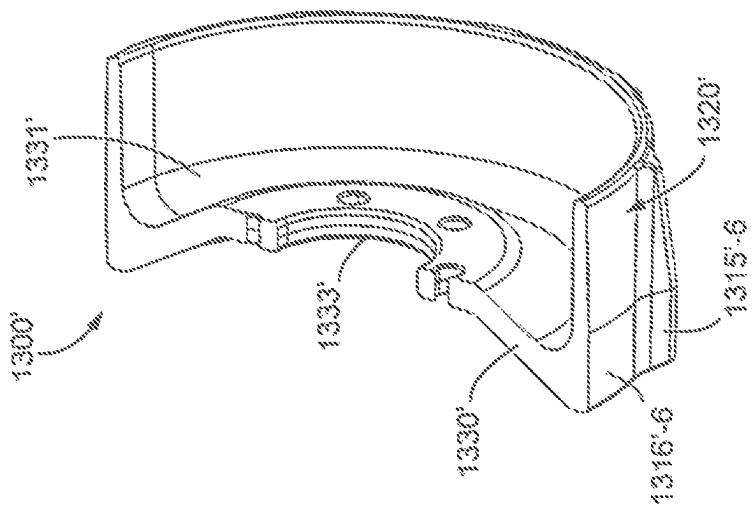


FIG. 124

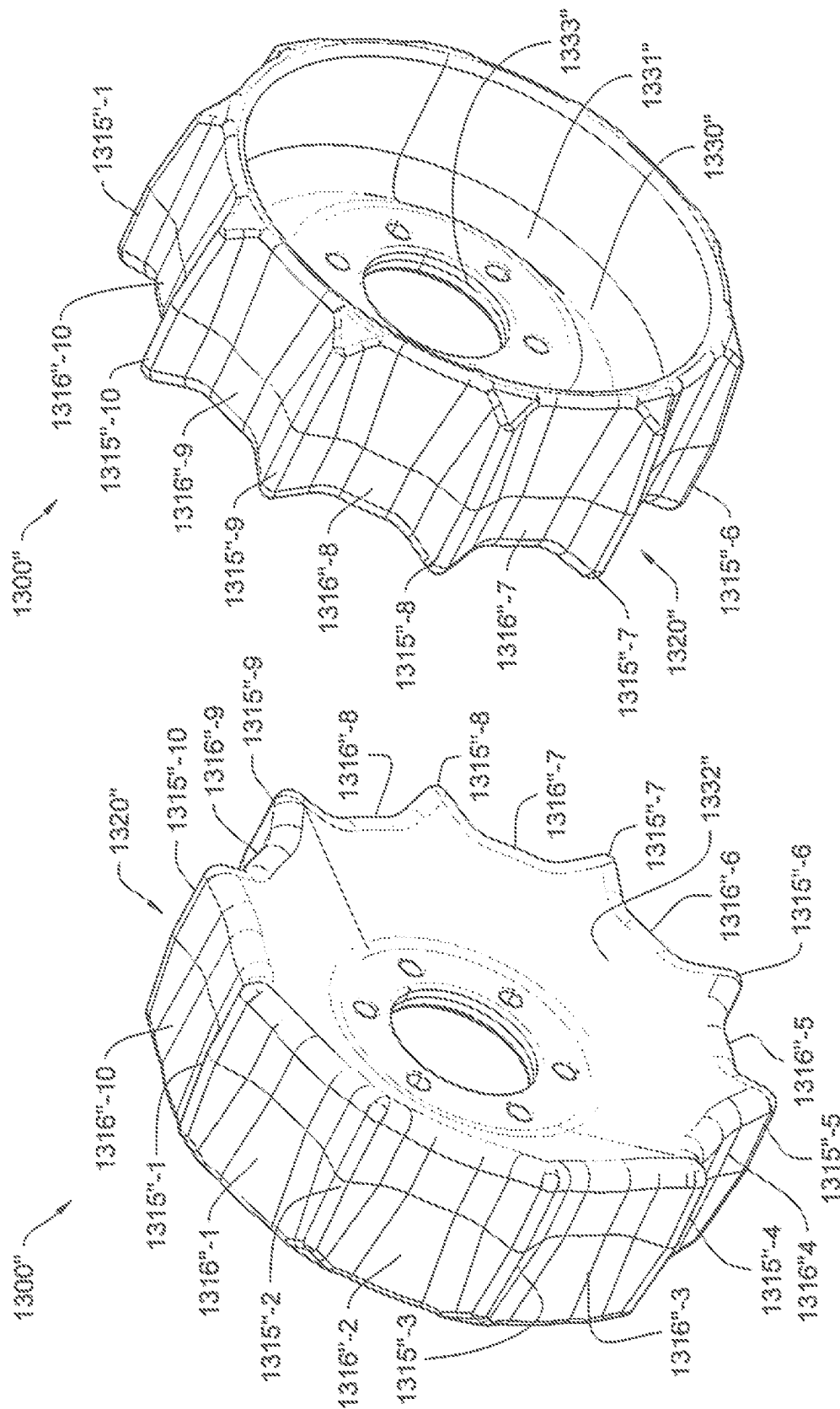


FIG. 125

FIG. 126

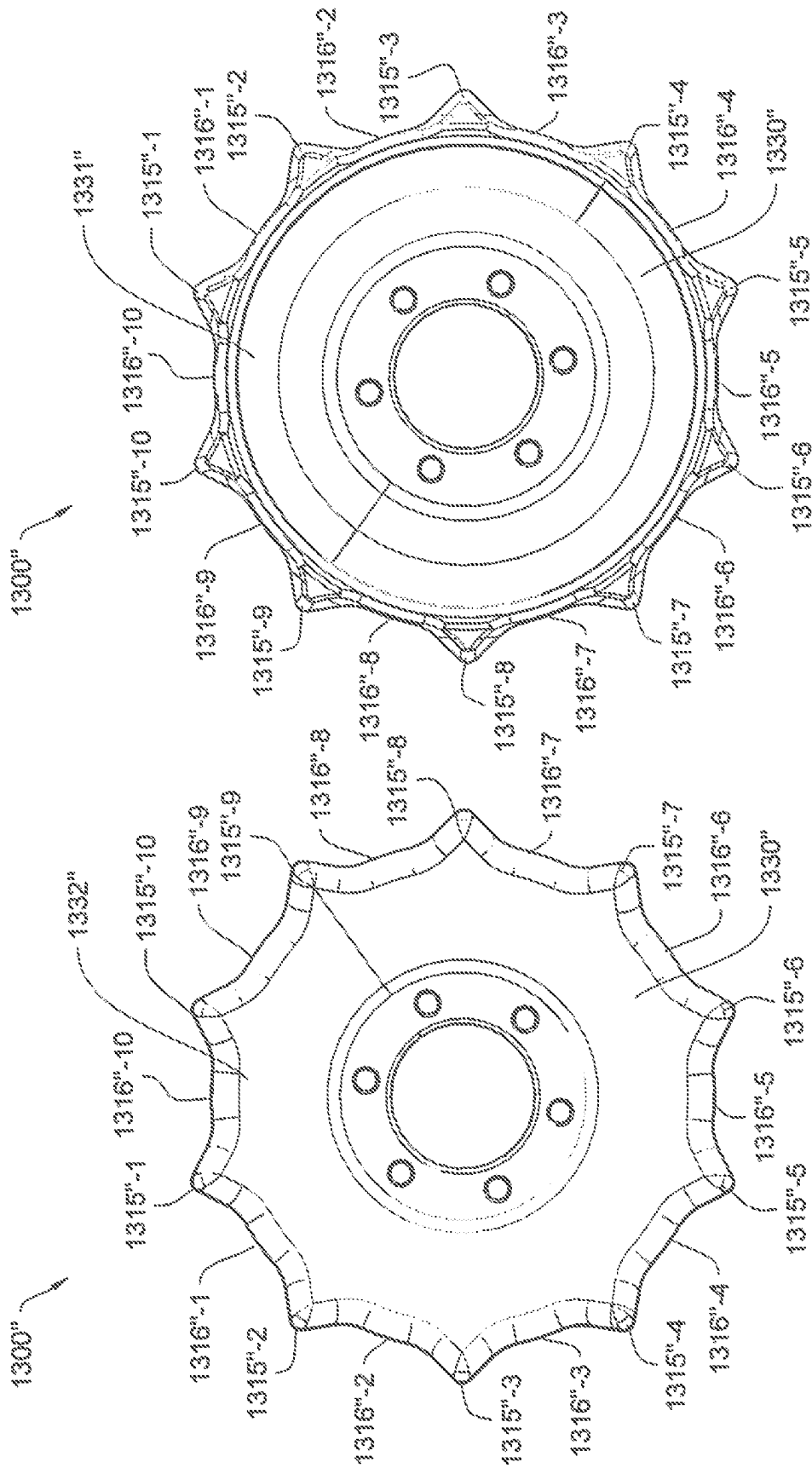
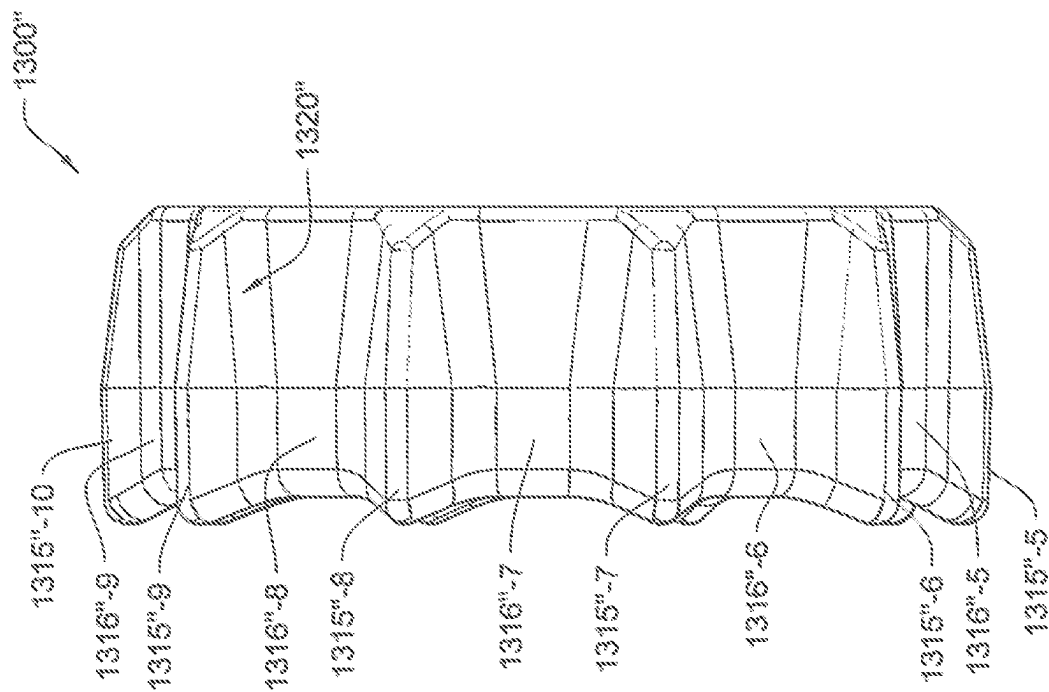
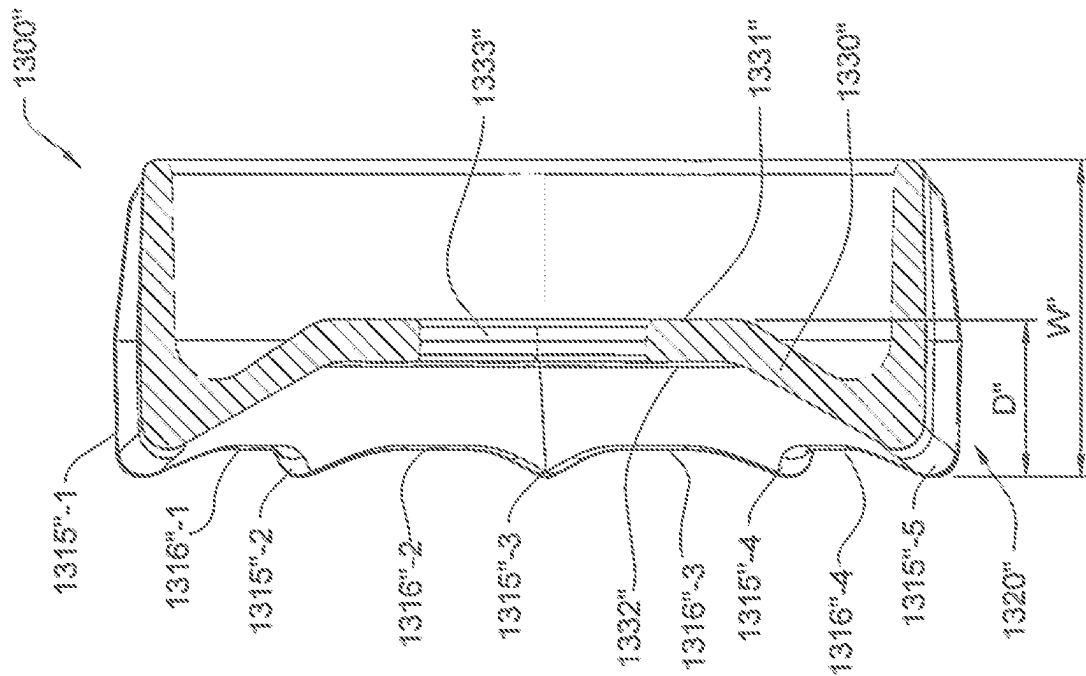


FIG. 127

FIG. 128



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WHEEL FOR CLOSING SYSTEM

BACKGROUND

It is well known that good seed-to-soil contact within the seed trench is a critical factor in uniform seed emergence and high yields. To obtain good seed-to-soil contact, once seeds are planted in a seed trench, the seed trench needs to be closed so that soil surrounds the seed.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a side elevation view of an embodiment of a row unit of an agricultural planter.

FIG. 2 is a diagram of a system for implementing operational control of the closing wheel assembly based on signals generated by the trench closing sensor.

FIG. 3 is a side elevation view of an embodiment of a trench closing assembly.

FIG. 4 is a top plan view of the trench closing assembly of FIG. 3.

FIG. 5 is a bottom plan view of the trench closing assembly of FIG. 3.

FIG. 6 is a perspective view of the trench closing assembly of FIG. 3.

FIG. 7 is a perspective view of the frame of the trench closing assembly of FIG. 3.

FIG. 8 is a bottom plan view of the frame of the trench closing assembly of FIG. 7.

FIG. 9 is a side elevation view of the frame of the trench closing assembly of FIG. 7.

FIG. 10 is a perspective view of the frame of the trench closing assembly of FIG. 7 with the actuator removed.

FIG. 11 is a bottom perspective view of the frame of the trench closing assembly of FIG. 10 with a swing arm removed.

FIG. 12A is top perspective view of the main frame of the trench closing assembly.

FIG. 12B is bottom perspective view of the main frame of the trench closing assembly.

FIG. 13 is a perspective view of the press wheel frame of the trench closing assembly.

FIG. 14 is perspective view of the transfer bar of the trench closing assembly.

FIG. 15 is a perspective view of the transfer bar and swing arms of the trench closing assembly.

FIG. 16 is a perspective view of the transfer bar and swing arms of FIG. 15 and further including the press wheel frame.

FIG. 17 is a side elevation view of a portion of the trench closing assembly.

FIG. 18 is a perspective view of portions of the trench closing assembly.

FIG. 19 is a perspective view of portions of the trench closing assembly.

FIG. 20 is a perspective view of portions of the trench closing assembly.

FIG. 21 is a side elevation view of a wheel according to one embodiment.

FIG. 22 is a perspective view of the wheel of FIG. 21.

FIG. 23 is a side elevation view of an embodiment of the spoke disk of the wheel of FIG. 21.

FIG. 24 is a perspective view of the spoke disk of FIG. 23 showing some of the flanges received within the notches of respective spoke arms.

FIG. 25 is a schematic illustration of a fluid system according to one embodiment.

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FIG. 26 is a schematic illustration of a fluid system according to another embodiment.

FIG. 27 is a perspective view of the trench closing assembly with a force sensor.

FIG. 28 is a side elevation view of the modified mounting arm with a position sensor.

FIG. 29 is a perspective view of an embodiment of a trench closing assembly.

FIG. 30 is a top plan view of the trench closing assembly of FIG. 29.

FIG. 31 is a bottom plan view of the trench closing assembly of FIG. 29.

FIG. 32 is a side elevation view of the trench closing assembly of FIG. 29.

FIG. 33 is a perspective view of the trench closing assembly of FIG. 29 with the actuator and a wheel removed.

FIG. 34 is a perspective view of the trench closing assembly of FIG. 33 with the actuator base removed.

FIG. 35 is a side elevation view of the trench closing assembly of FIG. 34 with the wheels removed.

FIG. 36 is a bottom perspective view of the trench closing assembly of FIG. 35.

FIG. 37 is a perspective view of the trench closing assembly of FIG. 35 with additional parts removed.

FIG. 38A is a side elevation view of the mounting arm.

FIG. 38B is a perspective view of the press wheel frame.

FIG. 39 is a perspective view of the trench closing assembly of FIG. 37 with additional parts removed.

FIG. 40 is a bottom perspective view of the trench closing assembly of FIG. 39.

FIG. 41 is a perspective view of the trench closing assembly of FIG. 39 with additional parts removed.

FIG. 42 is a bottom perspective view of the crossbar connected to the arms.

FIG. 43 is a side elevation view of FIG. 42.

FIG. 44 is a perspective view of a press wheel according to one embodiment.

FIG. 45 is a side elevation view of the press wheel of FIG. 44.

FIG. 46 is a side elevation view of the press wheel of FIG. 45 with the hub removed.

FIG. 47 is a side elevation view of a trench closing assembly with adjustable depth according to one embodiment.

FIG. 48 is a top plan view of the trench closing assembly of FIG. 47.

FIG. 49 is a schematic of a valve system.

FIG. 50 is a perspective view of an embodiment of a trench closing assembly.

FIG. 51 is a top plan view of the trench closing assembly of FIG. 50.

FIG. 52 is a bottom plan view of the trench closing assembly of FIG. 50.

FIG. 53 is a side elevation view of the trench closing assembly of FIG. 50.

FIG. 54 is a perspective view of the trench closing assembly of FIG. 50 with a wheel removed.

FIG. 55 is a perspective view of the trench closing assembly of FIG. 50 with the wheels, scrapper, and actuator removed.

FIG. 56 is a side elevation view of the trench closing assembly of FIG. 55.

FIG. 57 is a perspective view of the trench closing assembly of FIG. 55 with an arm removed.

FIG. 58 is a perspective view of the trench closing assembly of FIG. 57 with the connection bracket removed.

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FIG. 59 is a perspective view of a portion of the trench closing assembly of FIG. 50.

FIG. 60 is a perspective view of the trench closing assembly of FIG. 59 with the knob and force sensor removed.

FIG. 61 is a perspective view of the trench closing assembly of FIG. 50 with parts removed for viewing.

FIG. 62 is a side elevation view of the trench closing assembly of FIG. 61.

FIG. 63 is a perspective view of a portion of the trench closing assembly of FIG. 50.

FIG. 64 is a perspective view of an arm.

FIG. 65 is a perspective view of the connection bracket.

FIG. 66 is a perspective view of a packing wheel system with a force sensor of one embodiment.

FIG. 67 is a perspective view of a packing wheel system with a force sensor of one embodiment.

FIG. 68 is a perspective view of a packing wheel system with a force sensor of one embodiment.

FIG. 69 is a perspective view of a packing wheel system with a force sensor of one embodiment.

FIG. 70 is a perspective view of a trench closing assembly further including a soil leveler according to one embodiment.

FIG. 71 is a side elevation view of the trench closing assembly of FIG. 70.

FIG. 72 is a side elevation view of the soil leveler of FIG. 70.

FIG. 73 is a perspective view of the soil leveler of FIG. 72.

FIG. 74 is a bottom plan view of the soil leveler of FIG. 72.

FIG. 75 is a perspective view of a trench closing assembly further including a soil leveler according to one embodiment.

FIG. 76 is a side elevation view of the trench closing assembly of FIG. 75.

FIG. 77 is a perspective view of the soil leveler of FIG. 75.

FIG. 78 is a side elevation view of the soil leveler of FIG. 77.

FIG. 79 is a perspective view of a portion of the soil leveler of FIG. 75.

FIG. 80 is a left side elevation view of an embodiment of a trench closing assembly.

FIG. 81 is a top plan view of the trench closing assembly of FIG. 80.

FIG. 82 is a bottom plan view of the trench closing assembly of FIG. 80.

FIG. 83 is a top perspective view of the trench closing assembly of FIG. 80 with some parts removed.

FIG. 84 is a bottom perspective view of the trench closing assembly of FIG. 83.

FIG. 85 is a bottom perspective view of the trench closing assembly of FIG. 84 with some parts removed.

FIG. 86 is a bottom perspective view of the trench closing assembly of FIG. 85 with some parts removed.

FIG. 87 is a top perspective view of the trench closing assembly of FIG. 86.

FIG. 88 is a top perspective view of the frame of the trench closing assembly of FIG. 80.

FIG. 89 is a bottom perspective view of the frame of FIG. 88.

FIG. 90 is bottom plan view of the frame of FIG. 88.

FIG. 91 is top perspective view of the frame of the trench closing assembly of FIG. 80.

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FIG. 92 is a bottom perspective view of the frame of FIG. 91.

FIG. 93 is a top perspective view of the connection bracket of the frame of FIG. 91.

FIG. 94 is a bottom perspective view of the connection bracket of FIG. 93.

FIG. 95 is a top perspective view of the attachment bracket of the frame of FIG. 91.

FIG. 96 is a bottom perspective view of the attachment bracket of FIG. 95.

FIG. 97 is top perspective view of the mounting arm of the trench closing assembly of FIG. 80.

FIG. 98 is a bottom perspective view of the mounting arm of FIG. 97.

FIG. 99 is a front elevation view of the handle assembly of the trench closing assembly of FIG. 80.

FIG. 99 is a left side elevation view of the handle assembly of FIG. 98.

FIG. 101 is a perspective view of the shank of the handle assembly of FIG. 98.

FIG. 102 is a bottom perspective view of the pressure sensor of the handle assembly of FIG. 98.

FIG. 103 is a top perspective view of the pressure sensor of FIG. 102.

FIG. 104 is a perspective view of the handle assembly and mounting arm of the trench closing assembly of FIG. 80.

FIG. 105 is a side elevation view of a press wheel according to one embodiment.

FIG. 106 is a perspective view of the press wheel of FIG. 105.

FIG. 107 is a front elevation view of the press wheel of FIG. 105.

FIG. 108 is a perspective view of an embodiment of a fluid control assembly.

FIG. 109 is an exploded perspective view of the fluid control assembly of FIG. 108.

FIG. 110 is a top plan view of the fluid control assembly of FIG. 108.

FIG. 111 is a perspective view of the board of the fluid control assembly of FIG. 108.

FIG. 112 is a bottom perspective view of the housing of the fluid control assembly of FIG. 108.

FIG. 113 is a section elevation view of the housing of FIG. 112.

FIG. 114 is a bottom perspective view of the section of the housing of FIG. 113.

FIG. 115 is an enlarged perspective view of one embodiment of the closing wheel shown in FIG. 80.

FIG. 116 is a side elevation of the closing wheel of FIG. 115.

FIG. 117 is a perspective view of a trench closing assembly with an embodiment of a wheel which may be used as press wheels.

FIG. 118 is a perspective view of one of the wheels of FIG. 117 shown in isolation.

FIG. 119 is a front elevation view of the wheel of FIG. 118.

FIG. 120 is a rear elevation view of the wheel of FIG. 118.

FIG. 121 is another perspective view of the wheel of FIG. 118 with the hub removed.

FIG. 122 is a side elevation view of the wheel of FIG. 118.

FIG. 123 is a cross-section view of the wheel of FIG. 118.

FIG. 124 is a perspective view of the cross-sectional view of FIG. 123.

FIG. 125 is a perspective view of an outer side of another embodiment of a wheel which may be used as one of the press wheels for a trench closing assembly.

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FIG. 126 is a perspective view of the inner side of the wheel of FIG. 125.

FIG. 127 is an elevation view of the outer side of the wheel of FIG. 125.

FIG. 128 is an elevation view of the inner side of the wheel of FIG. 125.

FIG. 129 is a side elevation view of the wheel of FIG. 125.

FIG. 130 is a cross-section view of the wheel of FIG. 125.

DESCRIPTION

Referring now to the drawings, wherein like reference numerals designate identical or corresponding parts throughout the several views, FIG. 1 illustrates an embodiment of an agricultural planter row unit 200. The row unit 200 is comprised of a frame 204 pivotally connected to a toolbar 202 by a parallel linkage 206 enabling each row unit 200 to move vertically independently of the toolbar 202. The frame 204 operably supports one or more hoppers 208, a seed meter 210, a seed delivery mechanism 212, a downforce control system 214, a seed trench opening assembly 220, a trench closing assembly 250, an optional packer wheel assembly 260, and an optional row cleaner assembly 270. It should be understood that the row unit 200 shown in FIG. 1 may be for a conventional planter or the row unit 200 may be a central fill planter, in which case the hoppers 208 may be replaced with one or more mini-hoppers and the frame 204 modified accordingly as would be recognized by those of skill in the art.

The optional downforce control system 214 is disposed to apply lift and/or downforce on the row unit 200 such as disclosed in U.S. Publication No. US2014/0090585, which is incorporated herein in its entirety by reference. The downforce applied by downforce control system 214 can be determined by methods disclosed in U.S. Pat. No. 9,173, 339, which is incorporated herein in its entirety by reference.

The seed trench opening assembly 220 includes a pair of opening discs 222 rotatably supported by a downwardly extending shank member 205 of the frame 204. The opening discs 222 are arranged to diverge outwardly and rearwardly so as to open a v-shaped trench 10 in the soil 11 as the planter traverses the field. The seed delivery mechanism 212, such as a seed tube or seed conveyor, is positioned between the opening discs 222 to deliver seed from the seed meter 210 into the opened seed trench 10. The depth of the seed trench 10 is controlled by a pair of gauge wheels 224 positioned adjacent to the opening discs 222. The gauge wheels 224 are rotatably supported by gauge wheel arms 226 which are pivotally secured at one end to the frame 204 about pivot pin 228. A rocker arm 230 is pivotally supported on the frame 204 by a pivot pin 232. It should be appreciated that rotation of the rocker arm 230 about the pivot pin 232 sets the depth of the trench 10 by limiting the upward travel of the gauge wheel arms 226 (and thus the gauge wheels) relative to the opening discs 222. The rocker arm 230 may be adjustably positioned via a linear actuator 234 mounted to the row unit frame 204 and pivotally coupled to an upper end of the rocker arm 230. The linear actuator 234 may be controlled remotely or automatically actuated as disclosed, for example, in International Publication No. WO2014/186810, which is incorporated herein in its entirety by reference.

An optional downforce sensor 238 is configured to generate a signal related to the amount of force imposed by the gauge wheels 224 on the soil. In some embodiments the pivot pin 232 for the rocker arm 230 may comprise the downforce sensor 238, such as the instrumented pins dis-

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closed in U.S. Pat. No. 8,561,472, which is incorporated herein in its entirety by reference.

An optional seed meter 210 may be any commercially available seed meter, such as a finger-type meter or a vacuum seed meter. One exemplary vacuum seed meter is the VSet® meter, available from Precision Planting LLC, 23207 Townline Rd, Tremont, IL 61568.

An optional packer wheel assembly 260 comprises an arm 262 pivotally attached to the row unit frame 204 and extends rearward of the closing wheel assembly 250 and in alignment therewith. The arm 262 rotatably supports a packer wheel 264. An actuator 266 is pivotally attached at one end to the arm 262 and at its other end to the row unit frame 204 to vary the amount of downforce exerted by the packer wheel 264 to pack the soil over the seed trench 10.

An optional row cleaner assembly 270 may be the Clean-Sweep® system available from Precision Planting LLC, 23207 Townline Rd, Tremont, IL 61568. The row cleaner assembly 270 includes an arm 272 pivotally attached to the forward end of the row unit frame 204 and aligned with the trench opening assembly 220. A pair of row cleaner wheels 274 are rotatably attached to the forward end of the arm 272. An actuator 276 is pivotally attached at one end to the arm 272 and at its other end to the row unit frame 204 to adjust the downforce on the arm to vary the aggressiveness of the action of the row cleaning wheels 274 depending on the amount of crop residue and soil conditions.

Referring to FIG. 2, a monitor 300 is visible to an operator within the cab of a tractor pulling the planter. The monitor 300 may be in signal communication with a GPS unit 310, the trench closing assembly actuator 256 and the optional packer wheel assembly actuator 266 to enable operational control of the trench closing assembly 250 and the optional packer wheel assembly 260 based on the signals generated by trench closing sensors 1000, which are described in International Publication No. WO2017/197274, which is incorporated herein by reference in its entirety. Also as discussed later, the monitor 300 may be programmed to display operational recommendations based on the signals generated by the trench closing sensors 1000. The monitor 300 may also be in signal communication with the row cleaner actuator 276, the downforce control system 214, the depth adjustment actuator 234 to enable operational control of row cleaner assembly 270, the downforce control system 214 and the trench opening assembly 230, respectively.

Trench Closing Assembly

FIGS. 3 to 24 illustrate a trench closing assembly 250 according to one embodiment. The trench closing assembly 250 is adapted to be attached to row unit 200. Trench closing assembly 250 has a frame 251, an actuator 259, a pair of closing wheels 254, and optionally, a pair of press wheels 255. While illustrated with a pair of press wheels 255, a single press wheel 255 can be used. Actuator 259 can apply one force to frame 251, and this force can be divided between the closing wheels 254 and the press wheels 255.

Actuator 259 can be any actuator that can apply a force. Examples of actuators include, but are not limited to, pneumatic actuators, hydraulic actuators, electro-mechanical actuators, and electro-hydraulic actuators. In one embodiment, actuator 259 is a pneumatic actuator, such as an air bag or an air cylinder. A gas supply (not shown) can be connected to nozzle 1259 to supply gas (such as air) to actuator 259.

Referring to FIGS. 7 and 12A and 12B, frame 251 has a main frame 1200 having a connection bracket 1201 that is adapted to be connected to row unit 200. Optionally, one or more bolts 1209 extending through apertures 1207 (FIG.

12A) in the connection bracket **1201** for mounting the frame **251** to the row unit **200**. Extending up from connection bracket **1201** is actuator bracket **1210**. Actuator bracket **1210** in one embodiment has a first bracket arm **1211-1**, a second bracket arm **1211-2**, and a cross connector **1212** connecting first bracket arm **1211-1** and second bracket arm **1211-2**. Main frame **1200** has a first wing **1202-1** and a second wing **1202-2** extending laterally outward transverse to a direction of travel of trench closing assembly **250**. As best viewed in FIGS. **12A-12B**, extending downwardly from connection bracket **1201** is a first inner bracket **1204-1** and a second inner bracket **1204-2** to which the first wing **1202-1** and second wing **1202-2** are respectively attached. Extending downwardly from wings **1202-1** and **1202-2** at an end opposite to where the wings **1202-1**, **1202-2** attach to inner brackets **1204-1**, **1204-2** are outer brackets **1205-1** and **1205-2**, respectively. Extending downwardly from wings **1202-1** and **1202-2** and between inner brackets **1204-1**, **1204-2** and outer brackets **1205-1**, **1205-2** are middle brackets **1203-1** and **1203-2**, respectively.

As seen in FIG. **18**, an actuator base **1213** is connected to first base arm **1214-1** and **1214-2**. A first transfer arm **1271-1** is connected to first base arm **1214-1**, and a second transfer arm **1271-2** is connected to second base arm **1214-2**. Connected to first transfer arm **1271-1** and second transfer arm **1271-2** at an end opposite to where the first base arm **1214-1** and second base arm **1214-2** connect is a transfer block **1269** through which a hole **1268** extends. As shown in FIG. **11**, a transfer bar **1260** is disposed through the hole **1268** in the transfer block **1269**.

As seen in FIG. **14**, transfer bar **1260** has a longitudinal bar **1262** and a transverse bar **1261**. Transfer bar **1260** can be a unitary part, or (as shown) longitudinal bar is disposed through a bore in the transverse bar **1261**. Longitudinal bar **1262** is generally oriented parallel to the direction of travel of trench closing assembly **250**, and transverse bar **1261** is generally oriented transverse to the direction of travel. At each end of transverse bar **1261** are tabs **1263-1** and **1263-2**. Tabs **1263-1** and **1263-2** are for mating with respective swing arms **1220-1** and **1220-2**, via respective holes **1225-1** and **1225-2**, as best seen in FIG. **15**.

Transfer bar **1260** divides the force from actuator **259** between closing wheels **254** (**254-1**, **254-2**) and press wheels **255** (**255-1**, **255-2**). Closing wheels **254-1** and **254-2** are connected to swing arms **1220-1** and **1220-2**, respectively. Force applied to transfer bar **1260** is transferred through transverse bar **1261** to tabs **1263-1** and **1263-2**. Also, force is transferred to press wheel frame **1250** via longitudinal bar **1262**.

As seen in FIG. **13**, press wheel frame **1250** has a first arm **1251-1**, a second arm **1251-2**, a first cross brace **1253**, which can optionally have a hole **1255** for receiving longitudinal bar **1262**, a second cross brace **1252** having a hole **1254** for receiving longitudinal bar **1262**, and a mounting arm **1256** to which press wheels **255-1** and **255-2** are mounted through bracket **1258**. Press wheel frame **1250** is pivotably disposed between inner brackets **1204** (**1204-1**, **1204-2**) and outer brackets **1205** (**1205-1**, **1205-2**) about pivots **1206-1** and **1206-2**, respectively.

Swing arms **1220** (**1220-1**, and **1220-2**) are pivotably disposed between inner brackets **1204** (**1204-1**, **1204-2**) and outer brackets **1205** (**1205-1**, **1205-2**) about pivots **1206-1** and **1206-2**, respectively.

As illustrated in FIGS. **17** to **19**, first base arm **1214-1** and second base arm **1214-2** are pivotably connected to main frame **1200** through a first pivot arm **1280** and a second pivot arm **1285**. First pivot arm **1280** has a first arm **1281-1** and

a second arm **1281-2**. First pivot arm **1280** and second pivot arm **1285** are pivotably disposed between inner brackets **1204-1** and **1204-2**. First pivot arm **1280** pivots about pivot **1283**, and second pivot arm pivots about pivot **1286**.

As illustrated in FIGS. **6** and **7**, press wheels **255-1** and **255-2** are disposed on bracket **1258**. Bracket **1258** has a plurality of holes **1257** for adjustable mating to mounting arm **1256**. Wheels of different diameters can be attached to bracket **1258** or a different placement of wheels can be used to change the distribution of force with the adjustable mating.

Optionally, a scrapper **1221** (**1221-1**) can be included. Scrapper **1221** is attached to swing arm **1220** and is disposed to receive a closing wheel **254**. While shown with one scrapper **1221-1**, a scrapper **1221-2** (not shown) can be attached to swing arm **1220-2** similar to scrapper **1221-1** being connected to swing arm **1220-1**.

FIGS. **29-43** illustrate another trench closing assembly **250A** according to another embodiment. The trench closing assembly **250A** is adapted to be attached to row unit **200**. Trench closing assembly **250A** has a frame **251A**, an actuator **259**, a pair of closing wheels **254-1**, **254-2**, and optionally, a press wheel **255A**. As illustrated, the press wheel **255A** may comprise a pair of press wheels **255A-1**, **255A-2**, but a single press wheel (not shown) may be utilized. Actuator **259** can apply one force to frame **251A**, and this force can be divided between the closing wheels **254** and the press wheel **255A**.

Referring to FIG. **29**, frame **251A** has a main frame **1200A** having a connection bracket **1201A** that is adapted to be connected to row unit **200**. As in the previous embodiment, one or more bolts **1209** (FIG. **33**) can extend through apertures **1207** in connection bracket **1201A** for mounting the frame **251A** to the row unit **200**. Extending up from connection bracket **1201A** is actuator bracket **1210A**. In one embodiment, actuator bracket **1210A** has a first bracket arm **1211A-1**, a second bracket arm **1211A-2**, and a cross connector **1212A** connecting first bracket arm **1211A-1** and second bracket arm **1211A-2**. Actuator bracket **1210A** can be made from separate parts or as a single part. As best viewed in FIGS. **34** and **37**, extending downwardly from connection bracket **1201A** are first bracket **1204A-1** and second bracket **1204A-2** with cross-braces **1208A-1** and **1208A-2** extending first bracket **1204A-1** and second bracket **1204A-2**. Extending outward, transverse to the direction of travel, are optional stops **1229A**. Stop **1229A-1** is disposed on first bracket **1204A-1**, and stop **1229A-2** is disposed on second bracket **1204A-2**. Stop **1229A-1** cooperates with stops **1228A** (**1228A-1a** and **1228A-1b**) on first swing arm **1220A-1** and stops **1259A** (**1259A-1a** and **1259A-1b**) on first arm **1251A-1**. Stop **1229A-2** cooperates with stops **1228A** (**1228A-2a** and **1228A-2b**) on second swing arm **1220A-2** and stops **1259A** (**1259A-2a** and **1259A-2b**) on second arm **1251A-2**. The angle of rotation of swing arms **1220A** and first and second arms **1251A** can be limited.

As viewed in FIGS. **39** and **40**, extending through first bracket **1204A-1** is pivot **1206A-1** transverse to the direction of travel. Extending through second bracket **1204A-2** is pivot **1206A-2** transverse to the direction of travel. Pivots **1206A-1** and **1206A-2** allow for pivoting of swing arms **1220A-1** and **1220A-2**, respectively, first arm **1251A-1** and second arm **1251A-2**, respectively, and arms **1295A-1** and **1295A-2**, respectively.

Swing arms **1220A-1** and **1220A-2** are pivotably disposed about pivots **1206A-1** and **1206A-2**, respectively. Swing arms **1220A-1** and **1220A-2** are adjustable transverse to the

direction of travel along pivot **1206A-1** and pivot **1206A-2**, respectively. This allows for changing the width of spacing of closing wheels **254**.

Illustrated in FIG. **38A**, press wheel frame **1250A** has a first arm **1251A-1** and a second arm **1251A-2**. First arm **1251A-1** and second arm **1251A-2** pivot about pivots **1206A-1** and **1206A-2**, respectively. Mounting arm **1256A** is connected to both first arm **1251A-1** and second arm **1251A-2**. First arm **1251A-1** can optionally include stops **1259A-1a** and **1259A-1b** to limit the rotation of first arm **1251A-1** about pivot **1206A-1**. Second arm **1251A-2** can optionally include stops **1259A-2a** and **1259A-2b** to limit the rotation of second arm **1251A-2** about pivot **1206A-2** at stops **1229A-1** and **1229A-2**, respectively. As seen in FIG. **38B**, mounting arm **1256A** can have a plurality of holes **1257A** to provide adjustment for the distance of press wheels **255A** along the direction of travel.

As viewed in FIGS. **39** to **43**, also connected to press wheel frame **1250A** is arm **1290A**. Arm **1290A** connects with brackets **1214A-1** and **1214A-2**. Brackets **1214A-1** and **1214A-2** are connected to actuator base **1213A** (as seen in FIGS. **32** and **33**). As force is applied from actuator **259** through actuator base **1213A**, force is applied to both the closing wheels **254** and press wheels **255A**. Arm **1290A** is also connected to arms **1295A-1** and **1295A-2**. Arms **1295A-1** and **1295A-2** are disposed about pivots **1206A-1** and **1206A-2**, respectively. Crossbar **1260A** is connected to arms **1295A-1** and **1295A-2** and is disposed transverse to the direction of travel. As seen in FIG. **36**, crossbar **1260A** has a first end **1261A-1** disposed in opening **1225A-1** of swing arm **1220A-1** and a second end **1261A-2** disposed in opening **1225A-2** of swing arm **120A-2**.

In another embodiment, an angle sensor, which is described in International Publication No. WO2017/197274 as angle sensor **3006**, or in International Publication No. WO2014/066650 as angle sensor **280**, can be included. As illustrated in FIGS. **39** and **41**, the angle sensor is a Hall effect sensor **1400A** and a magnet **1401A**. While it can be installed on either the right or left side, Hall effect sensor **1400A** is illustrated as being disposed on second bracket **1204A-2** adjacent to arm **1295A-2**. A magnet **1401A** is disposed on arm **1295A-2**. In this embodiment, the angular rotation of arm **1295A-2**, which is connected to the entire assembly of the closing wheels **254** and press wheels **255A**, is measured as the average of both closing wheels **254** and both press wheels **255A**.

Optionally, a bracket **1299** can be included to route a line (not shown) through. The line could be for applying material into the furrow, such as fertilizer, herbicide, or insecticide. Bracket **1299** can be connected to crossbar **1260A** as seen in FIG. **40**.

As seen in FIG. **41**, there is an arm **1296A** pivotably disposed between brackets **1214A-1** and **1214A-2** about pivot **1293A** and between first bracket **1204A-1** and second bracket **1204A-2** about pivot **1295A**.

Optionally, a trailing arm (not shown), such as the bracket **132** and flap **130** from International Publication No. WO2014/066650, can be included. The optional trailing arm can be connected to mounting arm **1256A** at connection **1288A**.

Optionally, toe angle shim **1290A** (**1290A-1** and **1290A-2**) can be included to change the toe angle of closing wheels **254A** (**254A-1**, **254A-2**) by being disposed over axle **1291A** (**1291A-1**, **1291A-2**), respectively.

The trench closing assembly **250** or **250A** can be a single stage by not including press wheels **255**, **255A** and the associated press wheel frame **1250**, **1250A**.

In any of the embodiments, the distribution of force between the closing wheels **254** and press wheels **255** can be adjusted. In some embodiments, 80% of the force applied by actuator **259** is applied to the closing wheels **254** and 20% to the press wheels **255**. In another embodiment, 100% of the force can be applied to the closing wheels **254**.

FIGS. **47** and **48** illustrate a trench closing assembly **250B** according to another embodiment. In this embodiment, trench closing assembly **250B** has a main frame **251B** that is pivotably connected to row unit **200**. Disposed on row unit frame **204** is bracket **249** that extends over main frame **251B**. Disposed between bracket **249** and main frame **251B** is actuator **259** to apply a variable force to trench closing assembly **250B**. Closing wheels **254** (**254-1** and **254-2**) are disposed on main frame **251B**. Pivotably connected to main frame **251B** is secondary frame **252B**. Secondary wheels **255** (**255-1**, **255-2**), such as press wheels, are disposed on secondary frame **252B**. Extending upward from main frame **251B** is bracket **8251**, and extending upward from secondary frame **252B** is bracket **8252**. Connecting brackets **8251** and **8252** is depth adjuster **253B**. The relative angle between main frame **251B** and secondary frame **252B** is adjusted with depth adjuster **253B**. This allows secondary wheels **255** to act as gauge wheels for closing wheels **254**.

FIGS. **50-63** illustrate a trench closing assembly **250C** according to another embodiment. In this embodiment, trench closing assembly **250C** has a main frame **251C** that is pivotably connected to row unit **200**. Trench closing assembly **250C** has a frame **251C**, an actuator **259**, a pair of closing wheels **254-1**, **254-2**, and optionally, a press wheel **255C**. As illustrated, the press wheel **255C** may comprise a pair of press wheels **255C-1**, **255C-2**, but a single press wheel (not shown) may be utilized. Actuator **259** can apply one force to frame **251C**, and this force can be divided between the closing wheels **254** and the press wheel **255C**.

Referring to FIG. **50**, frame **251C** has a main frame **1200C** having a connection bracket **1201C** that is adapted to be connected to row unit **200**. As in a previous embodiment, one or more bolts **1209** (FIG. **33**) can extend through apertures **1207** in connection bracket **1201C** for mounting the frame **251C** to the row unit **200**. Extending up from connection bracket **1201C** is actuator bracket **1210C**. In one embodiment, actuator bracket **1210C** has a first bracket arm **1211C-1**, a second bracket arm **1211C-2**, and a cross connector **1212C** connecting first bracket arm **1211C-1** and second bracket arm **1211C-2**. Actuator bracket **1210C** can be made from separate parts or as a single part. As best viewed in FIGS. **55**, **58**, and **65**, extending downwardly from connection bracket **1201C** are first bracket **1204C-1** and second bracket **1204C-2** with cross-braces **1208C-1**, **1208C-2**, and **1208C-3** extending first bracket **1204C-1** and second bracket **1204C-2**. Extending outward, transverse to the direction of travel, are optional stops **1229C**. Stop **1229C-1** is disposed on first bracket **1204C-1**, and stop **1229C-2** is disposed on second bracket **1204C-2**. Stop **1229C-1** cooperates with stops **1228C** (**1228C-1a** and **1228C-1b**) on first swing arm **1220C-1**. Stop **1229C-2** cooperates with stops **1228C** (**1228C-2a** and **1228C-2b**) on second swing arm **1220C-2**. The angle of rotation of swing arms **1220C** can be limited.

As viewed in FIG. **61**, extending through first bracket **1204C-1** is pivot **1206C-1** transverse to the direction of travel. Extending through second bracket **1204C-2** is pivot **1206C-2** transverse to the direction of travel. Pivots **1206C-1** and **1206C-2** allow for pivoting of swing arms **1220C-1** and **1220C-2**, respectively.

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Swing arms **1220C-1** and **1220C-2** are pivotably disposed about pivots **1206C-1** and **1206C-2**, respectively. Swing arms **1220C-1** and **1220C-2** are adjustable transverse to the direction of travel along pivot **1206C-1** and pivot **1206C-2**, respectively. This allows for changing the width of spacing of closing wheels **254**.

As seen in FIG. **61**, an actuator base **1213C** is connected to first base arm **1214C-1** and second base arm **1214C-2**. First base arm **1214C-1** and second base arm **1214C-2** are pivotably connected to pivot **1223**. Also first transfer arm **1287-1** and second transfer arm **1287-2** are disposed about pivot **1223**. Disposed forward of pivot **1223** in a direction of travel is pivot **1224**. Pivot **1224** is disposed through swing arm **1220C-1** and swing arm **1220C-2**, and first transfer arm **1287-1** and second transfer arm **1287-2** are pivotably disposed about pivot **1224**. Disposed rearward of pivot **1223** in a direction of travel is pivot **1226**. Pivot **1226** is disposed through swing arm **1220C-1** and swing arm **1220C-2**, and first transfer arm **1287-1** and second transfer arm **1287-2** are pivotably disposed about pivot **1226**. Also disposed about pivot **1226** are first bracket **1284-1** and second bracket **1284-2**. Disposed between first bracket **1284-1** and second bracket **1284-2** is cross brace **1264**. As viewed in FIG. **63**, cross brace **1264** has a hole **1265** for bolt **1273** to be disposed through. As viewed in FIG. **53**, mounting arm **1256C** is connected between first bracket **1284-1** and second bracket **1284-2**. Mounting arm **1256C** has the same design as a previous embodiment. An arm **1296C** is disposed about pivot **1293C** between first base arm **1214C-1** and second base arm **1214C-2**. Arm **1296C** is also disposed about pivot **1222** between first bracket **1204C-1** and second bracket **1204C-2**.

As seen in FIGS. **56-63**, an adjustor **1272** is included for changing the relative position of closing wheels **254** (**254-1** and **254-2**) to press wheels **255C** (**255C-1** and **255C-2**). By changing the relative position, the percentage of the applied force from actuator **259** is changed between the closing wheels **254** (**254-1** and **254-2**) the press wheels **255C** (**255C-1** and **255C-2**). Adjustor **1272** has a bracket **1276** having a first arm **1279-1** a second arm **1279-2**, a cross-connector **1277**, and a hole **1278** through cross-connector **1277**. Bracket **1276** can be made as a unitary part or from separate parts. First arm **1279-1** and second arm **1279-2** are pivotably disposed about pivot **1227**. Pivot **1227** is disposed through swing arm **1220C-1** and swing arm **1220C-2**. Disposed through hole **1278** is bolt **1273**. Bolt **1273** is also disposed through hole **1265** in cross brace **1264**. Knob **1275** is disposed about bolt **1273** to adjust the position of bolt **1273**. Optionally, a retaining nut **1274** can be disposed at the end of bolt **1273** to retain knob **1275** on bolt **1273**. Optionally, force sensor **1298** can be disposed between knob **1275** and cross brace **1264** about bolt **1273**. The combined down force applied to closing wheels **254** (**254-1** and **254-2**) and press wheels **255C** (**255C-1** and **255C-2**) can be measured by force sensor **1298**. An example of force sensor **1298** is a load sensor, such as the Case IH Load Sensor, Part No. 725875, from Precision Planting, LLC.

As bolt **1273** is adjusted upwards by knob **1275**, pivot **1273** is pulled upwards by first arm **1279-1** and second arm **1279-2**, which raises swing arms **1220C** (**1220C-1** and **1220C-2**) under adjustor **1272**. As bolt **1273** is adjusted downwards by knob **1275**, pivot **1273** is lowered by first arm **1279-1** and second arm **1279-2**, which lowers swing arms **1220C** (**1220C-1** and **1220C-2**) under adjustor **1272**.

Wheel

FIGS. **21-24** illustrate a wheel **1300** according to one embodiment. Wheel **1300** can be used as press wheel **255**,

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255A or **255C**. Wheel **1300** comprises a spoke disk **1310** having a hub **1301** and a plurality of spokes **1302** (**1302-1** to **1302-12**) radially disposed about hub **1301**. Disposed at the end of a spoke **1302** is a cleat **1303** (**1303-1** to **1303-12**). Each spoke **1302** can be a two part spoke. FIG. **23** illustrates the spoke disk **1310** having a plurality of spoke arms **1311** (**1311-1** to **1311-12**). Each spoke arm **1311-1** to **1311-12** has a leading edge **1314** and a trailing edge **1315**. The radial end of each spoke arm **1311-1** to **1311-12** has a respective notch **1312-1** to **1312-12**. Received within each notch **1312-1** to **1312-12** is a respective flange **1313-1** to **1313-12**. FIG. **24** is a perspective view of the spoke disk **1310** showing flanges **1313-12** and **1313-3** disposed in the respective notches **1312-12** and **1312-3** of spoke arms **1311-12** and **1311-3**. As best viewed in FIGS. **21** and **22**, a respective cleat **1303-1** to **1303-12** is attached to each of the respective flanges **1313-1** to **1313-12**. As best viewed in FIG. **22**, the cleat **1303** can have an L shape. Referring to FIG. **21**, a forward edge **1304** of each of the cleats **1303** is shown aligned with the leading edge **1314** of the respective spoke arm **1311**, with a rearward end **1305** of each cleat **1303** extending rearwardly of the trailing end **1315** of the spoke arm **1311** substantially spanning the space between the adjacent radial spokes **1302**.

In one aspect, wheel **1300** can move soil from the sides of a seed trench to knit the soil together to increase the amount of closing of the seed trench.

FIGS. **44-46** illustrate another embodiment of a wheel **1300A**. Wheel **1300A** can be used as press wheel **255**, **255A** or **255C**. Wheel **1300A** comprises a spoke disk **1310A** and a hub **1301A**. Spoke disk **1310A** can be molded as a unitary part. Spoke disk **1310A** has plurality of spokes **1302A** (**1302A-1** to **1302A-10**). Connecting spokes **1302A** is a tread **1320A**. Tread **1320A** has a rib **1315A** (**1315A-1** to **1315A-10**) disposed at the radial end of spoke **1302A**. Between each rib **1315A**, there is a tread portion **1316A** (**1316A-1** to **1316A-10**). Tread portion **1316A** can extend through the entire width **W** of a rib **1315A**, or tread portion **1316A** can extend only a portion of the width **W** to leave a gap **1317A** (**1317A-1** to **1317A-10**).

FIGS. **117-124** illustrate wheel **1300'**. The wheel **1300'** may be used in place of a press wheel **255** (**255**, **255A**, **255C**, **255D**) or the wheel **1300** (**1300**, **1300A**, **1300B**), which are described above and in U.S. Provisional Patent Application No. 62/731,813 and International Application No. PCT/US2019/020452, both of which are incorporated herein in their entirety by reference. In FIG. **17**, the wheel **1300'** (**1300'-1**, **1300'-2**) is shown disposed on a trench closing assembly **250D**, describe above.

Turning to FIGS. **118-124**, wheel **1300'** has a central axis and a circumferential tread **1320'** coaxial with the central axis. The circumferential tread **1320'** extends axially between an inner edge and an outer edge defining an axial width **W'** (FIG. **123**) of the wheel **1300'**. The tread **1320'** may include a plurality of ribs **1315'** (**1315'-1** to **1315'-10**). While the wheel **1300'** is shown with **10** ribs **1315'** it should be appreciated that any number of ribs may be used. In one embodiment, the ribs **1315'** extend axially in a direction substantially parallel with the central axis. Between each rib **1315'** there is a tread web **1316'** (**1316'-1** to **1316'-10**). The tread web **1316'** may extend through the entire axial width **W'** between each of the ribs **1315'**. Alternatively, the tread web **1316'** may extend only a portion of the axial width **W'** between the ribs **1315'**, such as illustrated in the embodiments of wheels **1300A** and **1300B**.

The wheel **1300'** includes a wall member **1330'** joined with the circumferential tread **1320'**. The wall member **1330'** extends radially inward from the circumferential tread **1320'**

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to a central opening 1333' coaxial with the central axis. The wall member 1330' is continuous (i.e., solid) between the central opening 1333' and the circumferential tread 1320' for the reasons discussed later. The central opening 1333' receives a hub 1301' by which the wheel 1300' may be rotatably mounted to the frame of the trench closing assembly 250. The wall member 1330' has a first side 1331' (outer side) and a second side 1332' (inner side). The wall member 1330' extends axially outward from the inner edge of the circumferential tread 1320' toward the outer edge of the circumferential tread 1320' by an axial distance D' (FIG. 123), but the axial distance D' does not extend outward beyond a vertical plane that is perpendicular to the central axis at a midpoint of the axial width W'. In one embodiment, the axial distance D' is equal to or less than half of the axial width W'. In another embodiment, the axial distance D' is less than one half of the axial width W'. In another embodiment, the axial distance D' is equal to or greater than 10% of the axial width W'. As seen in FIG. 123, when viewed in cross-section, the circumferential tread together with the wall 1330' has a cross-sectional shape in the form of an uppercase Greek letter sigma (X). In one embodiment, by having the axial distance D' not extend beyond the vertical plane perpendicular to the central axis at a midpoint of the axial width, the portion of the tread 1320' that extends axially outward beyond the first side 1331' of the wall member 1330' is able to flex radially inward towards the central axis of the wheel 1300'. When the hub 1301' is installed, the center of the wheel 1300' is solid so that nothing can pass through the wall member 1330' from the second side 1332' to the first side 1331' or vice versa. Thus, when in muddy conditions, the solid wall of the wheel 1300' prohibits a flow of mud from one side of the wheel to other. It should be appreciated that if the wall member 1330' is spoked or has openings through the wall 1330' from one side of the wheel to the other, mud may become entrapped, which may require cleaning of the wheel to remove the mud.

FIGS. 125-130 illustrate another embodiment of a wheel 1300" which may be used in place of a press wheel 255 (255A, 255C, 255D) or any of the wheels 1300, 1300A, 1300B, 1300' referenced above. Similar to wheel 1300', wheel 1300" has a central axis and a circumferential tread 1320" coaxial with the central axis. The circumferential tread 1320" extends axially between an inner edge and an outer edge defining an axial width W" (FIG. 130) of the wheel 1300". The tread 1320" may include a plurality of ribs 1315" (1315"-1 to 1315"-10). While the wheel 1300" is shown with 10 ribs 1315" it should be appreciated that any number of ribs may be used. In one embodiment, the ribs 1315" extend axially in a direction substantially parallel with the central axis. Between each rib 1315" there is a tread web 1316" (1316"-1 to 1316"-10). The tread web 1316" may extend through the entire axial width W" between each of the ribs 1315". Alternatively, the web tread 1316" may extend only a portion of the axial width W" between the ribs 1315", such as illustrated in the embodiments of wheels 1300A and 1300B.

The wheel 1300" includes a wall member 1330" joined with the circumferential tread 1320". The wall member 1330" extends radially inward from the circumferential tread 1320" to a central opening 1333" coaxial with the central axis. The wall member 1330" is continuous (i.e., solid) between the central opening 1333" and the circumferential tread 1320" for the reasons discussed later. Similar to the previously described wheel 1300', the central opening 1333" of the wheel 1300" is adapted to receive a hub (not shown) by which the wheel 1300" may be rotatably mounted to the

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frame of the trench closing assembly 250. The wall member 1330" has a first side 1331" (outer side) and a second side 1332" (inner side). The wall member 1330" extends axially outward from the inner edge of the circumferential tread 1320" toward the outer edge of the circumferential tread 1320" by an axial distance D" (FIG. 130), but the axial distance D does not extend axially outward beyond a vertical plane that is perpendicular to the central axis at a midpoint of the axial width W". In one embodiment, the axial distance D" is equal to or less than half of the axial width W". In another embodiment, the axial distance D" is less than one half of the axial width W". In another embodiment, the axial distance D" is equal to or greater than 10% of the axial width W". As seen in FIG. 130, when viewed in cross-section, the circumferential tread together with the wall 1330" has a cross-sectional shape in the form of an uppercase Greek letter sigma (X). In one embodiment, by having the axial distance D" not extend beyond the vertical plane perpendicular to the central axis at a midpoint of the axial width, the portion of the tread 1320" that extends axially outward beyond the first side 1331" of the wall member 1330" is able to flex radially inward towards the central axis of the wheel 1300". When the hub is installed, the center of the wheel 1330" is solid so that nothing can pass through the wall 1330" from the second side 1332" to the first side 1331" or vice versa. Thus, when in muddy conditions, the solid wall of the wheel 1300" prohibits a flow of mud from one side of the wheel to other. It should be appreciated that if the wall member 1330" is spoked or has openings through the wall 1330" from one side of the wheel to the other, mud may become entrapped, which may require cleaning of the wheel to remove the mud.

Pressure Control

To control the flow of fluid (e.g., air) to actuator 259, a control valve 258 can be included. As illustrated in FIG. 25 for an eight row system, there can be a control valve 258 (258-1 to 258-8) associated with each actuator 259 (259-1 to 259-8). A fluid supply line 257 can supply fluid (e.g., air) from a fluid supply system. While shown schematically, control valve 258 can be disposed on the row unit 200, trench closing assembly 250, or on toolbar 202. Different locations are shown in FIG. 25 for illustrating different locations, but they can all be the same. Section control can also be used. FIG. 26 illustrates section control in which a control valve 258 (258-9 to 258-12) supplies fluid to two or more actuators 259 (259-1 to 259-8). While shown with one control valve 258 to two actuators 259, any amount of sectioning can be used up to having one control valve 258 supplying all actuators 259 (not shown). Each control valve 258 can be in signal communication with monitor 300 for controlling each valve 258. While illustrated for air, which can be vented to atmosphere, the fluid can be hydraulic, which can further include a return line (not shown). An example of control valve 258 is the ITV series (such as ITV 1051) electro-pneumatic valve from SMC Pneumatics. This electro-pneumatic valve has a solenoid supply valve for inlet air and a solenoid valve for exhaust to atmosphere. When air is needed to the actuator, the inlet air valve is open and the valve to atmosphere is closed. When air pressure at the actuator needs to be reduced, the inlet air valve is closed, and the valve to atmosphere is opened.

While a single control valve 258, such as the ITV 1051 valve, can be used, an equivalent valve system 258A can be used that is made from component parts. Valve system 258A is illustrated in FIG. 49. Valve system 258A is supplied with fluid (e.g., air) from line 9901 to inlet valve 9905. Inlet valve 9905 discharges to line 9903, which is connected to actuator

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259, exhaust valve 9910, and pressure sensor 9906. Pressure sensor 9906 is in signal communication with a circuit 9907, which is in signal communication with monitor 300. Exhaust valve 9910 discharges to atmosphere, such as through optional line 9902.

As is described in International Publication No. WO2017/197274, control of the amount of force to actuator 259 can be based on input from one or more of the trench closing sensor, the angle sensor, a force sensor 261, which is disposed on trench closing assembly 250, or position sensor 900. The control can be closed loop or open loop. Force sensor 261 can be disposed on trench closing assembly 250 at any location that measures the force on any part of trench closing assembly 250. In one embodiment, force sensor 261 is disposed to measure a force applied to press wheels 255. An example of a location is at location 9261-A or location 9261-B, which is illustrated in FIG. 27. For location 9261-A, a load sensing pin can be used. For location 9261-B, a Wheatstone bridge can be used.

For packing/firming wheels in general, a force sensor 261 can be a load pin installed on the axle, on the arm connecting the packing/firming wheel to the trench closing assembly 250 or row unit 200, at the connection of the packing/firming wheel arm to trench closing assembly 250 or row unit 200, or where a spring/actuator connects to the packing/firming wheel frame. FIGS. 66-69 illustrate various locations for force sensors 261 on different press wheel systems 9255A, 9255B, 9255C, and 9255D. Each press wheel system 9255A, 9255B, 9255C, and 9255D has a press wheel 255D and mounting arms 9256-1 and 9256-2. Press wheel system 9255A, 9255B, 9255C and 9255D have a connection bracket 9259 for connection to row unit 200 or to closing system 250. A load sensor 9258 can be disposed between connection bracket 9259 and row unit 200 or closing system 250. A load sensing pin 9257 can be disposed on axle 9253 or as shown in FIG. 67 on axles 9253-1 or 9253-2. A force sensor 9254, such as a Wheatstone bridge, can be disposed on arm 9256 (9256-1 or 9256-2) or as shown in FIG. 67 on arm 9256 or shank 9260. In FIGS. 67 and 68, a force sensor 9262 can be disposed where spring 9250 is attached to frame 9251 or 9251A. In FIG. 66, a force sensor 9262 can be disposed at connection 9249.

Force sensors 261, 9258, 9257, 9254, 9262 are in signal communication with monitor 300.

In another embodiment, instead of a force sensor 261, a position sensor 900 can be used. Referring to FIG. 28, modified mounting arm 1256A has a first section 1256A-1 and a second section 1256A-2 connected with hinge 901 and with biasing member 902 (such as a spring) disposed between first section 1256A-1 and second section 1256A-2. Position sensor 900 includes a transmitter 904 and a receiver 903. It is understood that the positions of transmitter 904 and receiver 903 can be switched. An example of the transmitter 904 and receiver 903 are a magnet and Hall effect sensor. As downward forces are applied to modified mounting arm 1256A, biasing member 902 is compressed, and the distance between transmitter 904 and receiver 903 is reduced. Position sensor 900 is in signal communication with monitor 300.

Soil Leveler

FIGS. 70-74 illustrate a soil leveler 8000 according to one embodiment. As illustrated, soil leveler 8000 is attached to mounting arm 1256C of trench closing assembly 250C. Soil leveler has a first bracket 8020-1 and a second bracket 8020-2 for attaching to mounting arm 1256C. Pivotably connected to first bracket 8020-1 and second bracket 8020-2 through pivot 8030 are first arm 8010-1 and second arm

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8010-2. Optionally, to stabilize first arm 8010-1 and second arm 8010-2 is cross brace 8015 connecting first arm 8010-1 and second arm 8010-2. To adjust a relative angle between the first arm 8010-1 and second arm 8010-2 and first bracket 8020-1 and second bracket 8020-2, each arm 8010 (8010-1 and 8010-2) have a slot 8011 (8011-1 shown, and 8011-2 not shown). A fastener 8019 is disposed through slots 8011 and through first bracket 8020-1 and second bracket 8020-2. Each arm 8010 (8010-1 and 8010-2) have a space 8015 (8015-1 and 8015-2) having notches 8016 (8016-1 and 8016-2), respectively. Plate 8045 is engagable with notches 8016 (8016-1 and 8016-2). Attached to plate 8045 and disposed about fastener 8019 is biasing element 8040 (such as a spring). Moving plate 8045 to different notches 8016 (8016-1 and 8016-2) adjusts the amount of biasing force applied to arms 8010 (8010-1 and 8010-2). Attached to arms 8010 (8010-1 and 8010-2) is flap 8060. Flap 8060 can be a unitary part, or as illustrated, flap 8060 has plate 8061 attached to arms 8010 (8010-1 and 8010-2) and flap portion 8065 attached to plate 8061. Optionally, flap portion 8065 can have a serrated edge 8066 for engaging the soil. Serrated edge 8066 can be sloped upwards from edges 8062 (8062-1, 8062-2) to a center 8063 of flap portion 8065.

FIGS. 75-79 illustrate another soil leveler 8000A. As illustrated, soil leveler 8000A is attached to mounting arm 1256C of trench closing assembly 250C. Soil leveler 8000A has a first bracket 8120-1 and second bracket 8120-2 for attaching to mounting arm 1256C. Connected to first bracket 8120-1 and second bracket 8120-2 are first arm 8110-1 and second arm 8110-2. As shown, first arm 8110-1 and second arm 8110-2 are disposed between first bracket 8120-1 and second bracket 8120-2, but first arm 8110-1 and second arm 8110-2 are disposed outside of first bracket 8120-1 and second bracket 8120-2. Arms 8110 (8110-1 and 8110-2) have a first section 8112 (8112-1 and 8112-2) for connecting to first bracket 8120-1 and second bracket 8120-2. Extending laterally outward (transverse to the direction of travel) is a lateral segment 8111 (8111-1 and 8111-2). Disposed downwardly from lateral segment 8111 (8111-1 and 8111-2) is leg 8113 (8113-1 and 8113-2). Disposed in leg 8113 (8113-1 and 8113-2) is a hole 8114 (8114-1 and 8114-2). While not shown, a drag line extends from hole 8114-1 to hole 8114-2 and has a length to drag across the ground. Examples of the drag line include, but are not limited to, a chain, a wire, a cable, or a rope. Each bracket 8120 (8120-1 and 8120-2) have an adjustment slot 8121 (8121-1 and 8121-2) through which fastener 8122 is disposed, and fastener 8122 is disposed through first section 8112 (8112-1 and 8112-2) of arms 8110 (8110-1 and 8110-2). Fastener 8123 is also disposed through each bracket 8120 (8120-1 and 8120-2) and each first section 8112 (8112-1 and 8112-2) of arms 8110 (8110-1 and 8110-2). The relative angle between the brackets 8120 (8120-1 and 8120-2) and the arm 8110 (8110-1 and 8110-2) can be adjusted by pivoting about fastener 8123. To adjust a distance between holes 8114 (8114-1 and 8114-2) and brackets 8120 (8120-1 and 8120-2), first sections 8112 (8112-1 and 8112-2) can optionally have a plurality of holes 8115 for adjusting a position of arms 8110 (8110-1 and 8110-2).

FIGS. 80-104 illustrate a trench closing assembly 250D according to another embodiment. In this embodiment, trench closing assembly 250D has a main frame 251D that is connected to row unit 200. Trench closing assembly 250D has a frame 1520, an actuator 259, a pair of closing wheels 254D-1, 254D-2, and optionally, a press wheel 255D.

FIGS. 115 and 116 are enlarged perspective and side elevation views, respectively, of the closing wheels 254D-1,

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254D-2 of the trench closing assembly 250D. The closing wheels 254D-1, 254D-2 are mirror images of one other and therefore only one closing wheel is shown in FIGS. 115 and 116 generally designated by reference number 254D. In this embodiment, the closing wheels 254D comprise a generally dish-shaped body 510 in the form of a disc blade with a circumferential edge 512 and having a concave surface 514 on one side and a convex surface 516 on the opposing side. As best viewed in FIG. 80, the closing wheels 254D-1, 254D-2 are mounted so as to be disposed on each side of the open furrow with their convex sides 516 oriented inward toward the open furrow. The closing wheels 254D-1, 254D-2 are also oriented such that they angle outwardly upward (i.e., their respective circumferential edges 512 are closer to one another or they converge toward one another in the direction of the soil surface). Additionally, the closing wheels 254D-1, 254D-2 are mounted at an angle with respect to one another fore and aft (i.e., in the direction of travel) such that their respective circumferential edges 512 are closer to one another toward the rear than toward the forward direction of travel. Thus, due to their orientation and the convex shape toward the open furrow, as the row unit is drawn through the field causing the closing wheels to rotate through the soil, the closing wheels act to push the soil inward toward the open furrow, thus "closing" or filling the furrow with soil to cover the previously deposited seed.

The circumferential edge 512 of the dish-shaped body 510 may be continuous or the circumferential edge 512 may include a series of radially spaced notches 518 that are cut or otherwise formed around the outer circumferential periphery resulting in a series of radially spaced teeth or spikes 520. The notches 518 may be formed, so that each tooth or spike 520 curves or sweeps opposite the direction of rotation as indicated by arrow 511. It should be appreciated that this rearward swept orientation will reduce the amount of soil thrown by the teeth 520 because the teeth rotate out of the soil more vertically than if the teeth were straight. To improve penetration into the soil, the convex side 516 may be ground down, beveled or otherwise formed to taper toward the circumferential edge 512 as best viewed in FIG. 116, so the circumferential edge 512 is thinner or sharper for easier penetration into the soil.

The dish shaped body 510 may have a generally flat or planar central region 522 with bolt holes 524 and a central aperture 526 for mounting to a hub or spindle as shown in FIG. 80.

The closing wheel 524D may be fabricated by any suitable means. One exemplary manner of fabrication is to cut the body 510 out of flat plate steel having a generally uniform thickness to produce a wheel blank. The blank may then be placed in a forming die and pressed to the desired dish-shape. The dish-shaped blank may then undergo further processing, such as cutting the notches 518, if desired, to form teeth 520 around the outer circumferential periphery. The outer circumferential periphery may then be ground or tapered to produce a thinner or sharper outer circumferential edge. Alternatively the notches or teeth may be cut into the blank before being pressed to the desired dish shape.

As illustrated, the press wheel 255D may comprise a pair of press wheels 255D-1, 255D-2, but a single press wheel (not shown) may be utilized. Actuator 259 can apply one force between frame 251D and frame 1520, and this force can be divided between the closing wheels 254 and the press wheel 255D.

Turning to FIGS. 91-96, frame 251D has a connection bracket 1540 and an attachment bracket 1550. Connection bracket 1540 can have one or more bolts 1209D extending

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through apertures 1207D in connection bracket 1540 for mounting the frame 251D to the row unit 200. Connection bracket 1540 (as with connection brackets 1200 and 1201A) can be varied to mate attachment to different styles of row units. Connection bracket 1540 has a first side 1542-1, a second side 1542-2, and a plate 1543 disposed between first side 1542-1 and second side 1542-2. First side 1542-1 and second side 1542-2 each have a post 1541-1 and 1541-2, respectively, protruding perpendicularly outward.

Attachment bracket 1550 connects to connection bracket 1540. Attachment bracket 1550 provides a common structure for mounting other parts while connection bracket 1540 has a varied structure to mate with different styles of row units. Attachment bracket 1550 has a first side 1552-1, a second side 1552-2, a crossbar 1557 disposed between first side 1552-1 and second side 1552-2, and plate 1553 disposed between first side 1552-1 and second side 1552-2, and plate 1553. First side 1552-1 and second side 1552-2 each have a u-shaped opening 1555-1 and 1555-2, respectively, for connection to posts 1541-1 and 1541-2, respectively. First side 1552-1 and second side 1552-2 have openings 1558-1 and 1558-2, respectively, for accepting pivots 1529-1 and 1529-2, respectively. Attachment bracket 1550 can be secured to connection bracket with fastener 1549. Optionally, attachment bracket 1550 can also have an opening 1501 disposed in first side 1552-1 or second side 1552-2 for accepting a pin. While connection bracket 1540 is illustrated with separate parts it may be fabricated as a unitary part.

Optionally, as illustrated in FIGS. 82 and 84, a guard 1590 can be disposed on frame 1520 ahead of closing wheels 254D-1, 254D-2. Guard 1590 can prevent rocks, rootballs, or other trash from approaching closing wheels 254D-1, 254D-2. The height of guard 1590 can be adjusted by changing the placement of arms 1591-1 and 1591-2 on frame 1520.

Turning to FIGS. 97-98, mounting arm 1570 has a first side 1571-1, a second side 1571-2, a plate 1574 disposed between first side 1571-1 and second side 1571-2, a hole 1573 through plate 1574 for passage of sleeve 1603, axle hubs 1575-1 and 1575-2 for mounting press wheels 255D-1 and 255D-2, respectively, and, optionally, a handle cradle 1579.

Turning to FIGS. 83-90, frame 1520 is pivotally connected to frame 251D through pivots 1529-1 and 1529-2. Frame 1520 has a first side 1526-1 and a second side 1526-2, a plate 1525 connecting first side 1526-1 and second side 1526-2, a connection arm 1527 extending rearwardly along the direction of travel from plate 1525, a cross-brace 1528-1 connecting connection arm 1527 to first arm 1526-1, a cross-brace 1528-2 connecting connection arm 1527 to second arm 1526-2. First side 1526-1 and second side 1526-2 have openings 1521-1 and 1521-2 for disposing about pivots 1529-1 and 1529-2, respectively. Mounting arm 1570 can attach to frame 1520 at connections 1522-1a and 1522-2a or to 1522-1b and 1522-2b. The plurality of connections allows the distance between closing wheels 254D-1, 254D-2, and press wheel 255D (255D-1, 255D-2) to be changed. There can be one connection 1522 or a plurality of connections 1522. As the mounting arm 1570 position is changed, there are also corresponding connections 1523-a and 1523-b (matching in number to connection 1522) for connecting handle assembly 1600 to frame 1520. Frame 1520 also has connections 1524 (1524-1a, 1524-1b, 1524-2a, and 1524-2b) for mounting closing wheels 254D-1, 254D-2. While there can be one set of connections 1524, the plurality of connections 1524 allow for forward and back placement of the closing wheels 254D-1, 254D-2 on frame 1520, or

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closing wheels **254D-1**, **254D-2** can be offset from each other with one closing wheels **254D-1**, **254D-2** being mounted to a forward location (the “a” position) or to a rear location (the “b” position). As illustrated, closing wheels **254D-1**, **254D-2** are offset from each other. Optionally, frame **1520** can have an opening **1502** in either the first side **1526-1** or second side **1526-2** (shown in **1526-2**) for accepting a pin. Frame **1520** can be raised to allow opening **1501** and opening **1502** to align for accepting a pin (not shown). This allows trench closing assembly **250D** to be raised for transport or when closing is not needed. While illustrated on one side, openings **1501** and **1502** can be disposed on both sides.

Actuator **259** is disposed between plate **1525** and plate **1553** to apply a force to plate **1553** to cause frame **1520** to pivot and apply pressure to closing wheels **254D-1** and **254D-2**.

Turning to FIGS. **99-104**, handle assembly **1600** is illustrated. Handle assembly **1600** has a sleeve **1603** having a first diameter **1620** and a second diameter **1621**. Second diameter **1621** is small enough to be disposed through load sensor **1610**, and first diameter **1620** is large enough so that it cannot pass through load sensor **1610**. Sleeve **1603** has a bracket **1622** (u-shaped bracket as illustrated or any other shape) for accepting handle **1609**. Disposed on sleeve **1603** below bracket **1622** are bevel washers **1601** and **1602**. Bevel washer **1601** and bevel washer **1602** are disposed with their concave surfaces facing each other. This allows bevel washers **1601** and **1602** to flex to absorb shocks experienced by trench closing assembly **250D** to prevent overloading of load sensor **1610**. Before bevel washers **1601** and **1602** reach maximum flex, first diameter **1620** will contact plate **1604** to limit the travel. Load sensor **1610** is a pancake load sensor. Load sensor **1610** has a hole **1612** for passage of sleeve **1603**. Disposed on the underside of load sensor **1610** are a plurality of feet **1611** to allow load sensor **1610** to flex and measure force. Load sensor **1610** can be disposed on mounting arm **1570** directly, or as shown, a plate **1604** can be disposed between load sensor **1610** and mounting arm **1570**. Optionally, a washer **1607** can be disposed about sleeve **1603** under mounting arm **1570**. Handle assembly **1600** is connected to frame **1520** at connection **1523-a** or **1523-b** with a bracket **1606** (u-shaped bracket) and a bolt **1605** connecting bracket **1606** with sleeve **1603**. In the horizontal position, handle **1609** locks handle assembly **1600** in place against mounting arm **1570**. In the vertical position, handle **1609** releases handle assembly from engagement with mounting arm **1570**. Bolt **1605** can be adjusted to set a vertical placement of mounting arm **1570** relative to frame **1520**.

Load sensor **1610** can be connected to a network directly through a plug (not shown) having a CAN processor to allow direct communication over a CAN network. The CAN processor can communicate pressure readings and provide control signals over the CAN network. Alternatively, load sensor **1610** can be connected to a control module (either an on-row module, or a module controlling a plurality of row) to communicate pressure readings that are then processed by the control module.

FIGS. **105-107** illustrate another embodiment of a wheel **1300B**. Wheel **1300B** can be used as press wheel **255**, **255A**, **255C** or **255D**. Wheel **1300B** comprises a spoke disk **1310B** and a hub **1301B**. Spoke disk **1310B** can be molded as a unitary part. Spoke disk **1310B** has plurality of spokes **1302B** (**1302B-1** to **1302B-10**). Connecting spokes **1302B** is a tread **1320B**. Tread **1320B** has a rib **1315B** (**1315B-1** to **1315B-10**) disposed at the radial end of spoke **1302B**.

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Between each rib **1315B**, there is a tread portion **1316B** (**1316B-1** to **1316B-10**). Tread portion **1316B** can extend through the entire width A of a rib **1315B**, or tread portion **1316B** can extend only a portion of the width A to leave a gap **1317B** (**1317B-1** to **1317B-10**). Wheel **1300B** can be similar to wheel **1300A**. Tread portion **1316B** can be flexible to allow tread portion **1316B** to deflect inward towards hub **1301B**. To limit the amount of flex of tread portion **1316B**, stops **1318B** can be disposed radially outward from spoke disk **1310B** between spokes **1302B**. The height of stops **1318B** (as a percentage of the distance between spoke disk **1310B** and tread portion **1316B**) and the width of stops **1318B** (as a percentage of the distance between spokes **1302B**) can be varied to regulate the amount of flex permitted for tread portion **1316B** so that mud that is built up on tread portion **1316B** will come off of tread portion **1316B**.

FIGS. **108** to **114** illustrate a fluid control assembly **6800** according to one embodiment. Fluid control assembly **6800** controls the flow of fluid to and from actuator **259**. Fluid control assembly has a housing **6801**. There is a fluid inlet port **6802** from a fluid source (not shown). In one embodiment, the fluid can be air, but other fluids can be used. There is an outlet port **6803** on housing **6801** to return fluid to the fluid source, or in the case of air, outlet port **6803** can vent to atmosphere. Fluid control assembly **6800** has a conduit **6804** on housing **6801** to provide fluid communication to actuator **259**. Fluid control assembly **6800** has a port **6809** disposed through housing **6801** to allow signal communication with communication port **6890**.

Turning to FIGS. **109-111** with the housing **6801** removed, board **6891** can be seen having communication port **6890** disposed on board **6891**. Board **6891** contains circuitry (not shown) to control fluid control assembly **6800**. A first valve **6810** and a second valve **6820** are disposed on board **6891**. First valve **6810** and second valve **6820** can be identical. Examples of these valves are 2-way pneumatic valves from Asco Valve, Inc., 160 Park Avenue, Florham Park, NJ, 07932. These valves can operate normally closed.

First valve **6810** has an inlet **6811**, which is in fluid communication with inlet port **6802**, and an outlet **6812**, which is in fluid communication with conduit **6804**. Second valve **6820** has an inlet **6821**, which is in fluid communication with conduit **6804**, and an outlet **6822**, which is in fluid communication with outlet port **6803**. Seals **6889**, such as o-rings, can seal inlet **6811**, outlet **6812**, inlet **6821**, and outlet **6822**.

Also disposed in fluid control assembly **6800** is a pressure sensor **6830**, which is in data communication with board **6891**. While pressure sensor **6803** can be disposed anywhere in fluid control assembly **6800**, it is illustrated as being disposed on board **6891**. Pressure sensor **6830** is in fluid communication with conduit **6804** through conduit **6808**. An example of pressure sensor **6803** is a Honeywell board mounted pressure sensor.

Turning to FIGS. **112-114**, fluid communication of conduit **6804** is illustrated. Conduit **6804** is in fluid communication outlet **6802** through port **6816**. Conduit **6804** is in fluid communication with pressure sensor **6830** through conduit **6808**. Conduit **6804** is in fluid communication with inlet **6821** through port **6825**. Outlet **6822** is in fluid communication with outlet port **6803** through port **6826**. Inlet port **6802** is in fluid communication with inlet **6811** through port **6815**. Seals **6889** can be disposed in ports **6815**, **6816**, **6825**, and **6826**.

Communication port **6890** can be any port used in various types of signal/data communication. Examples include, but are not limited to, Computer Area Network (CAN) port,

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USB, Ethernet, or RS-232. All processing of signals and control can be done on board **6891**, or signals can be sent to monitor **300** or a remote controller (not shown) for processing and control to return a signal to fluid control assembly **6800** to control valves **6810** and **6820**. Closed loop control can be used to control the pressure in actuator **259** to a selected value that is set by an operator. The selected value can be a selected amount of pressure for actuator **259**, or the selected value can be a selected position for trench closing assembly **250**, **250A**, **250B**, **250C** or **250D**, such as from position sensor **900**. As the hardness of soil changes, more or less pressure is needed to maintain the same amount of closing. In harder soils, an increase in pressure may be needed to obtain the same amount of closing, and in softer soils, a decrease in pressure may be used.

In another embodiment, pressure to actuator **259** can be controlled based on an amount of closing as measured by a trench closing sensor. Trench closing sensors are described in International Patent Publication No. WO2017/197274, or in U.S. Application Nos. 62/586,397; 62/640,279; and 62/672,993, all of which are incorporated herein by reference in their entireties.

In operation, when valves **6810** and **6820** are closed, pressure in actuator **259** can be measured by pressure sensor **6830**. If additional pressure is needed in actuator **259**, valve **6810** can be opened (with valve **6820** closed) to place inlet port **6802** in fluid communication to conduit **6804** to actuator **259**. If too much pressure is in actuator **259**, valve **6820** can be opened (with valve **6810** closed) to place outlet port **6803** in fluid communication to conduit **6804** to actuator **259**.

Fluid control assembly **6800** can be disposed anywhere on trench closing assembly **250**, **250A**, **250B**, **250C**, **250D** or on row unit **200**.

In another embodiment, a feed forward control method is provided to adjust the downforce control system **214** based on a force applied to the actuator **259**. If a change in force applied by the actuator **259** is applied, the same amount of change or a portion thereof can also be made to the downforce control system **214**. This can be done to balance the force on the row unit **200**. For example, if an increase of 50 units of force is to be applied by the actuator **259**, then a signal can be sent to also increase the force applied by the downforce control system **214** by an additional 50 units of force. In other embodiments, the change of force to the downforce control system **214** can be the opposite to the change in force applied by the actuator **259**. For example, if an increase of 50 units of force is to be applied by the actuator **259**, then a signal can be sent to decrease the force applied by the downforce control system **214** by 50 units of force. In either embodiment, the amount of change to the downforce control system **214** can be less than, equal to, or greater than the change in force applied by the actuator **259**.

In another embodiment, there can be just one sensor **10000** per row to control both the actuator **259** and the downforce control system **214**. Sensor **10000** is not separately illustrated, but it is a reference to any of the following sensors: downforce sensor **238**, force sensor **261**, force sensor **1298**, force sensor **9258**, force sensor **9257**, force sensor **9254**, force sensor **9262**, position sensor **900**, angle sensor **3006**, trench closing sensor **1000**, or load sensor **1610**. Because of the close proximity of the closing system **250** (or its alternatives) to the opening system **220**, the soil is going to be approximately the same in terms of one or more of hardness, moisture, texture, etc. Setting the force for one system (i.e., the closing system **250** or the opening system **220**) will dictate a proportional amount of force

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needed for the other system (i.e., the opening system **220** or the closing system **250**). For example, if the downforce sensor **238** measures a change in force indicating that the soil is harder and sends a signal to the downforce control system **214** to increase the downforce to the row unit **200**, the same signal can be sent to the actuator **259** to also increase the downforce applied by the actuator **259**. The amount of force can be the same or a portion thereof. The force applied to the downforce control system **214** and the actuator **259** can be proportional to each other. By way of example, the absolute amount of force needed to open a 2 inch deep trench in the soil is greater than the amount of force needed to open a 1 inch deep trench in the soil, but the amount of force required to close the 2 inch deep trench as compared to the 1 inch deep trench may not be the same percentage change as the force needed to open the respective trenches. The system can thus adjust one system relative to the other system for a given depth.

In another embodiment, instead of or in conjunction with any of the various sensors (i.e., force sensor **261**, force sensor **1298**, force sensor **9258**, force sensor **9257**, force sensor **9254**, force sensor **9262**, position sensor **900**, angle sensor **3006**, trench closing sensor **1000** or load sensor **1610**), the force applied by the actuator **259** can be obtained from a downforce prescription map. An example of a downforce prescription map is described in U.S. Patent Application Publication No. 2012/0186503, which is incorporated herein in its entirety by reference. The amount of downforce applied can be prescribed based on one or more of hardness, texture, moisture, organic matter, or any other soil data layer that impacts the force needed to open or close a seed trench.

The following are non-limiting Examples.

Example 1: A wheel comprising: a central axis; a circumferential tread coaxial with the central axis, the circumferential tread extending axially between an inner edge and an outer edge defining an axial width of the wheel; a wall member joined with the circumferential tread, the wall member being continuous between the circumferential tread and a central opening, the wall member extending axially outward from the inner edge of the circumferential tread toward an outer edge of the circumferential tread by an axial distance, the axial distance not extending beyond a vertical plane that is perpendicular to the central axis at a midpoint of the axial width, the circumferential tread and the wall member together defining a shape in the form of an upper-case Greek letter sigma.

Example 2: The wheel of claim Example 1, wherein the axial distance is equal to or less than one half of the axial width.

Example 3: The wheel of any of any preceding Example, wherein the axial distance is between 10% and one half of the axial width.

Example 4: The wheel of any preceding Example, further comprising a plurality of ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.

Example 5: The wheel of Example 2, further comprising a plurality of ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.

Example 6: The wheel of Example 3, further comprising a plurality of ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.

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Example 7: The wheel of any preceding Example, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward.

Example 8: The wheel of Example 4, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward.

Example 9: The wheel of Example 5, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward.

Example 10: The wheel of Example 6, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward.

Example 11: A trench closing system for a row unit of an agricultural planter, the row unit having a row unit frame supporting an opener disk for opening a seed trench in a soil surface as the row unit travels in a forward direction of travel, the trench closing system comprising: a main frame supported by and extending rearwardly from the row unit frame; a frame member pivotally supported from said main frame; a pair of closing wheels each rotatably supported by said frame member with a first one of the pair of closing wheels disposed on a first side of the seed trench and a second one of the pair of closing wheels disposed on a second side of the seed trench, said pair of closing wheels cooperating with one another to close the opened seed trench with soil as the row unit travels in the forward direction of travel; an actuator supported between said main frame and said frame member such that said actuator is adapted to apply a down force on said frame member; a mounting arm pivotally mounted to said frame member, said mounting arm supporting a press wheel rearward of said closing wheels, wherein the press wheel is the wheel of any of Examples 1 to 10.

Various embodiments of the invention have been described above for purposes of illustrating the details thereof and to enable one of ordinary skill in the art to make and use the invention. The details and features of the disclosed embodiments are not intended to be limiting, as many variations and modifications will be readily apparent to those of skill in the art.

The invention claimed is:

1. A unitary molded wheel adapted to be rotatably mounted to a mounting arm for packing soil over a seed trench, the unitary molded wheel comprising:
 - a central axis;
 - a circumferential tread coaxial with the central axis, the circumferential tread defining an axial width of the wheel;
 - a wall member unitarily molded with the circumferential tread, the wall member extending radially inward from the circumferential tread to a central opening coaxial with the central axis, the central opening adapted to receive a hub by which the unitary molded wheel is rotatably mounted to the mounting arm, wherein an outer side edge of the unitary molded wheel is oriented away from the mounting arm and an inner side edge of the unitary molded wheel is oriented toward the mounting arm, the wall member being continuous between the circumferential tread and the central opening, the wall member extending axially outward from the inner side edge of the unitary molded wheel toward the outer side edge of the unitary molded wheel by an axial distance, the axial distance not extending beyond a vertical plane that is perpendicular to the central axis at a midpoint of the axial width;

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whereby, with the unitary molded wheel mounted to the hub, the continuous wall member prevents soil passing through the unitary molded wheel as the unitary molded wheel rotates over the seed trench packing the soil.

2. The unitary molded wheel of claim 1, wherein the axial distance is equal to or less than one half of the axial width.

3. The unitary molded wheel of claim 1, wherein the axial distance is between 10% and one half of the axial width.

4. The unitary molded wheel of claim 1, further comprising a plurality of ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.

5. The unitary molded wheel of claim 2, further comprising a plurality of ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.

6. The unitary molded wheel of claim 3, further comprising a plurality of ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.

7. The unitary molded wheel of claim 1, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward toward the central axis.

8. The unitary molded wheel of claim 4, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward toward the central axis.

9. The unitary molded wheel of claim 5, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward toward the central axis.

10. The unitary molded wheel of claim 6, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward toward the central axis.

11. A trench closing system for a row unit of an agricultural planter, the row unit having a row unit frame supporting an opener disk for opening a seed trench in a soil surface as the row unit travels in a forward direction of travel, the trench closing system comprising:

a main frame supported by and extending rearwardly from the row unit frame relative to the forward direction of travel;

a frame member pivotally supported from the main frame; a first closing wheel rotatably supported by the frame member on a first side of the opened seed trench;

a second closing wheel rotatably supported by the frame member on a second side of the opened seed trench, the first and second closing wheels cooperating with one another to close the opened seed trench with soil as the row unit travels in the forward direction of travel;

an actuator supported between the main frame and the frame member such that the actuator is adapted to apply a down force on the frame member;

a mounting arm pivotally mounted to the frame member, the mounting arm supporting a unitary molded press wheel rearward of the first and second closing wheels, the unitary molded press wheel comprising:

a central axis;

a circumferential tread coaxial with the central axis, the circumferential tread defining an axial width of the press wheel;

a wall member unitarily molded with the circumferential tread, the wall member extending radially inward from the circumferential tread to a central opening

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coaxial with the central axis, the central opening adapted to receive a hub by which the unitary molded press wheel is rotatably mounted to the mounting arm, wherein an outer side edge of the unitary molded press wheel is oriented away from the mounting arm and an inner side edge of the unitary molded press wheel is oriented toward the mounting arm, the wall member being continuous between the circumferential tread and the central opening, the wall member extending axially outward from the inner side edge of the unitary molded press wheel toward the outer side edge of the unitary molded press wheel by an axial distance, the axial distance not extending beyond a vertical plane that is perpendicular to the central axis at a midpoint of the axial width;

whereby, with the unitary molded press wheel mounted to the hub, the continuous wall member prevents soil passing through the unitary molded press wheel as the unitary molded press wheel rotates over the seed trench packing the soil.

12. The trench closing system of claim 11, wherein the axial distance is equal to or less than one half of the axial width.

13. The trench closing system of claim 11, wherein the axial distance is between 10% and one half of the axial width.

14. The trench closing system of claim 11, wherein the unitary molded press wheel further includes a plurality of

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ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.

15. The trench closing system of claim 12, wherein the unitary molded press wheel further includes a plurality of ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.

16. The trench closing system of claim 13, wherein the unitary molded press wheel further includes a plurality of ribs disposed about the circumferential tread, the ribs extending axially in a direction parallel with the central axis.

17. The trench closing system of claim 11, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward toward the central axis.

18. The trench closing system of claim 14, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward toward the central axis.

19. The trench closing system of claim 15, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward toward the central axis.

20. The trench closing system of claim 16, wherein a portion of the circumferential tread extending outward beyond the axial distance is capable of flexing radially inward toward the central axis.

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